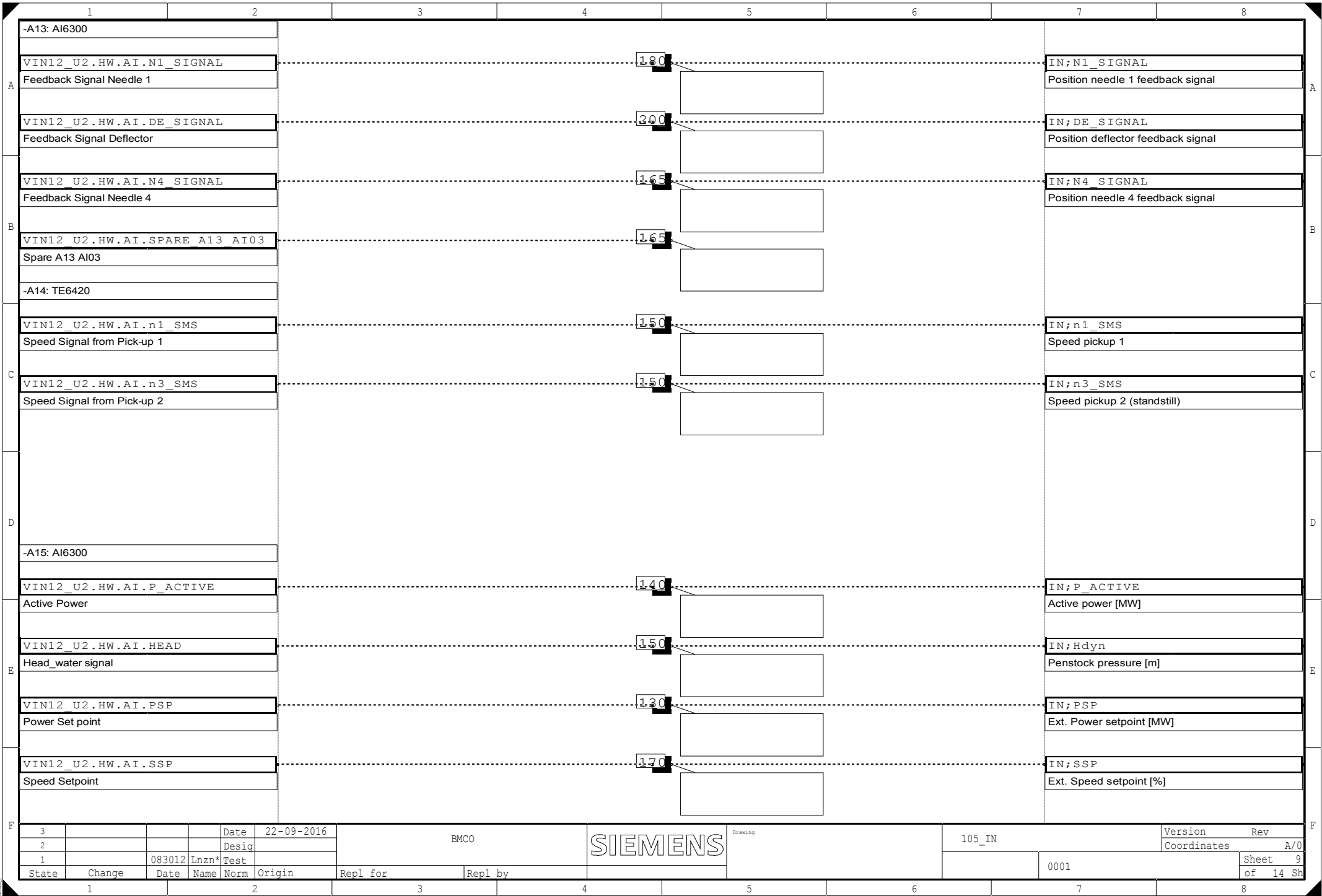
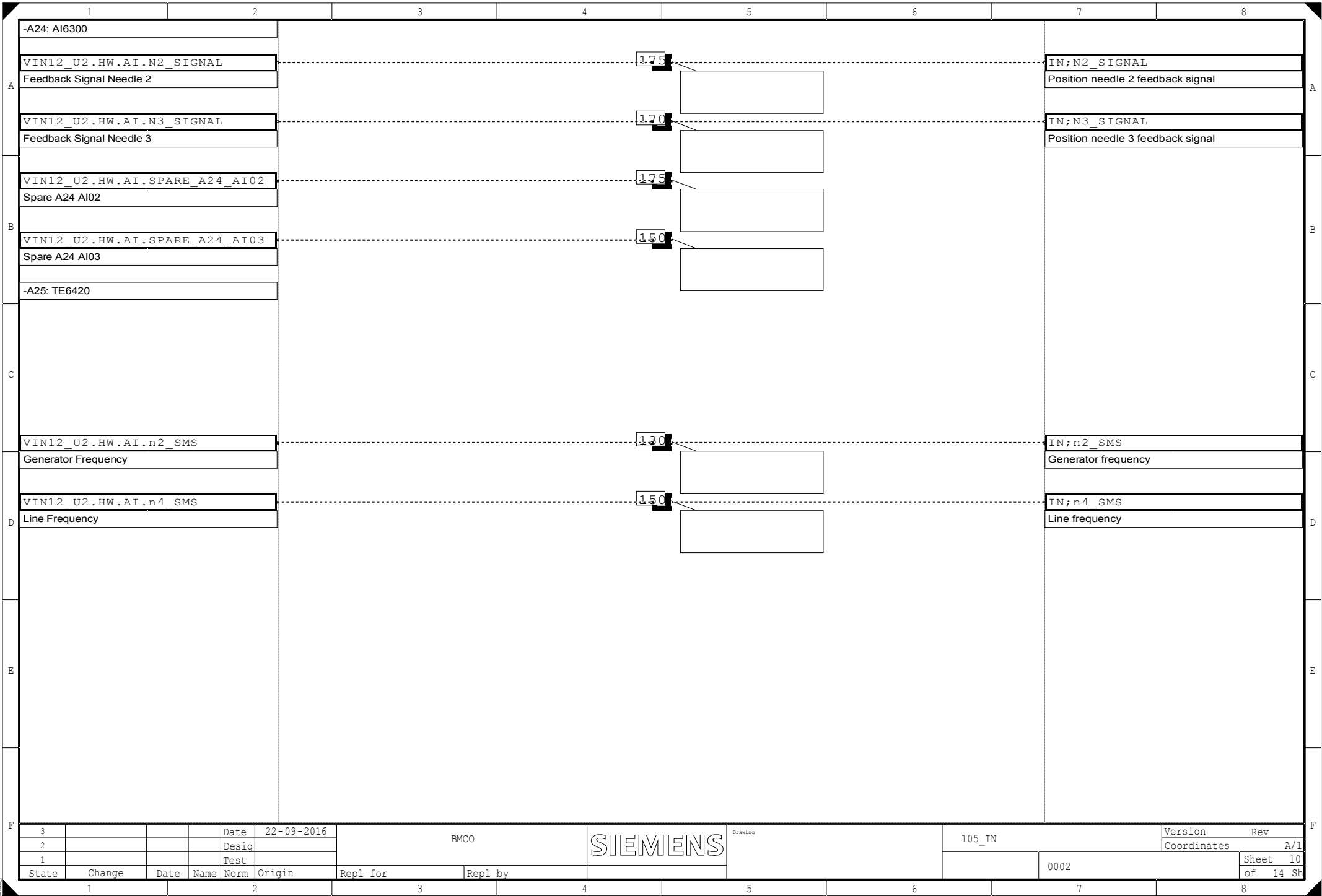
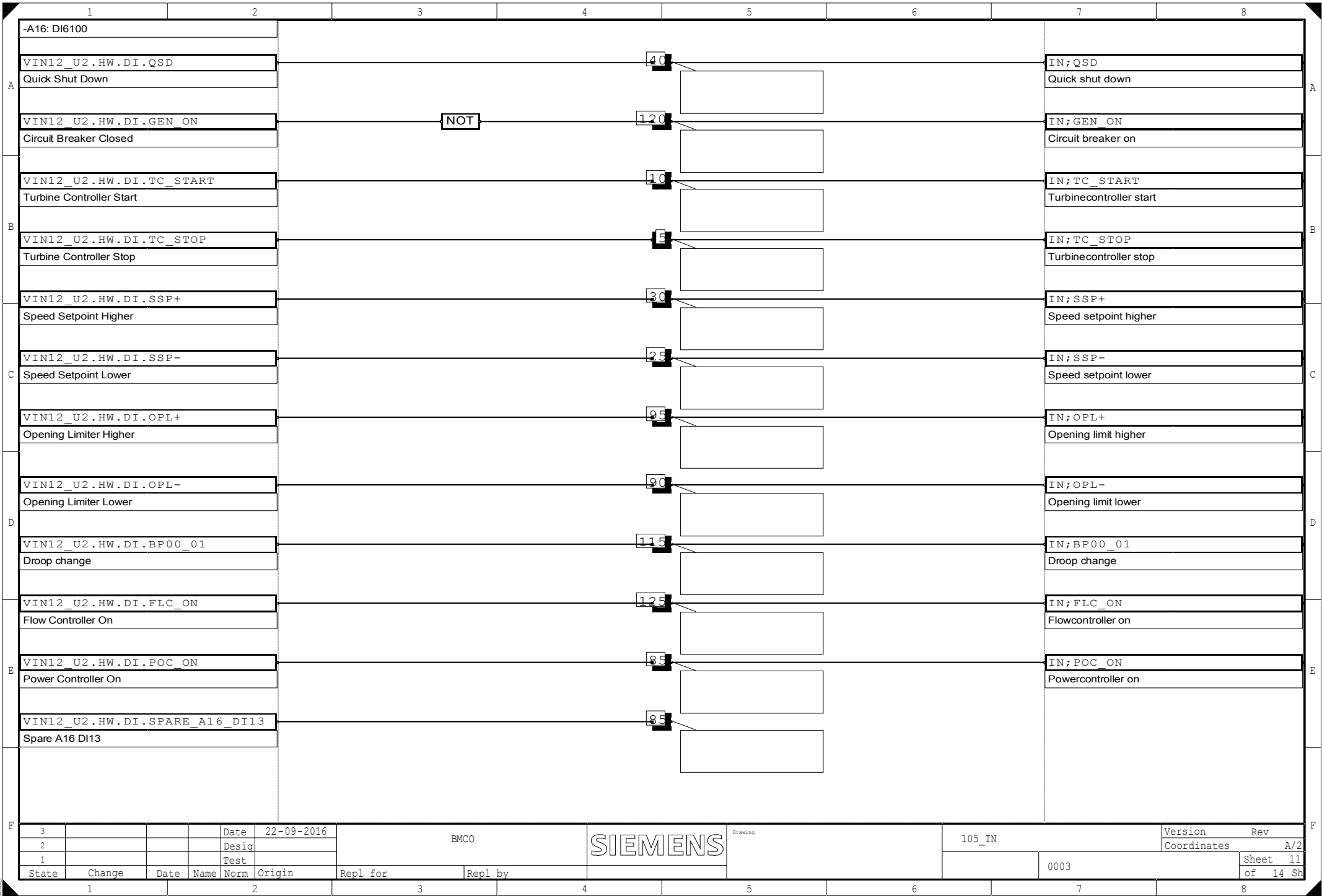




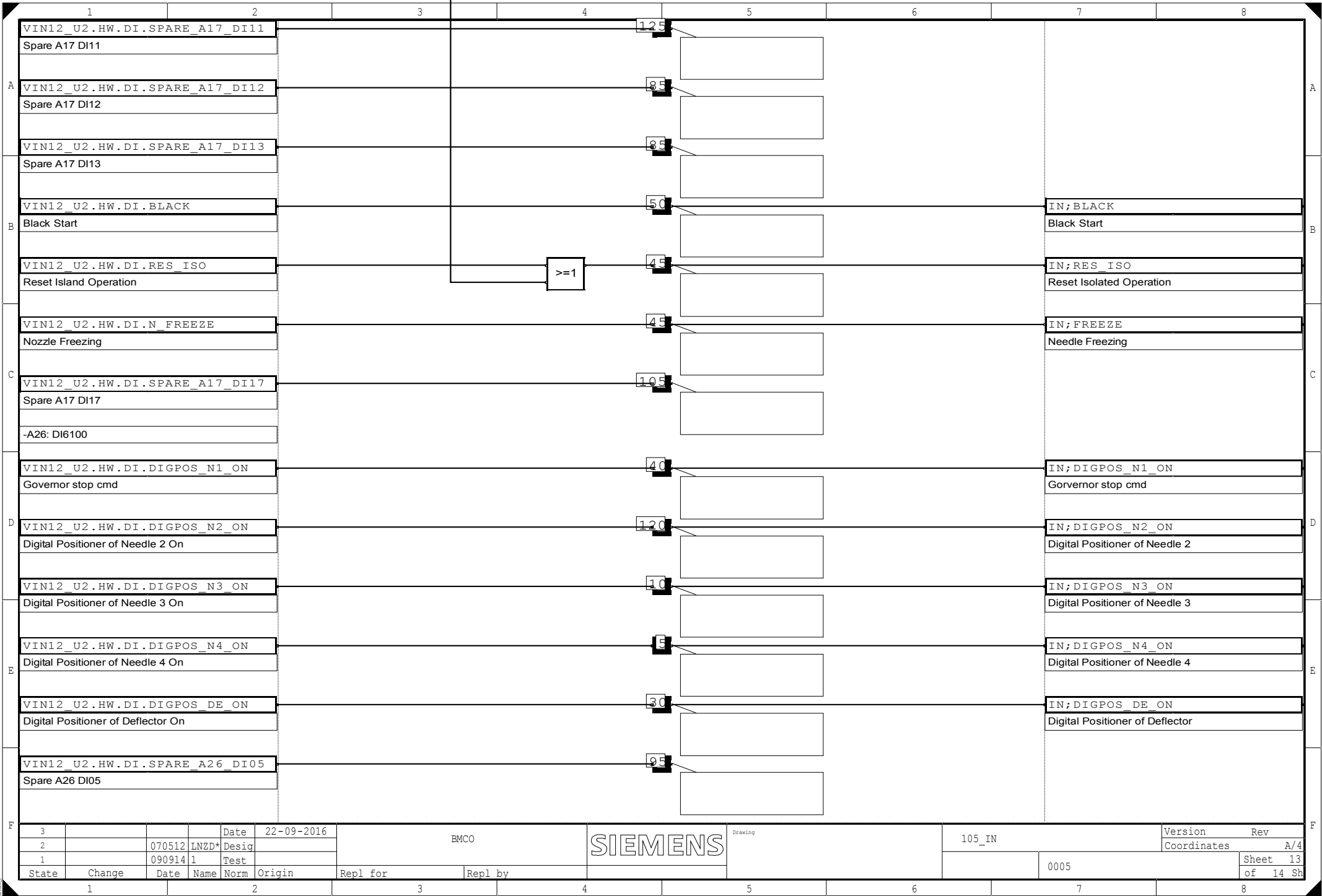


3				Date	22-09-2016	BMCO		SIEMENS		Drawing		105_IN		Version	Rev
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1				Test											
State	Change	Date	Name	Norm	Origin	Repl for	Repl by							0002	Sheet 10 of 14 Sh



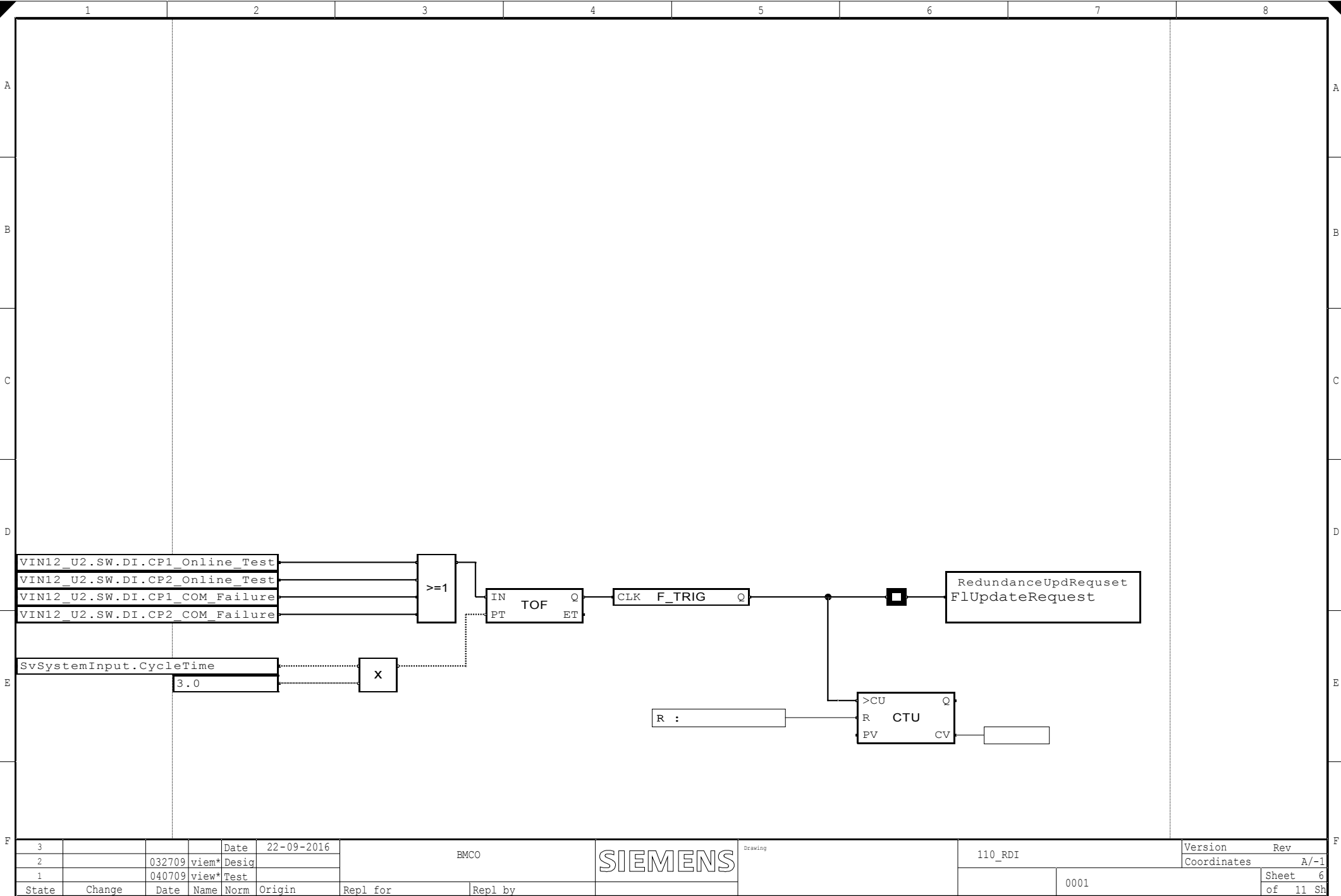
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1				Test									
State	Change	Date	Name	Norm	Origin	Repl for	Repl by			0003	Sheet 11	of 14 Sh	

1			2			3			4			5			6			7			8		
A	VIN12_U2.HW.DI.PSP/FSP+										50								IN;PSP/FSP+				
	Setpoint Higher																		Power/flow setpoint higher				
A	VIN12_U2.HW.DI.PSP/FSP-										45								IN;PSP/FSP-				
	Setpoint Lower																		Power/flow setpoint lower				
B	VIN12_U2.HW.DI.SP_SET										45								IN;SP_SET				
	Set Setpoint																		Set Setpoint (enable Setpoint)				
B	VIN12_U2.HW.DI.LOCAL										105								IN;LOCAL				
	Turbine Controller Local Mode																		Operation from local panel				
	-A17: DI6100																						
C	VIN12_U2.HW.DI.SY_MAN										40								IN;SY_MAN				
	Manual Synchronisation On																		Manual Synchronisation on				
C	VIN12_U2.HW.DI.PC_ON										120								IN;PC_ON				
	Primary Control On																		Primary Control on				
D	VIN12_U2.HW.DI.OFF_LINE										10								IN;OFF_LINE				
	OFF_LINE																		Curcuit breaker OPEN				
D	VIN12_U2.HW.DI.SNL										5								IN;SNL				
	Speed No Load (Deload to 0MW)																		Speed No Load (Deload to 0MW)				
E	VIN12_U2.HW.DI.LINE_CH										30								IN;LINE_CH				
	Line charging																		Line charging				
E	VIN12_U2.HW.DI.SPARE_A17_DI05										25												
	Spare A17 DI05																						
F	VIN12_U2.HW.DI.RESET										95								IN;RESET				
	Reset after Turbine Governor Trip																		Reset several FLIP FLOP				
F	VIN12_U2.HW.DI.N_DISCHARG										90								IN;N_DISCHARG				
	Depressurize Ring Pipe																		Dischg distributionpipe				
F	VIN12_U2.HW.DI.PRE_LTU										115								IN;PRE_LTU				
	Preparation for Load Take Up																		Preparation for Load Take Up				
3			Date			22-09-2016			BMCO			Drawing			105_IN			Version			Rev		
2			070512			LNZD*Desig												Coordinates			A/3		
1			091312			AdminTest															Sheet 12		
State			Change			Date			Name			Norm			Origin			Repl for			Repl by		
1						2						3									0004		
																					Sheet 12		
																					of 14 Sh		
1			2			3			4			5			6			7			8		

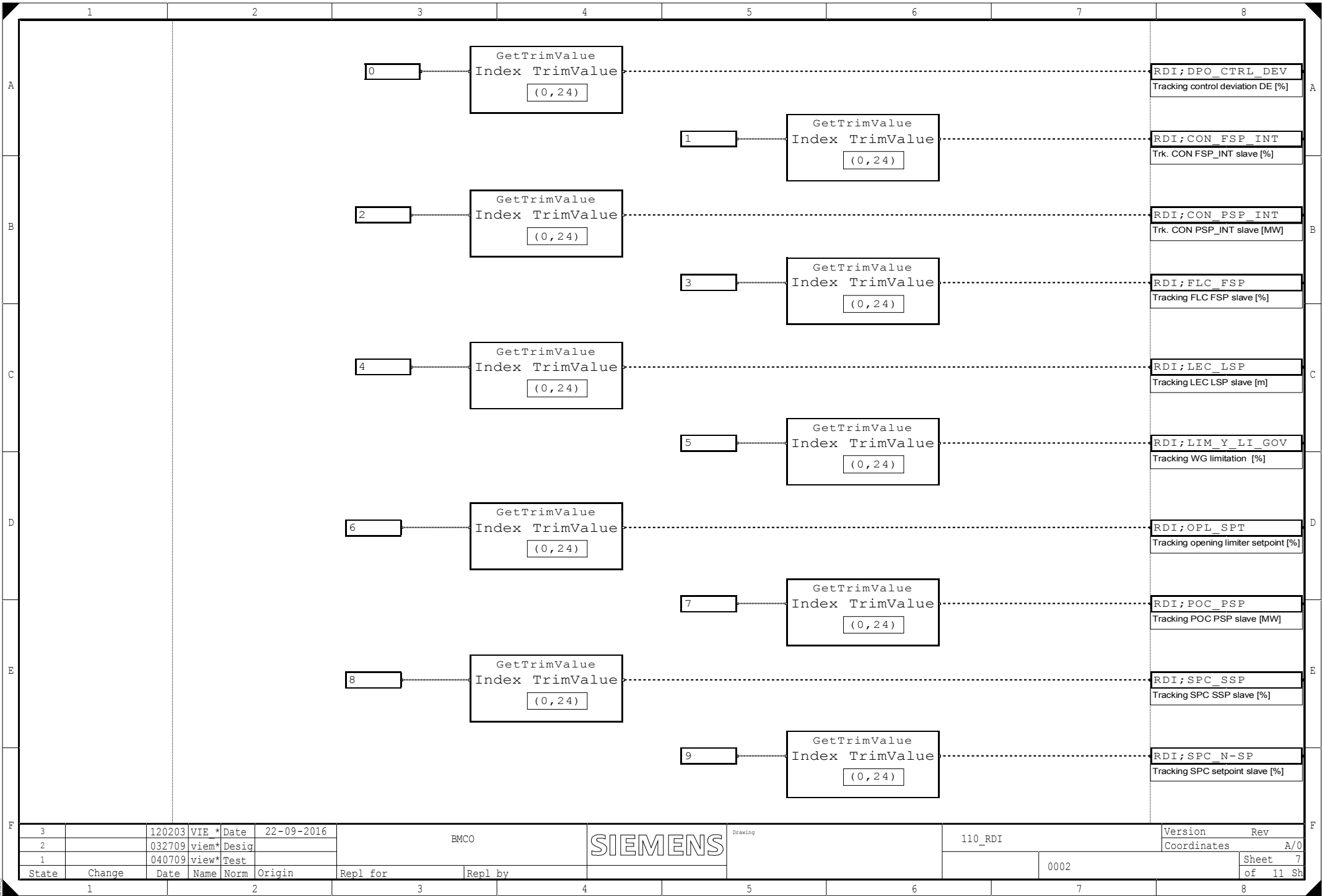


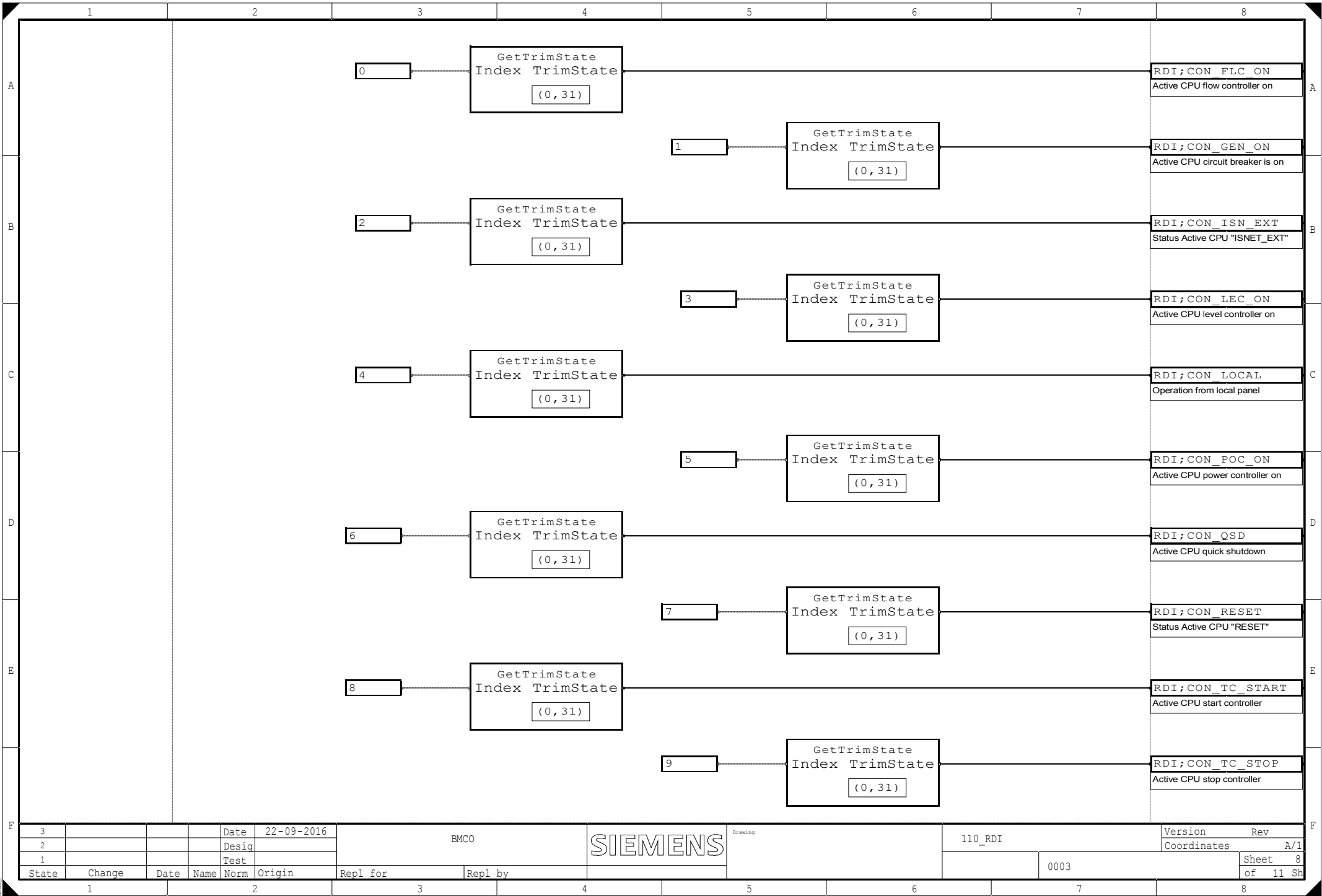
3			Date	22-09-2016	EMCO		SIEMENS	Drawing	105_IN		Version		Rev
2		070512	LNZD*	Desig							Coordinates		A/4
1		090914	1	Test							Sheet 13		
State	Change	Date	Name	Norm	Origin	Repl for	Repl by			0005			of 14 Sh

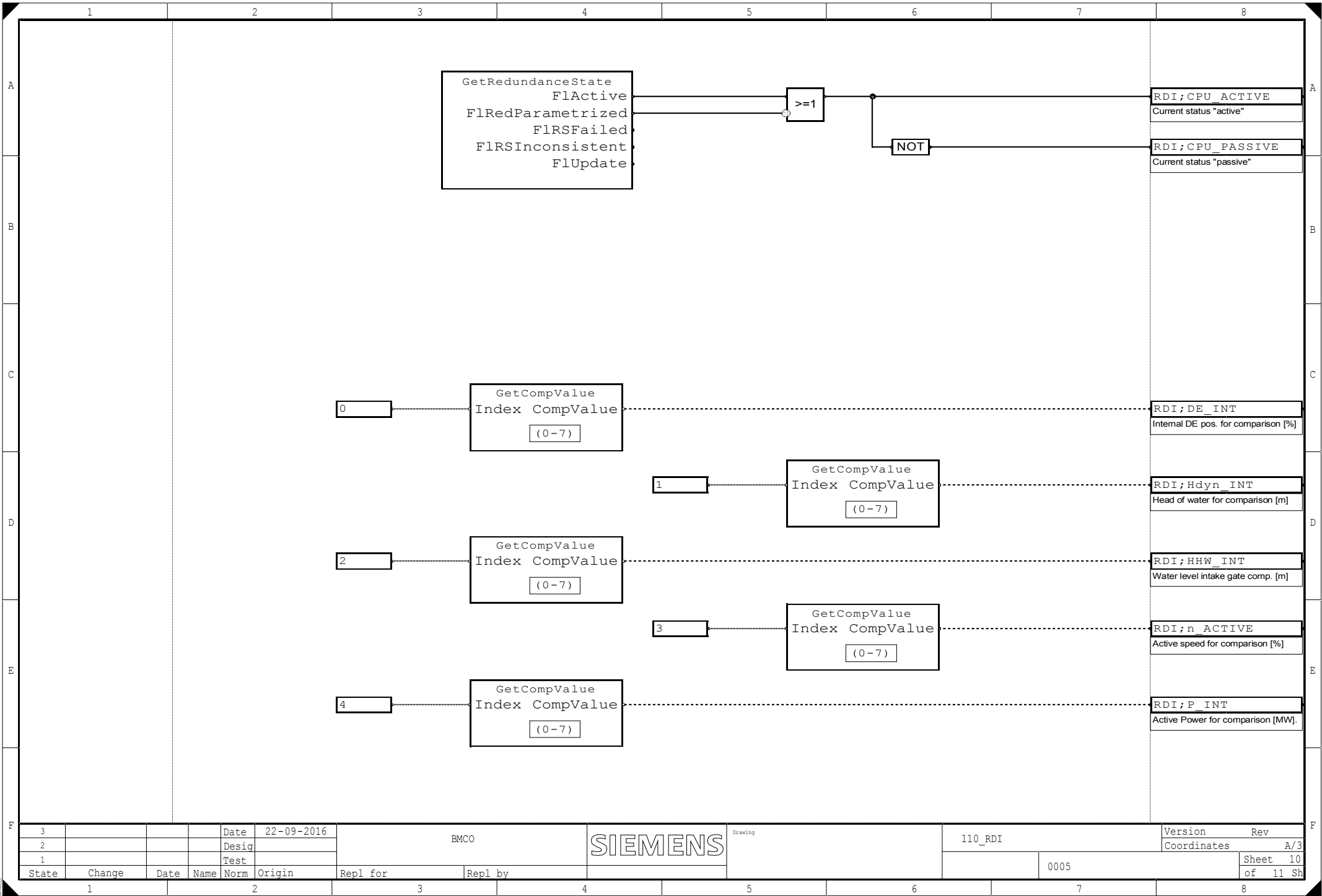
	1	2	3	4	5	6	7	8					
A	VIN12_U2.HW.DI.SPARE_A26_DI06			90									
	Spare A26 DI06												
	VIN12_U2.HW.DI.SPARE_A26_DI07			115									
	Spare A26 DI07												
	VIN12_U2.HW.DI.SPARE_A26_DI10			125									
	Spare A26 DI10												
B	VIN12_U2.HW.DI.SPARE_A26_DI11			85									
	Spare A26 DI11												
	VIN12_U2.HW.DI.SPARE_A26_DI12			85									
	Spare A26 DI12												
C	VIN12_U2.HW.DI.F_LCC			50			IN;F_LCC						
	Load Circuit Controls Fault						Load Circuit Controls Fault						
	VIN12_U2.HW.DI.SPARE_A26_DI14			45									
	Spare A26 DI14												
	VIN12_U2.HW.DI.F_MCB			45			IN;F_FUSE						
	MCB Tripped						MCB Tripped						
D	VIN12_U2.HW.DI.F_PS1			45			IN;F_PS1						
	Power Supply 1 Fault						Power Supply 1 Fault						
	VIN12_U2.HW.DI.F_PS2			105			IN;F_PS2						
	Power Supply 2 Fault						Power Supply 2 Fault						
E													
F													
	3			Date	22-09-2016	BMC0		SIEMENS	Drawing	105_IN		Version	Rev
	2			Desig								Coordinates	A/5
	1		070512	LNZD*	Test							Sheet 14	
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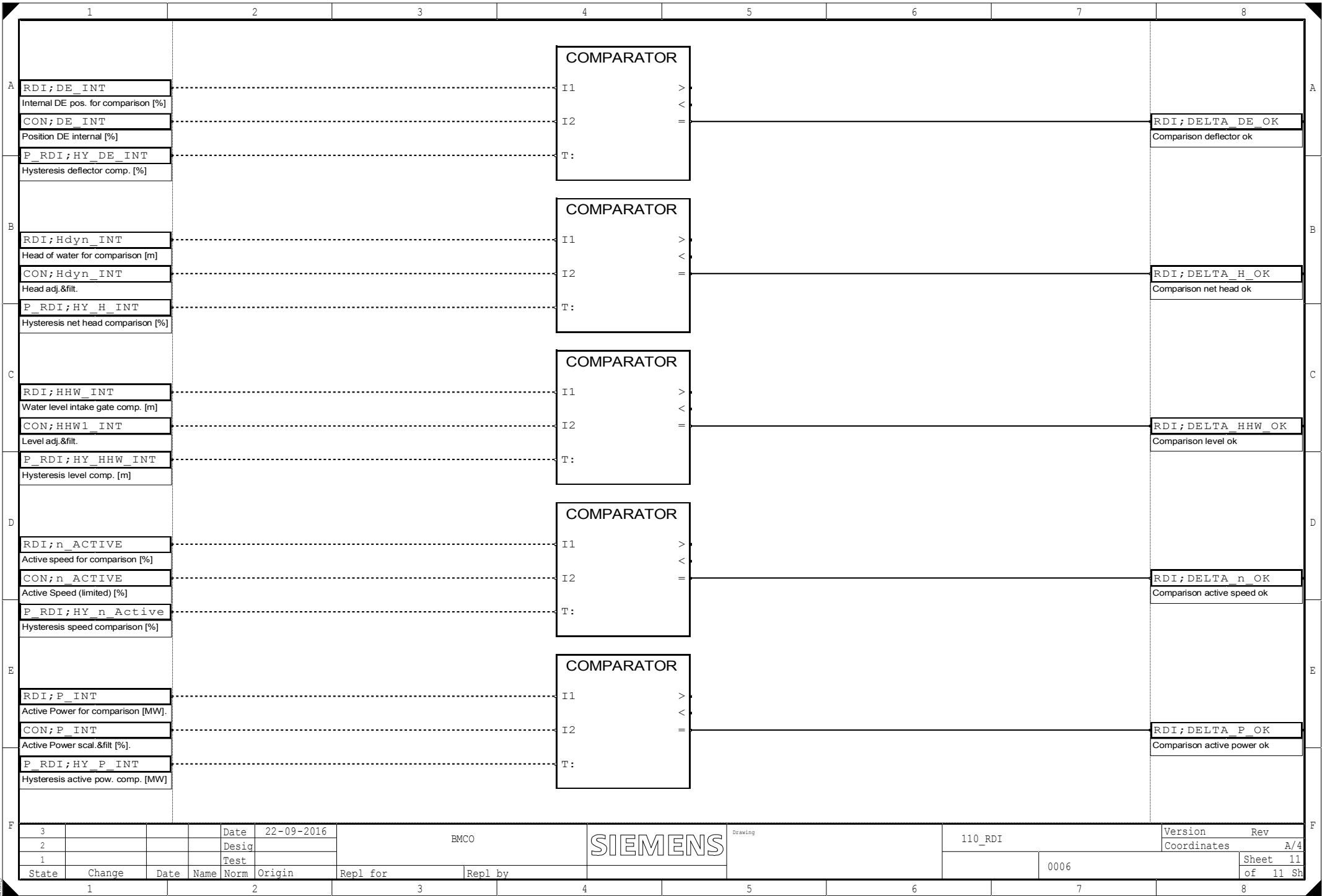
3			Date	22-09-2016	EMCO		SIEMENS	Drawing	110_RDI		Version		Rev
2		032709	viem*	Desig							Coordinates		A/-1
1		040709	view*	Test									Sheet
State	Change	Date	Name	Norm	Origin	Repl for	Repl by			0001	of 11 Sh		

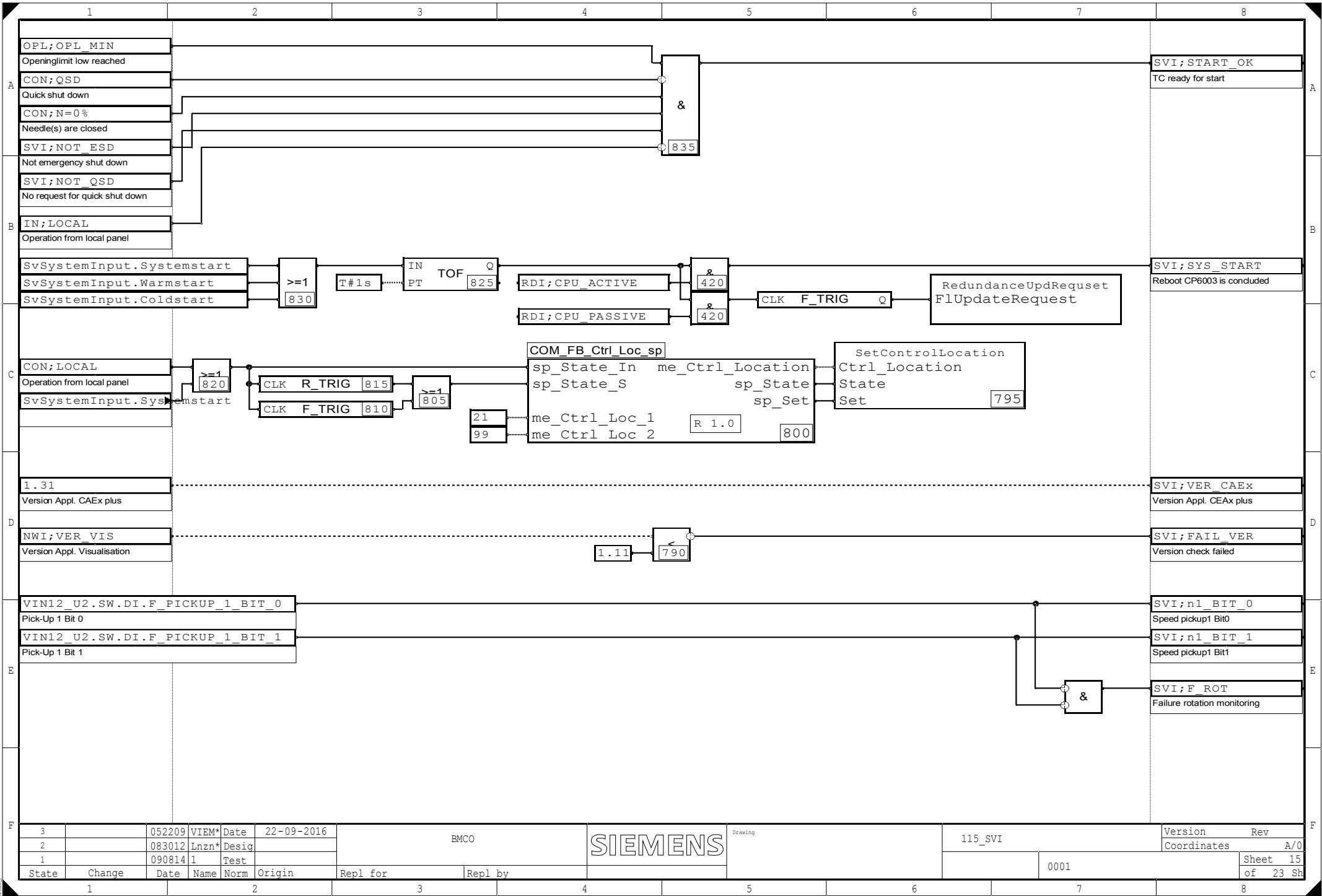


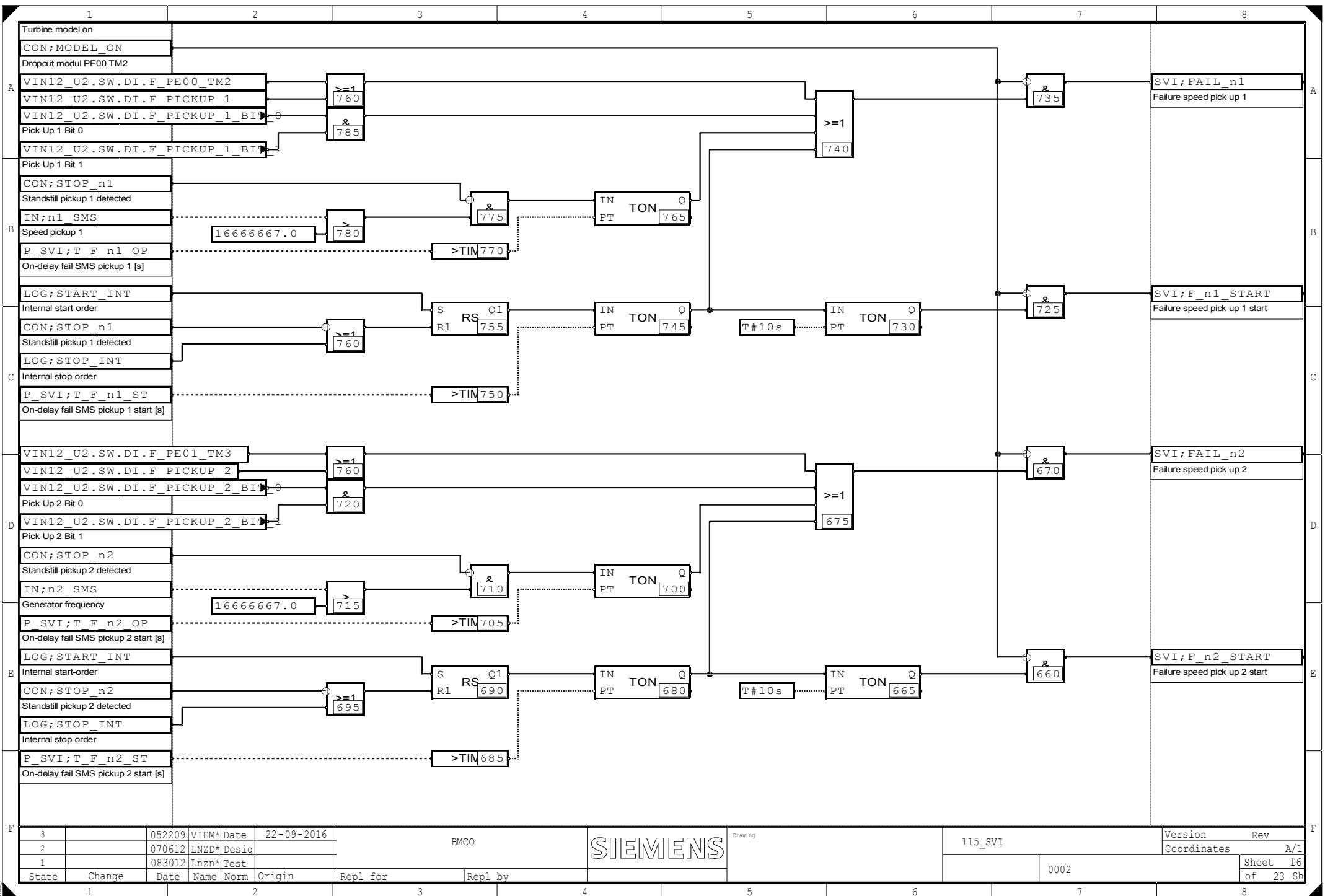


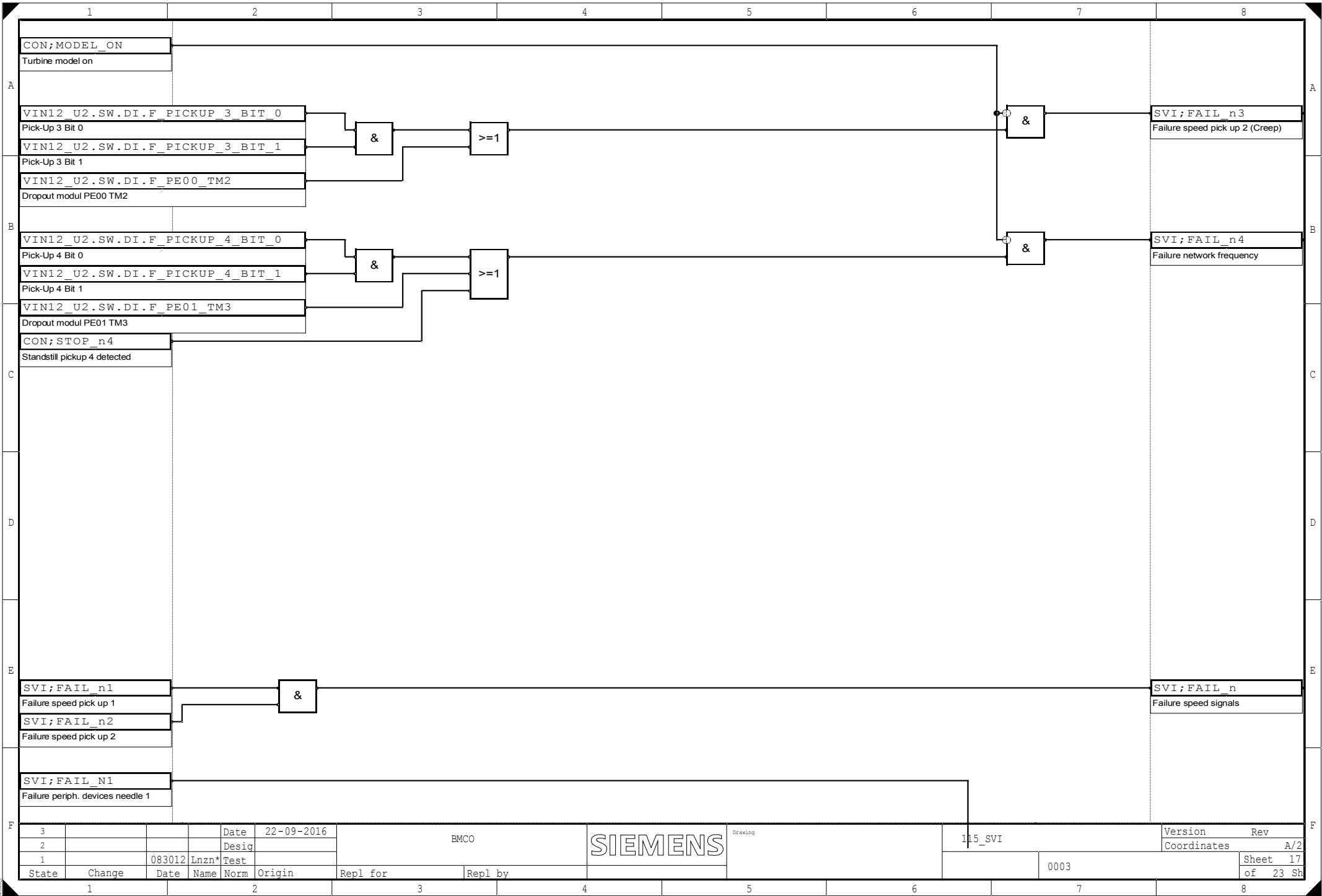


3				Date	22-09-2016	BMCO		SIEMENS	Drawing	110_RDI		Version	Rev
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1				Test									Sheet 10
State	Change	Date	Name	Norm	Origin	Repl for	Repl by				0005		of 11 Sh

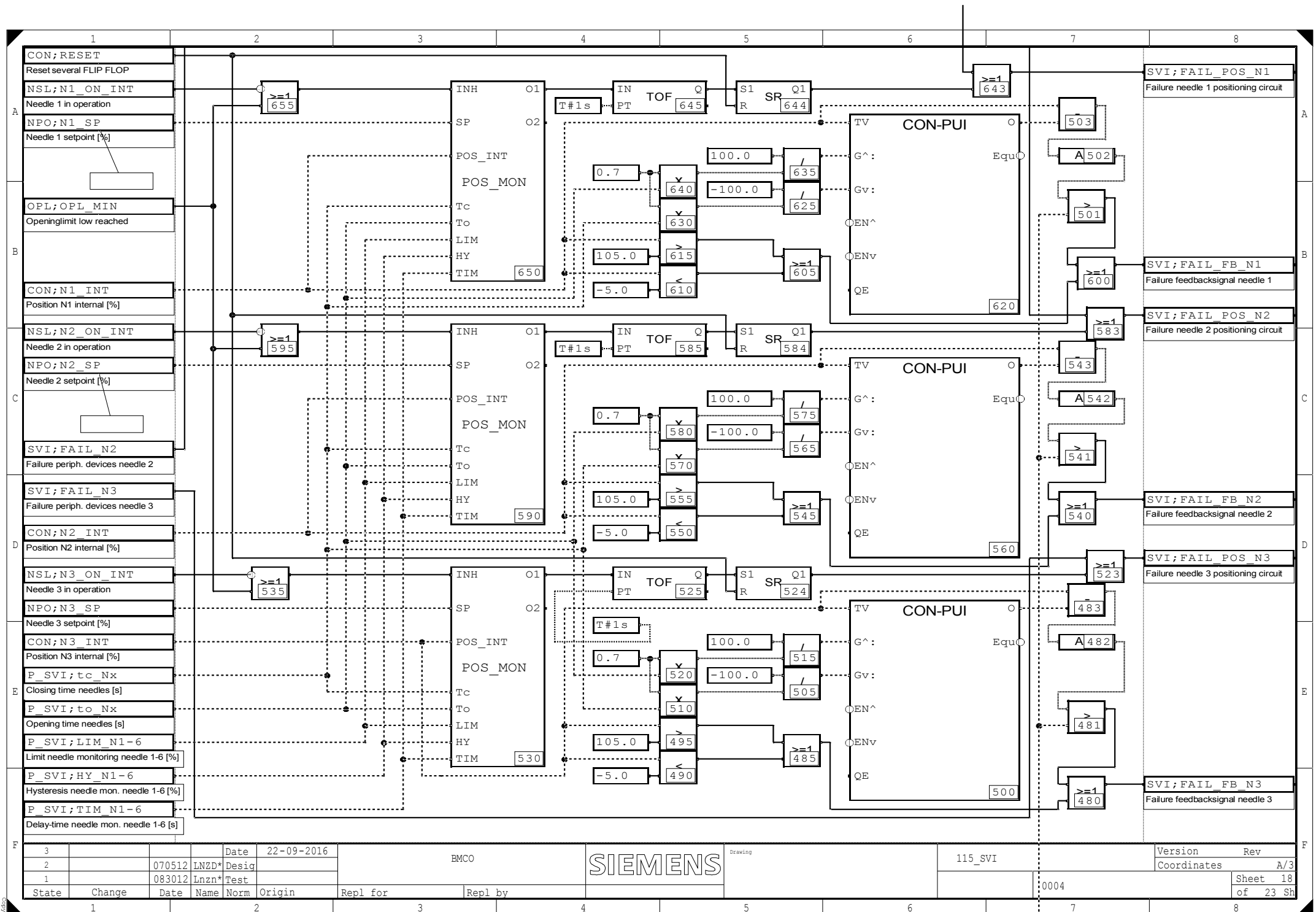


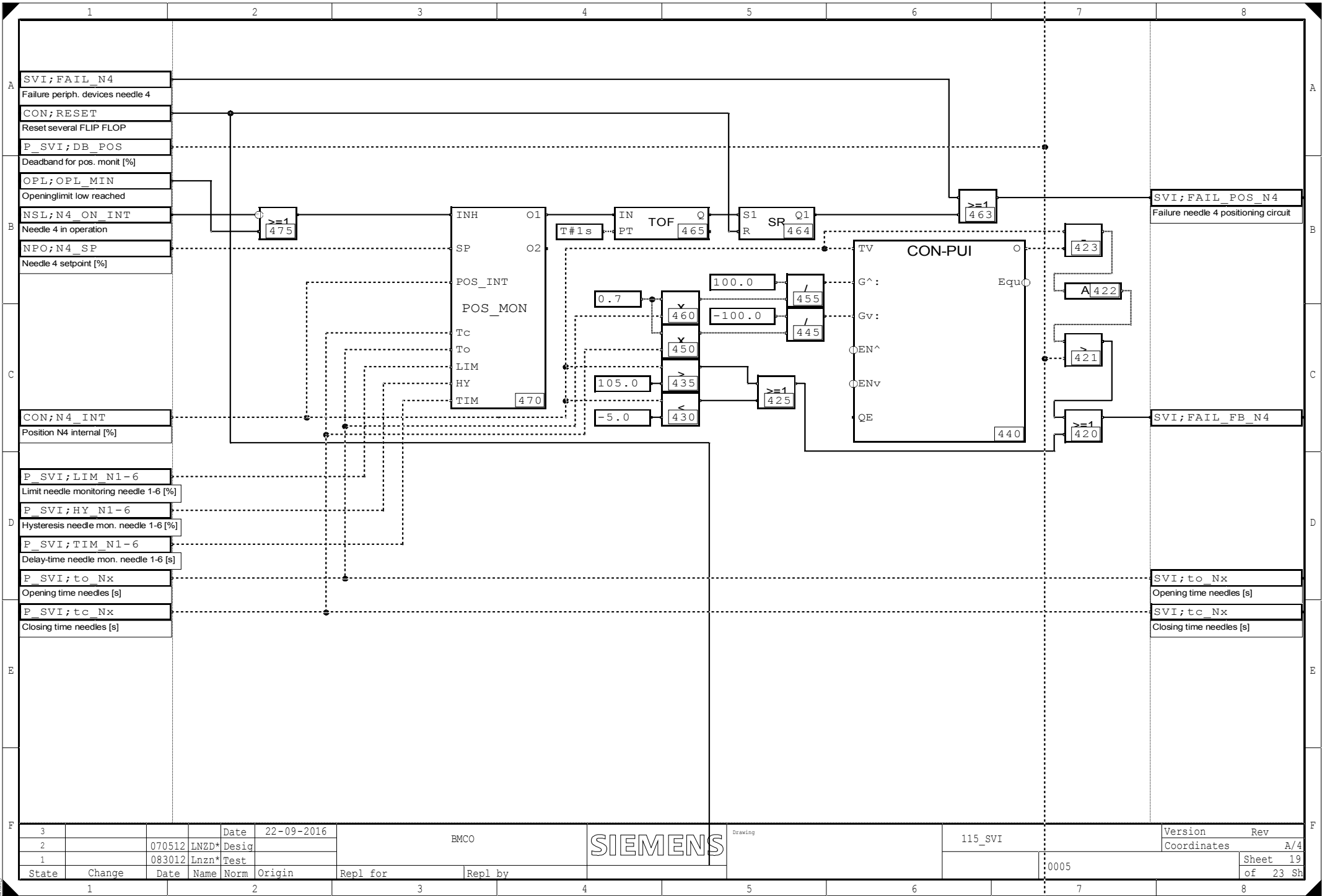






3			Date	22-09-2016	BMCO		Drawing	115_SVI		Version	Rev
2			Desig							Coordinates	A/2
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State	Change	Date	Name	Norm	Origin	Repl for	Repl by			of	23 Sh



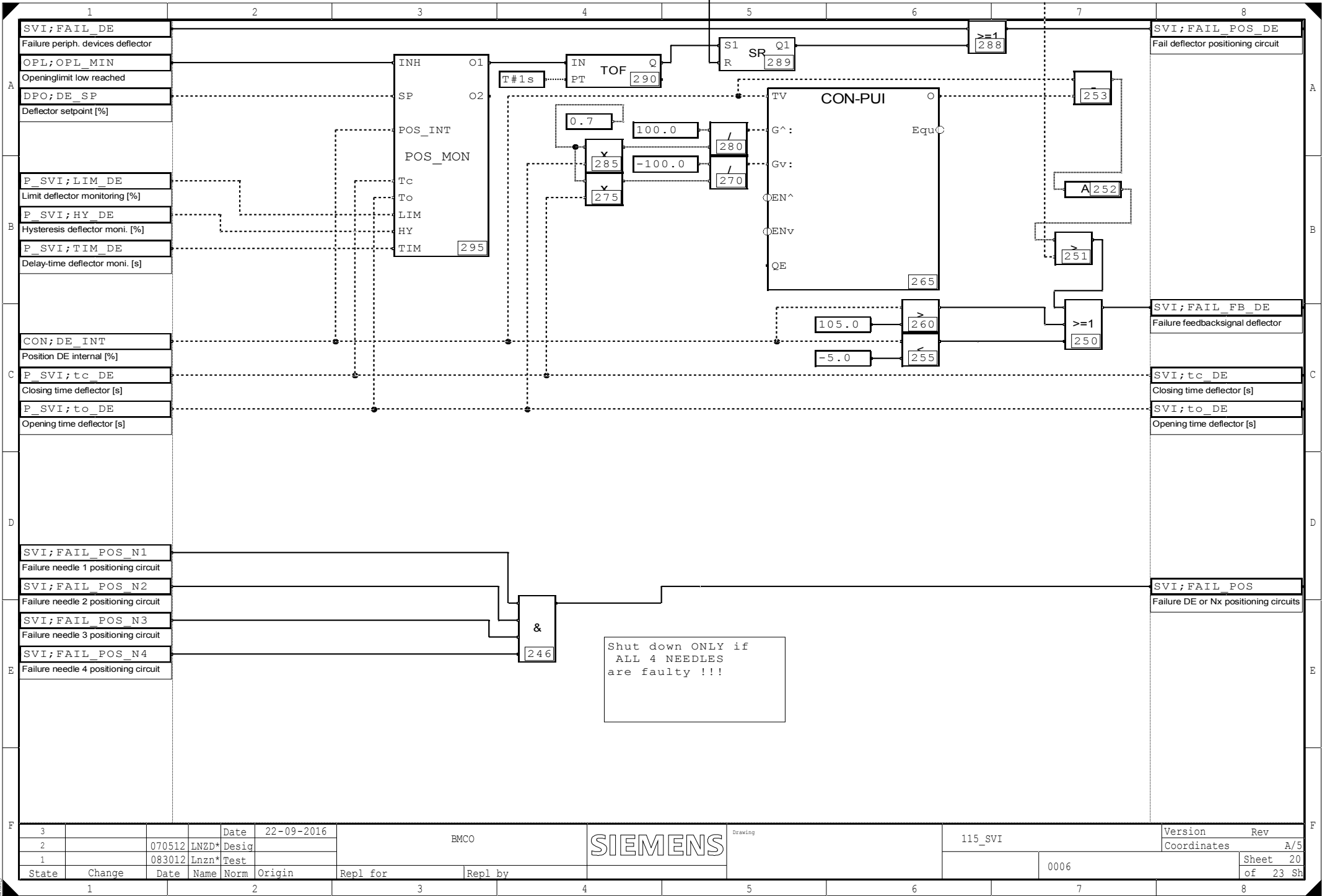


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2		070512	LNZD*	Desig
1		083012	Lnzn*	Test
State	Change	Date	Name	Norm

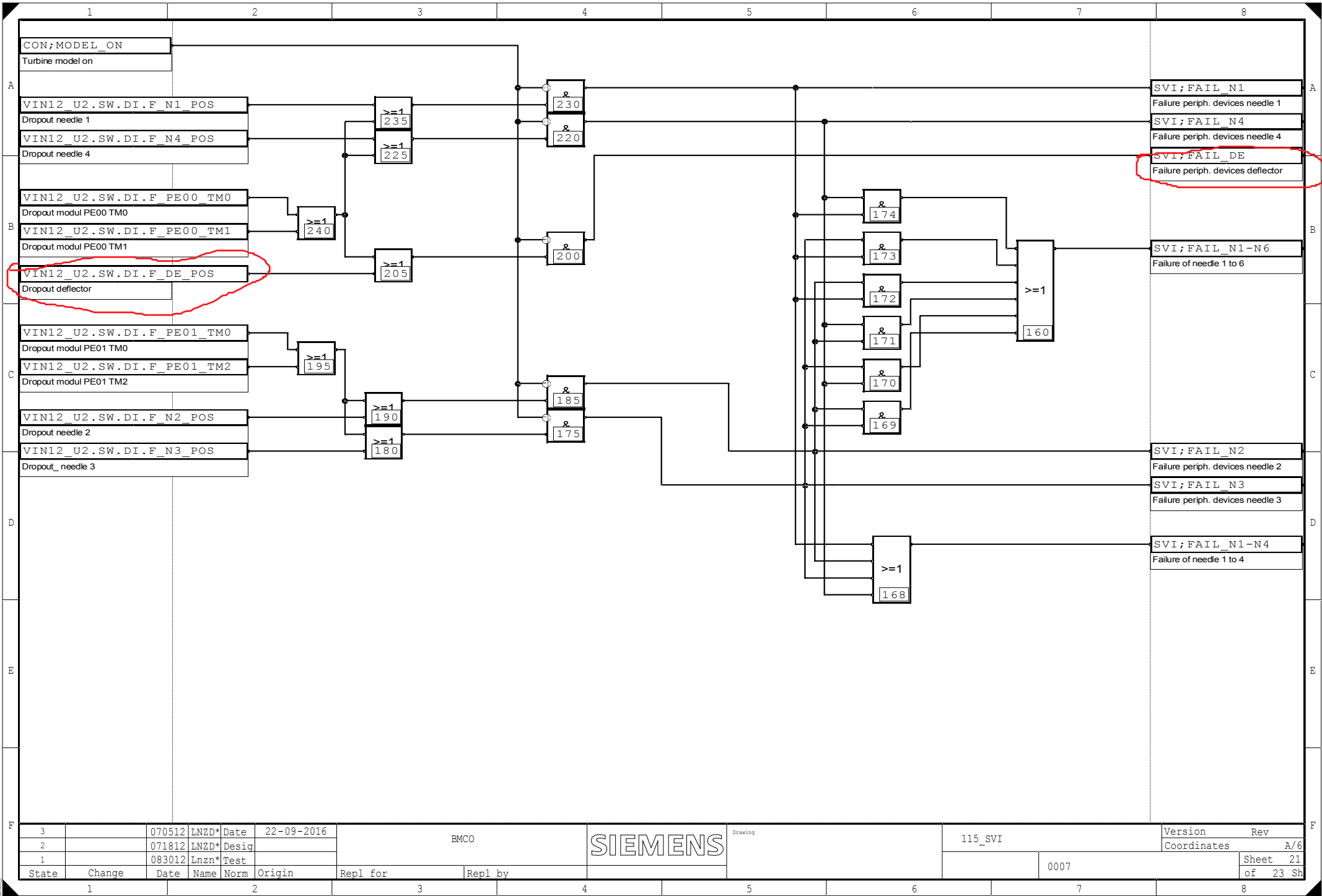
Origin	Repl for	Repl by

Drawing	115_SVI	Version	Rev
		Coordinates	A/4
	0005	Sheet	19
		of	23 Sh

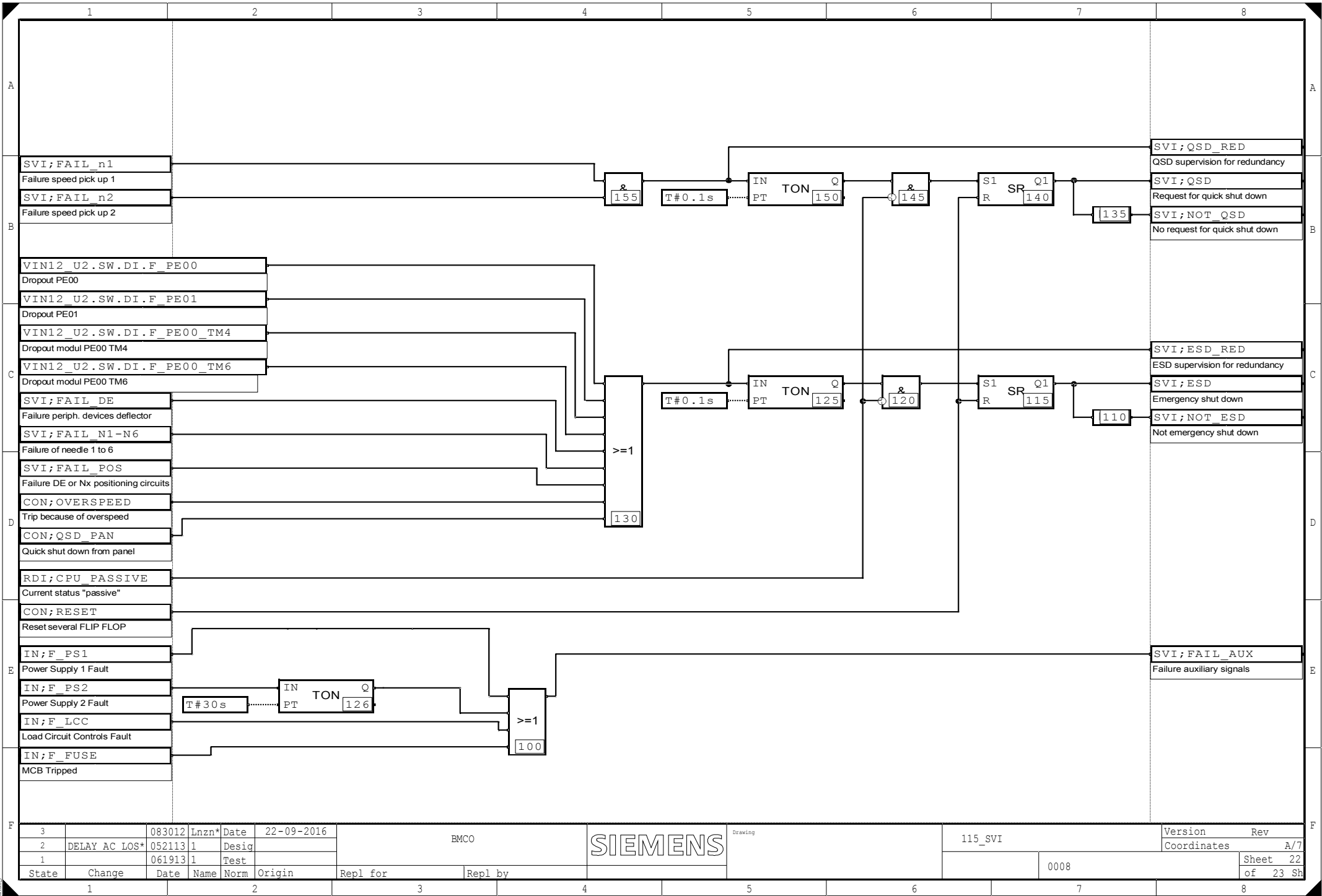
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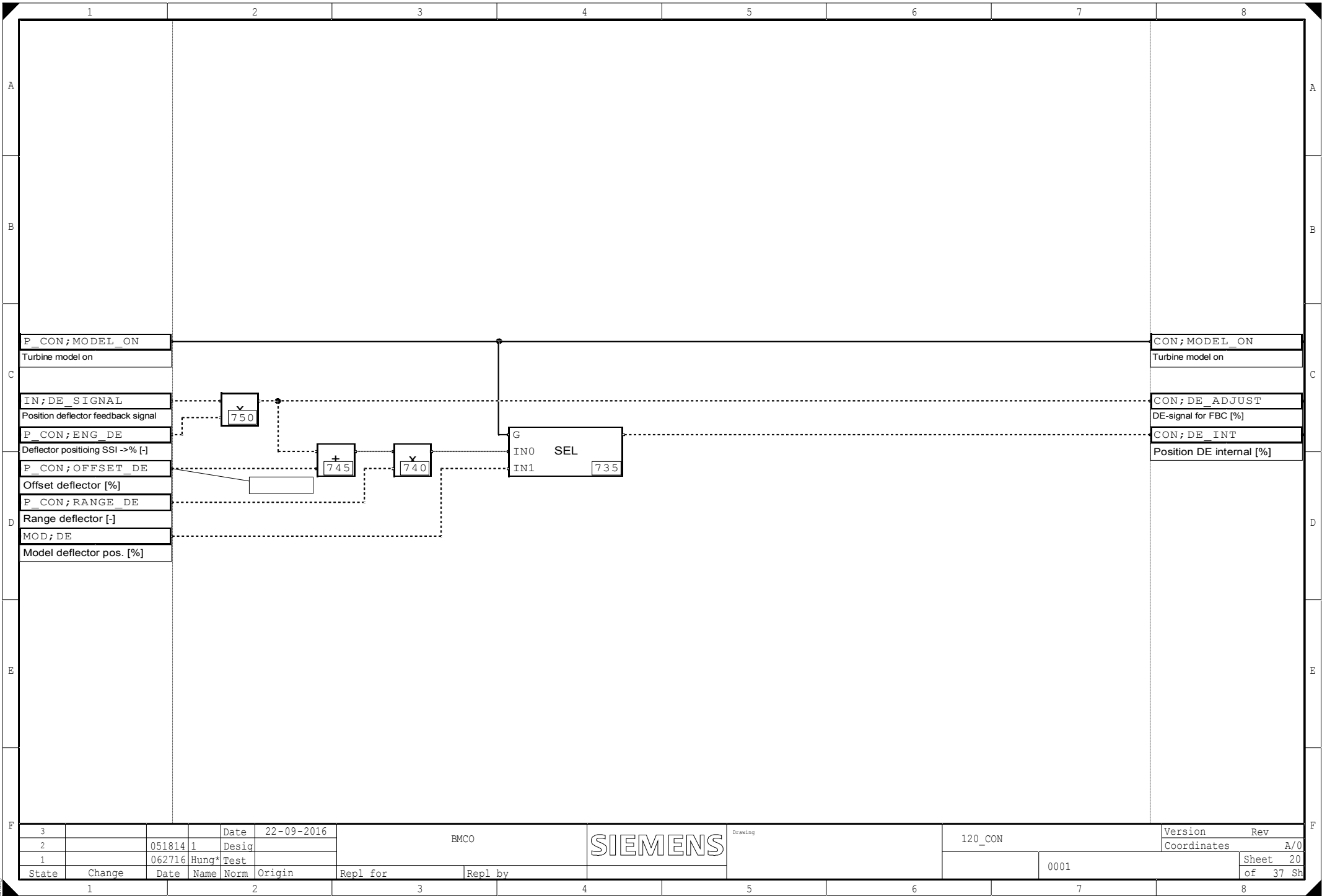


3			Date	22-09-2016	EMCO		SIEMENS	Drawing	115_SVI		Version	Rev
2		070512	LNZD*	Desig								Coordinates
1		083012	Lnzn*	Test								Sheet
State	Change	Date	Name	Norm	Origin	Repl for	Repl by		0006	of 23 Sh		

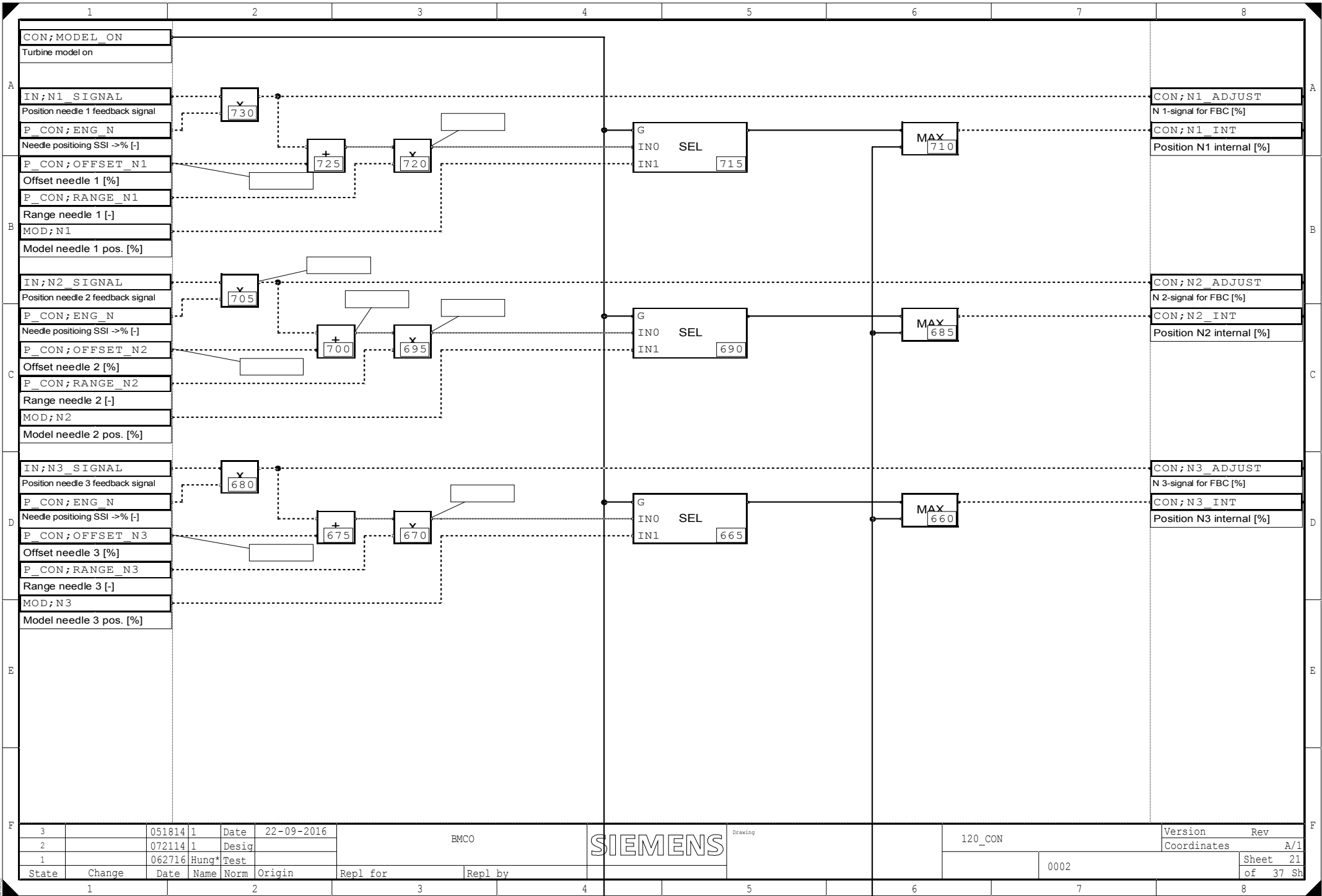


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2		071812	LNZD*	Desig								Coordinates	A/6
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State	Change	Date	Name	Norm	Origin	Repl for	Repl by				0007	of	23 Sh

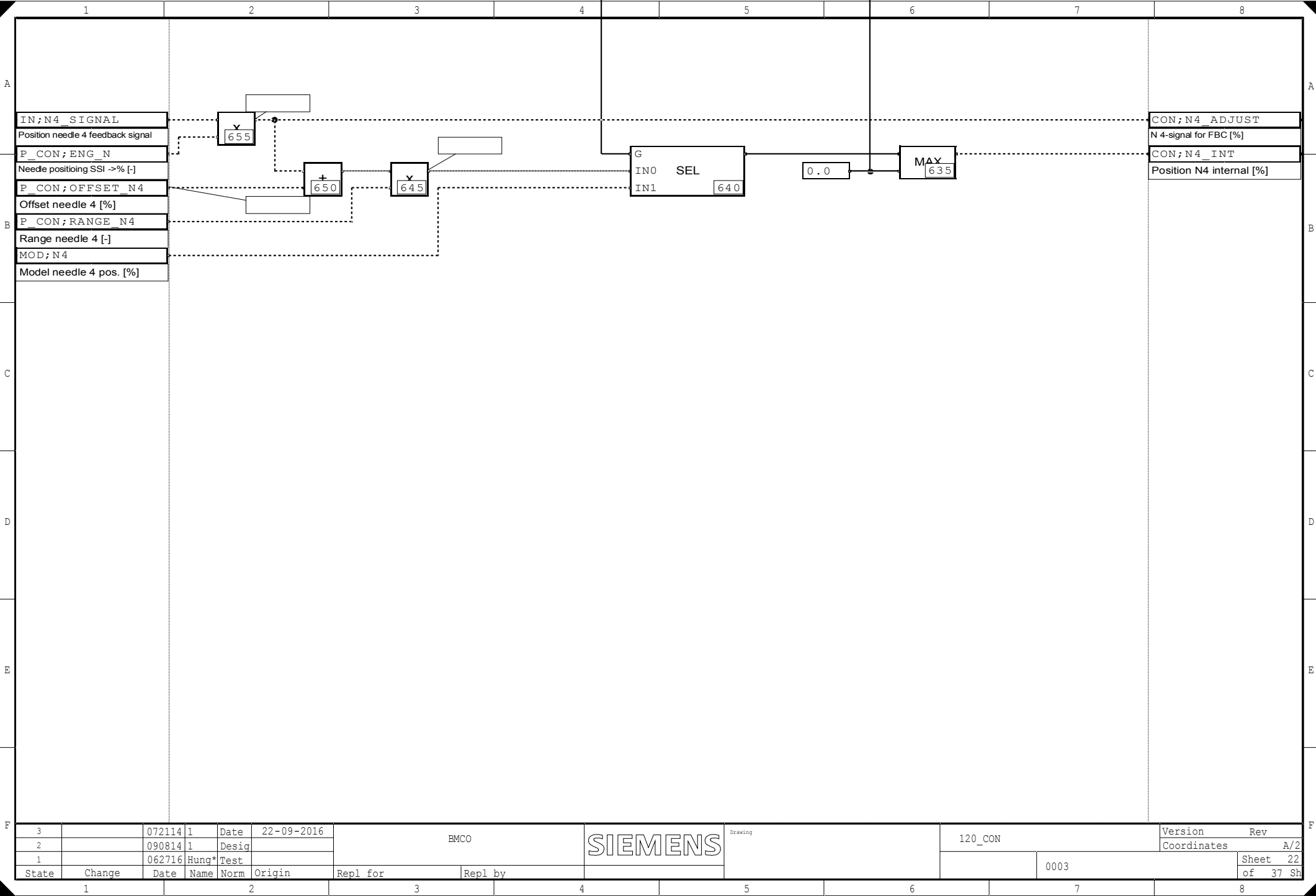




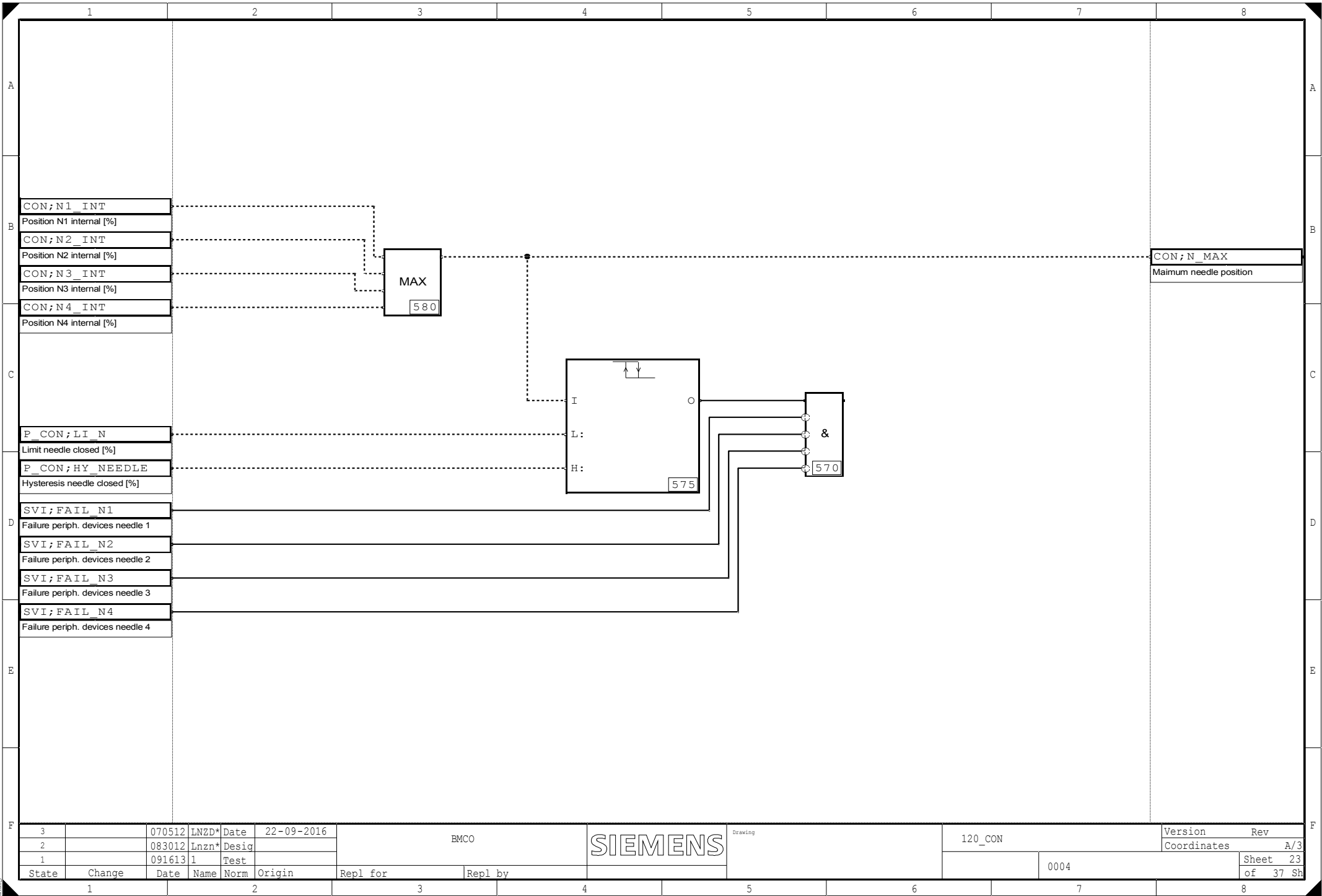
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2		051814	1	Desig							Coordinates	A/0
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State	Change	Date	Name	Norm	Origin	Repl for	Repl by			0001	of	37 Sh



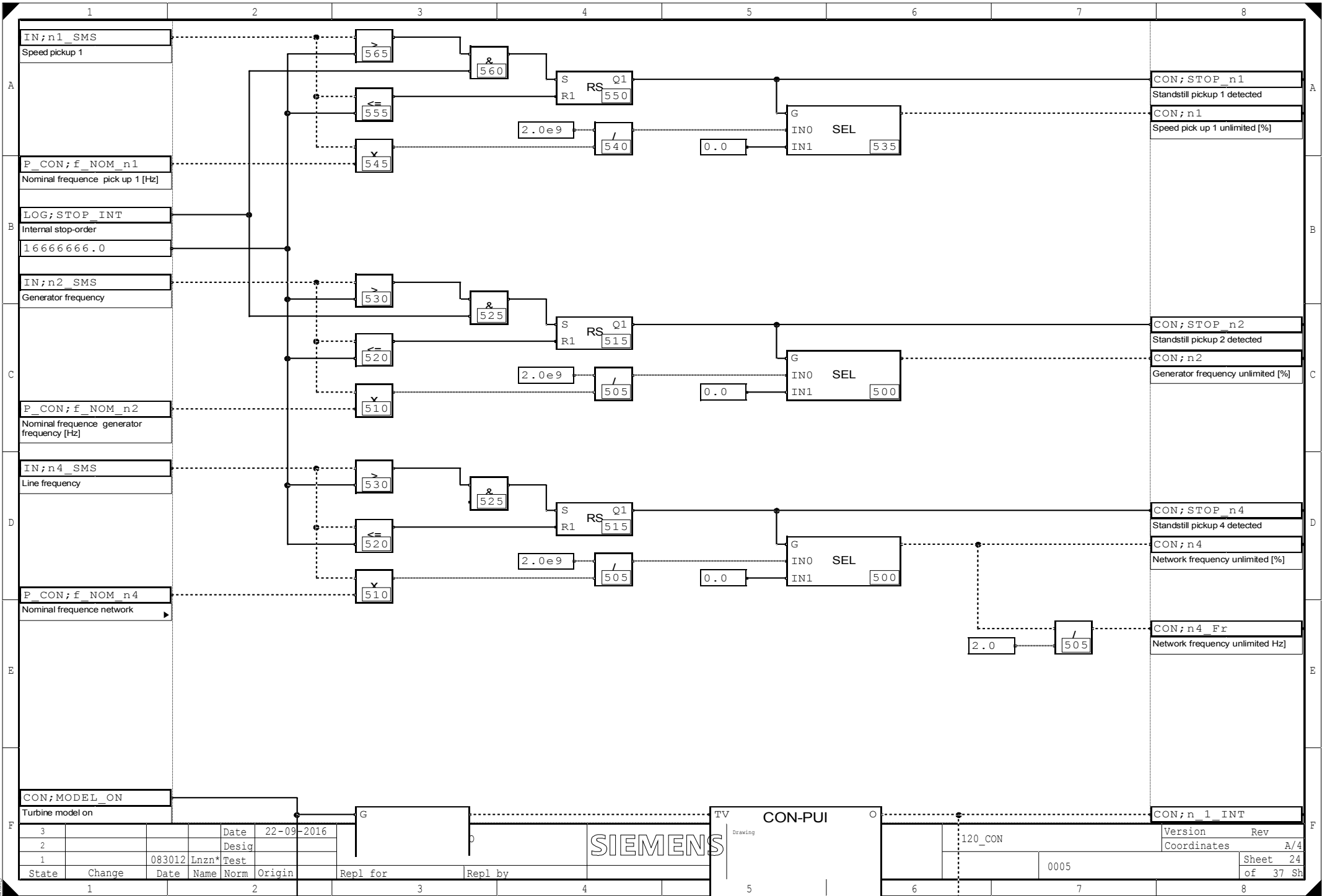
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1		062716	Hung*	Test									
State	Change	Date	Name	Norm	Origin	Repl for	Repl by			0002		Sheet 21	of 37 Sh



3		072114	1	Date	22-09-2016	BMC0		SIEMENS		Drawing		120_CON		Version	Rev
2		090814	1	Desig										Coordinates	A/2
1		062716	Hung*	Test											
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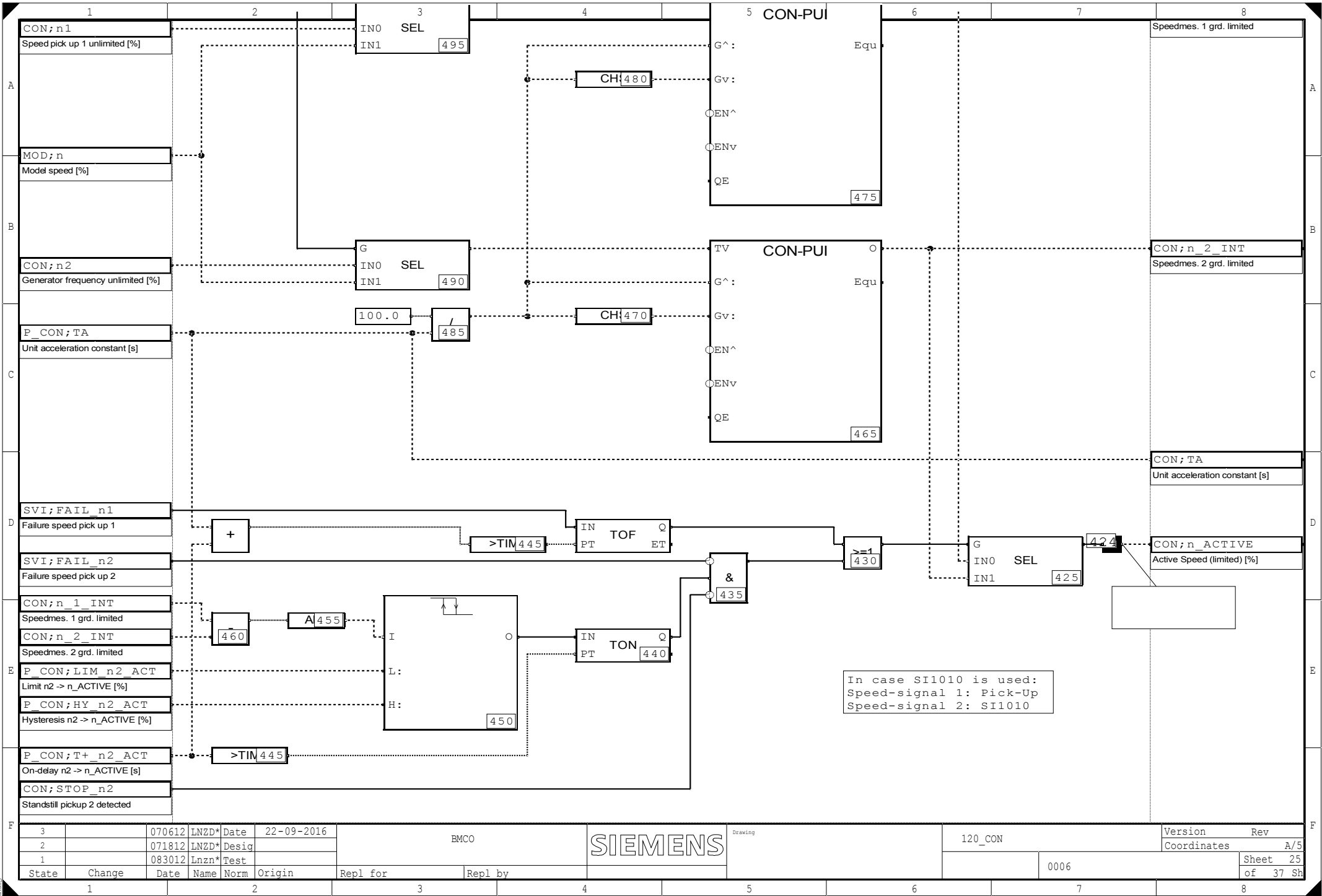


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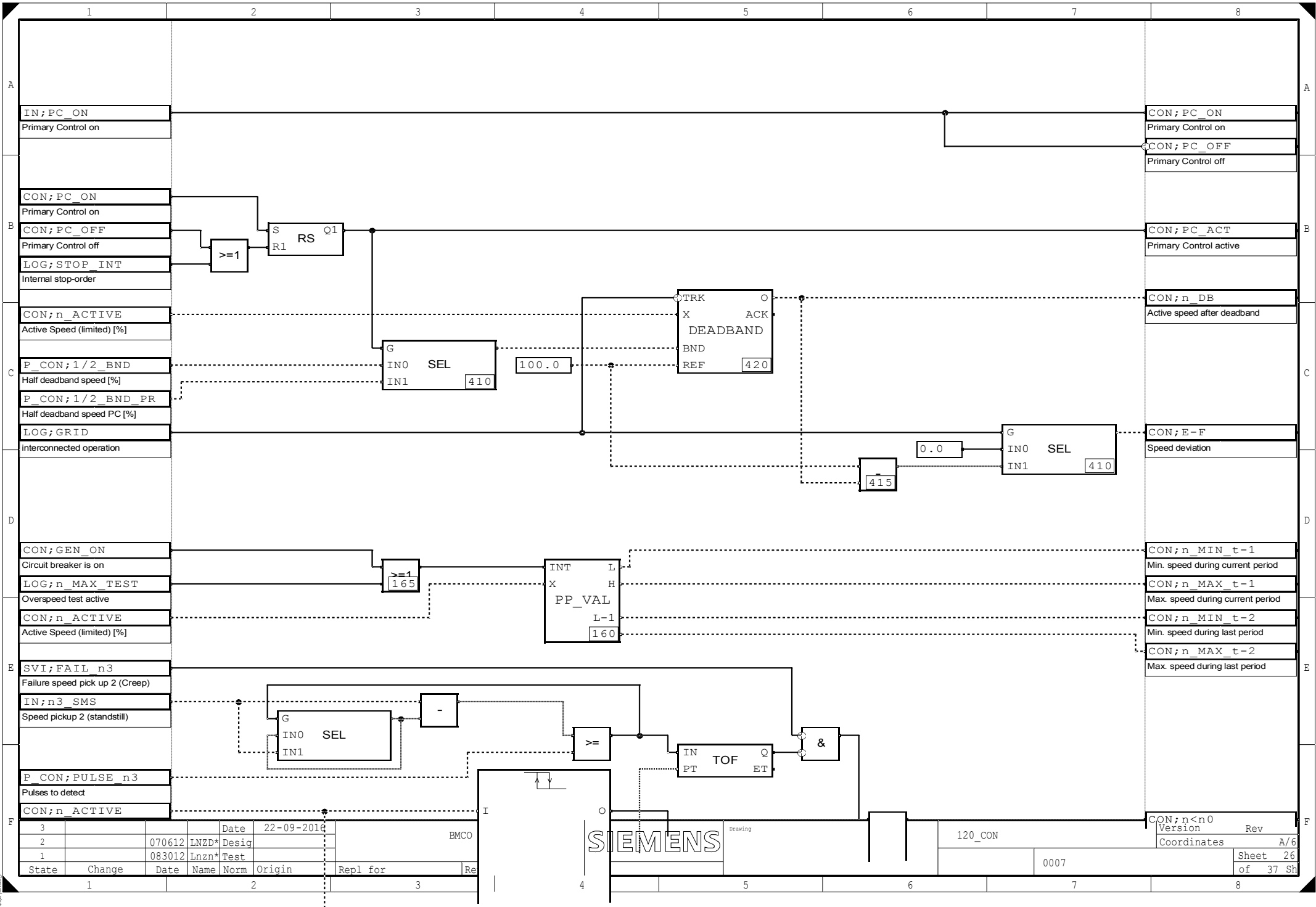


CON;MODEL_ON									
Turbine model on									
3			Date	22-09-2016					
2			Desig						
1		083012	Lnzn*	Test					
State	Change	Date	Name	Norm	Origin	Repl for	Repl by		

CON-PUI				CON; n1_INT			
Drawing				Version			
				Rev			
				Coordinates			
				A/4			
				Sheet			
				24			
				of 37 Sh			



3		070612	LNZD*	Date	22-09-2016	BMCO		SIEMENS	Drawing	120_CON		Version	Rev		
2		071812	LNZD*	Desig								Coordinates	A/5		
1		083012	Lnzn*	Test								Sheet	25		
State	Change	Date	Name	Norm	Origin	Repl for	Repl by				0006			of	37 Sh



3			Date	22-09-2016		
2		070612	LNZD*	Desig		BMC0
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State	Change	Date	Name	Norm	Origin	Repl for

SIEMENS

Drawing

120_CON

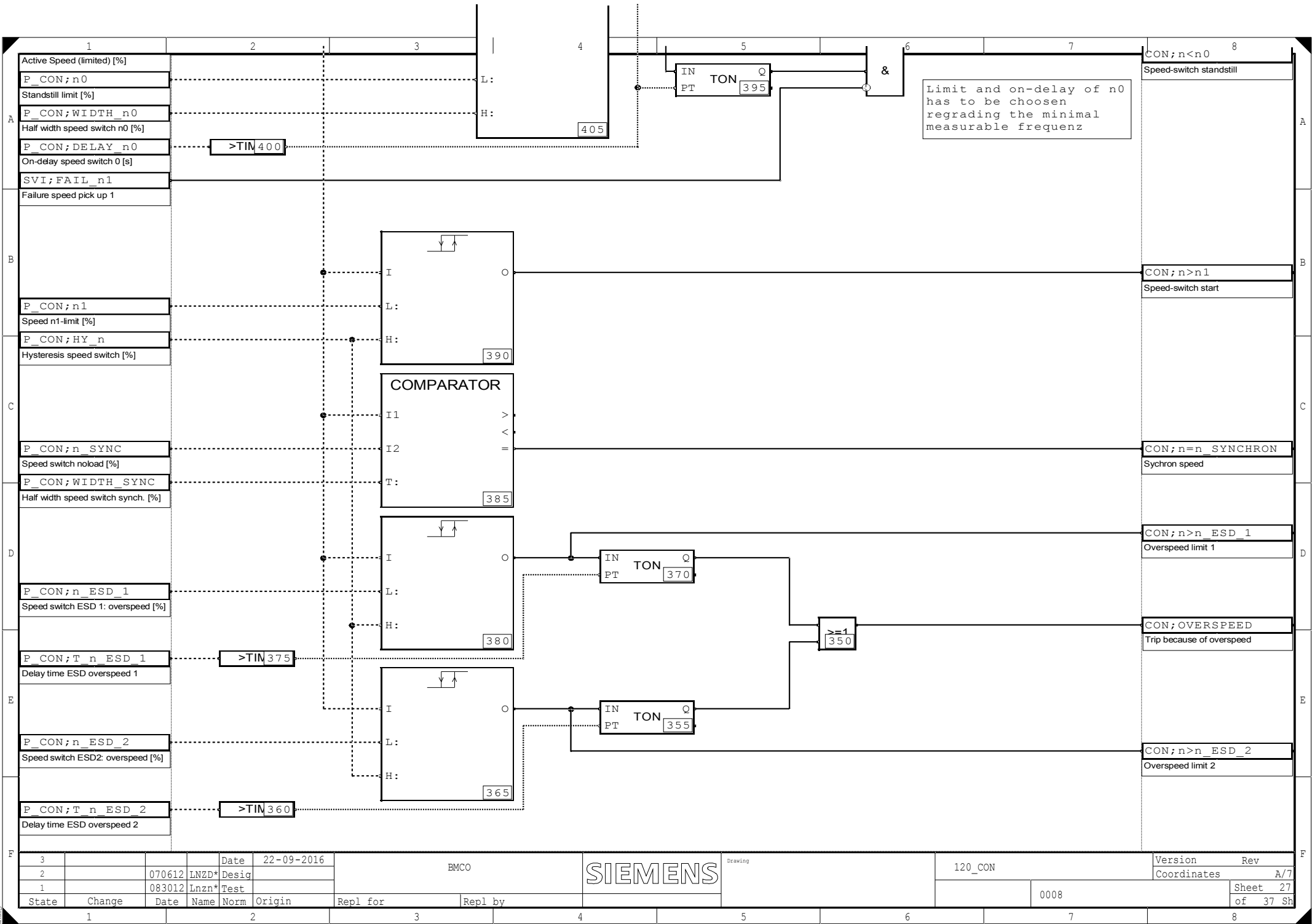
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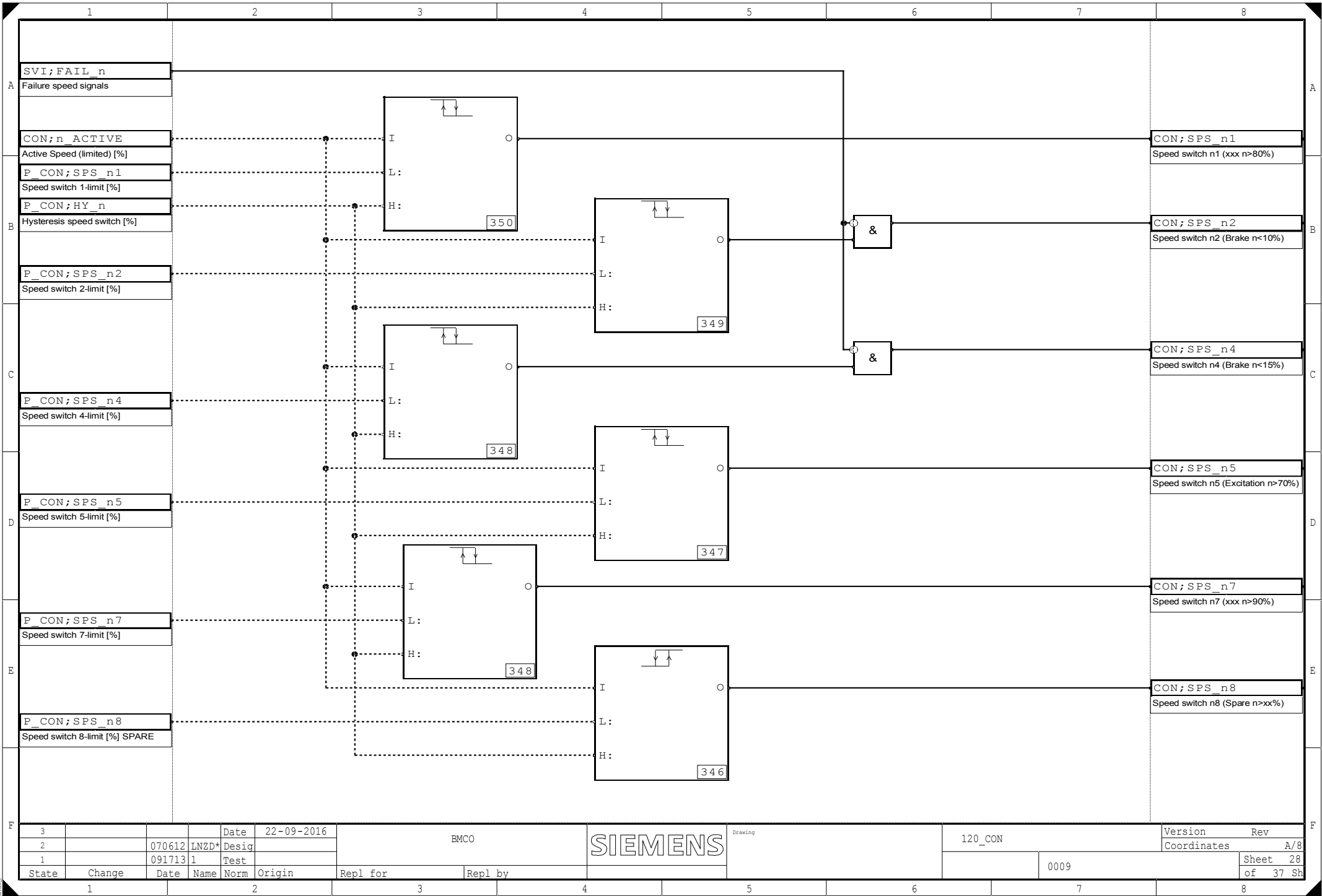
Rev
A/6

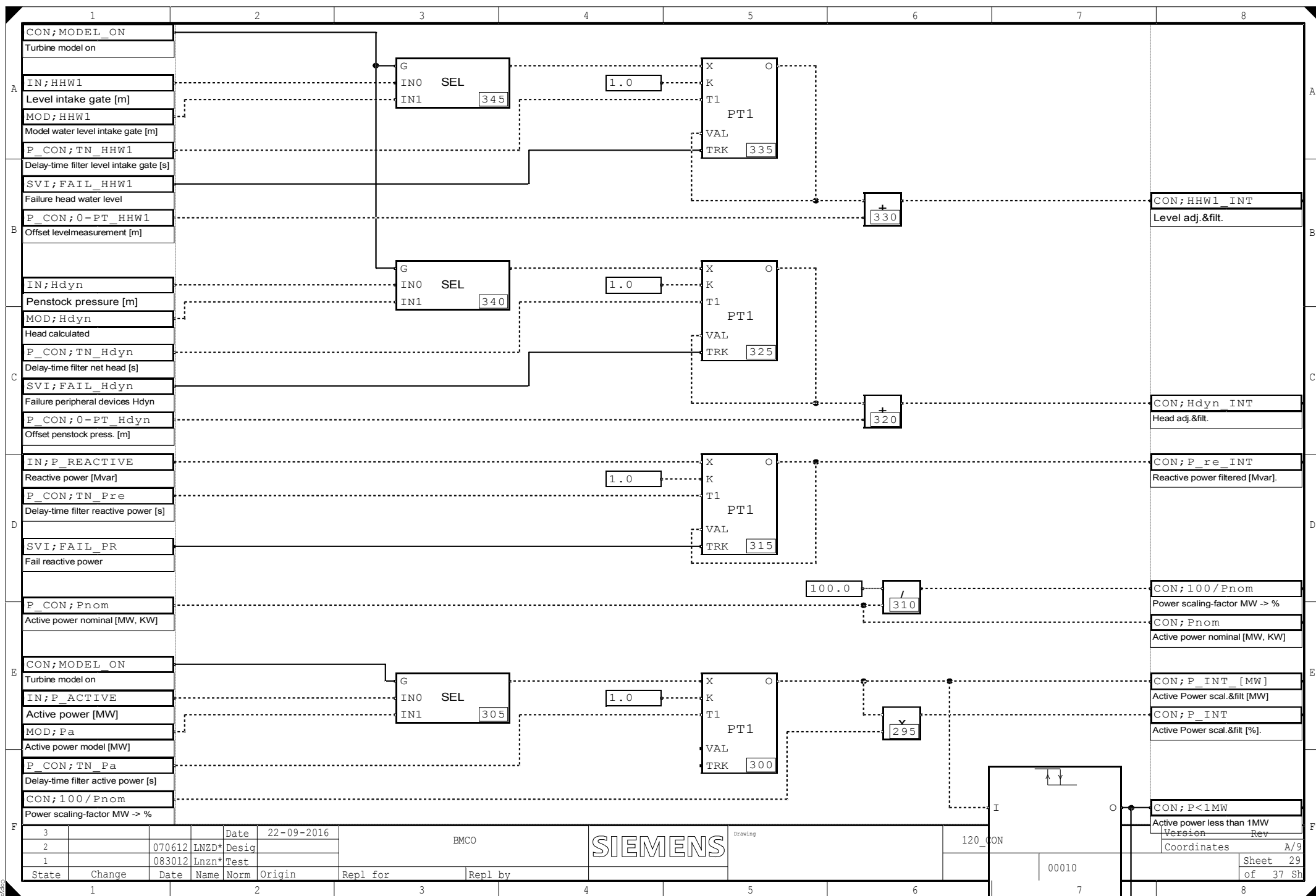
Sheet 26
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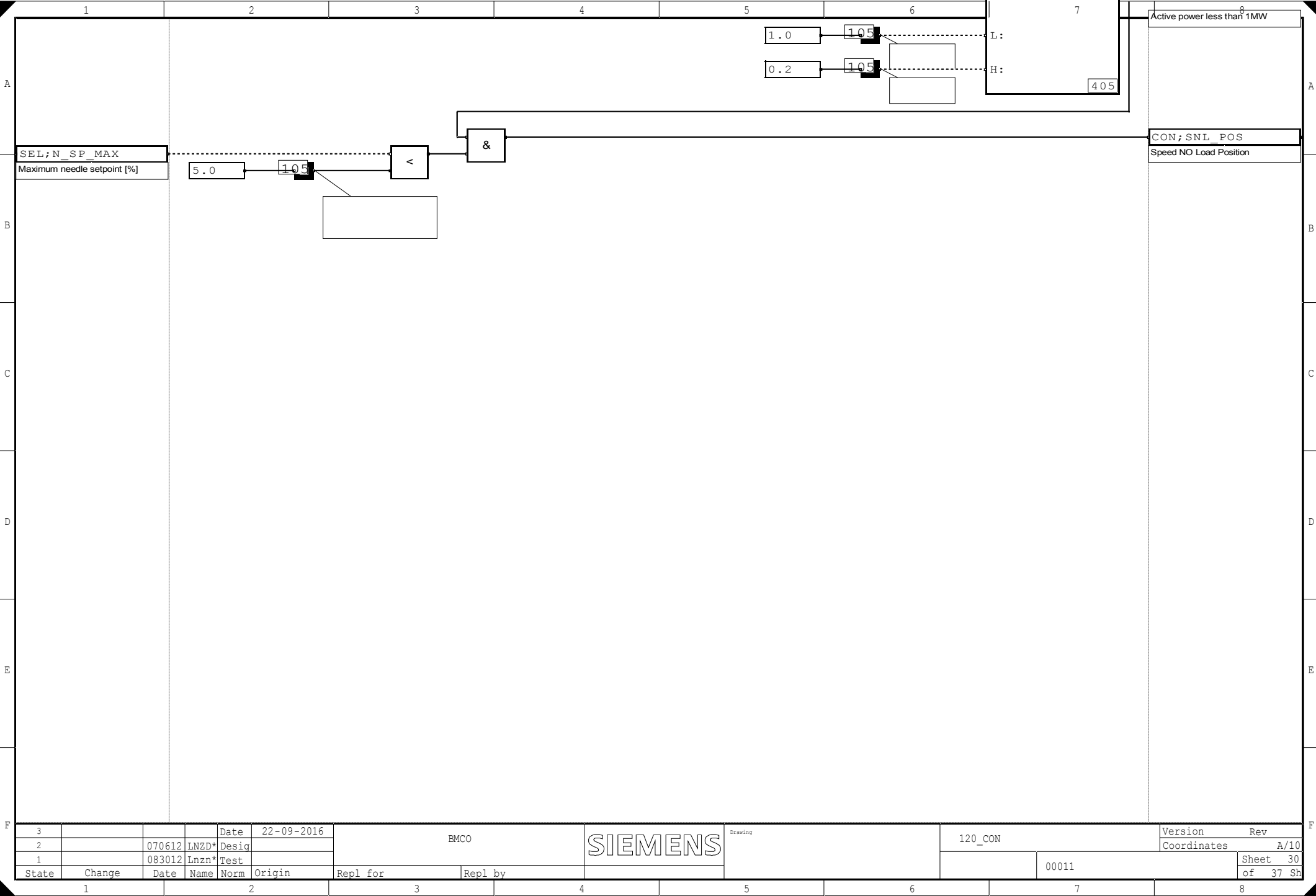
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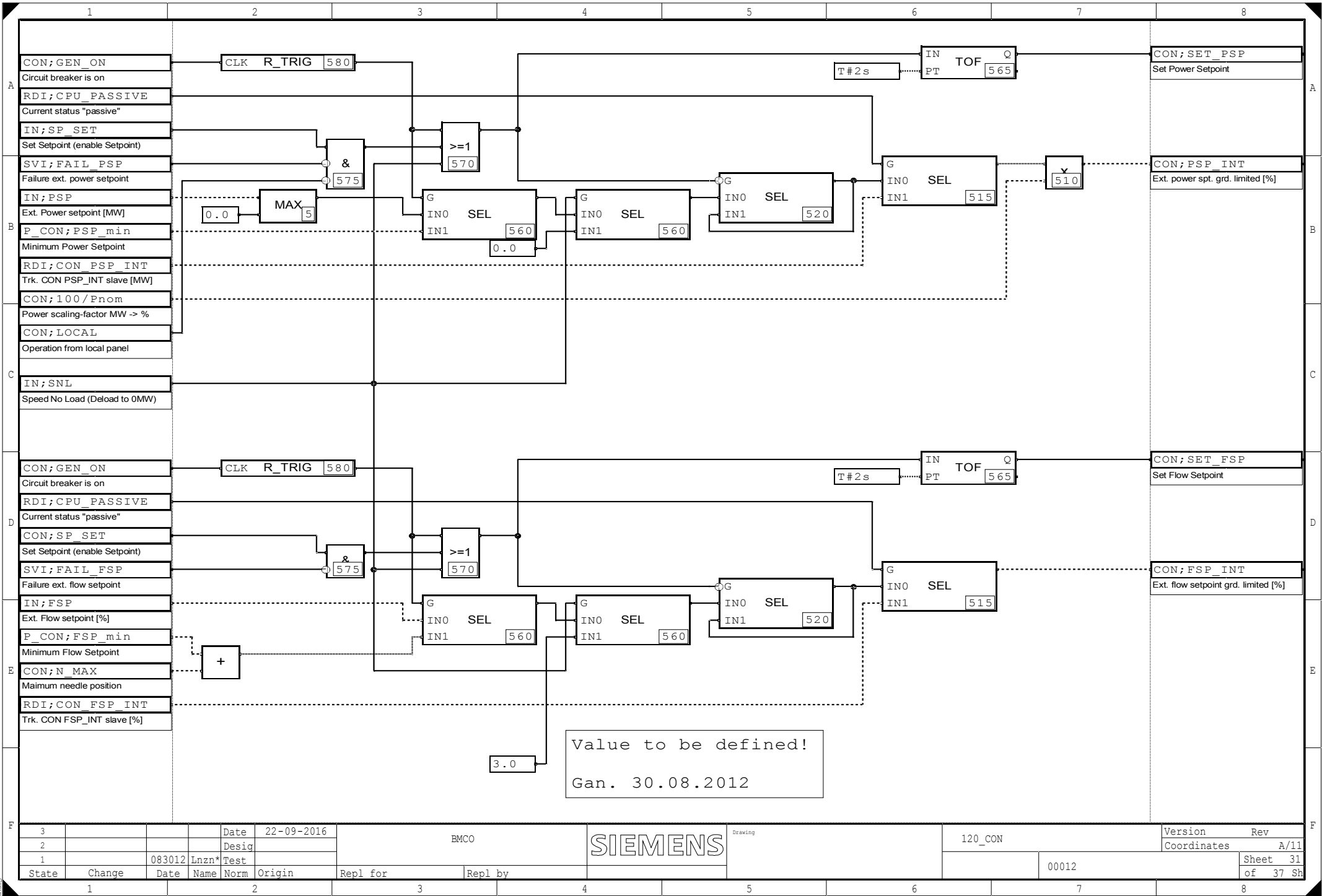
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State	Change	Date	Name	Norm							



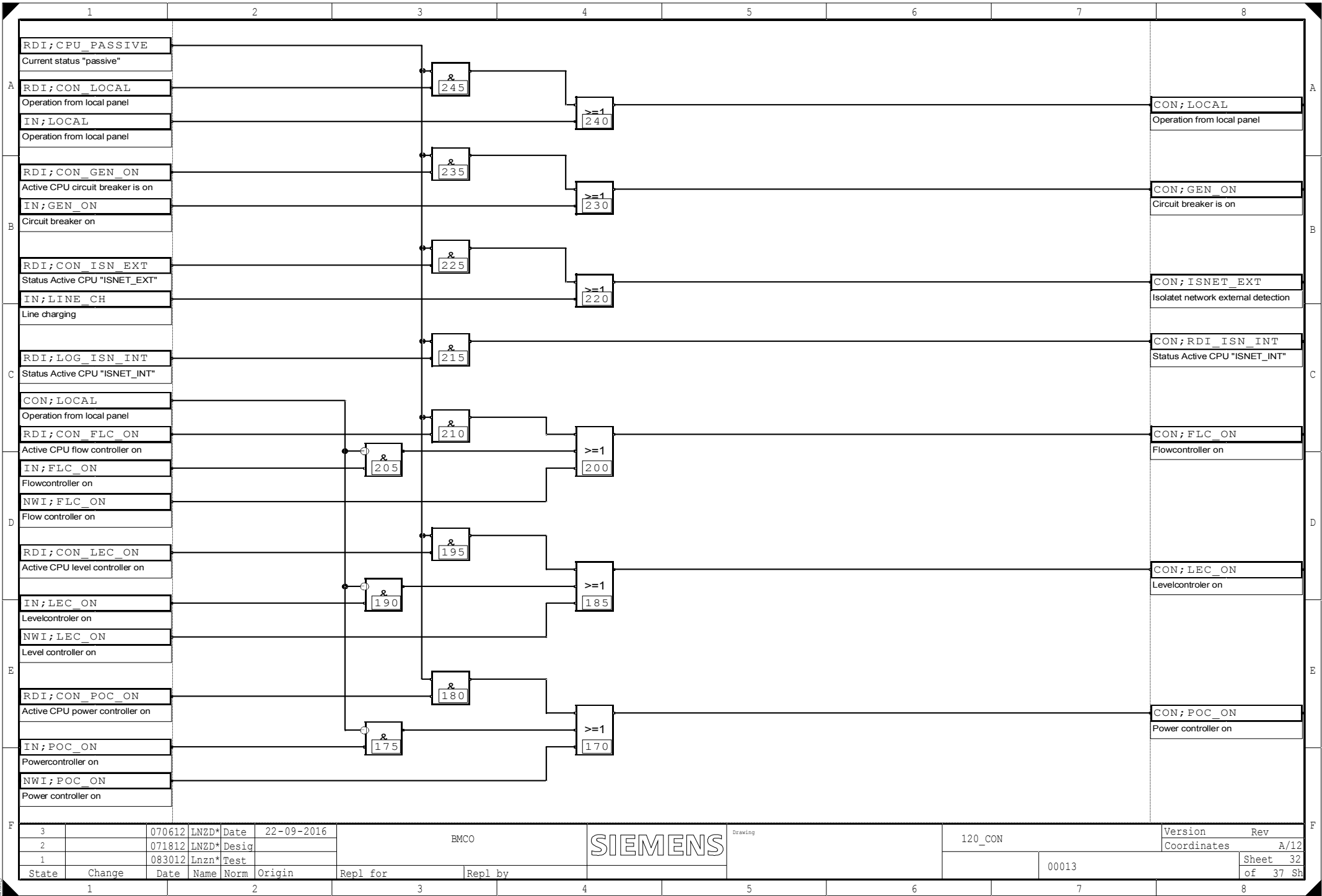




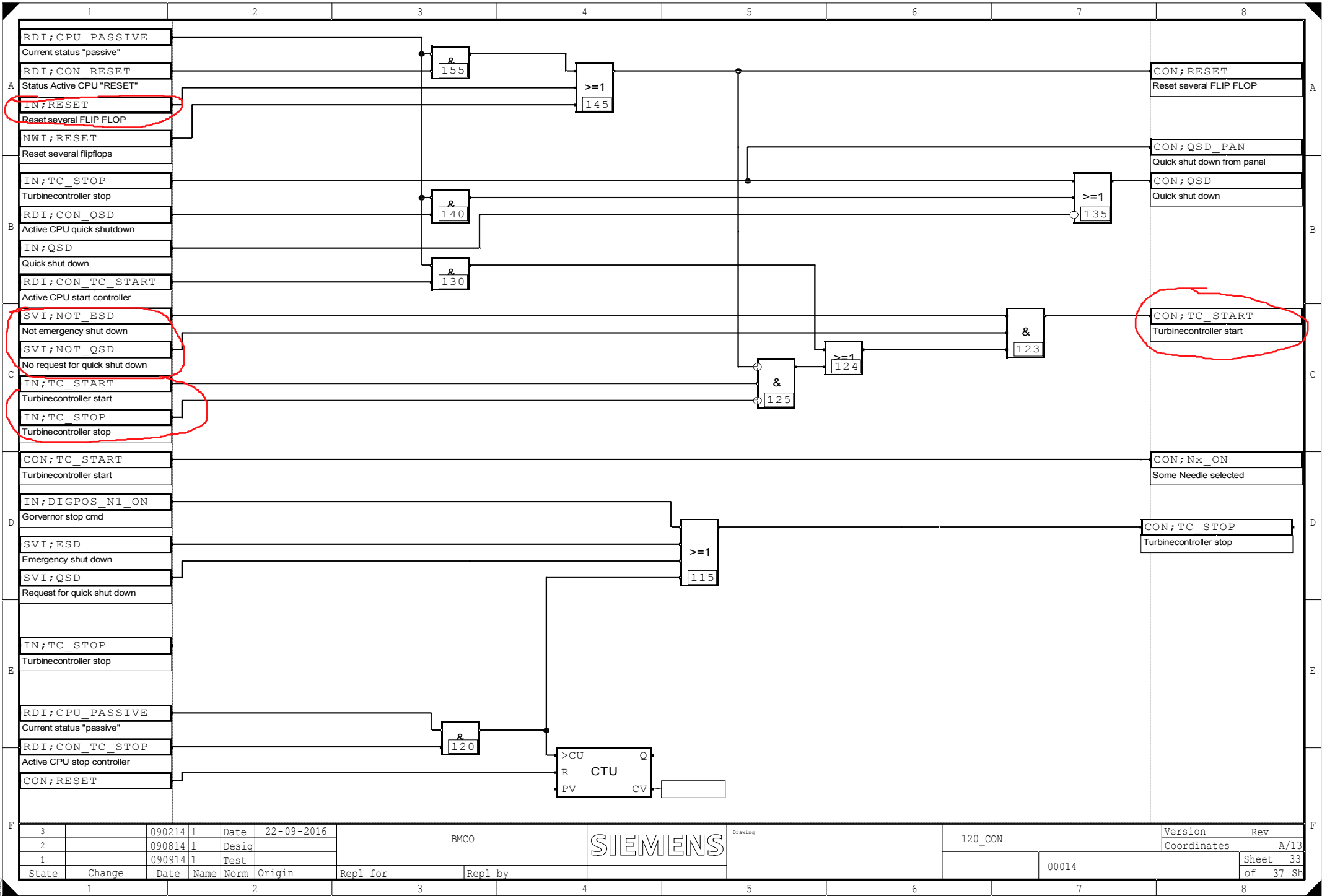
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State	Change	Date	Name	Norm										



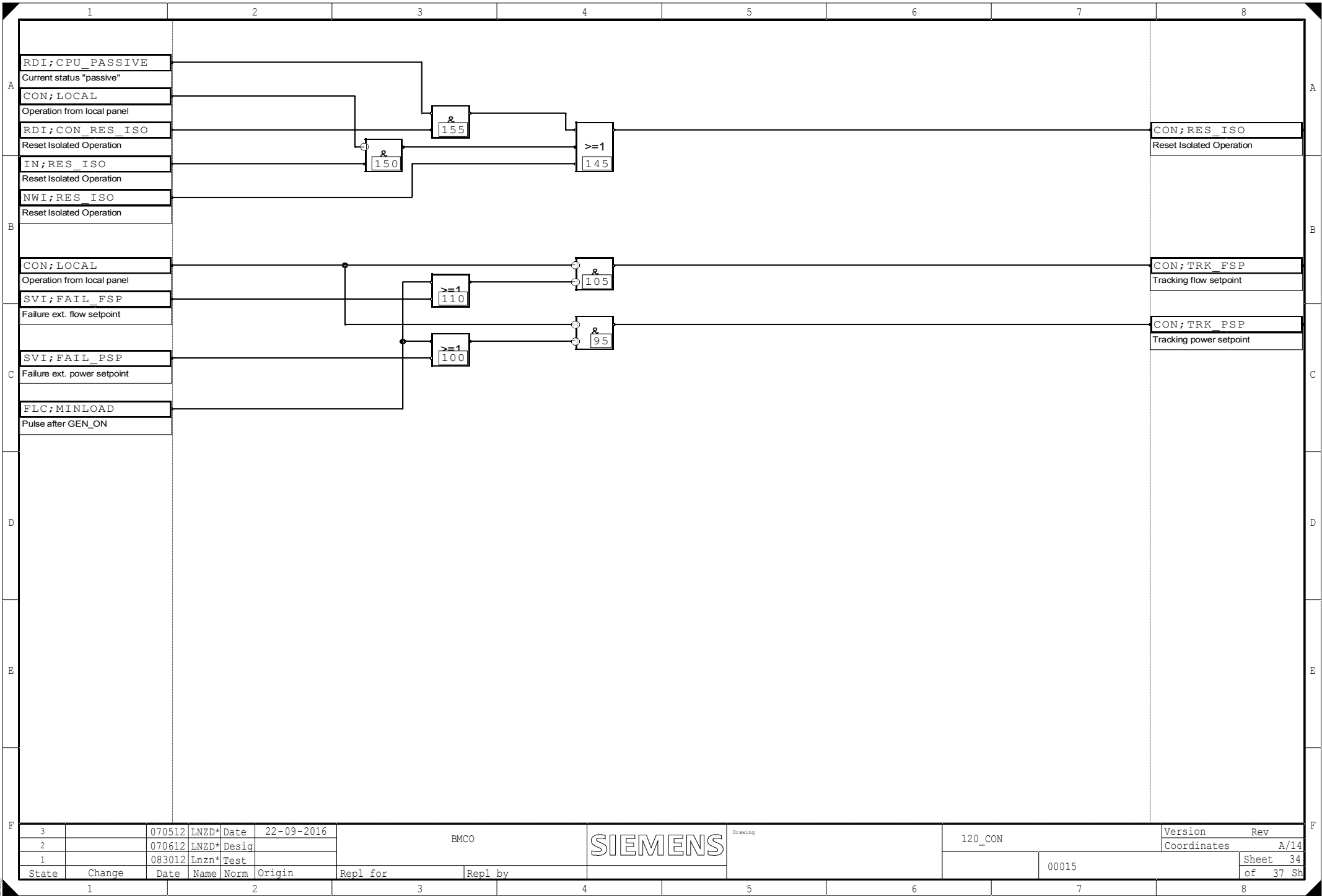
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State	Change	Date	Name	Norm	Origin	Repl for	Repl by			00012	of 37 Sh	

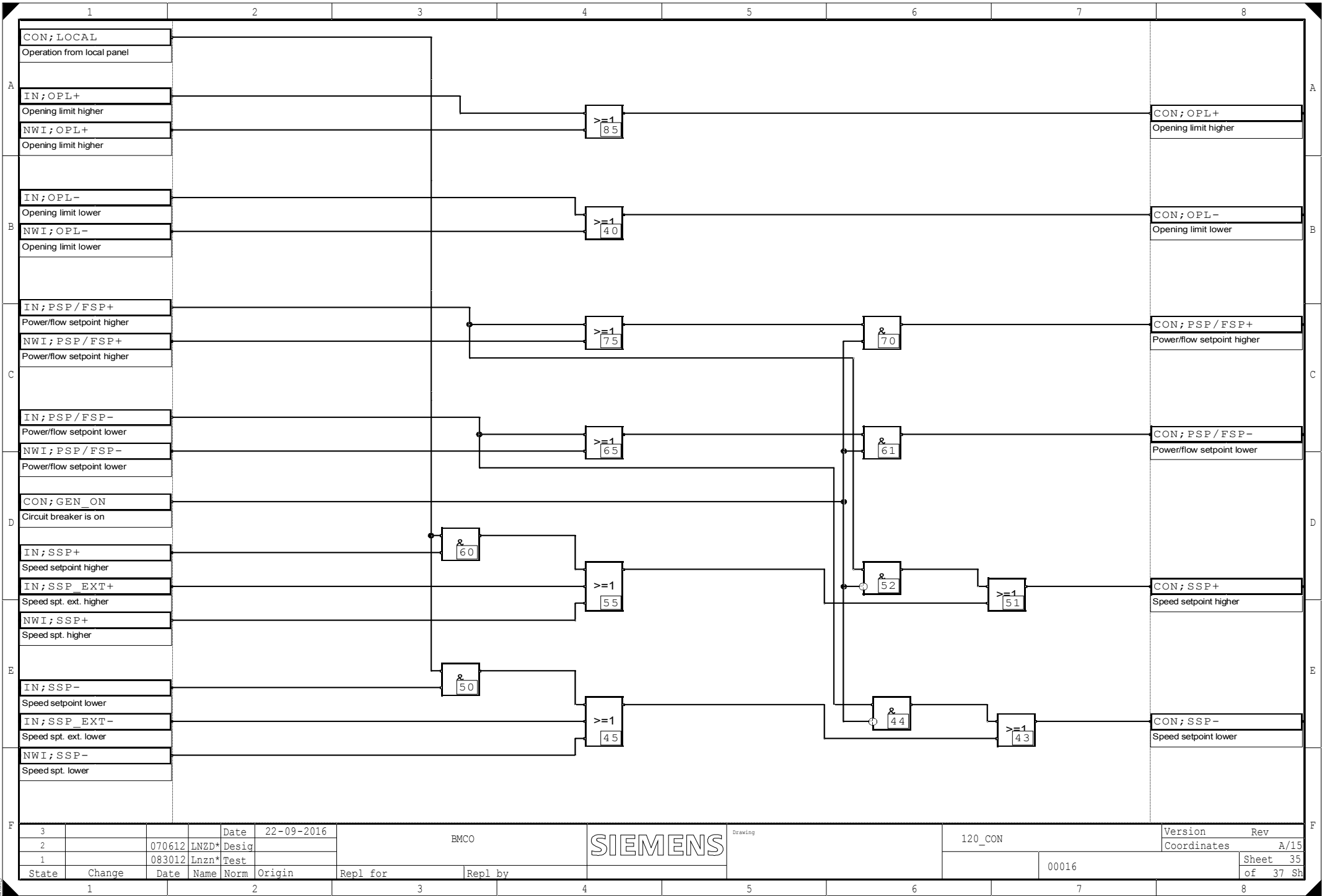


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State	Change	Date	Name	Norm	Origin	Repl for	Repl by					00013		Sheet 32		of 37 Sh

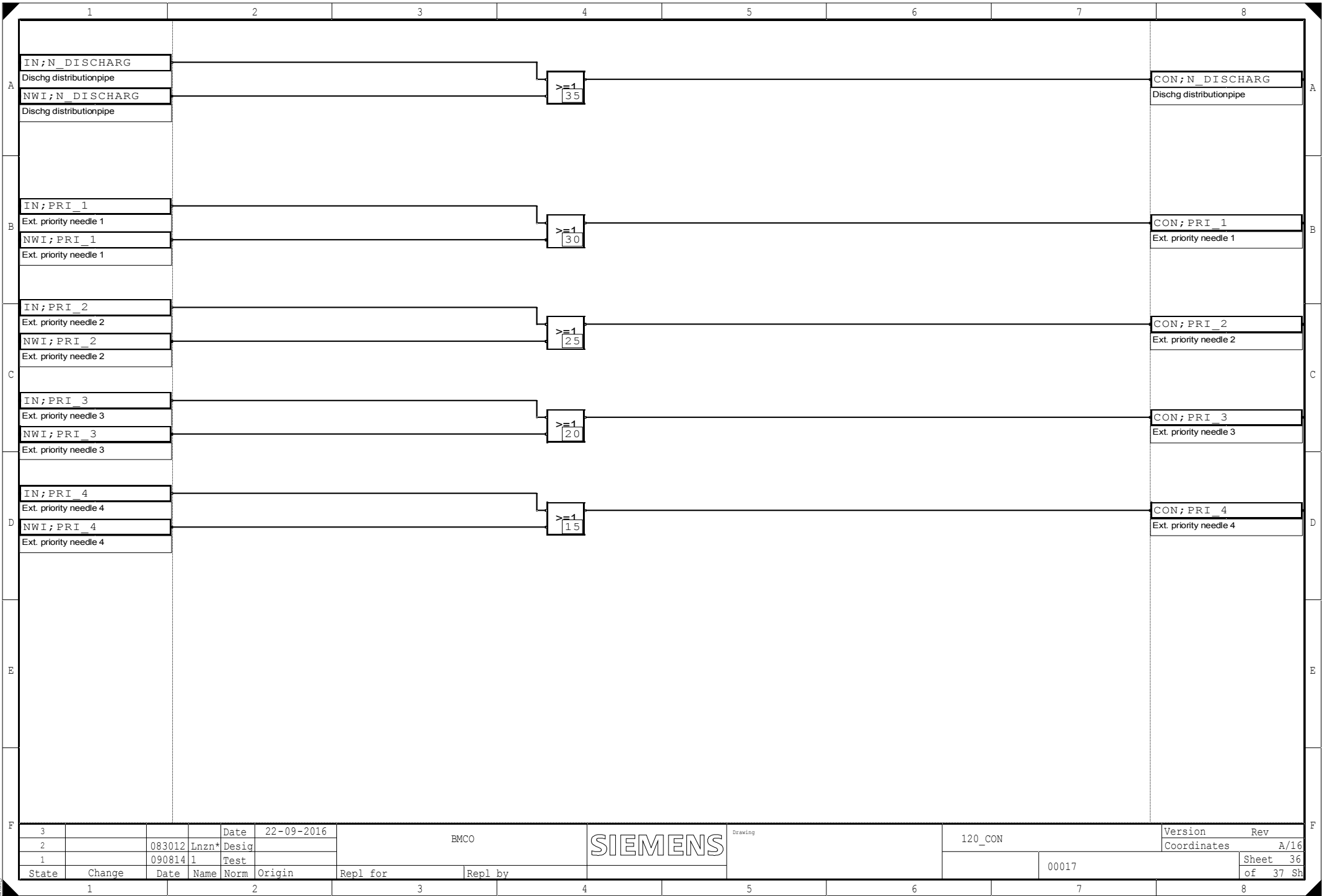


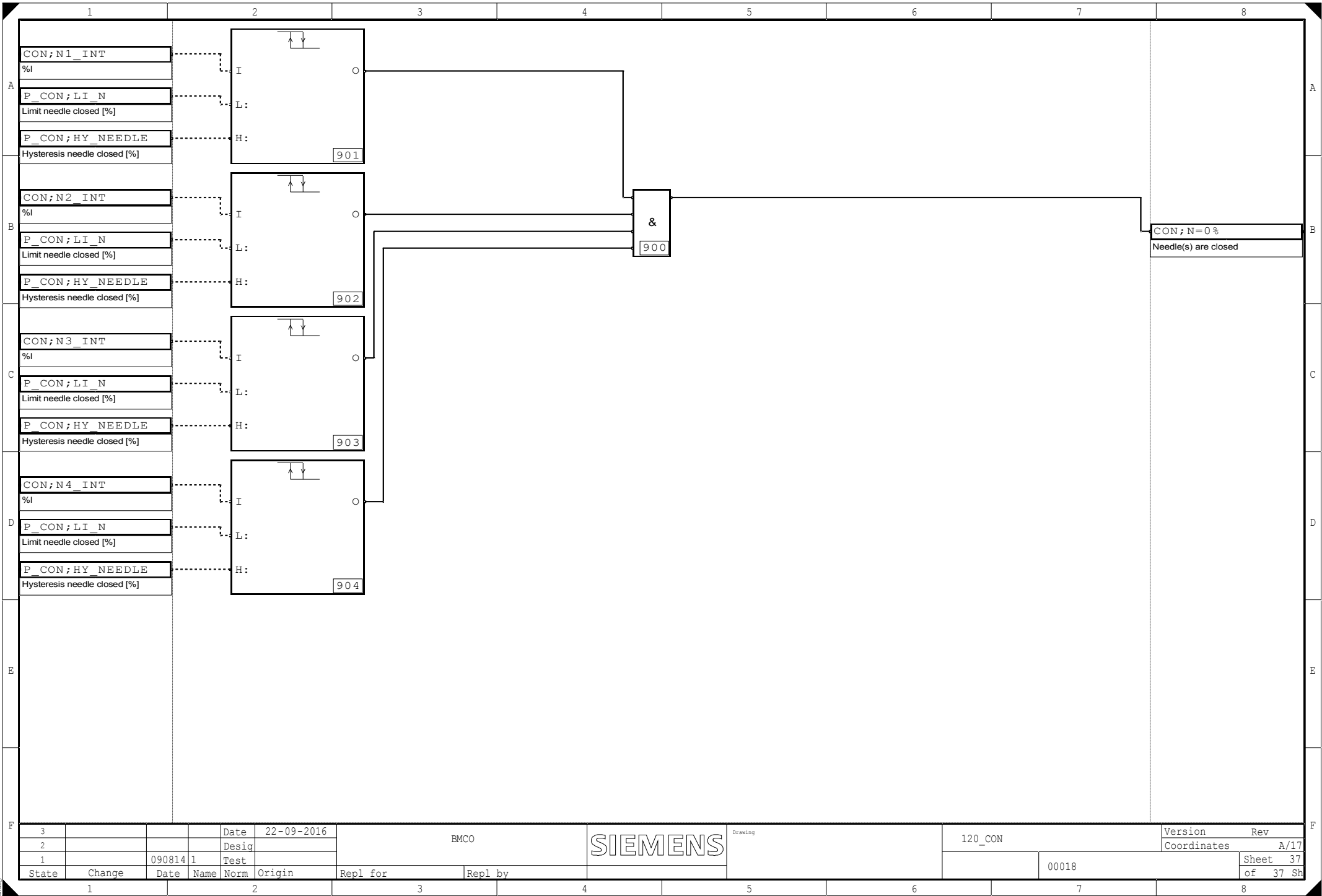
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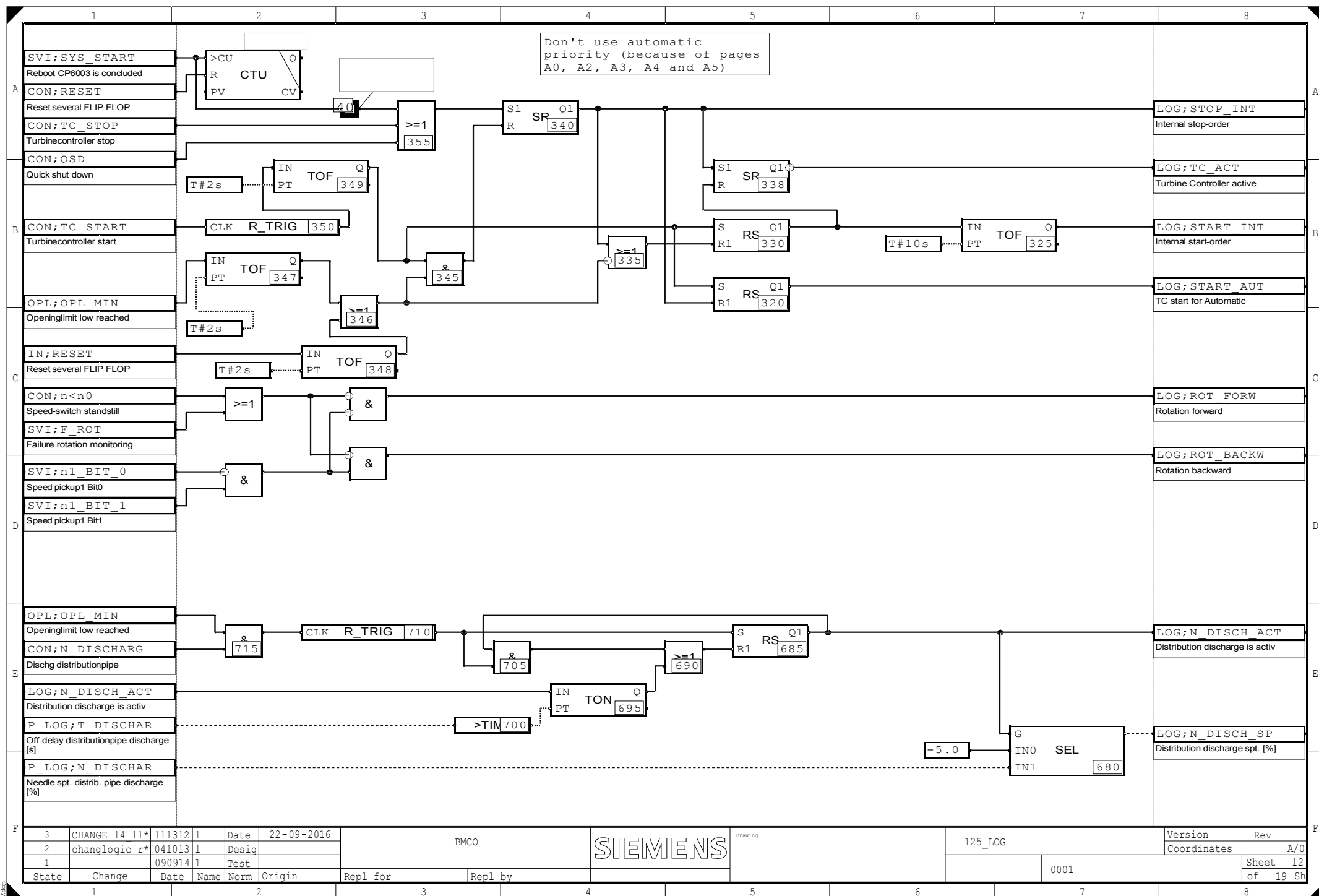


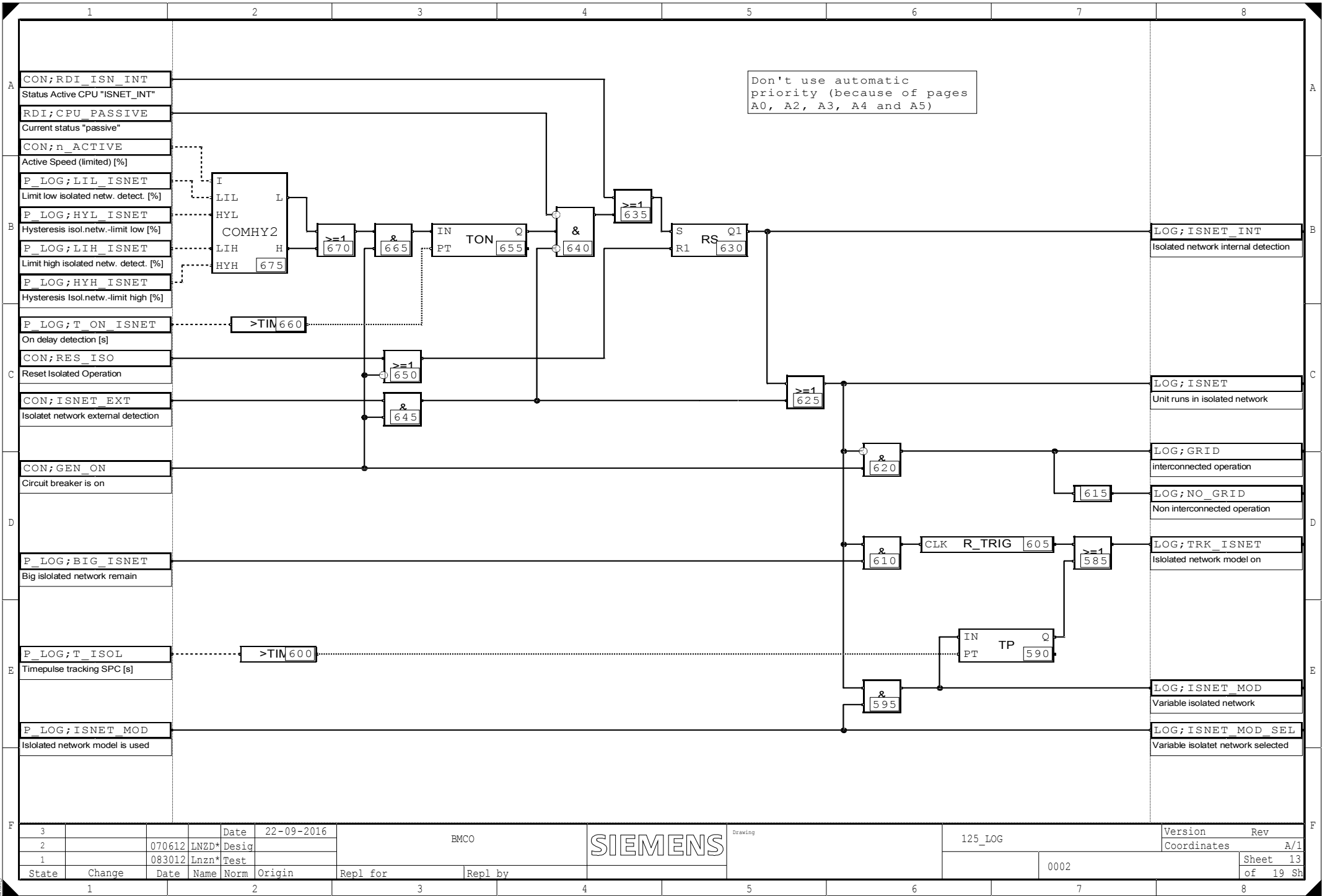
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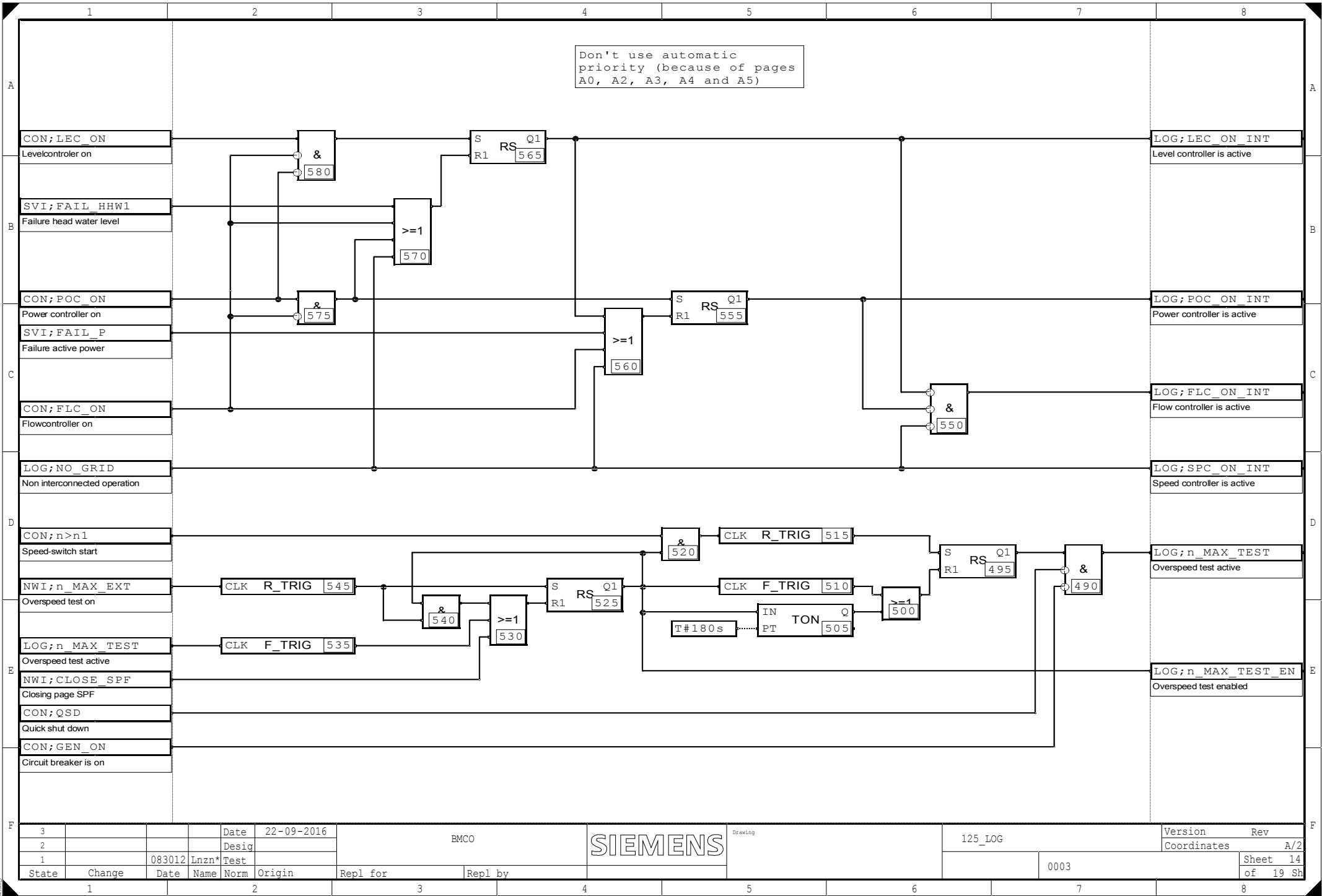


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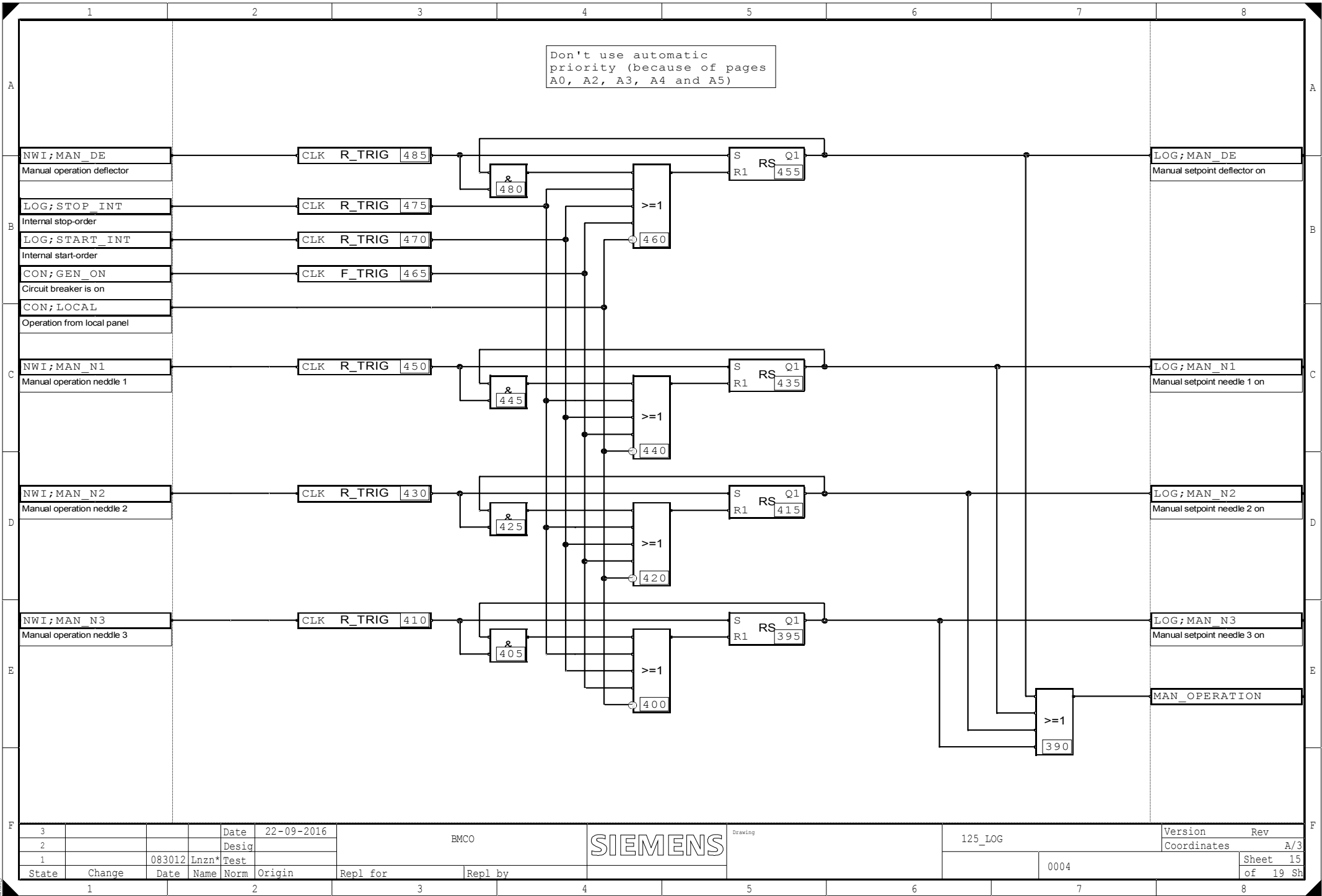




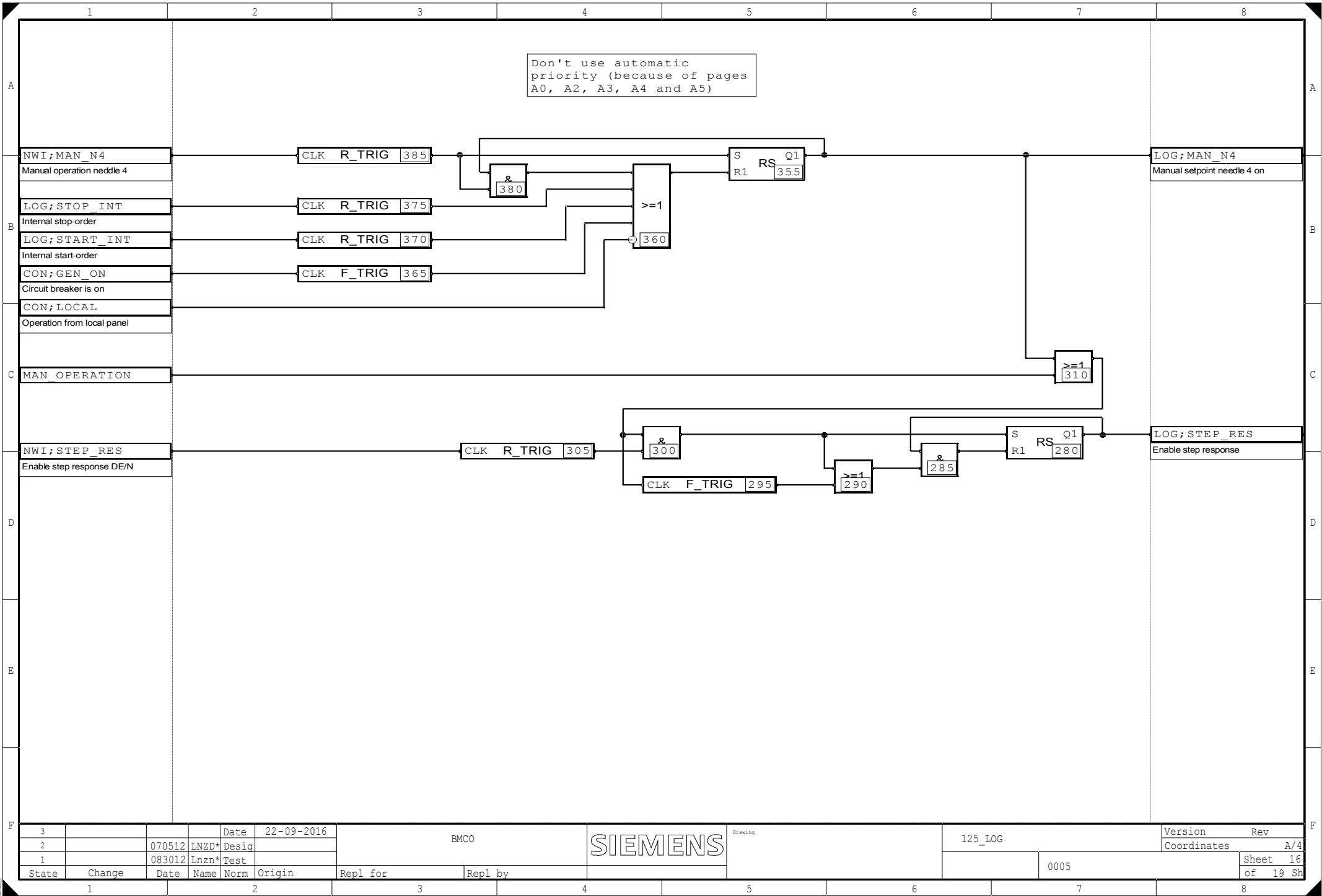
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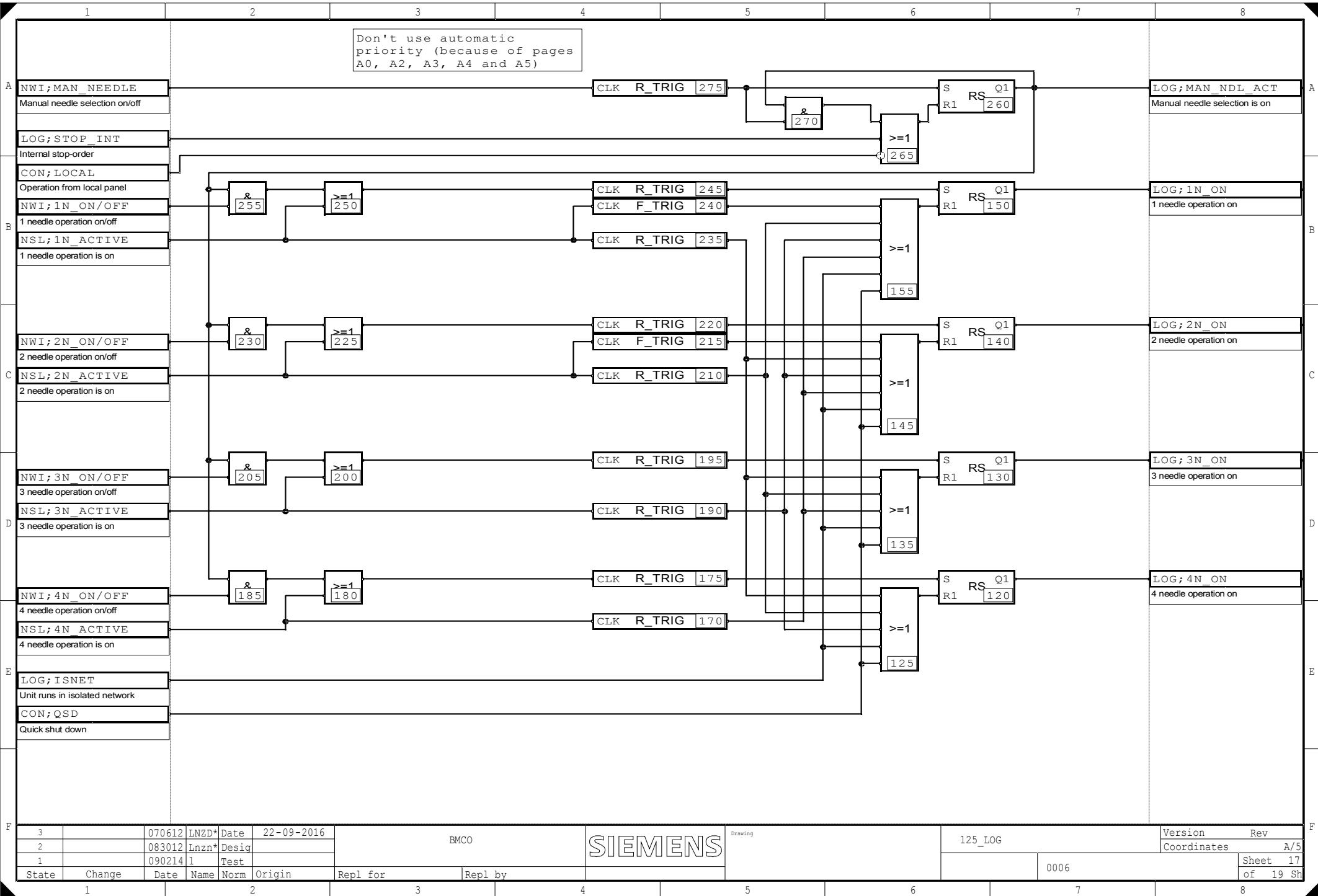
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State	Change	Date	Name	Norm	Origin	Repl for	Repl by			0003	Sheet 14	of 19 Sh

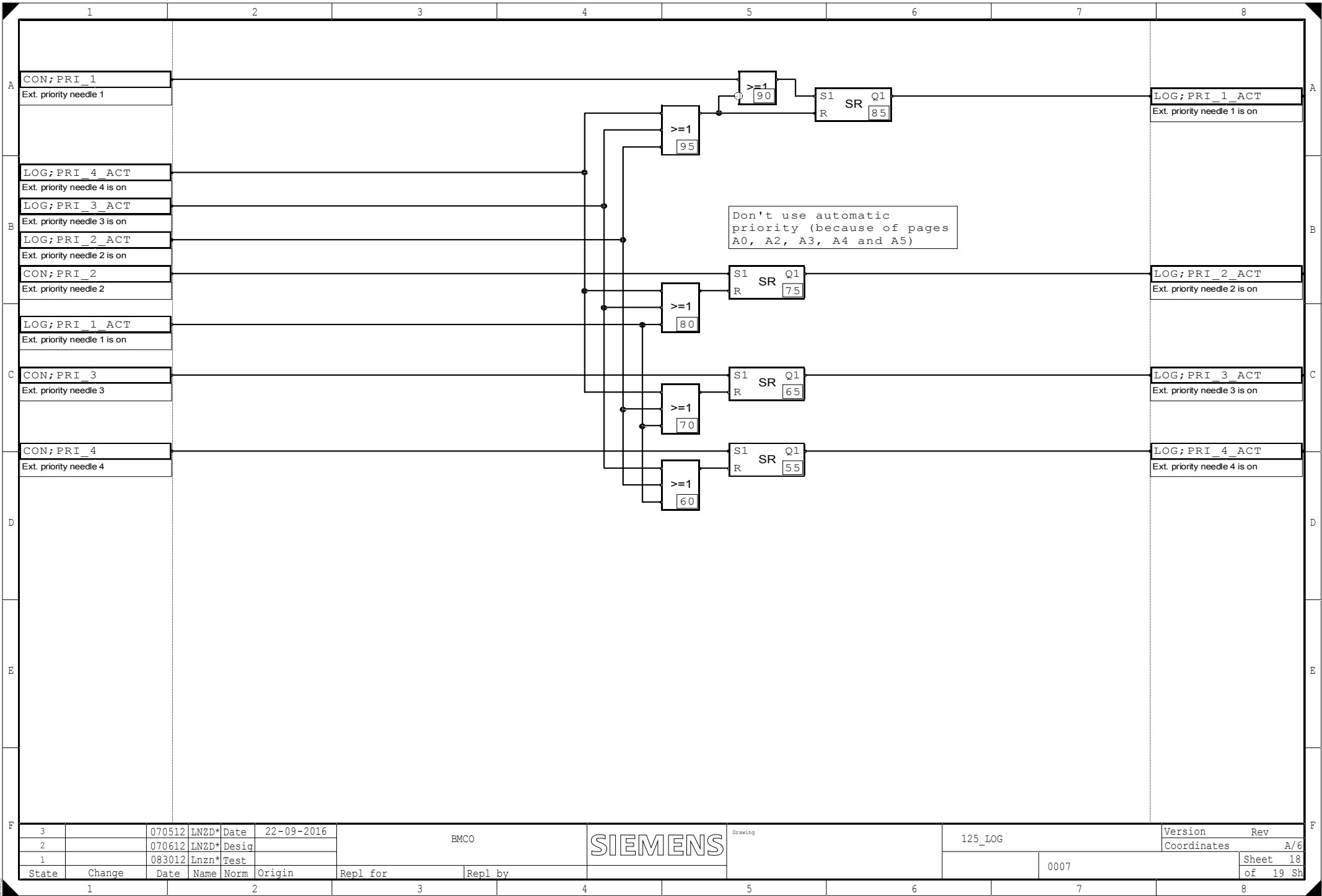


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1			083012	Lnzn*	Test										Sheet		15														
State		Change		Date	Name	Norm	Origin	Repl for		Repl by				0004		of		19 Sh													
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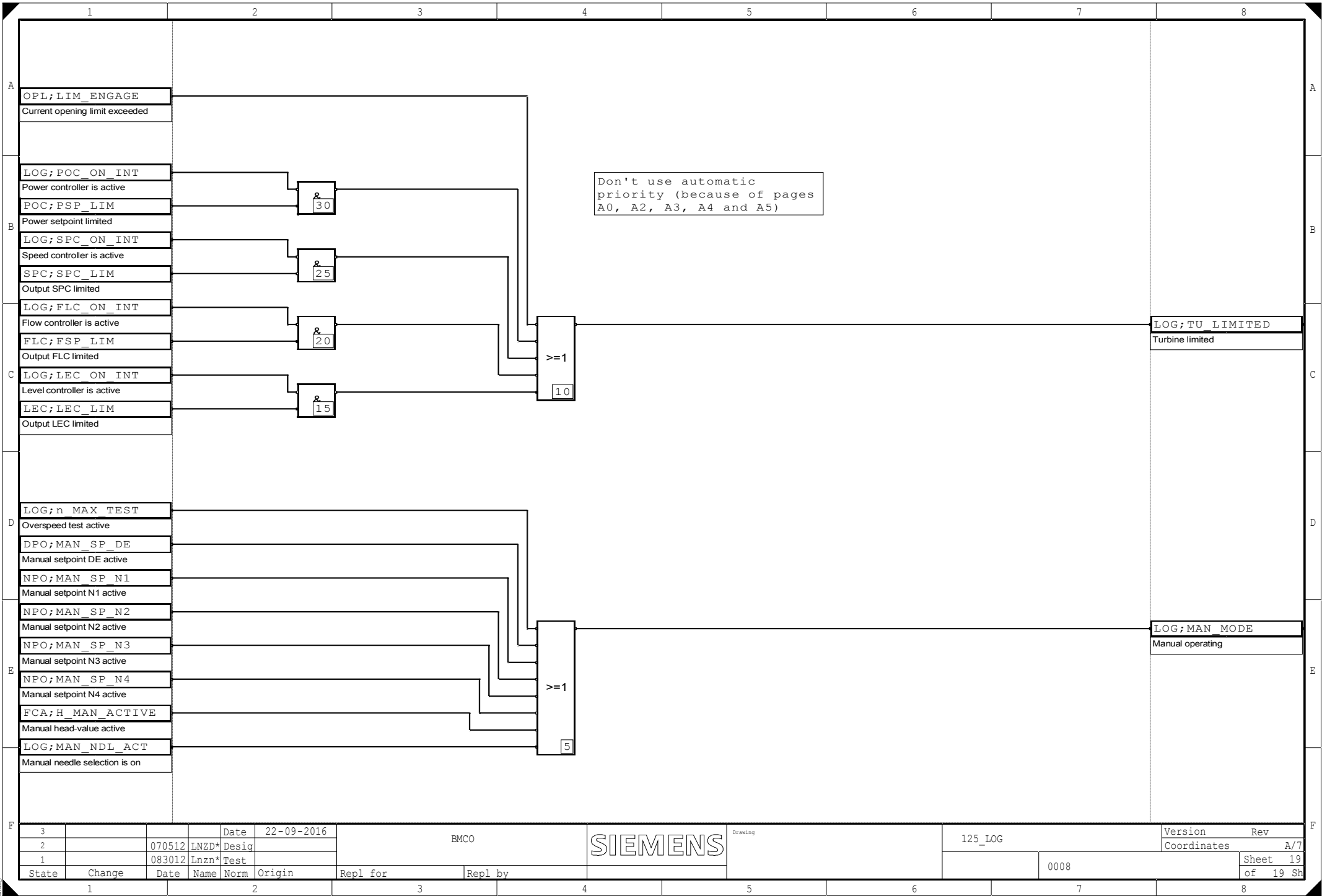


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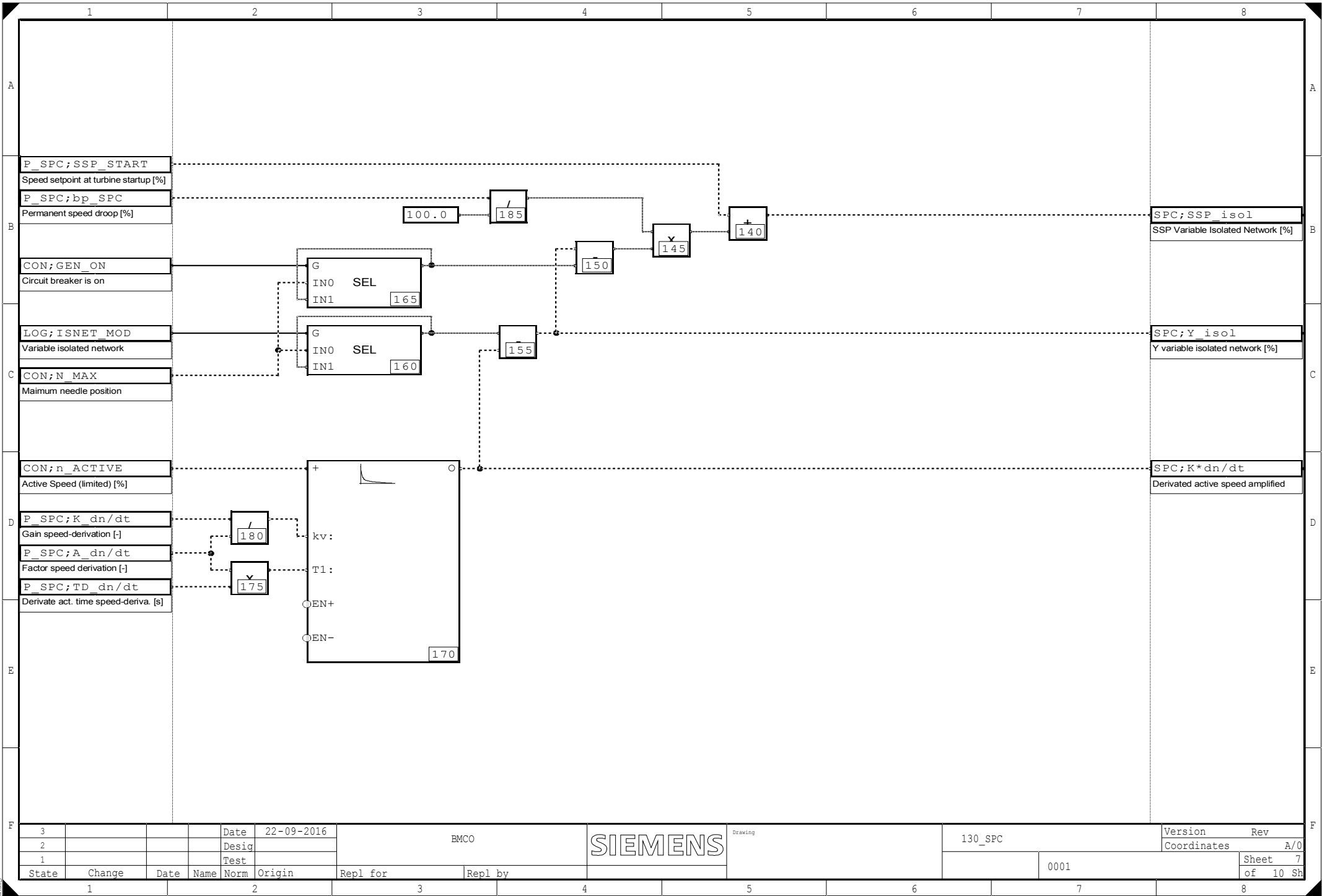




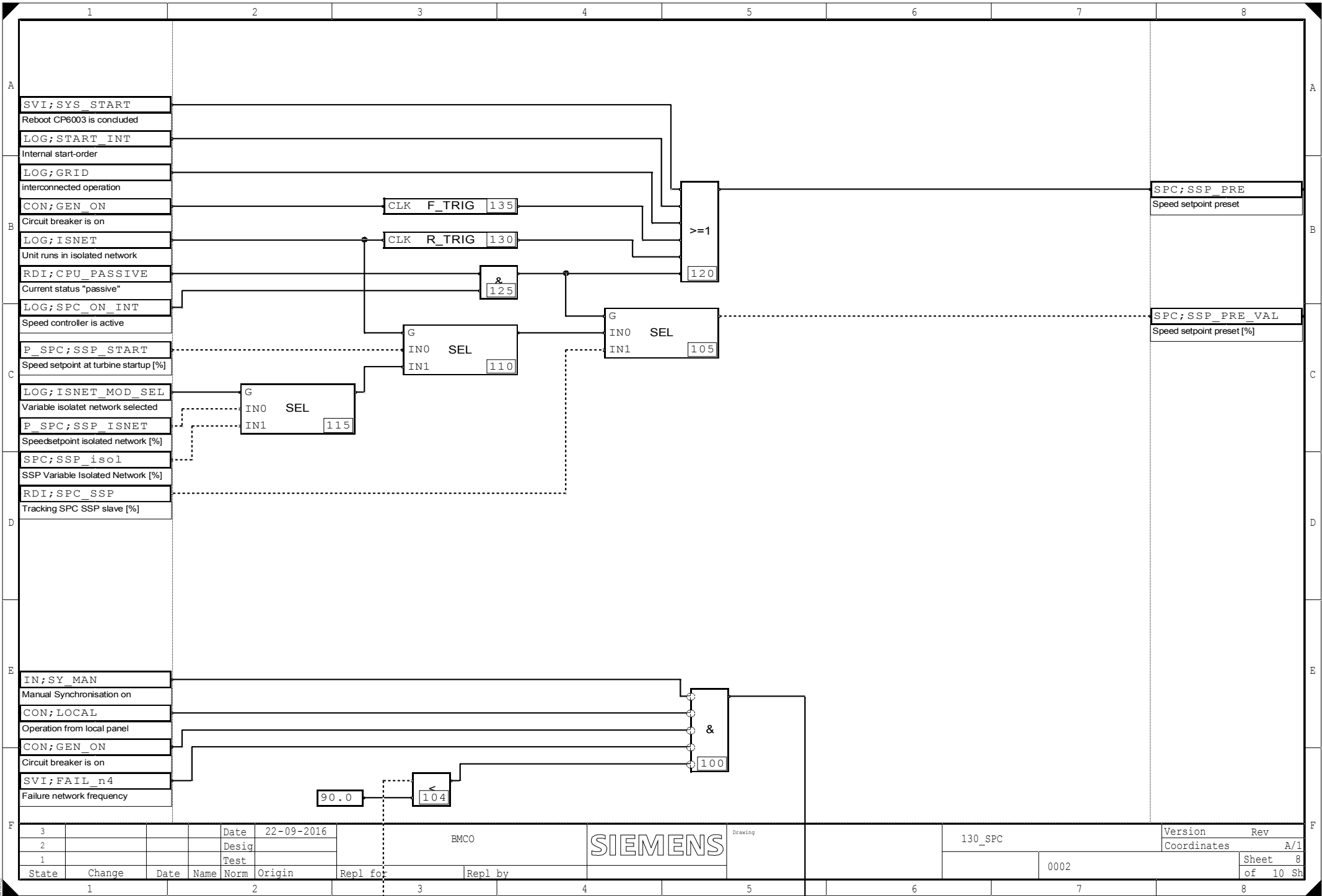
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2		070612	LNZD*	Desig								Coordinates	A/6
1		083012	Lnzn*	Test								Sheet	18
State	Change	Date	Name	Norm	Origin	Repl for	Repl by				0007	of 19	Sh



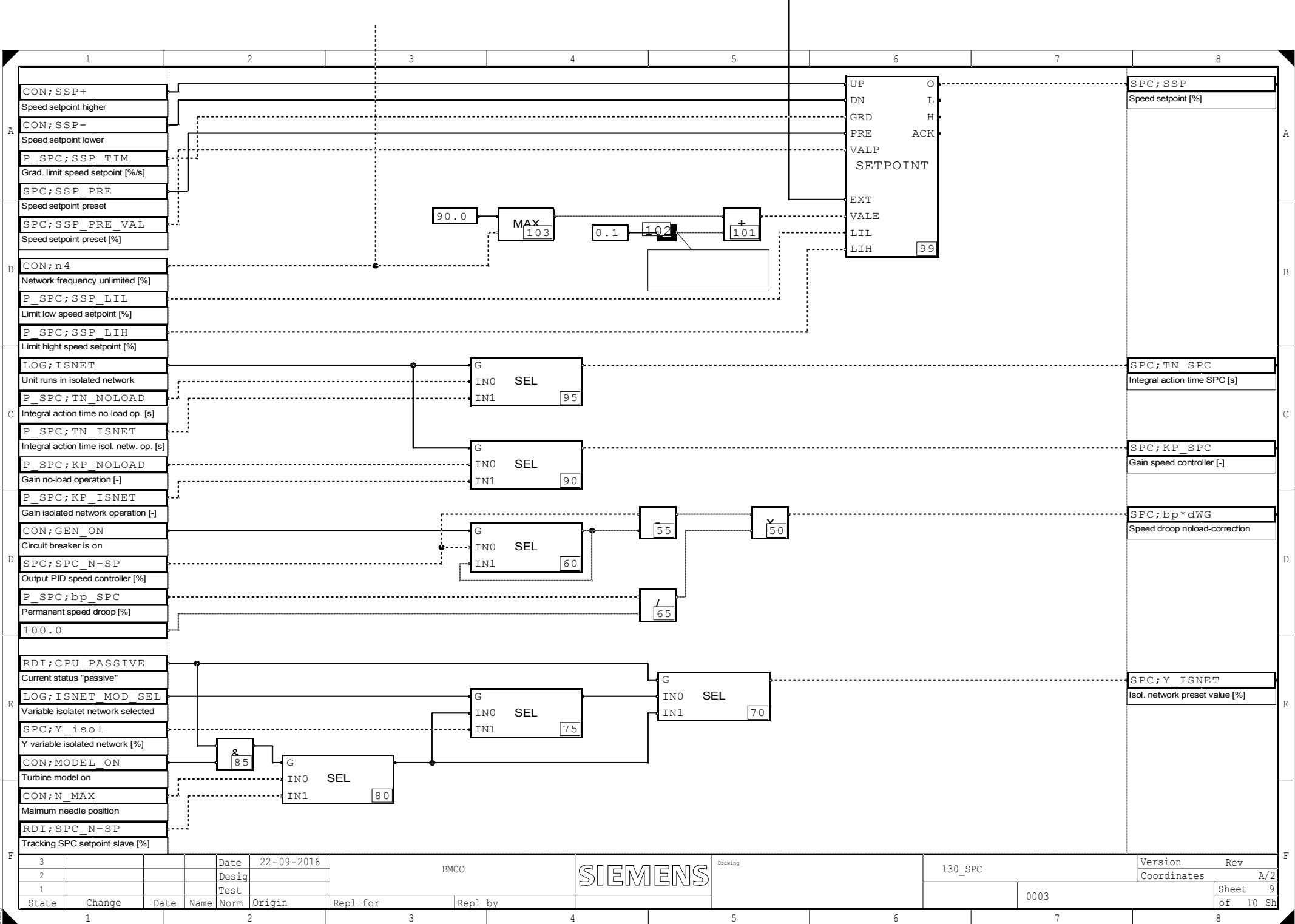
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State	Change	Date	Name	Norm	Origin	Repl for	Repl by			of	19 Sh

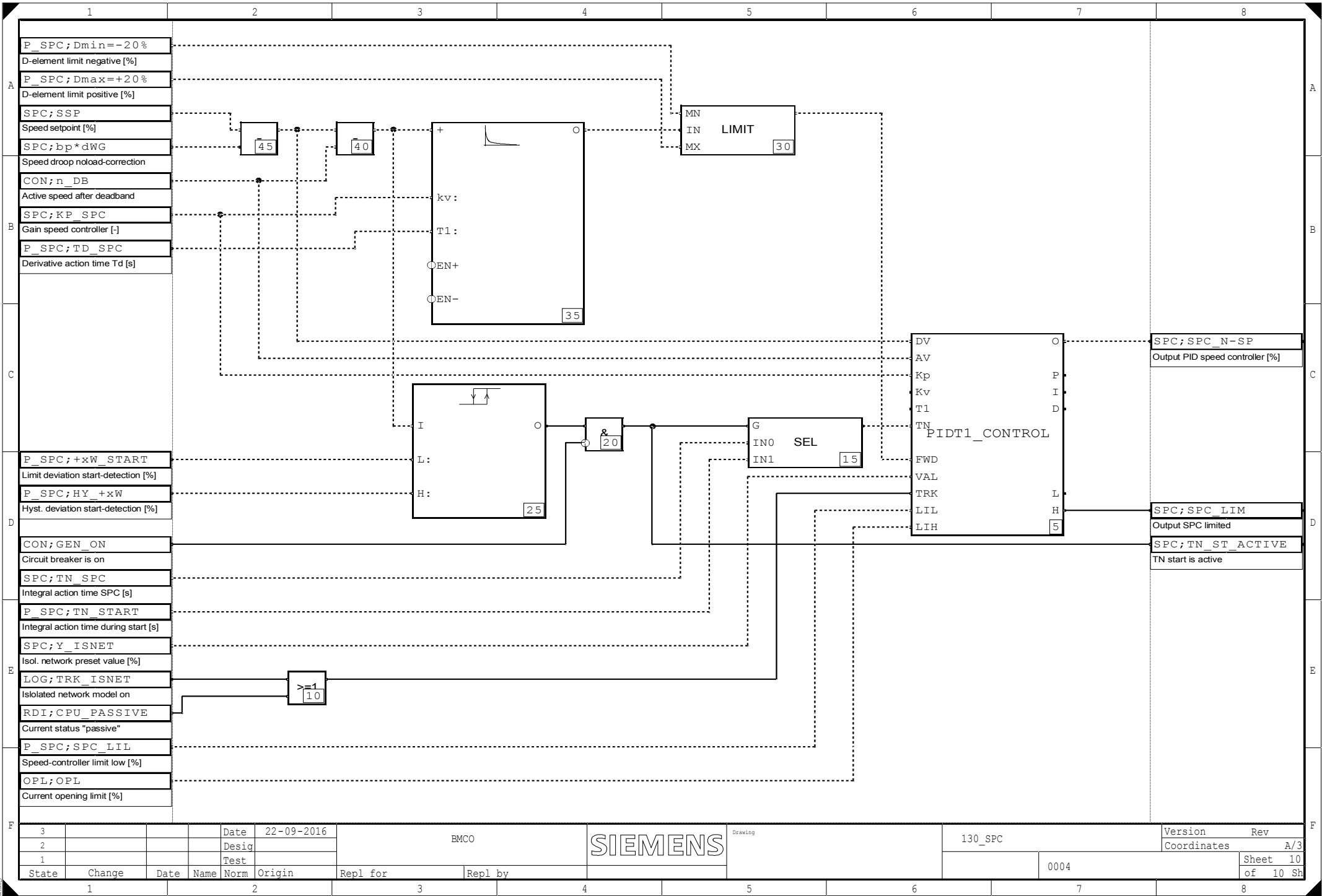


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State	Change	Date	Name	Norm	Origin	Repl for	Repl by					0001		Sheet	7
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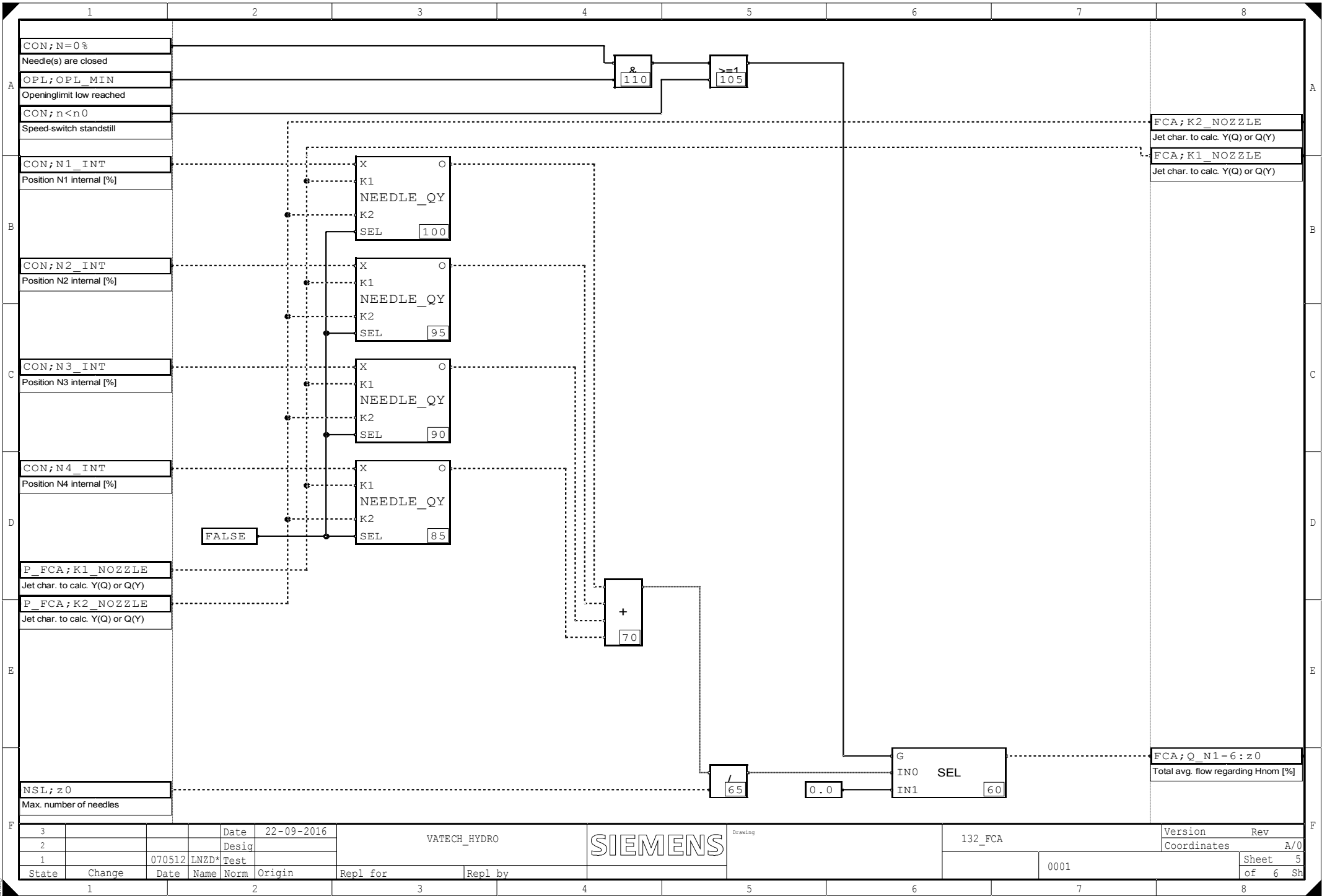


3			Date	22-09-2016	BMC0		SIEMENS	Drawing	130_SPC		Version	Rev
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State	Change	Date	Name	Norm	Origin	Repl for	Repl by		0002		of 10 Sh	

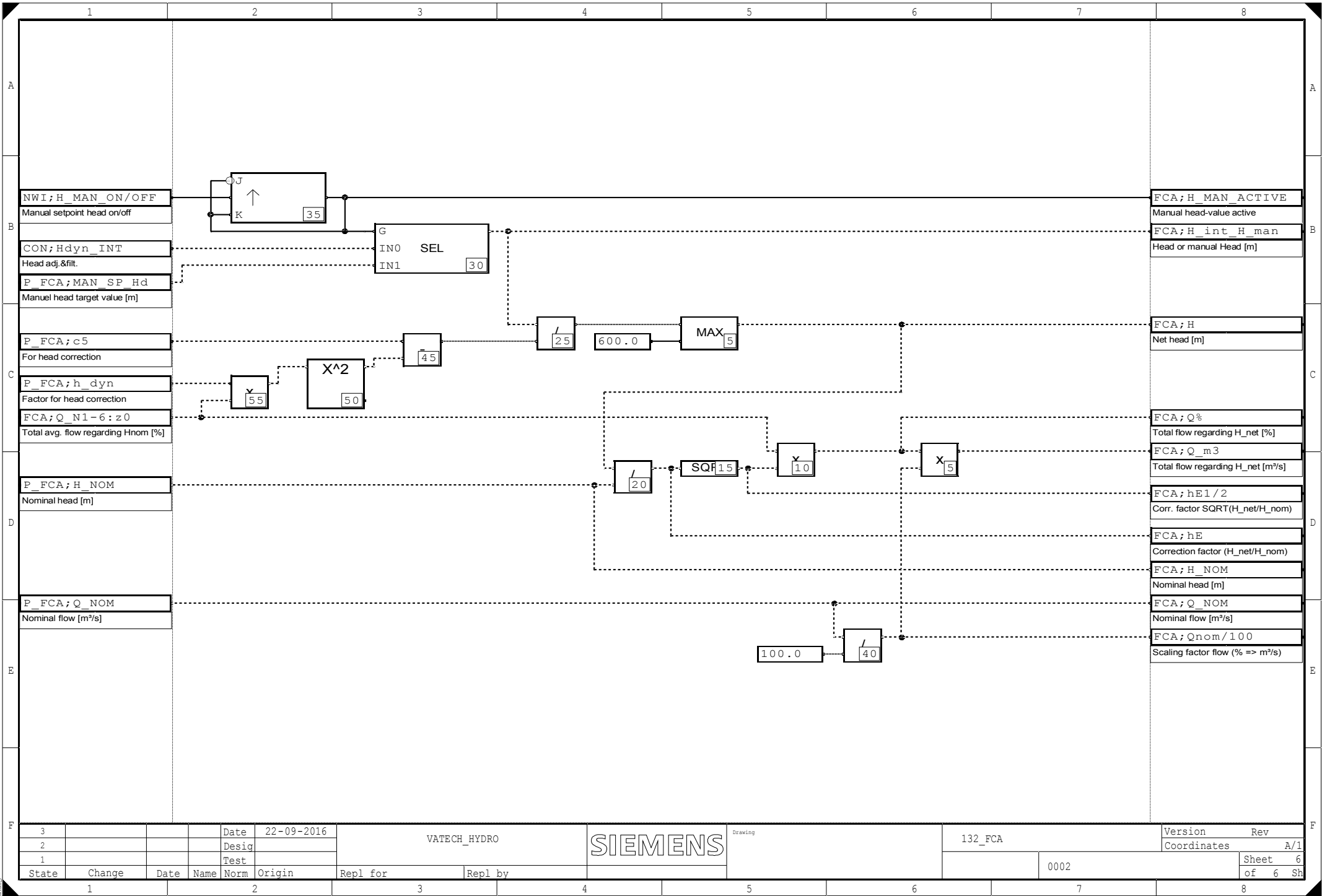




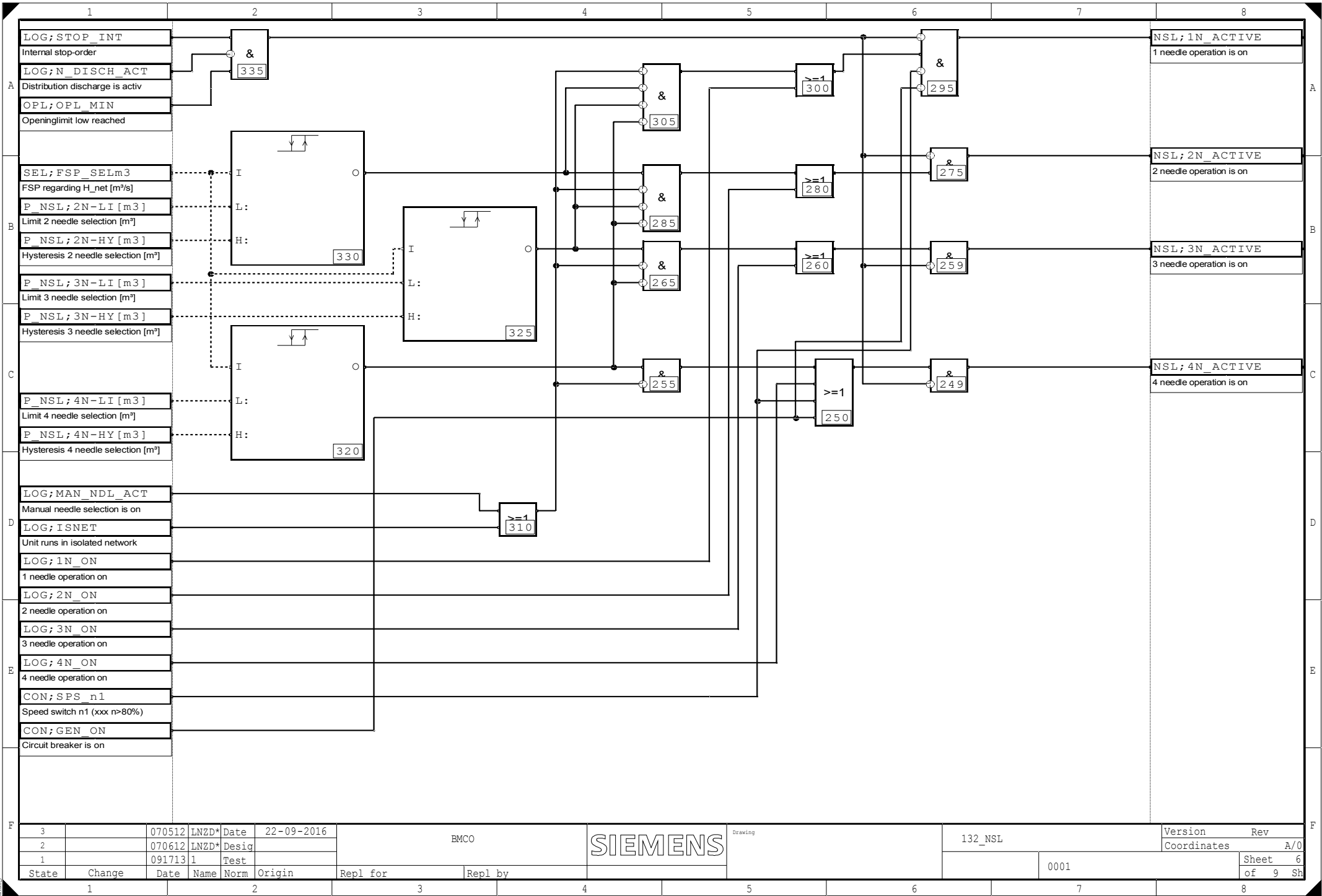
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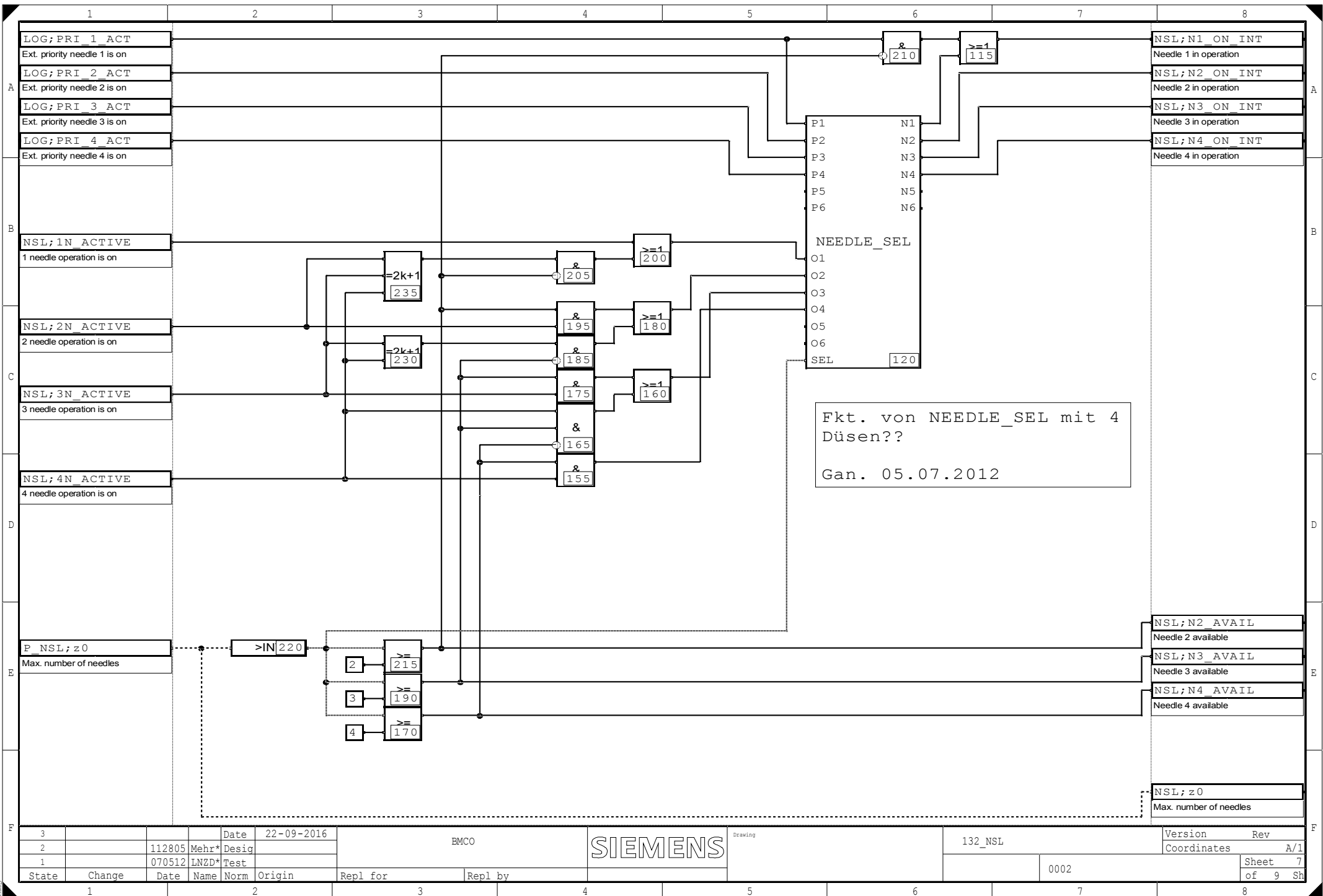
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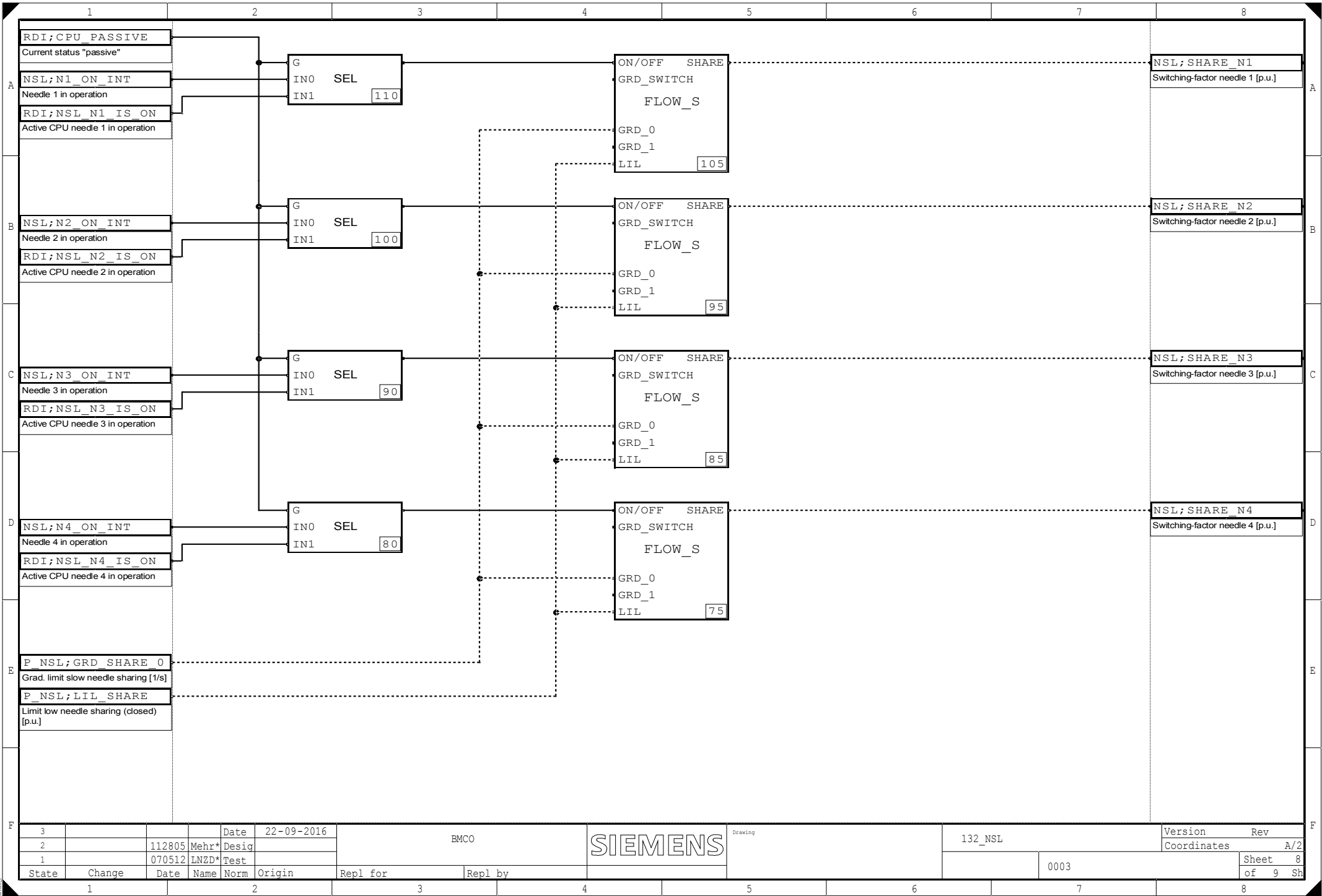


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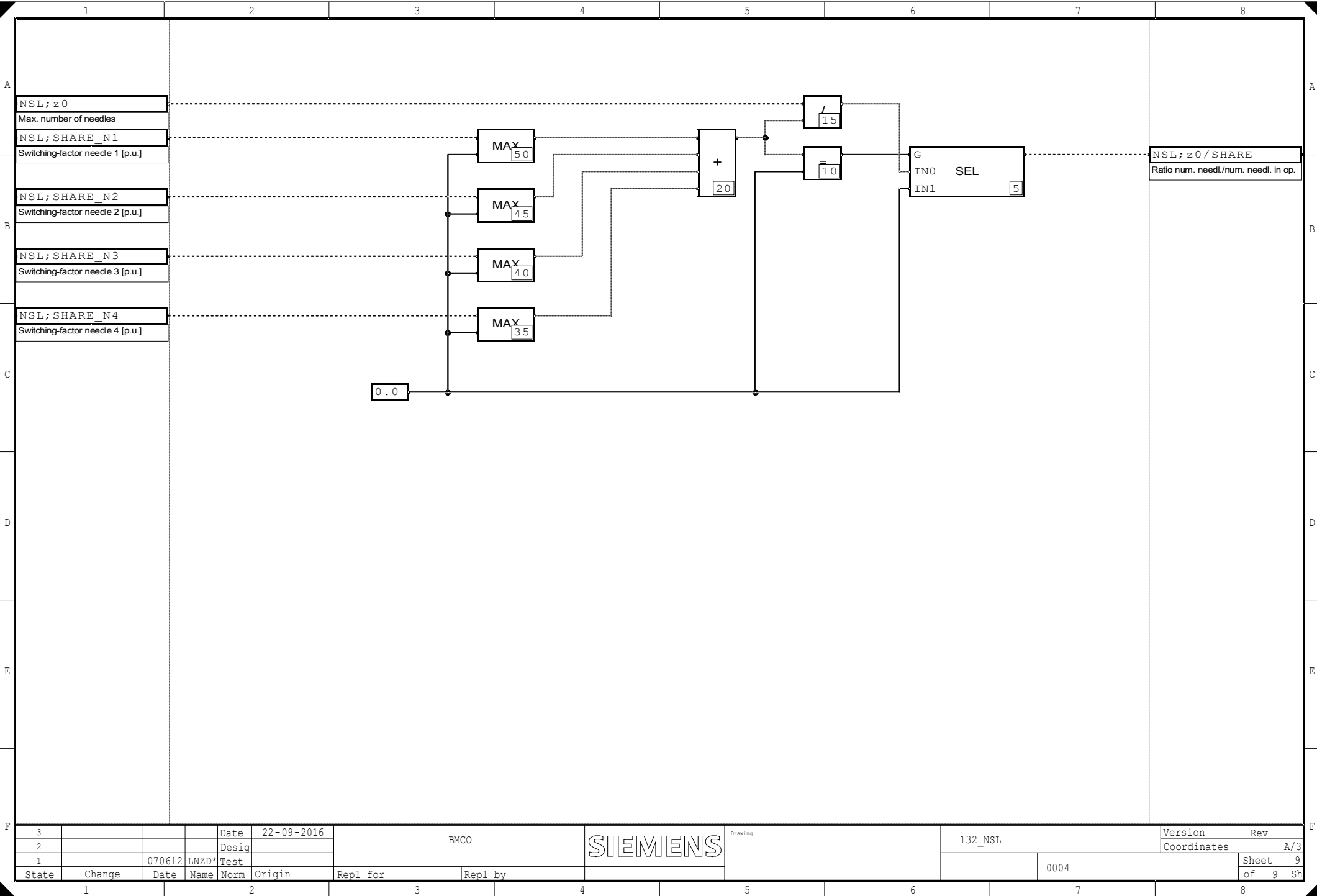


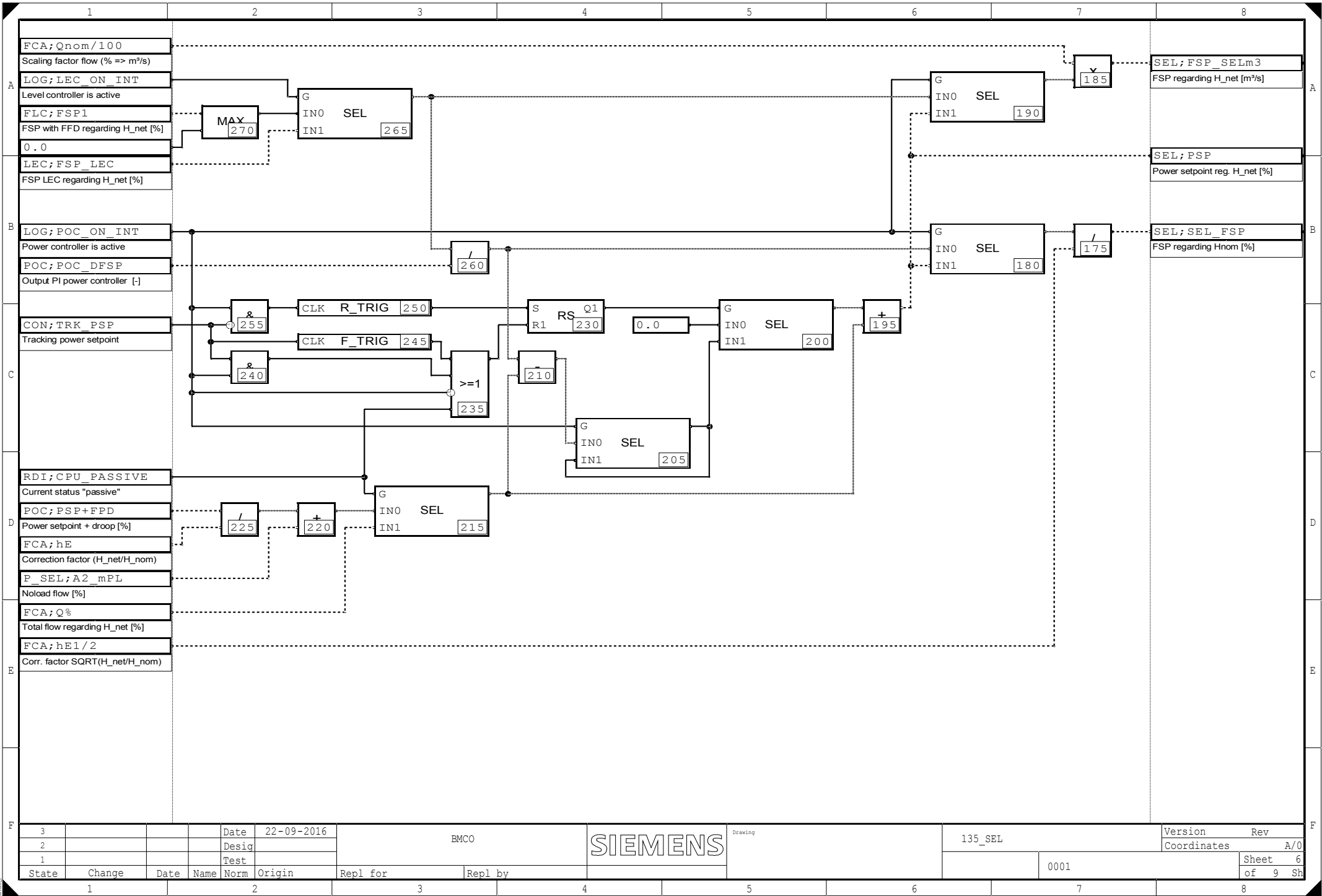
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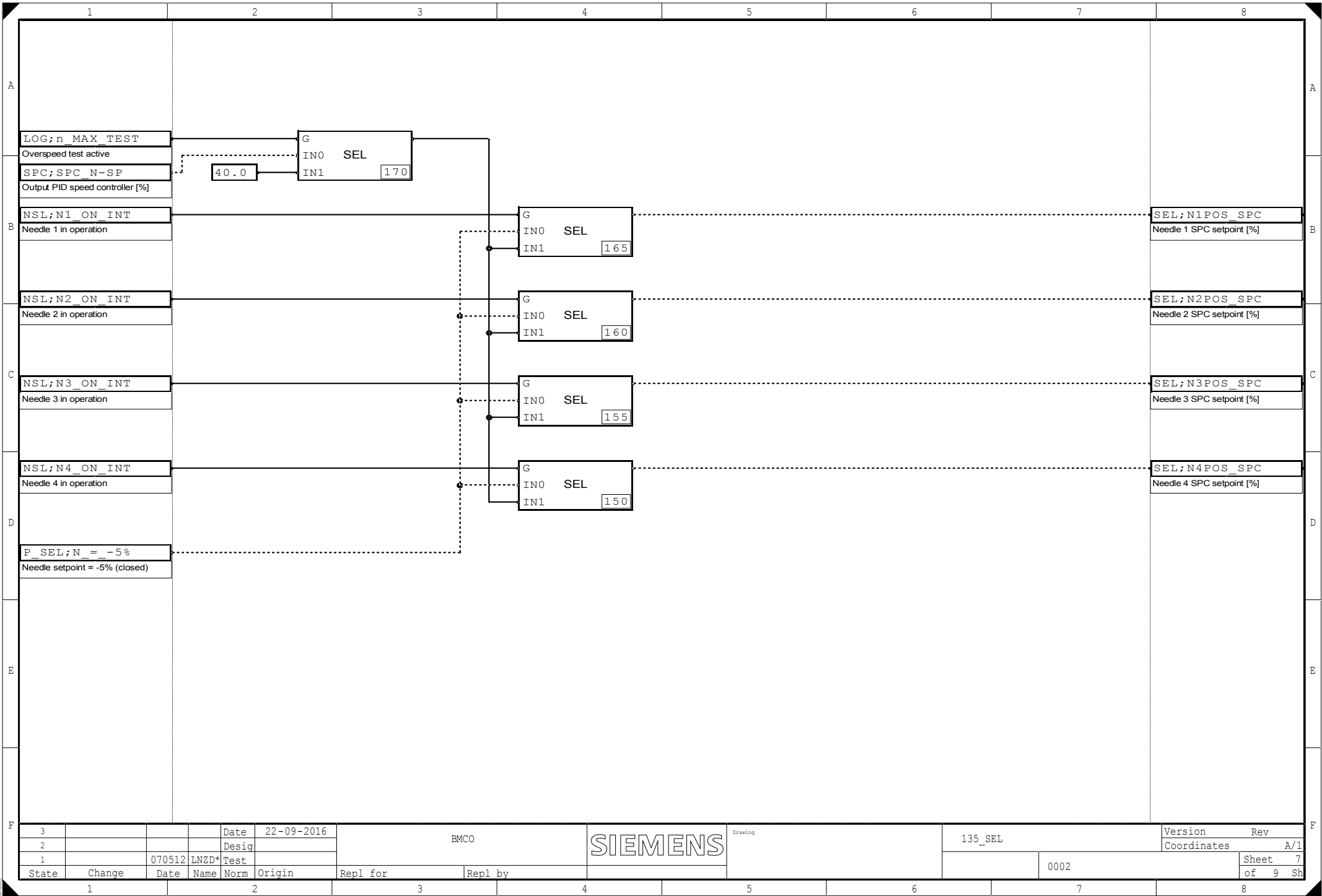


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State	Change	Date	Name	Norm						of 9	Sh

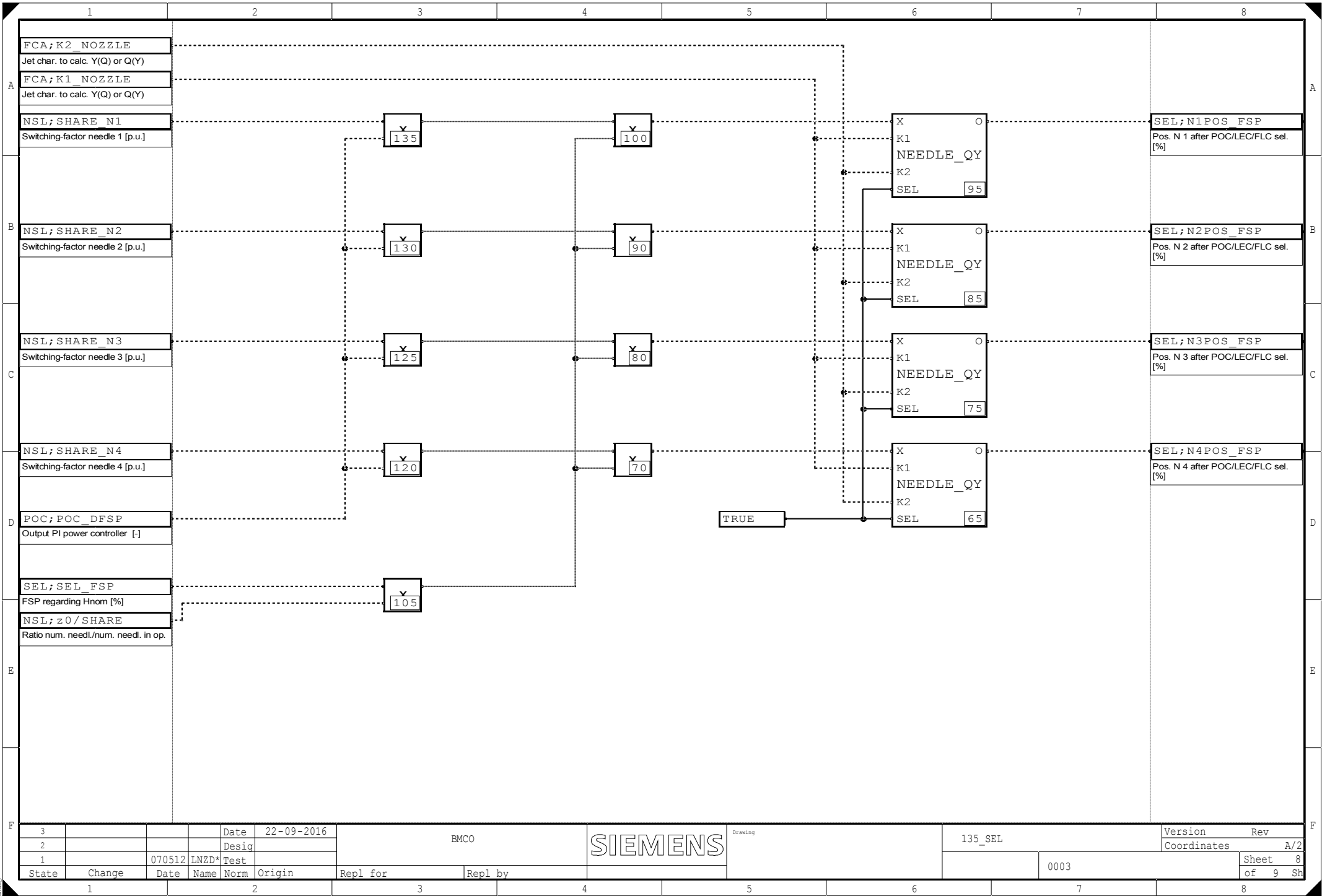




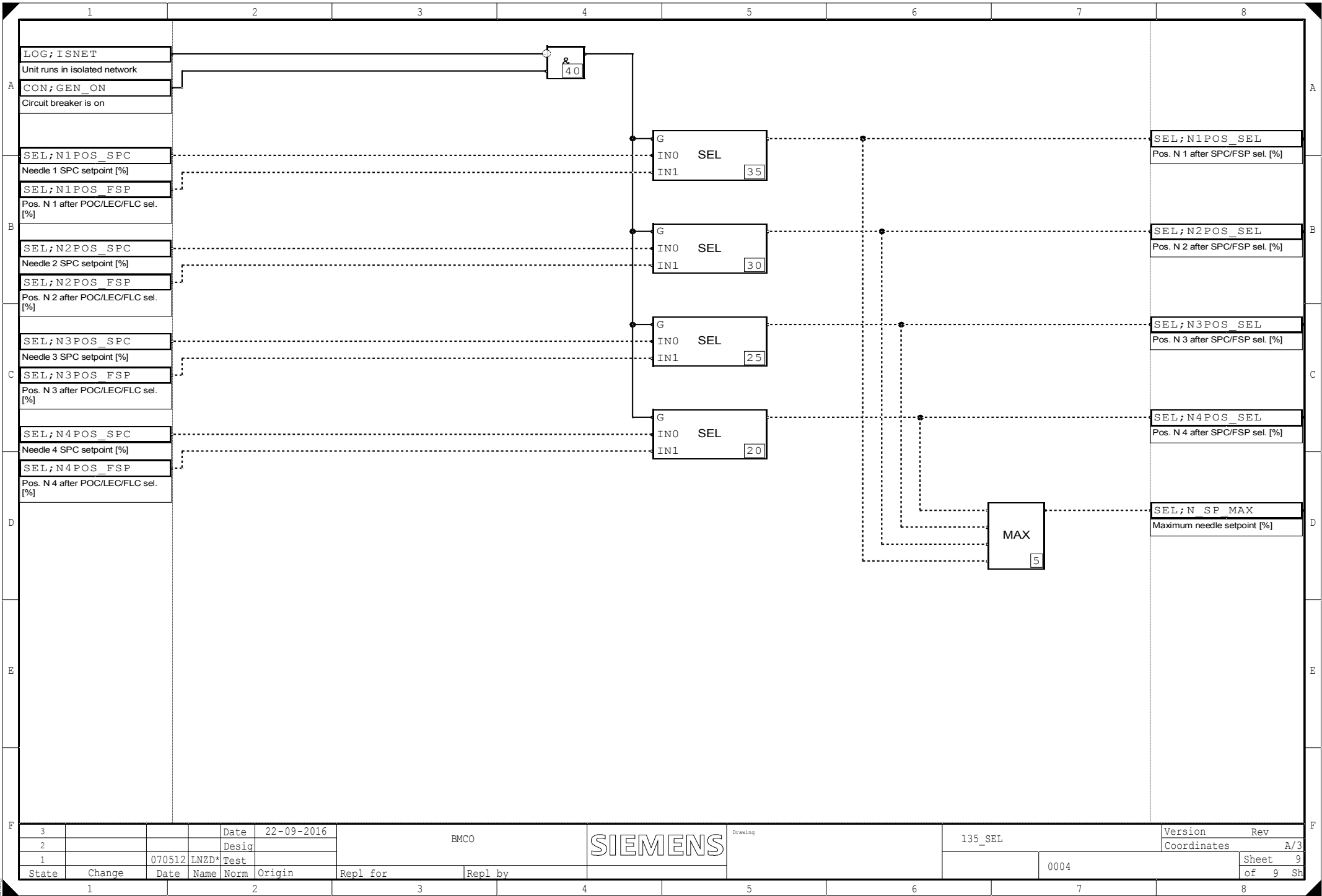
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State	Change	Date	Name	Norm	Origin	Repl for	Repl by				0001	Sheet	6
1												of 9	Sh



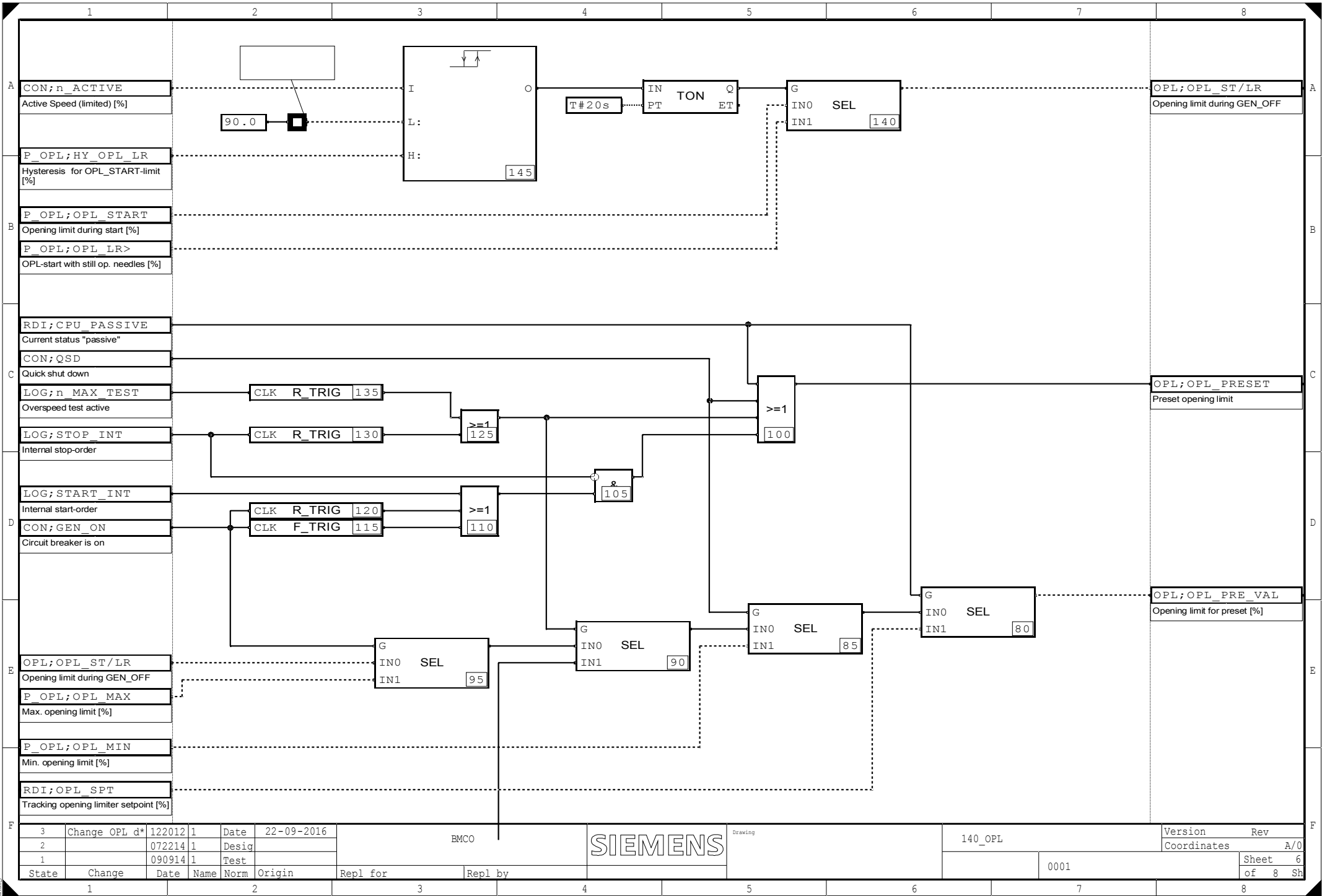
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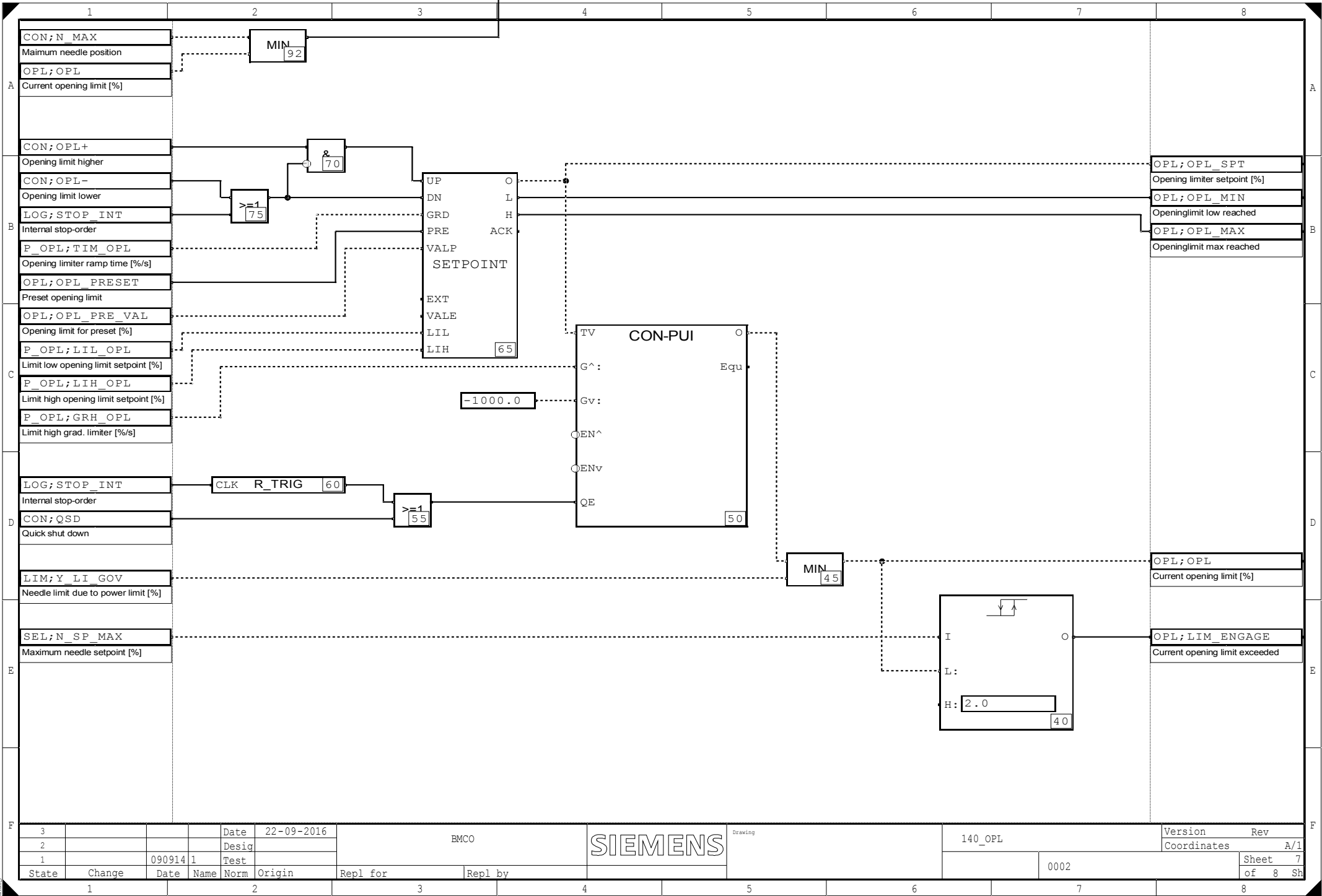


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State	Change	Date	Name	Norm						of	9 Sh

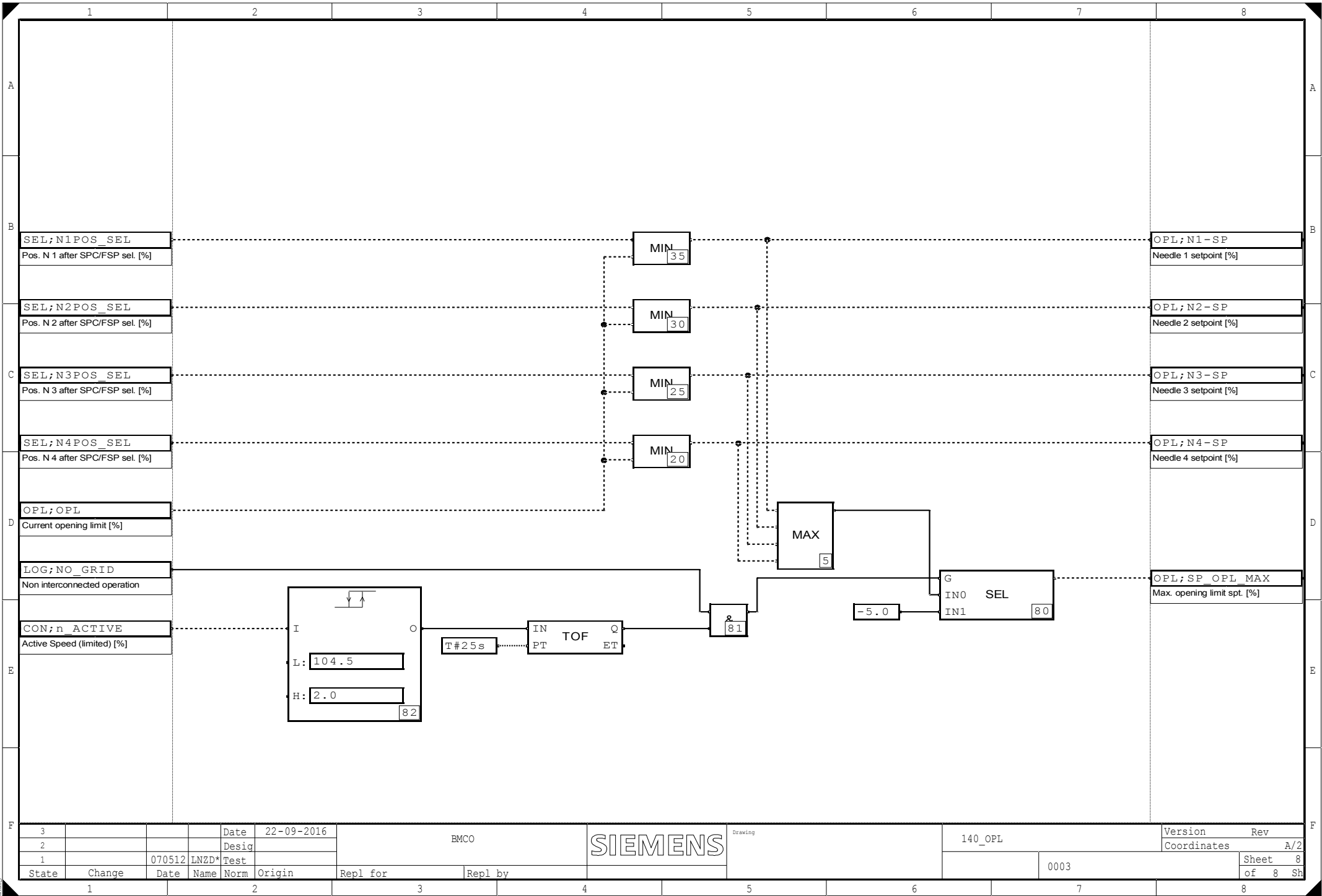


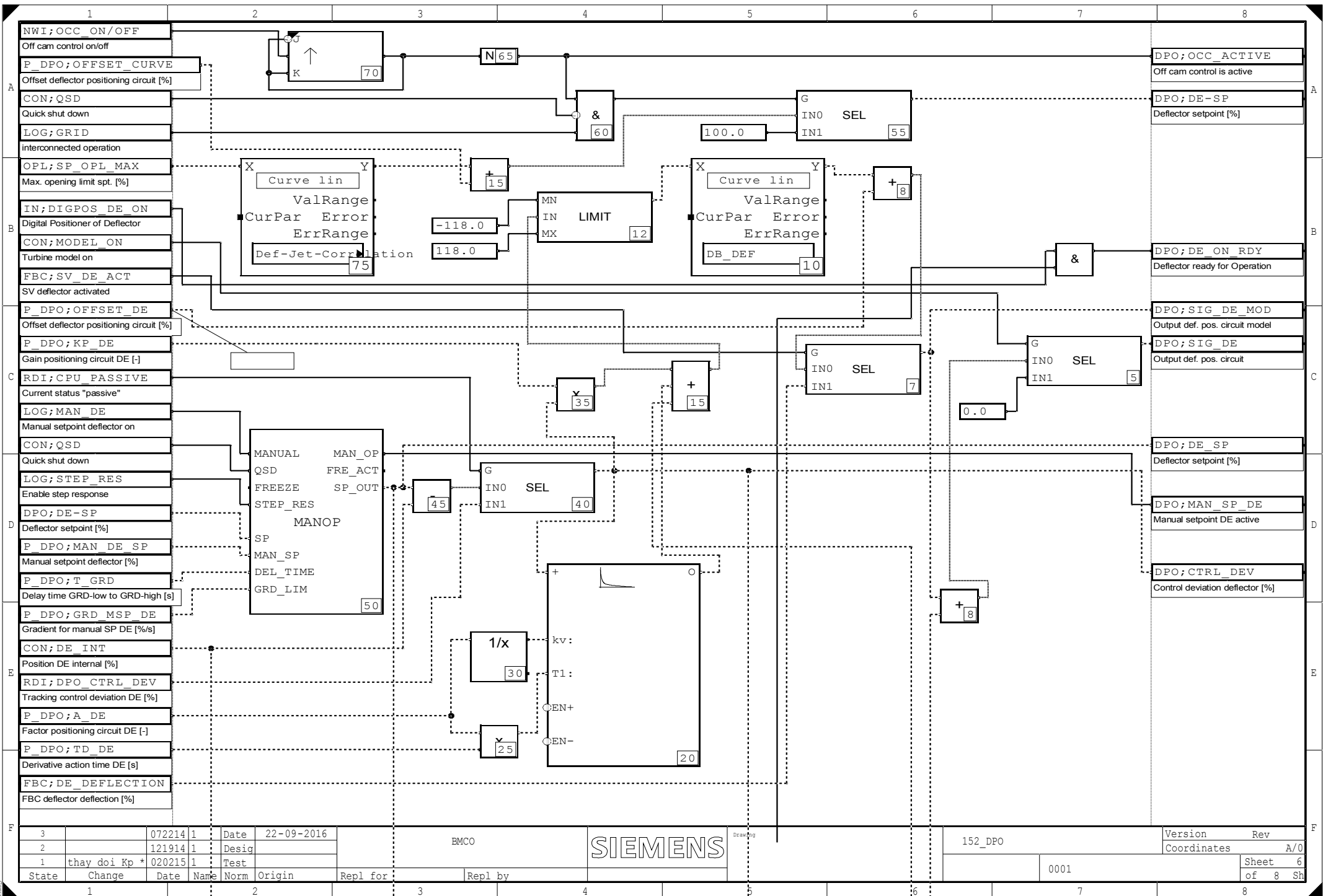
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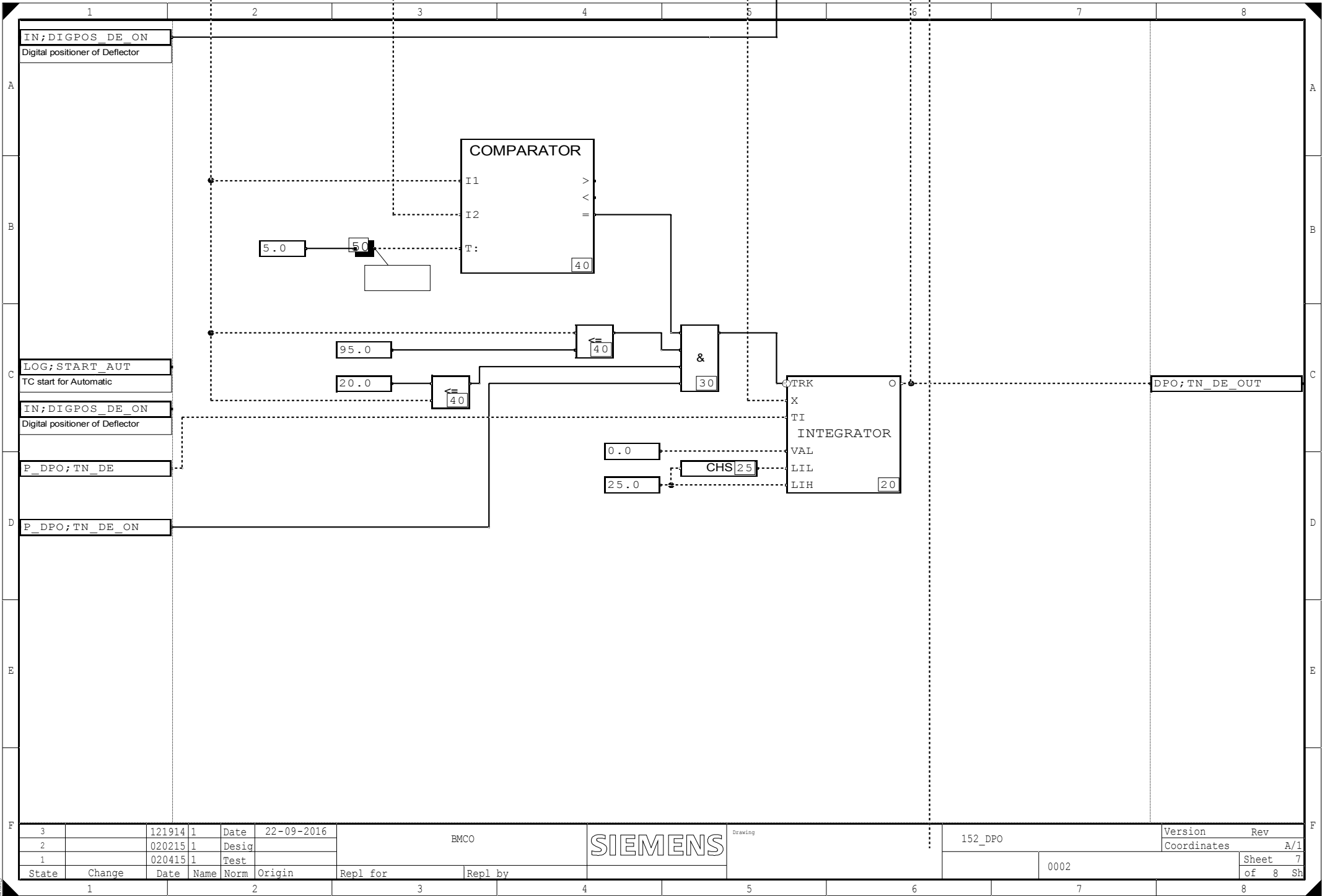


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State	Change	Date	Name	Norm	Origin	Repl for	Repl by			0002	of 8 Sh			

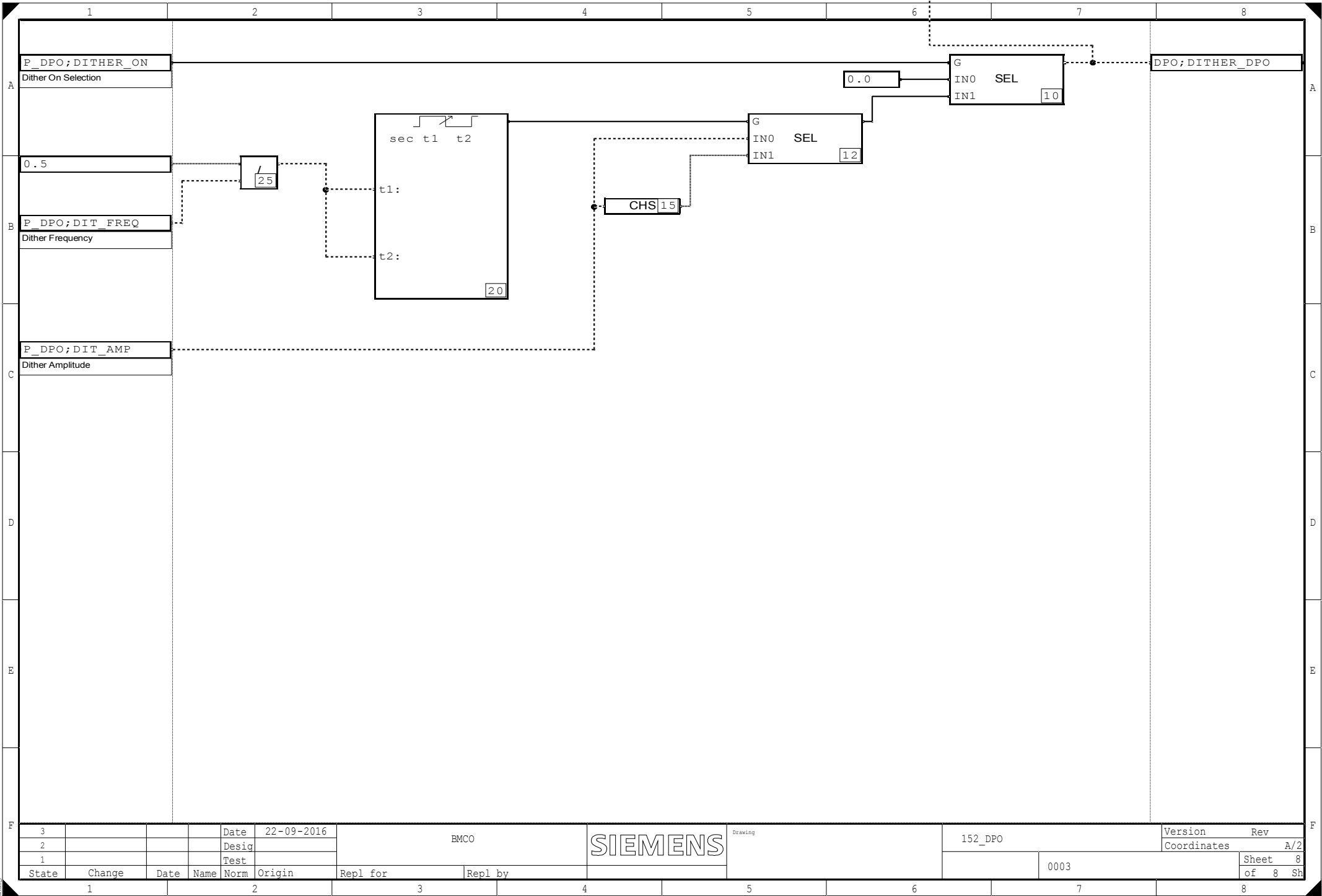




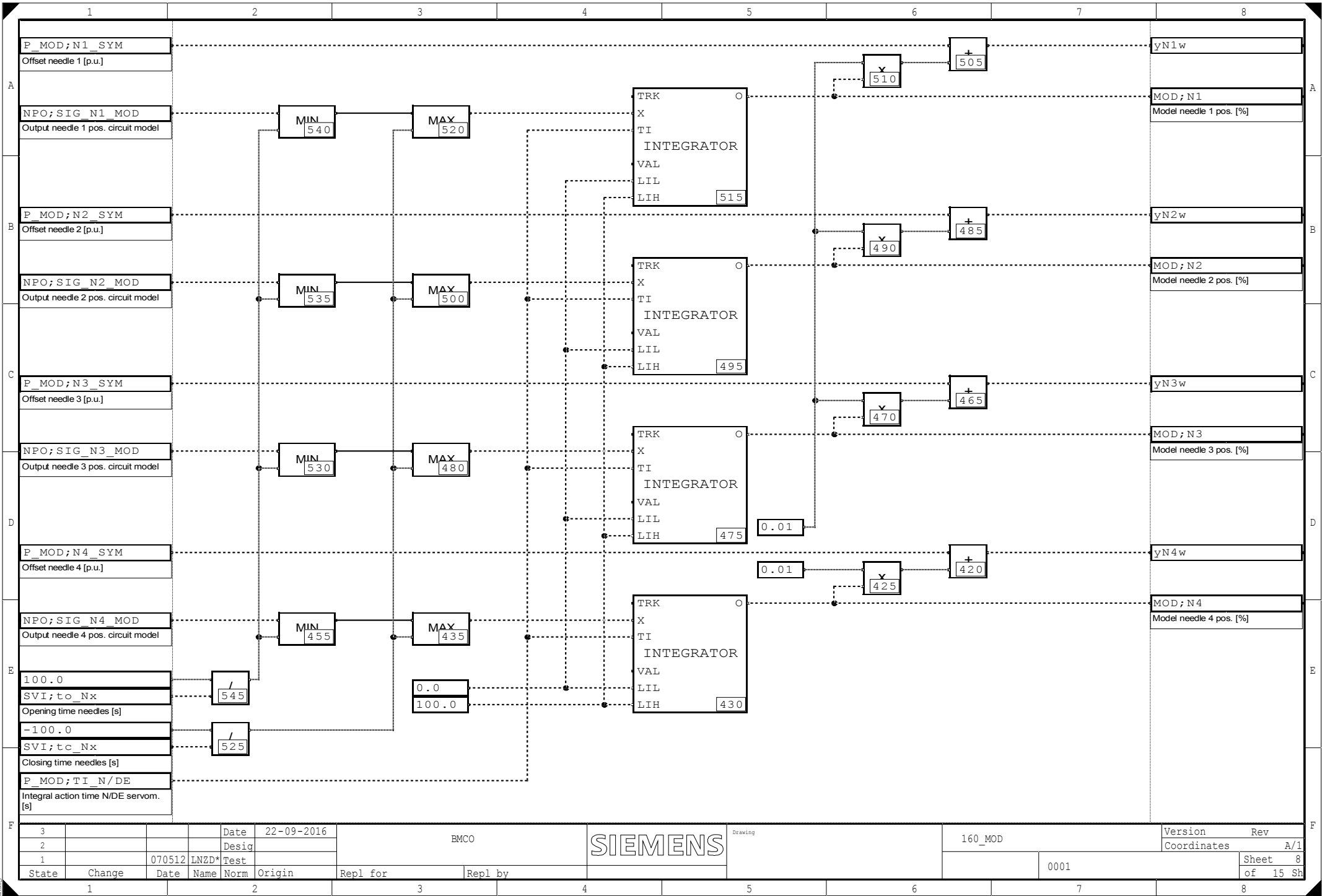
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1		thay doi Kp *		020215	1	Test											Sheet		6		
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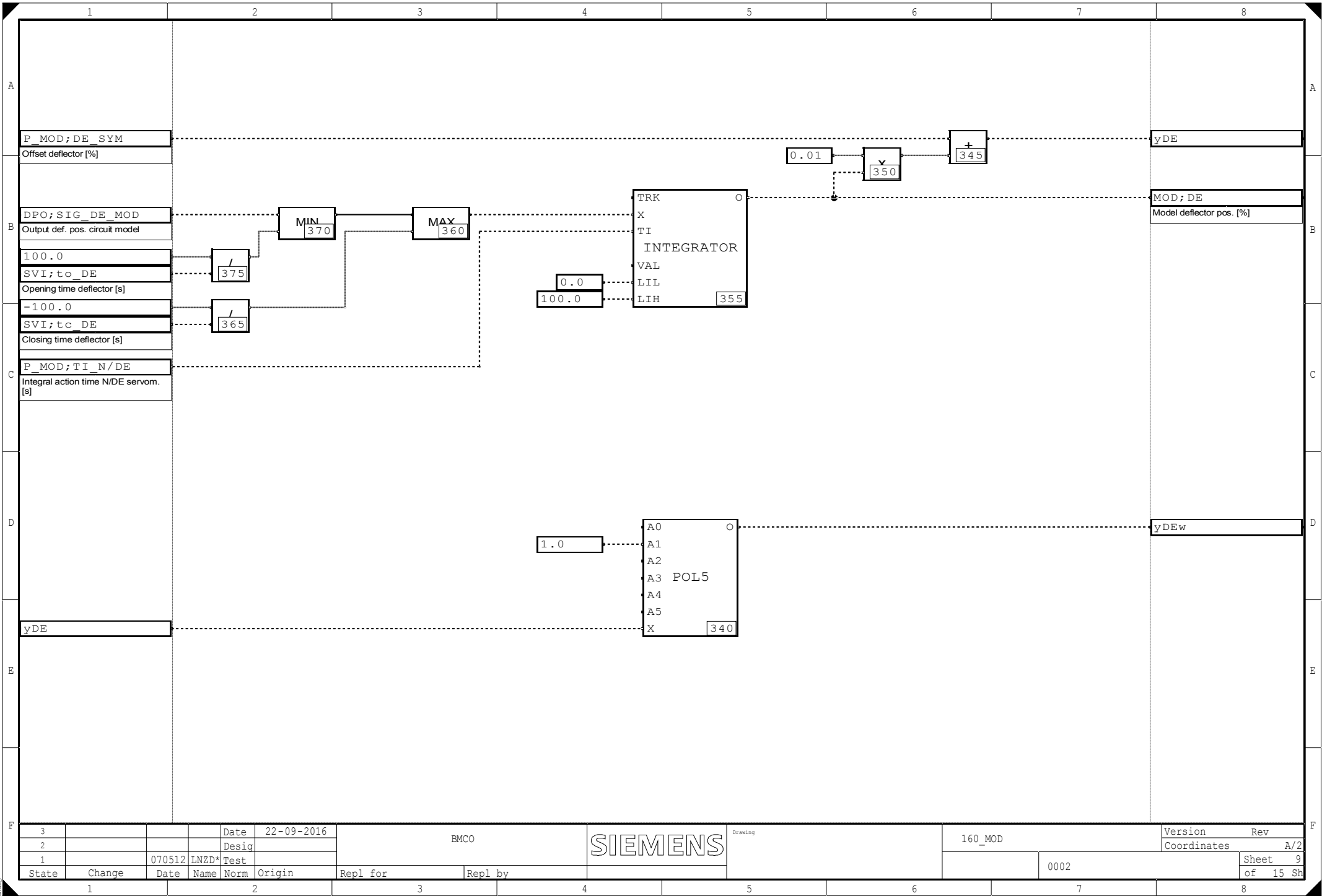


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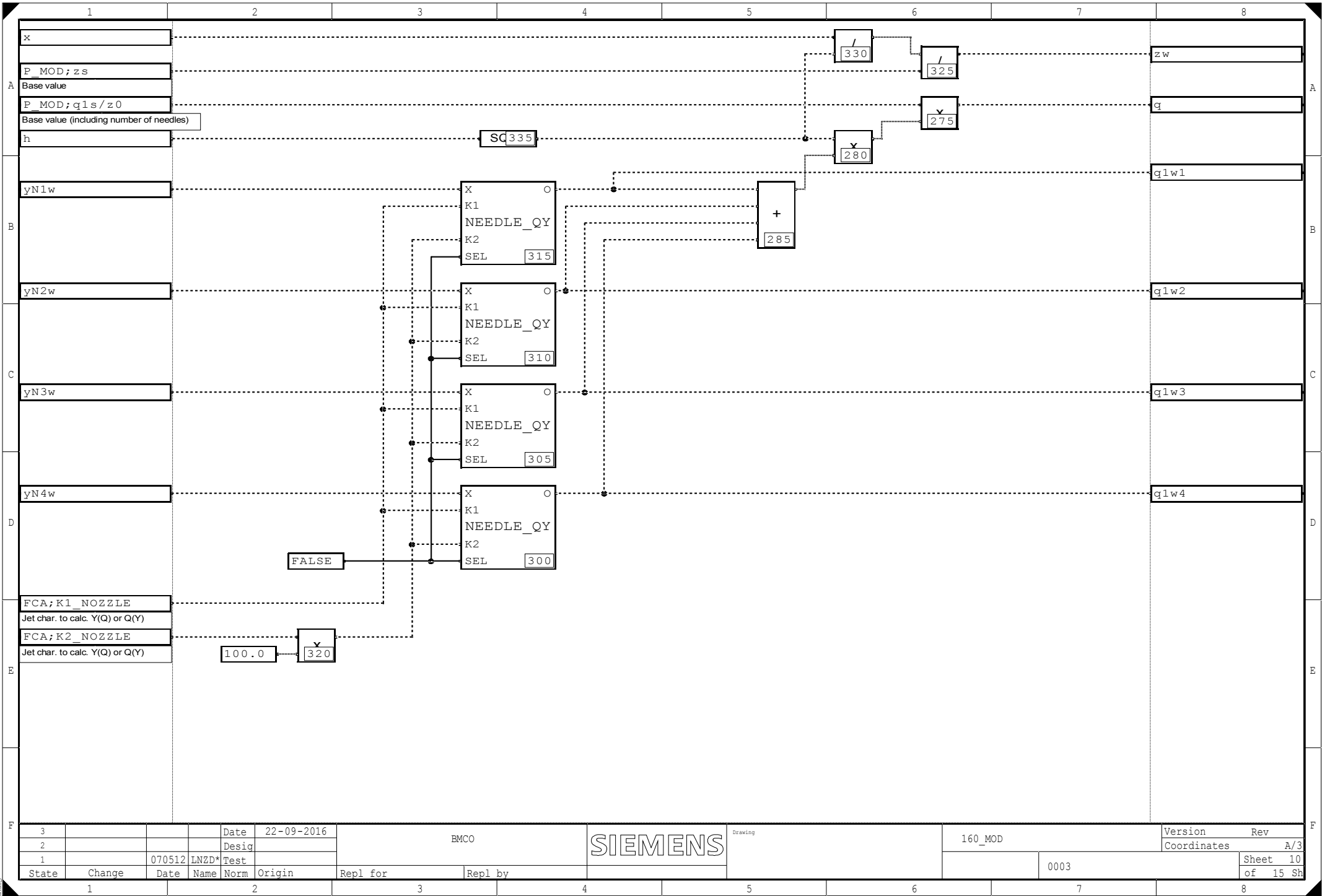


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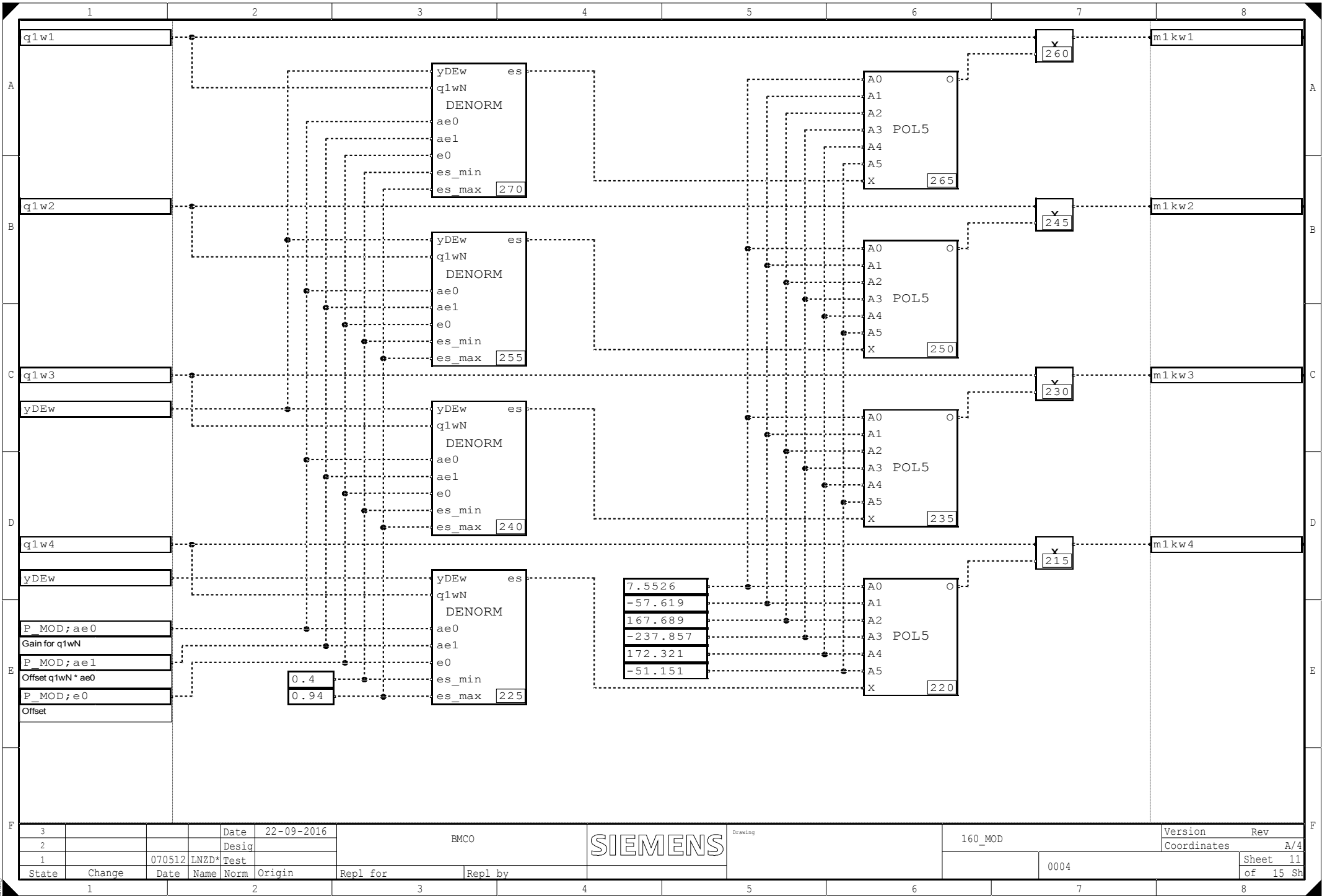




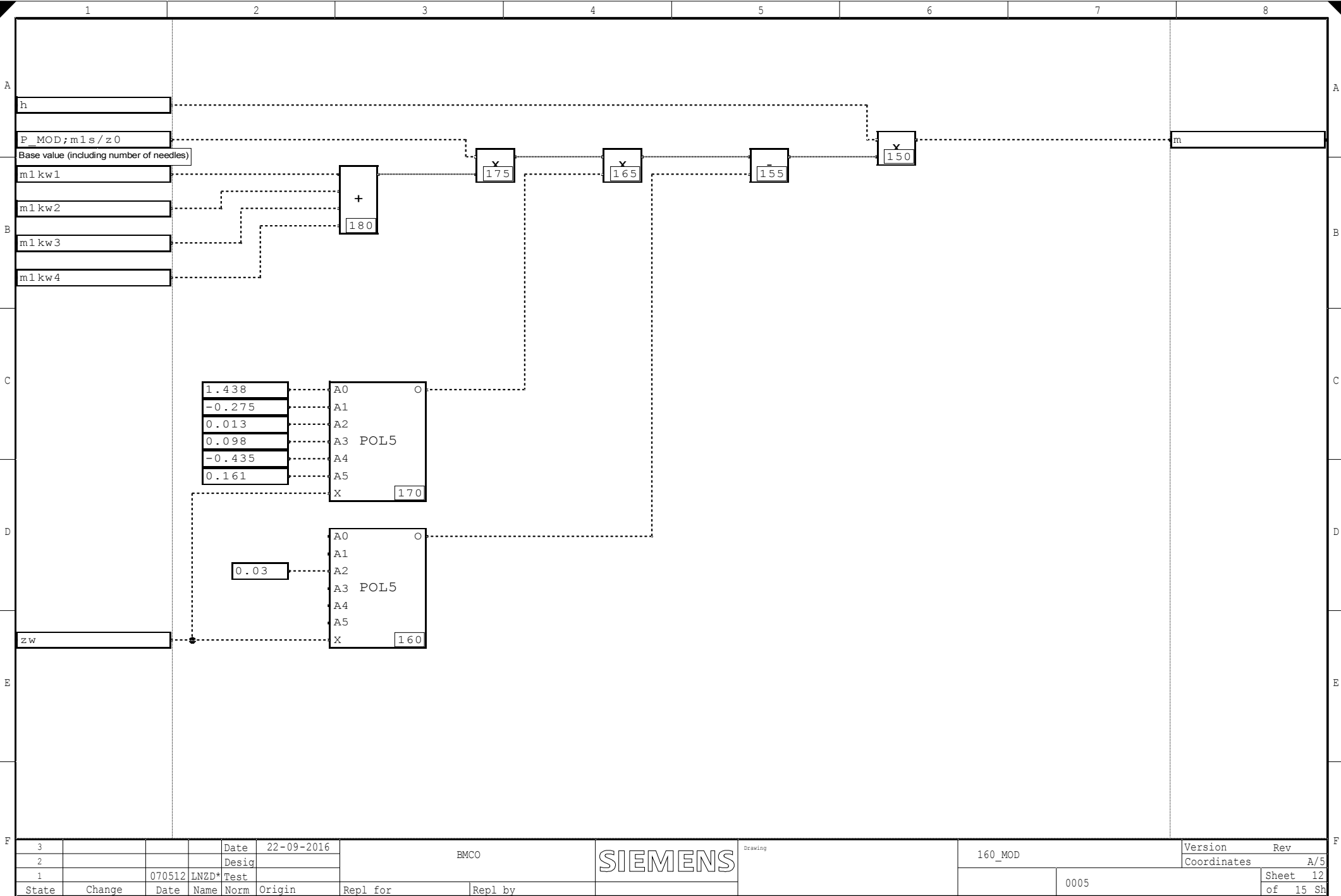
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State	Change	Date	Name	Norm	Origin	Repl for	Repl by			0002	of 15 Sh	



3			Date	22-09-2016	BMC0		Drawing	160_MOD		Version	Rev
2			Desig							Coordinates	A/3
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State	Change	Date	Name	Norm	Origin	Repl for	Repl by			of	15 Sh

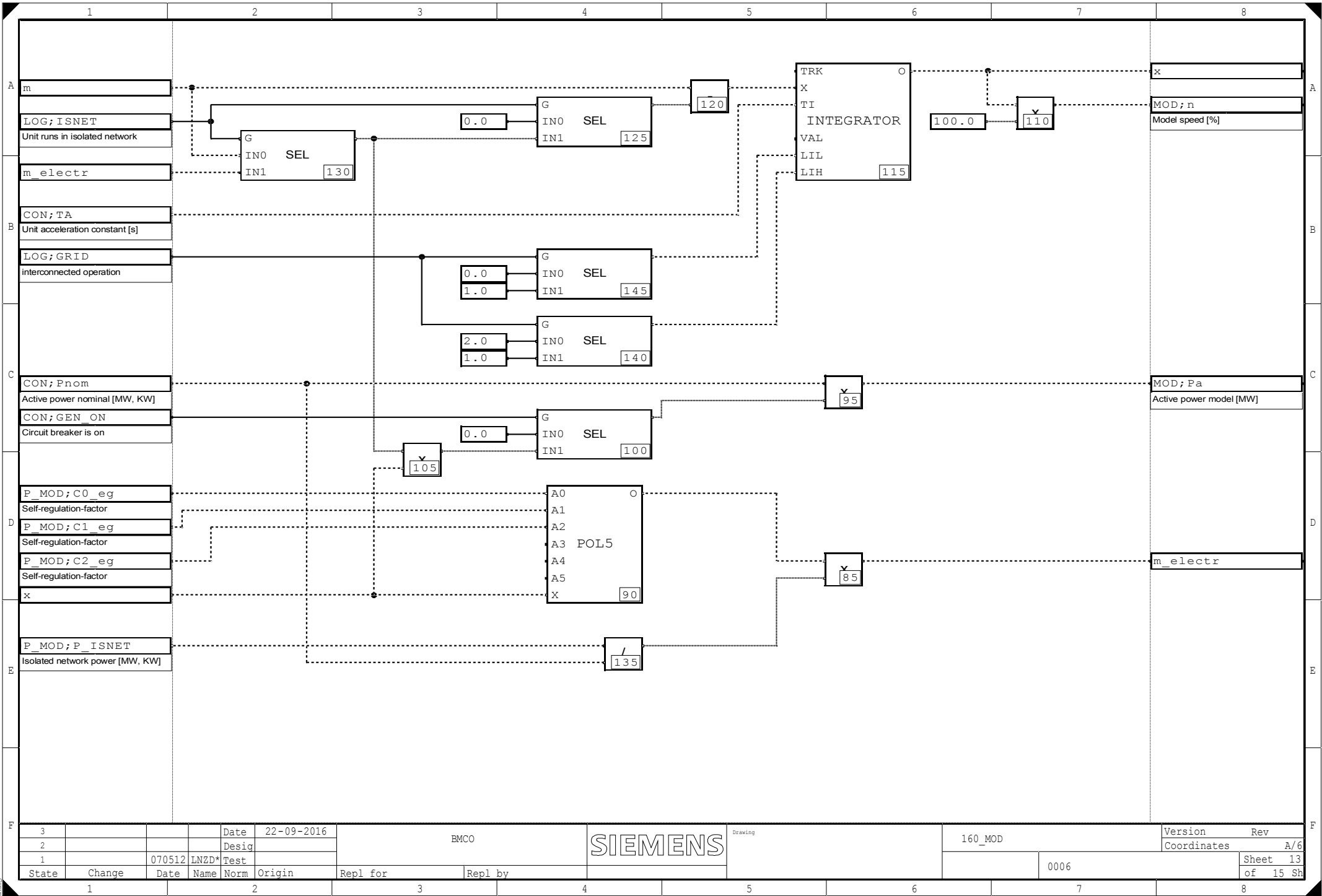


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												8	



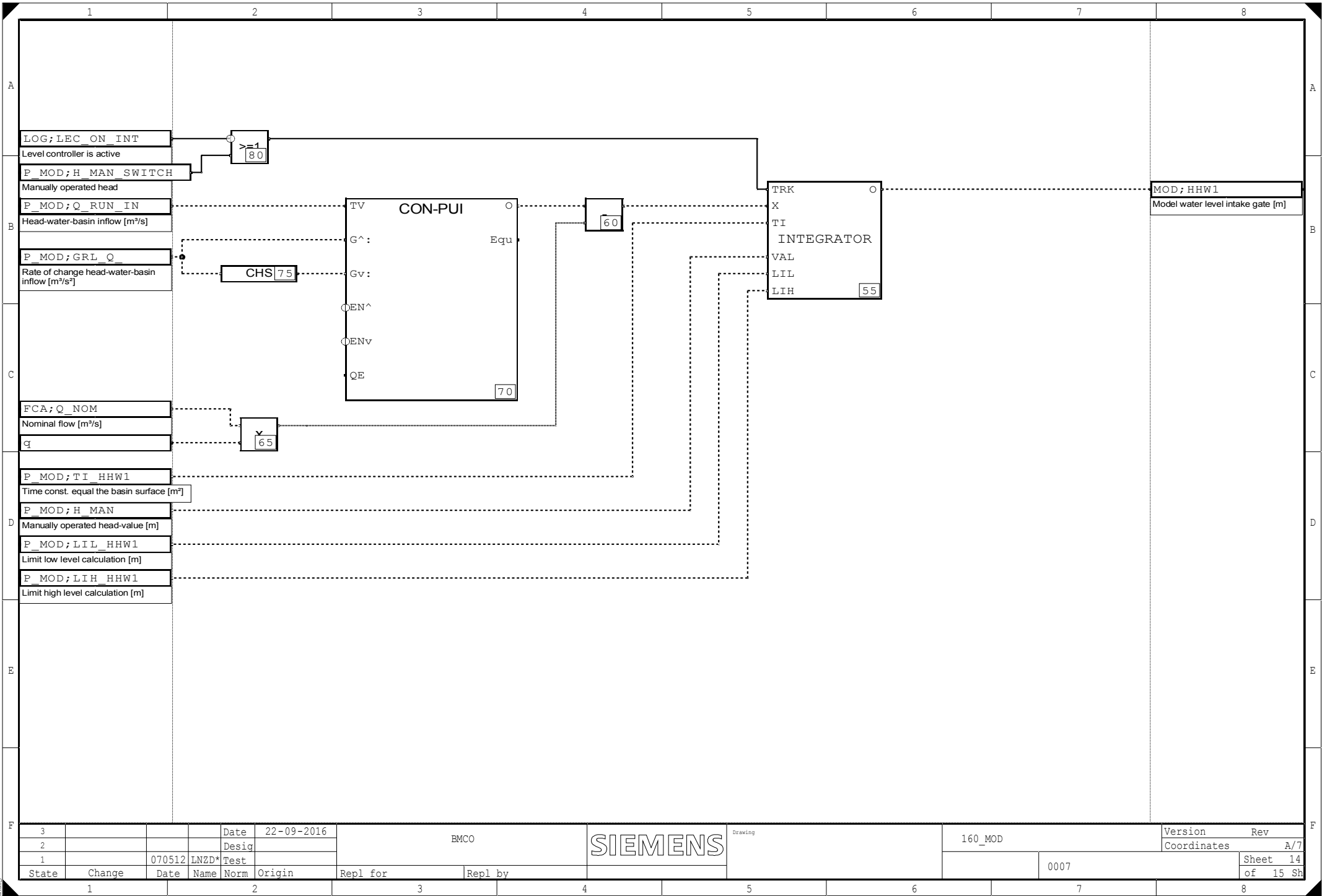
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3				Date	22-09-2016	BMCO		SIEMENS	Drawing	160_MOD		Version	Rev
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State	Change	Date	Name	Norm	Origin	Repl for	Repl by			of	15 Sh		

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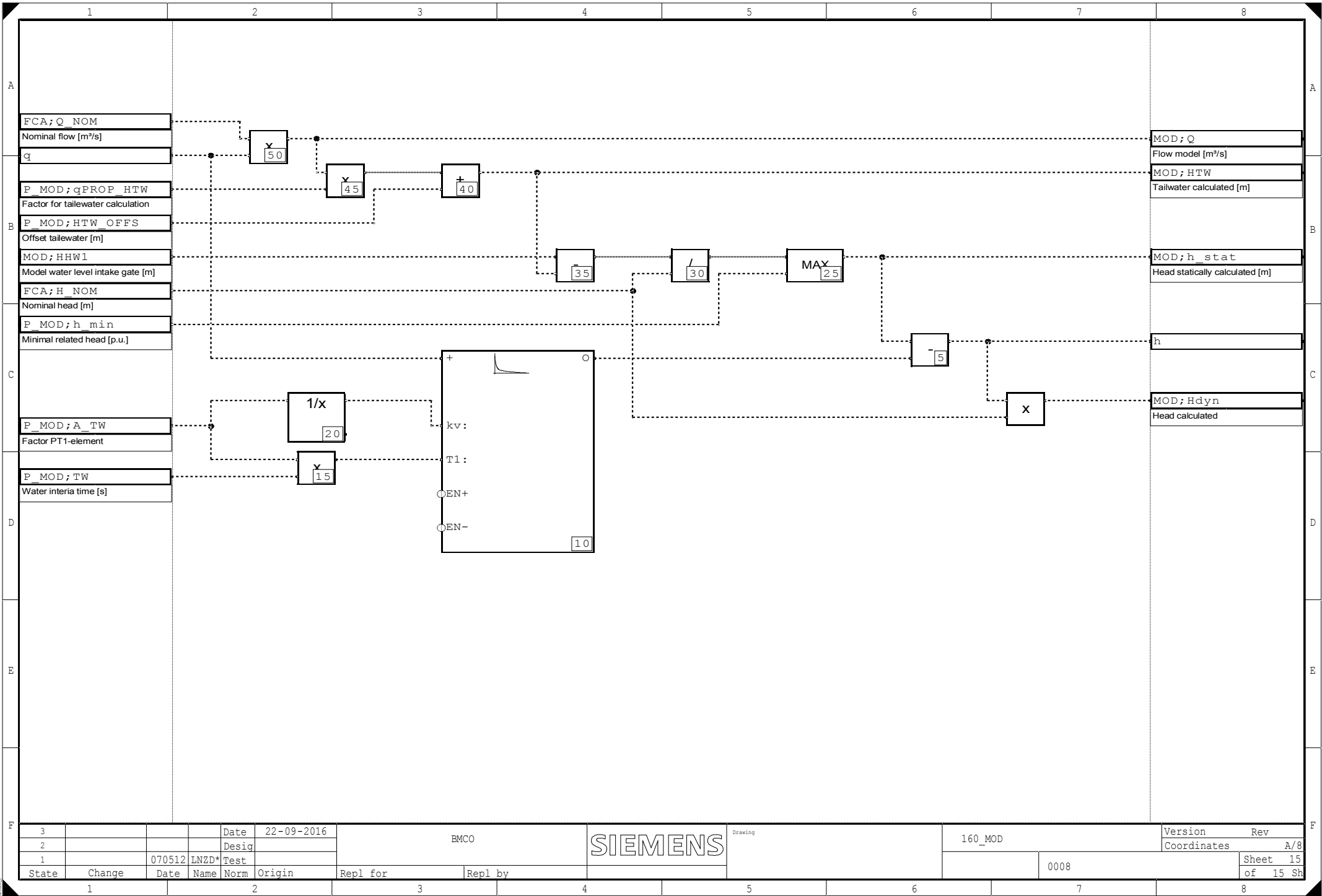


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2			Desig	
1		070512	LNZD*	Test
State	Change	Date	Name	Norm

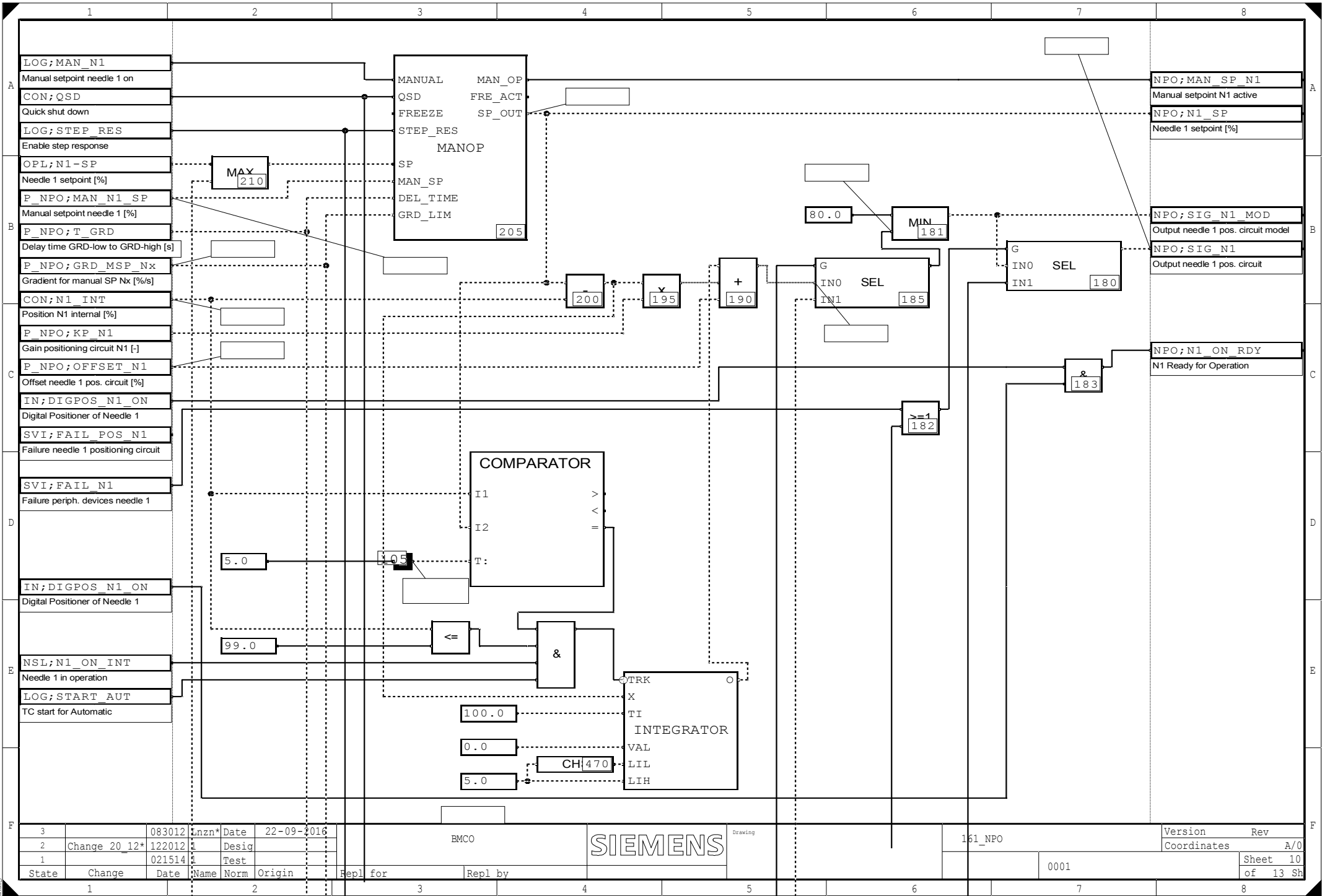
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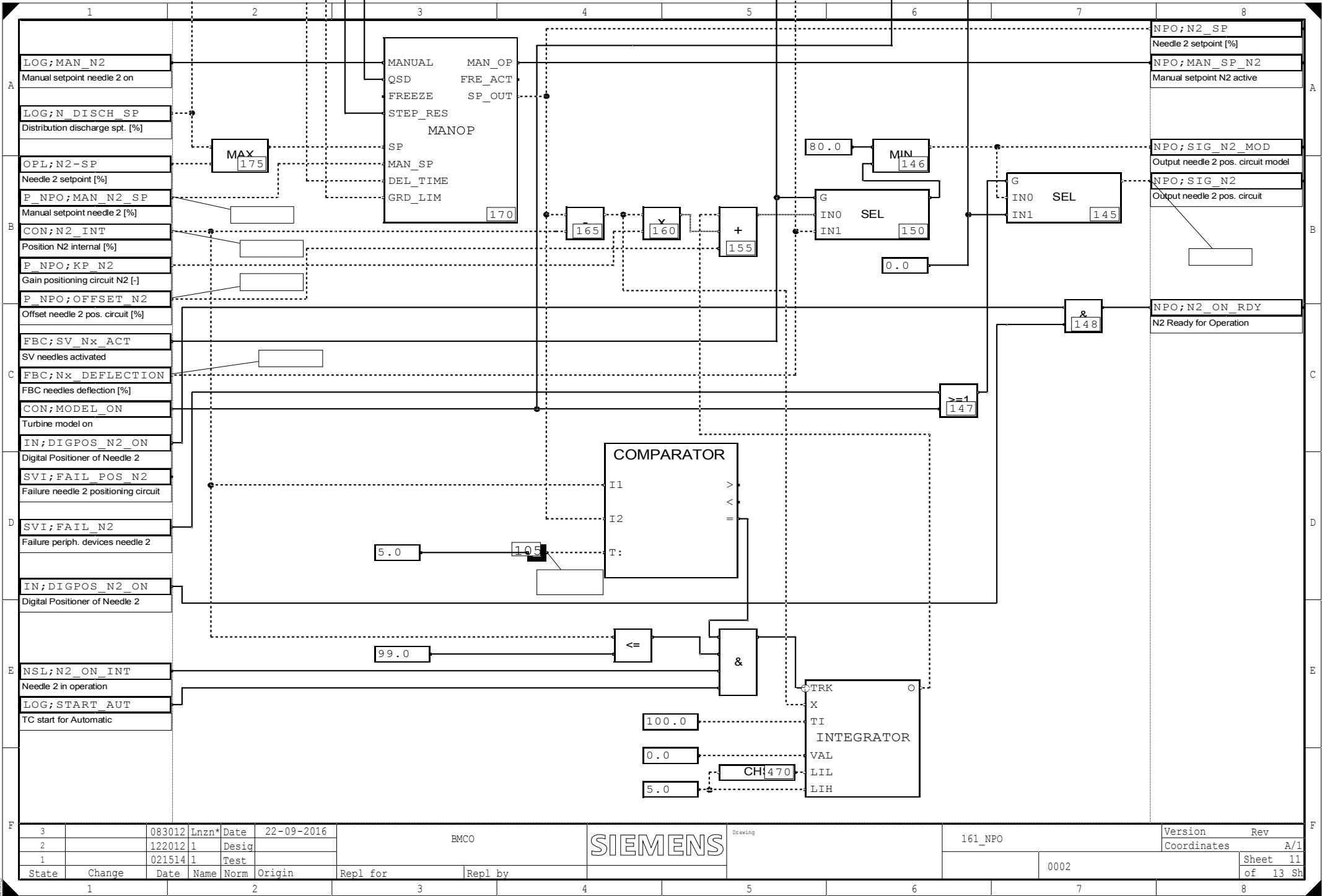
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	0007	Sheet	14
		of	15 Sh



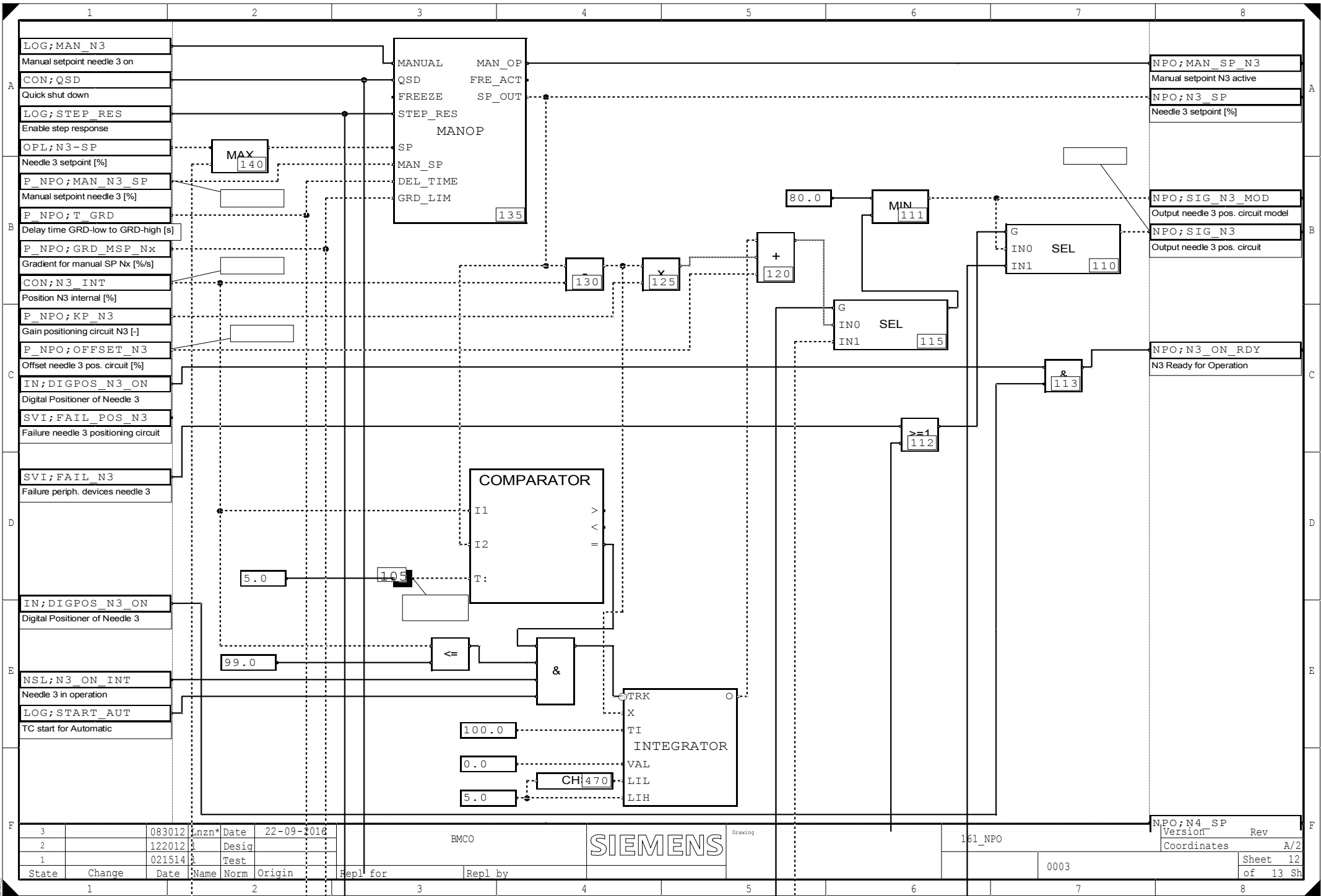
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State	Change	Date	Name	Norm						of	15 Sh



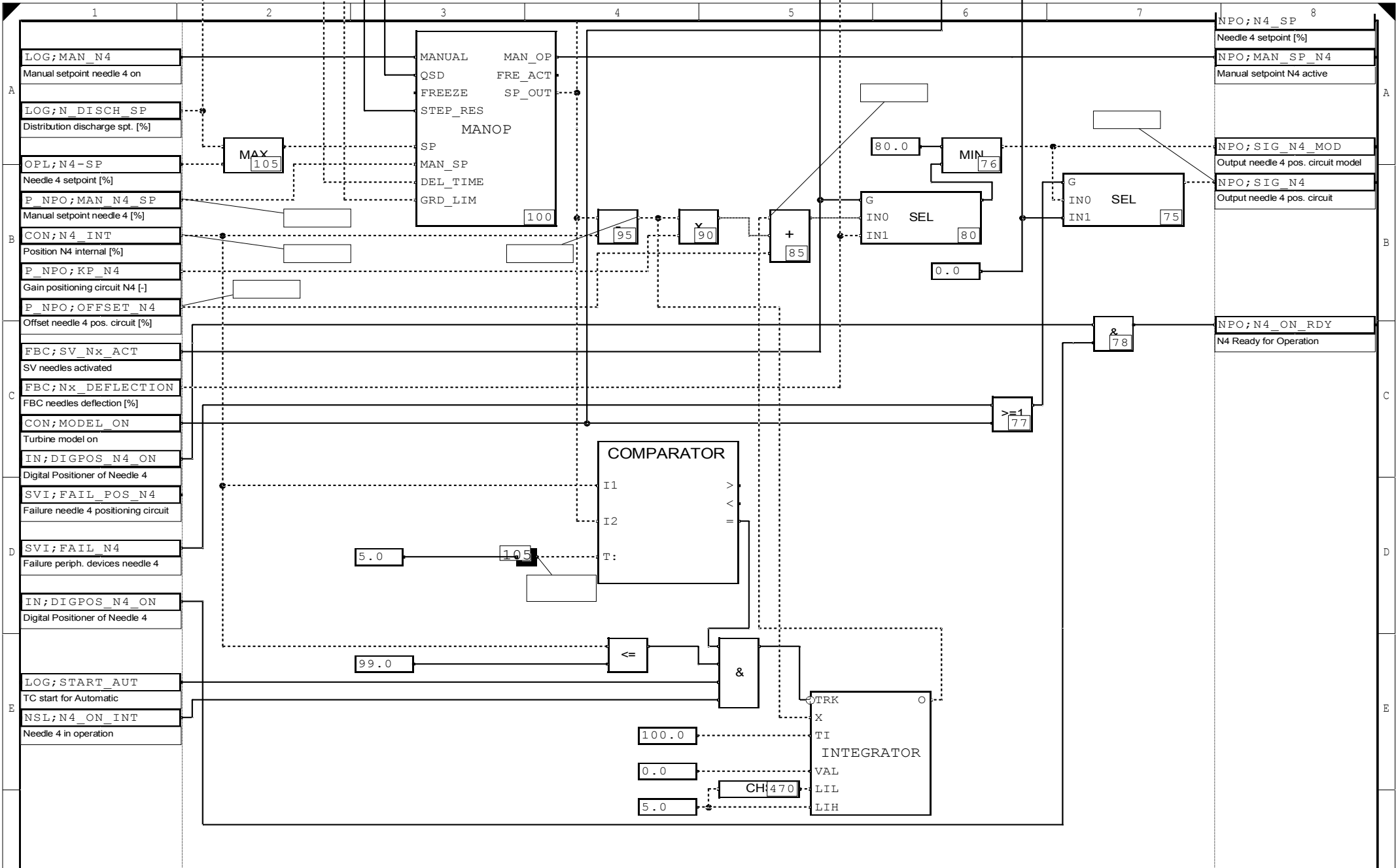
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State	Change	Date	Name	Norm	Origin									
Repl for						Repl by	0001							Sheet 10
														of 13 Sh



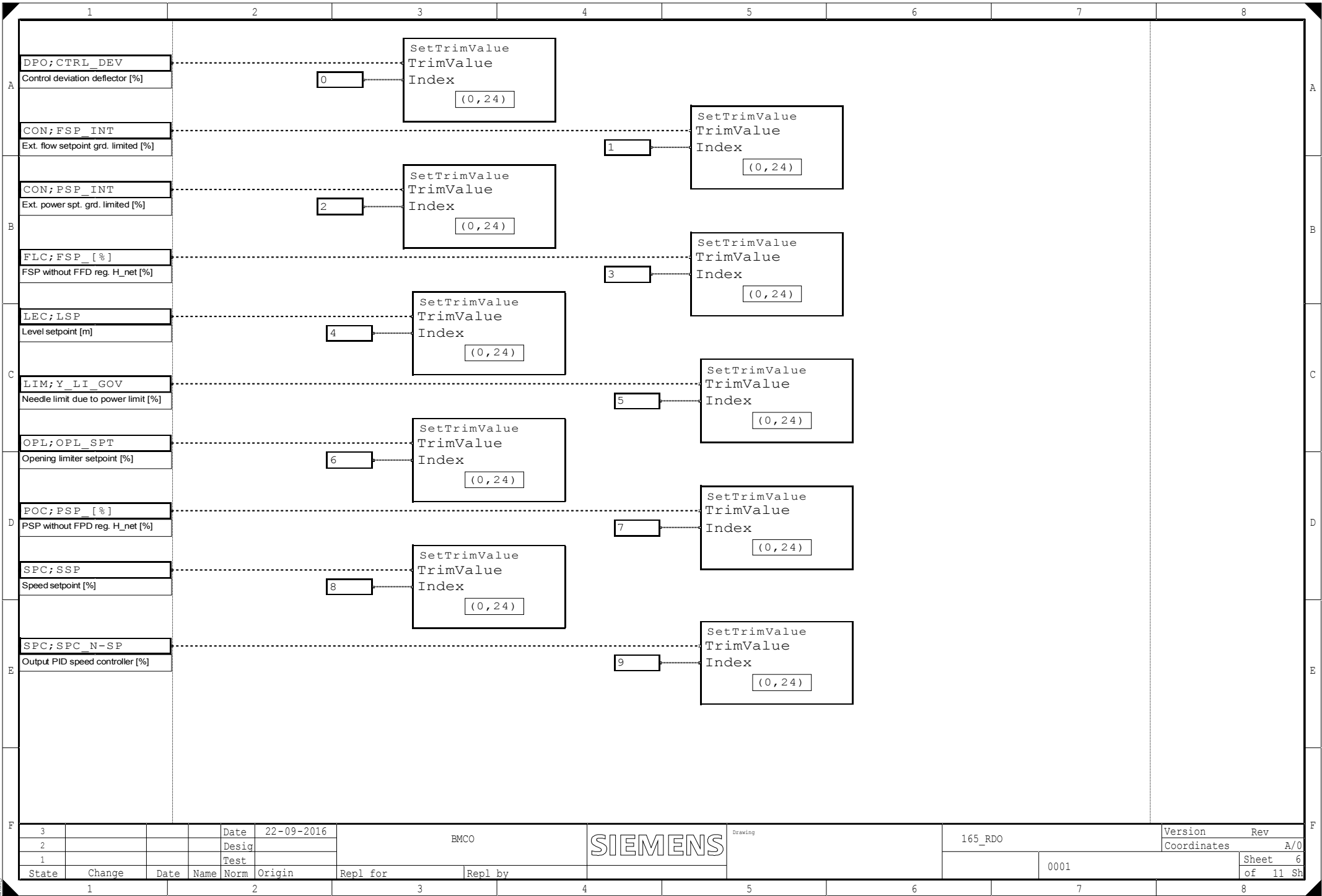
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State	Change	Date	Name	Norm	Origin	Repl for	Repl by				0002		Sheet 11 of 13 Sh



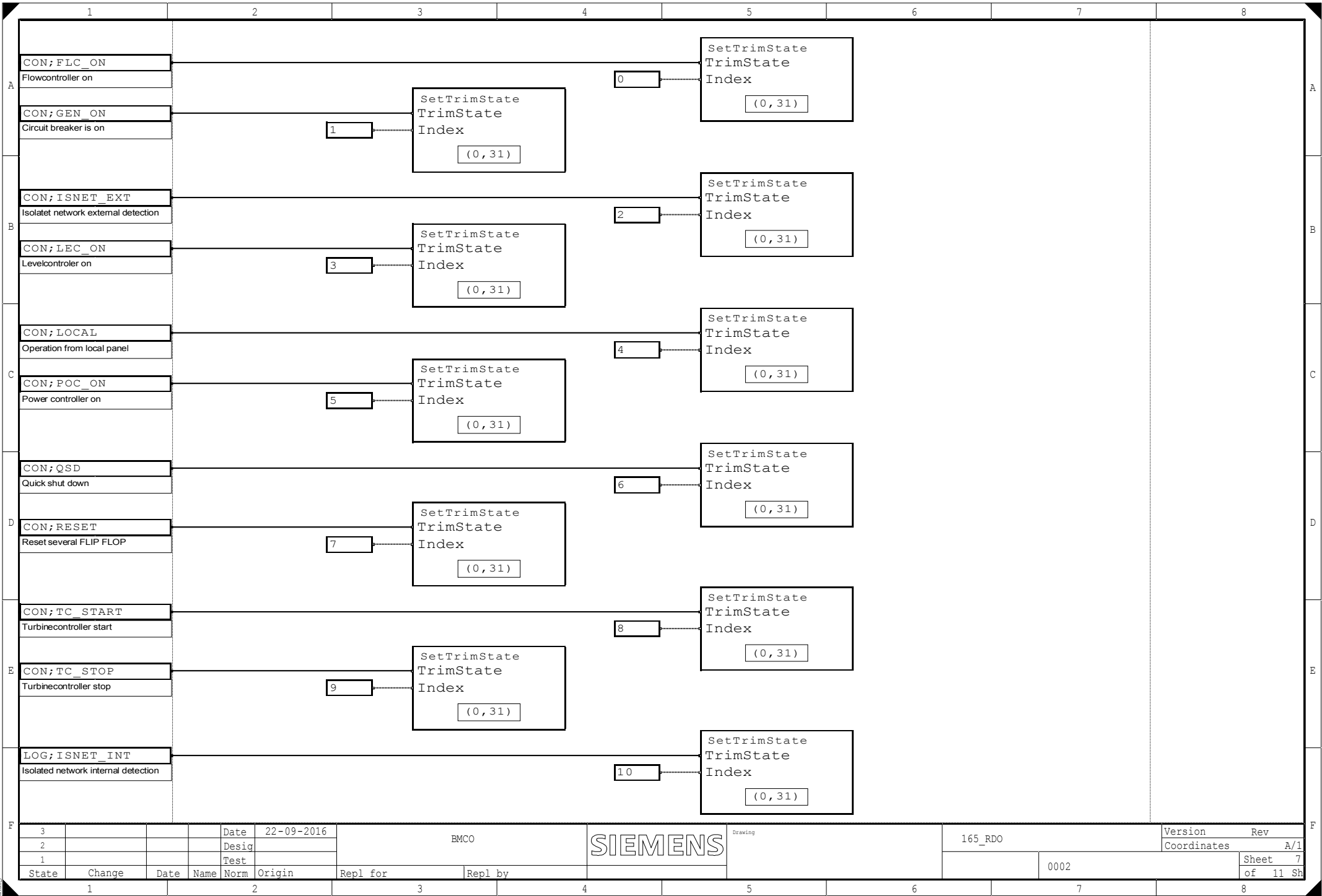
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1	021514	1	Test									Sheet 12
State	Change	Date	Name	Norm	Origin	Repl for	Repl by			0003		of 13 Sh
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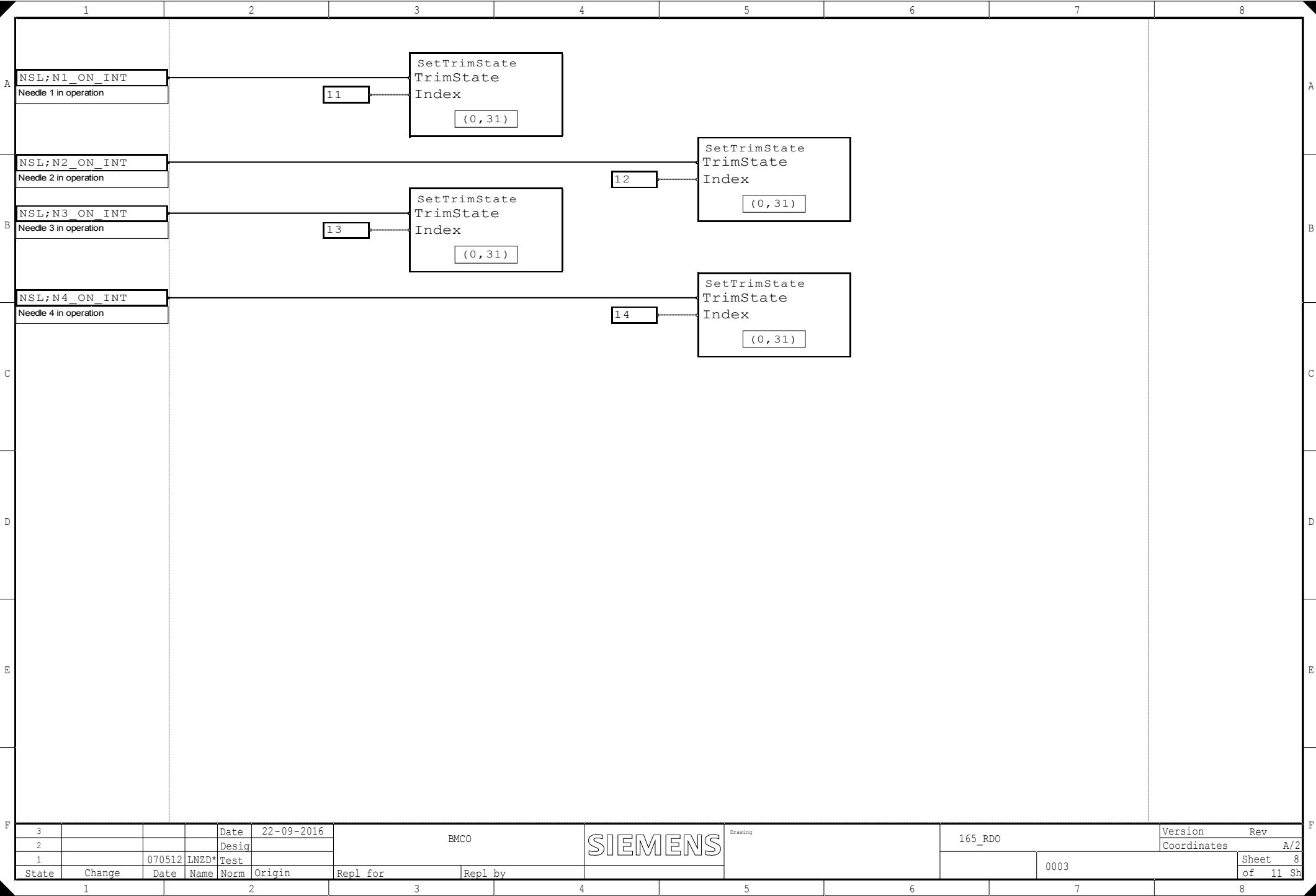


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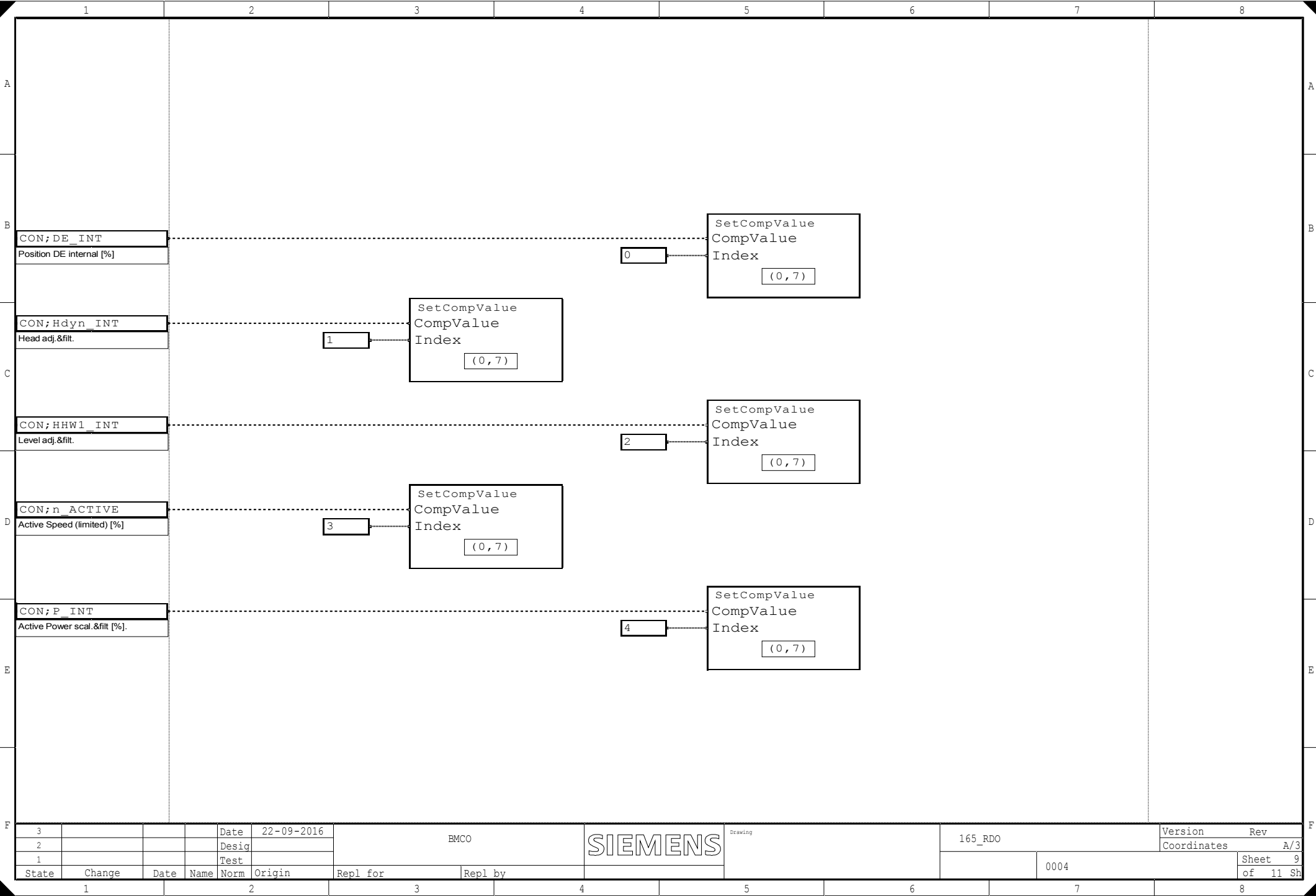


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1												of	11 Sh

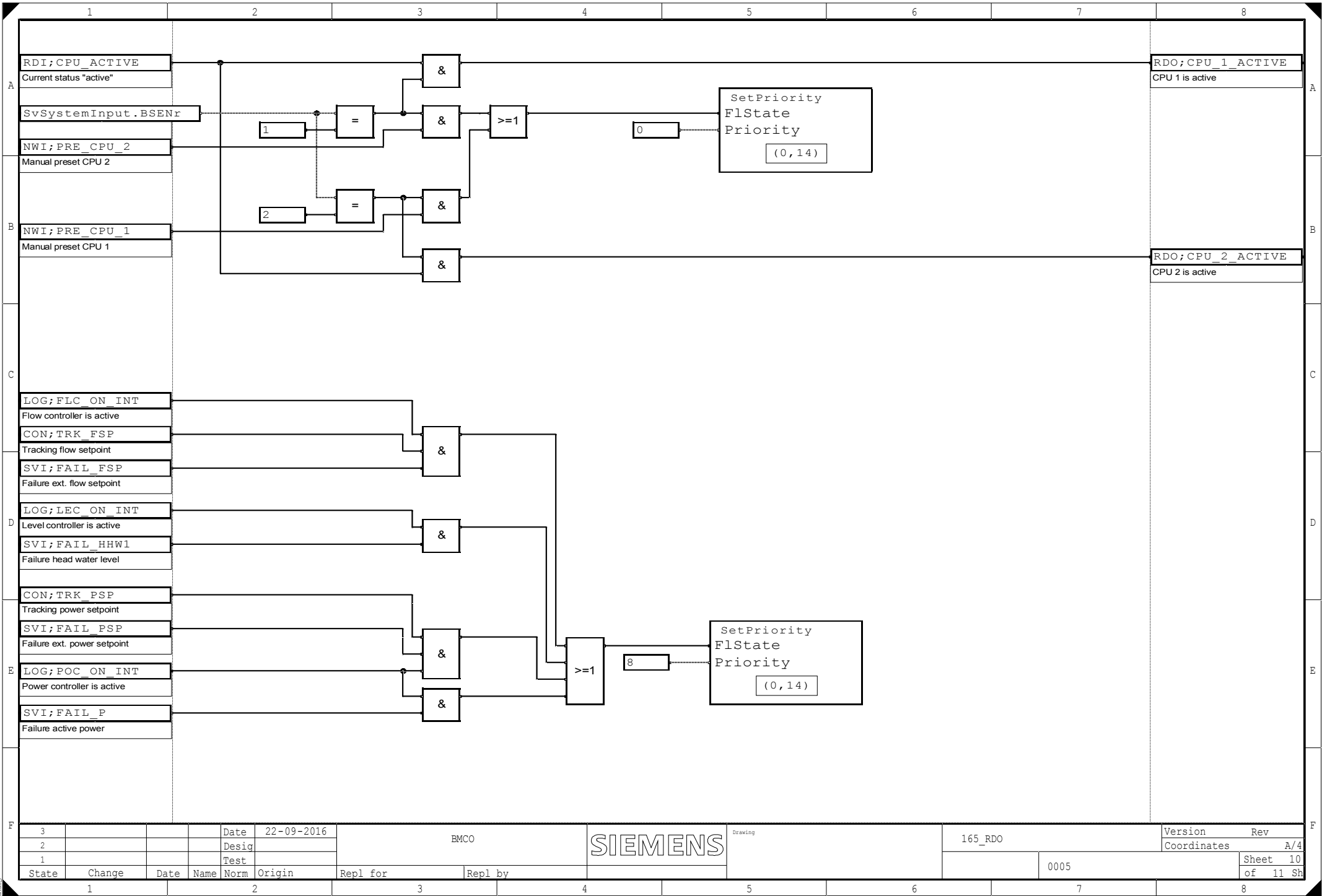




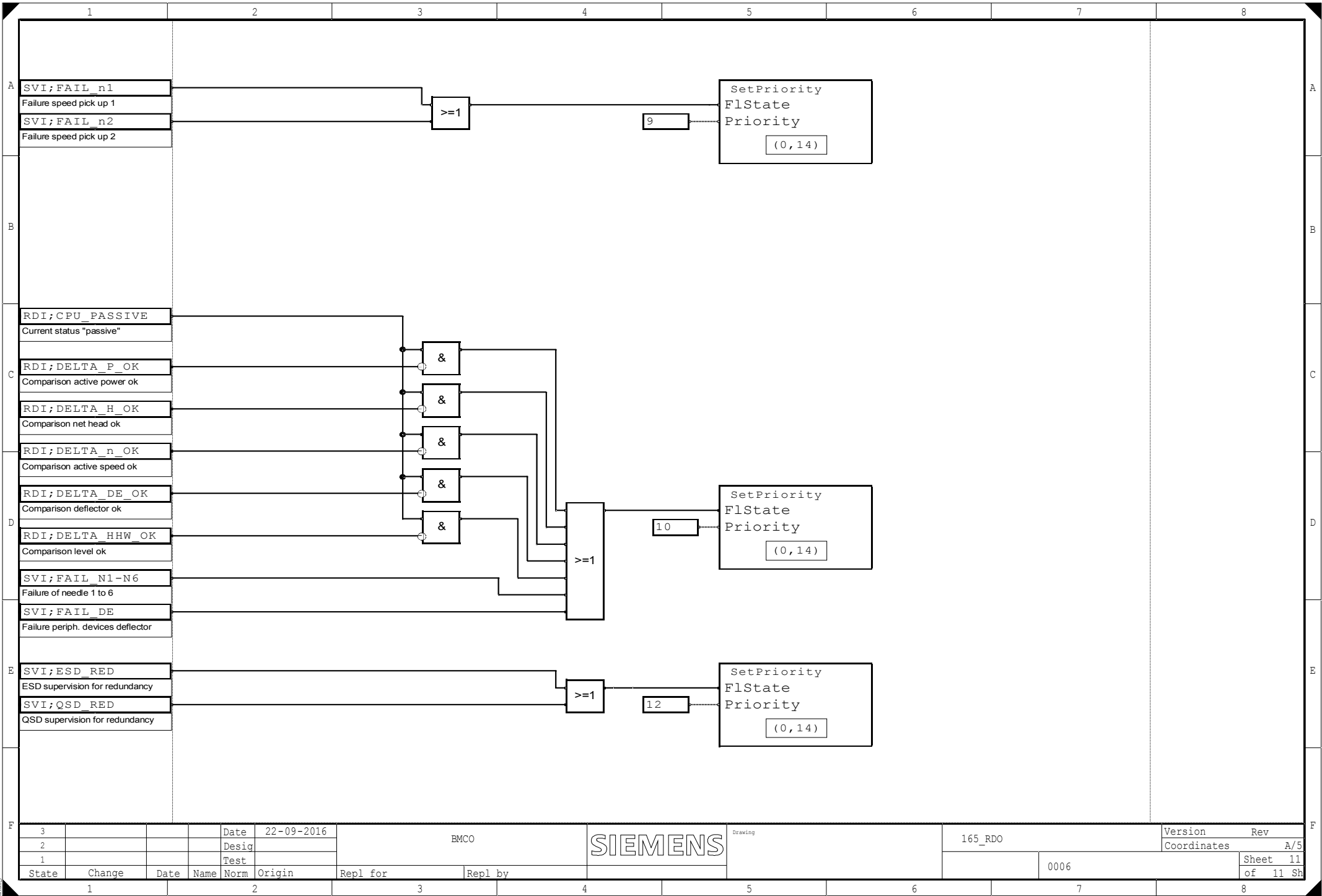
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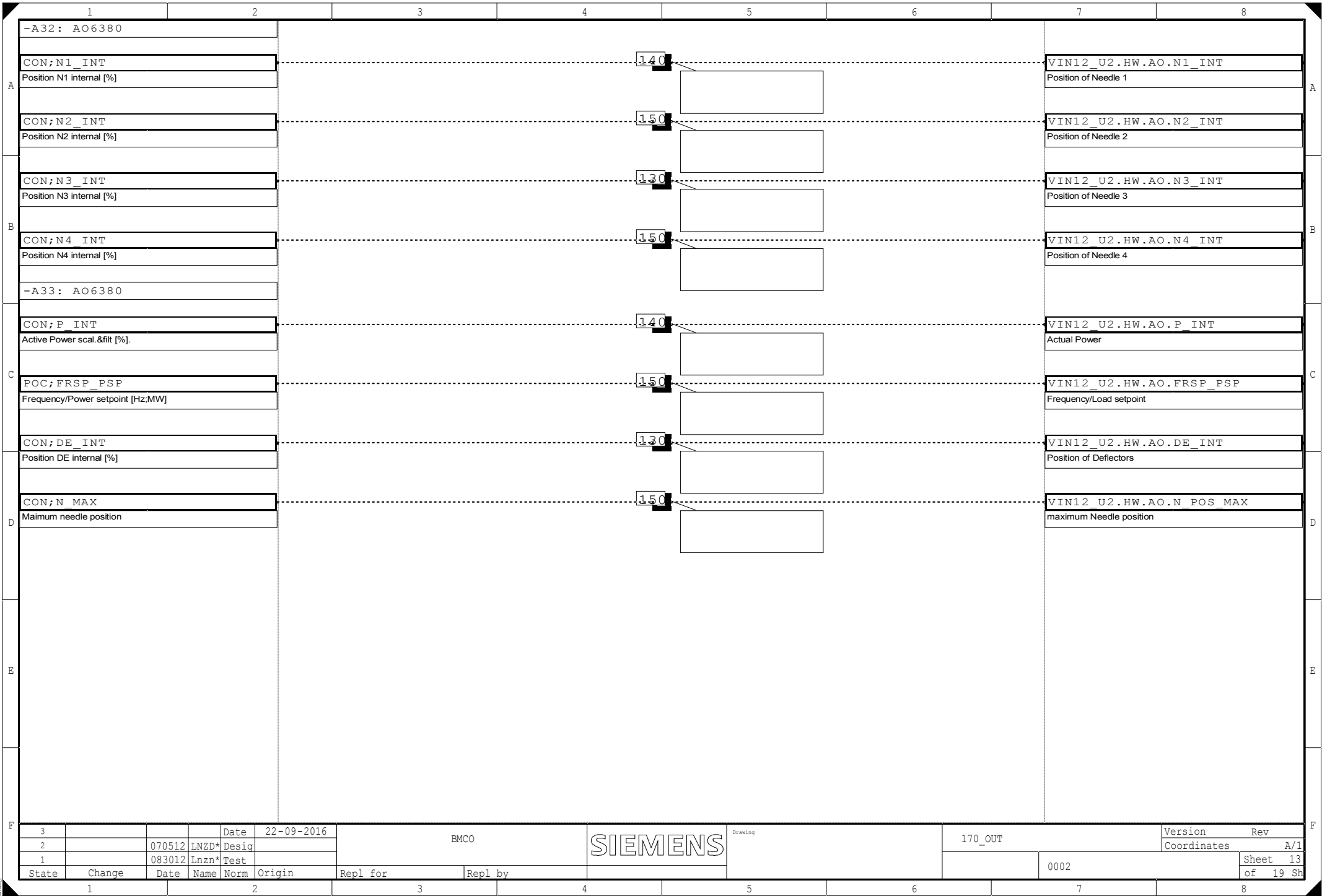
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1				Test								Sheet	9
State	Change	Date	Name	Norm	Origin	Repl for	Repl by				0004	of	11 Sh
1		2		3		4		5		6		7	

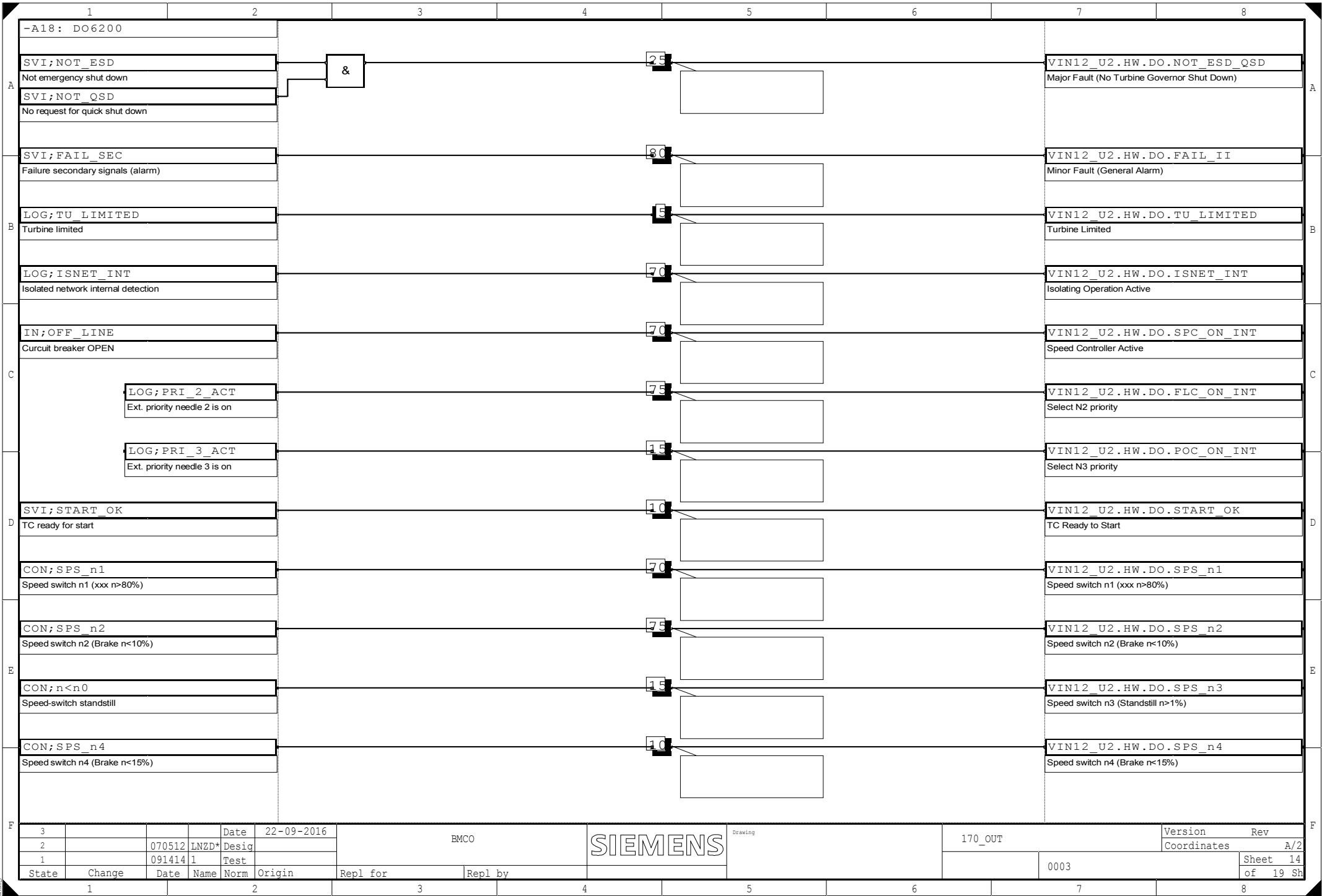


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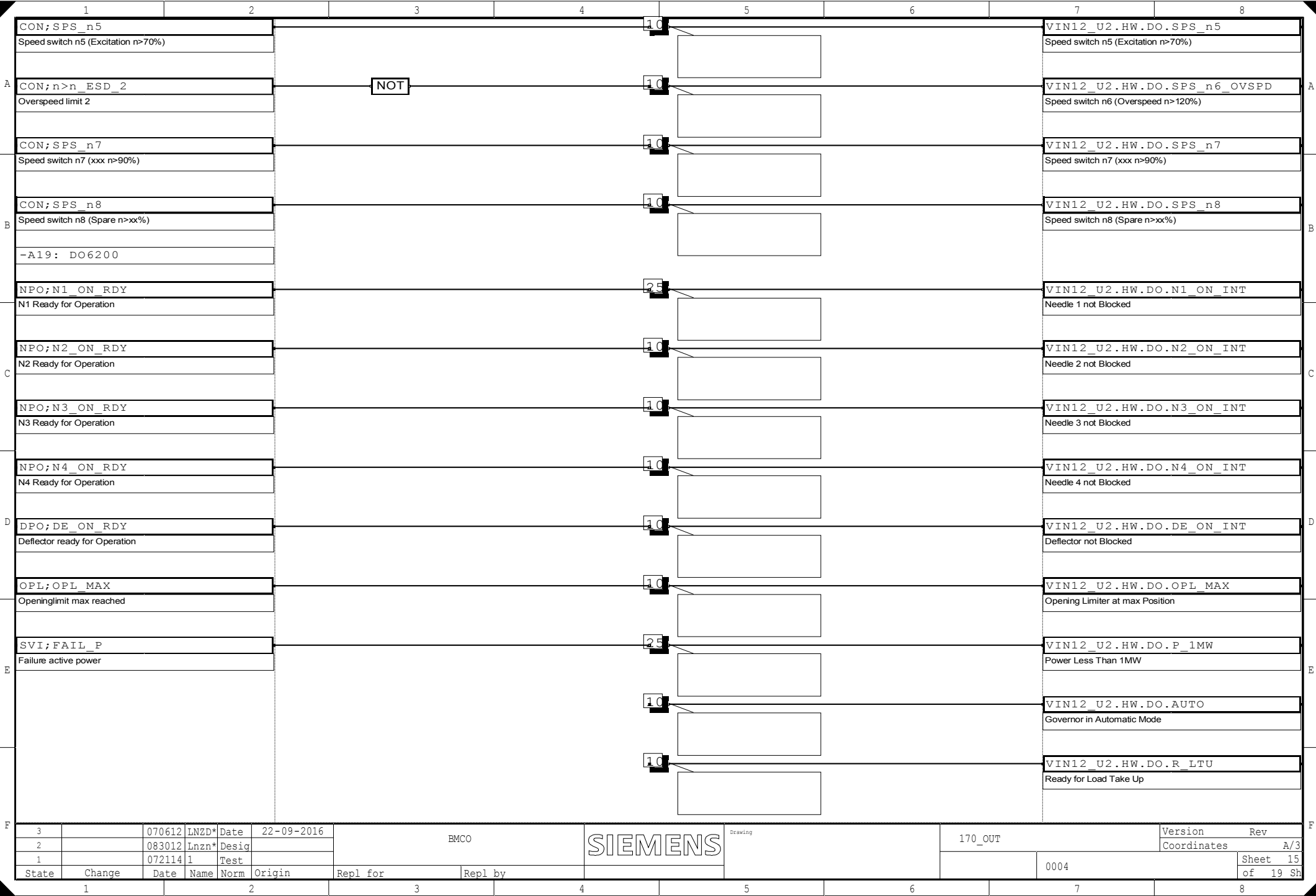
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1				Test									
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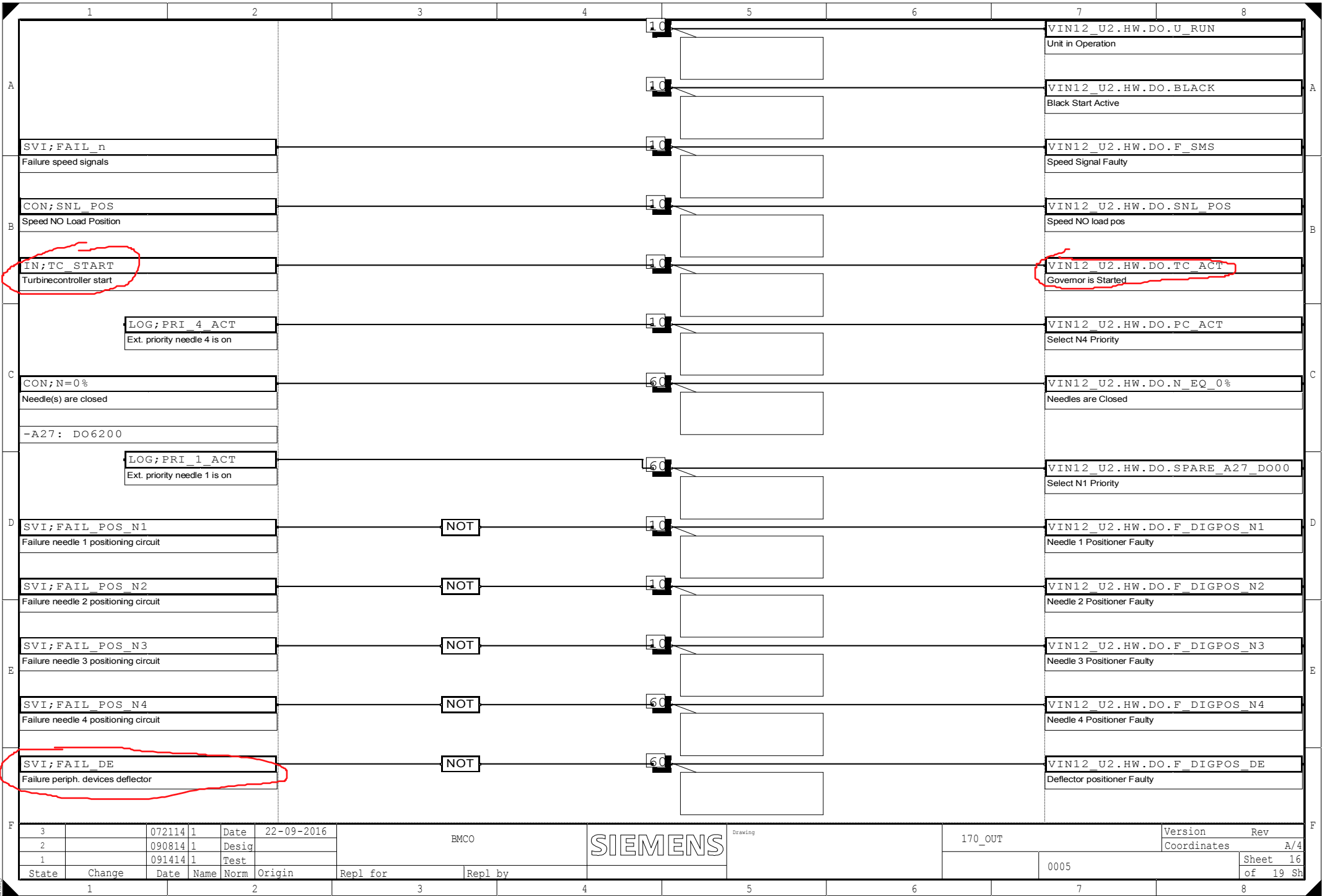




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State	Change	Date	Name	Norm	Origin	Repl for	Repl by			0003	of 19 Sh			

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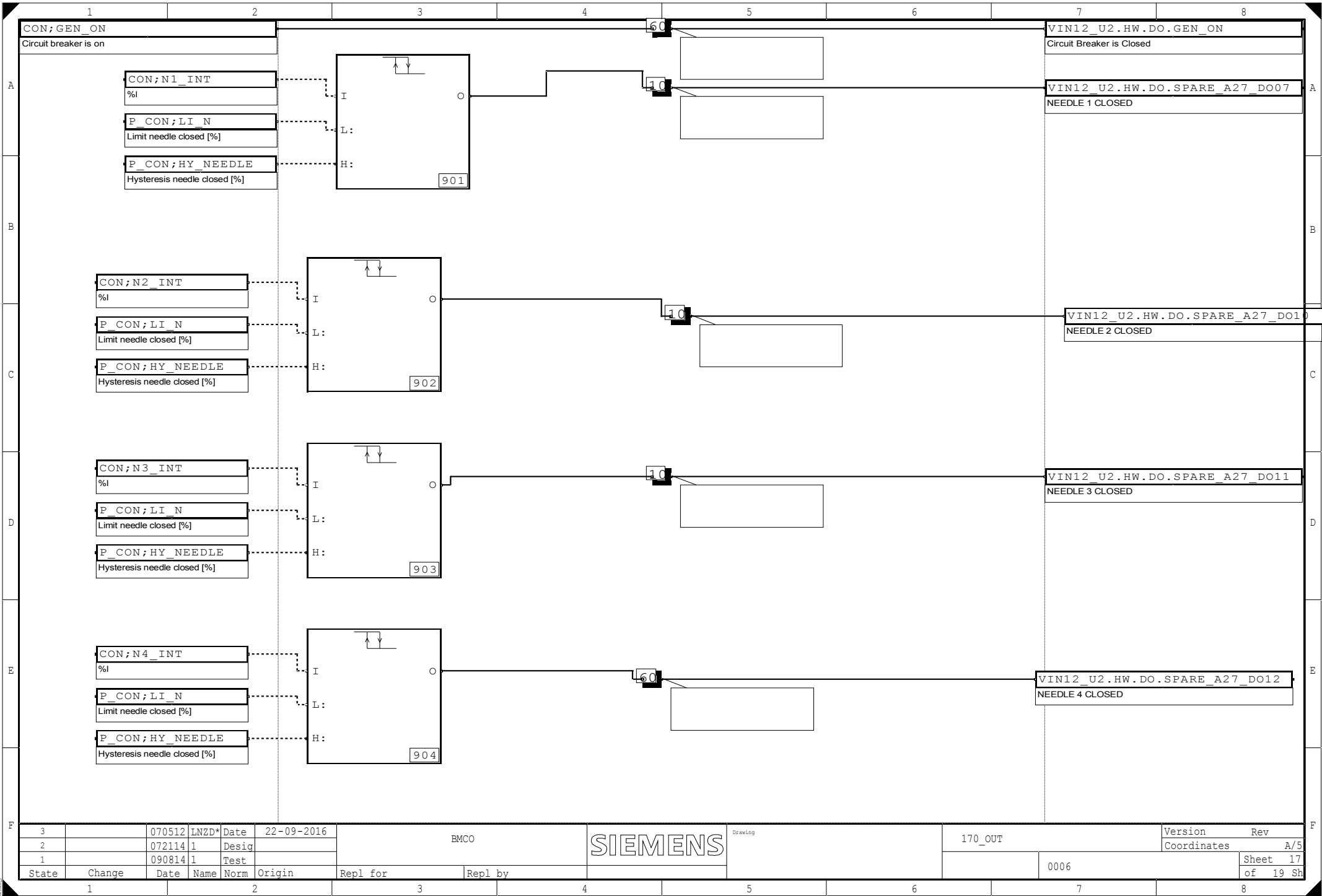




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3		072114	1	Date	22-09-2016	BMCO		Drawing	170_OUT		Version		Rev
2		090814	1	Desig							Coordinates		A/4
1		091414	1	Test									Sheet 16
State	Change	Date	Name	Norm	Origin	Repl for	Repl by			0005			of 19 Sh



3		070512	LN2D*	Date	22-09-2016	BMCO		SIEMENS	Drawing	170_OUT		Version		Rev	
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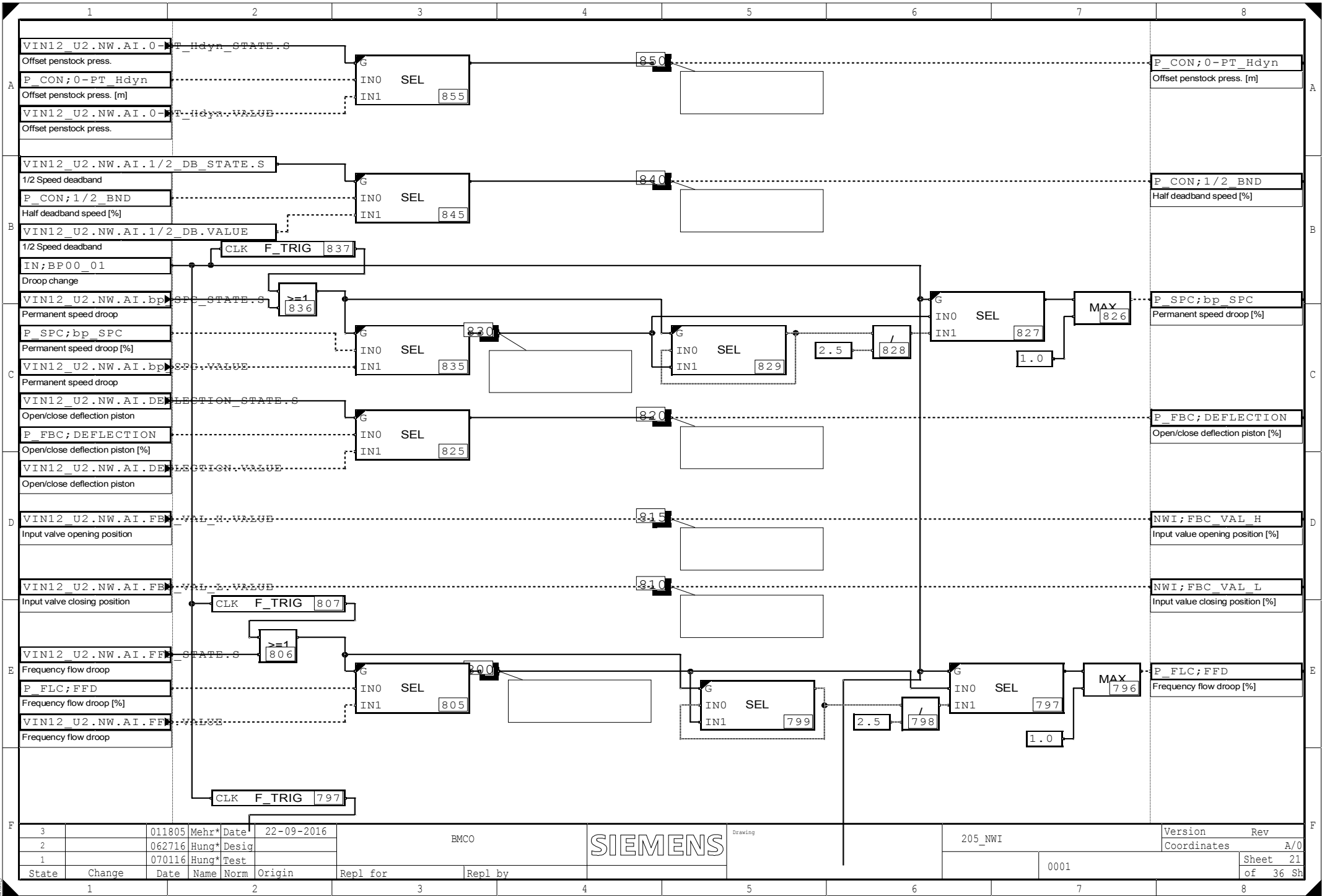
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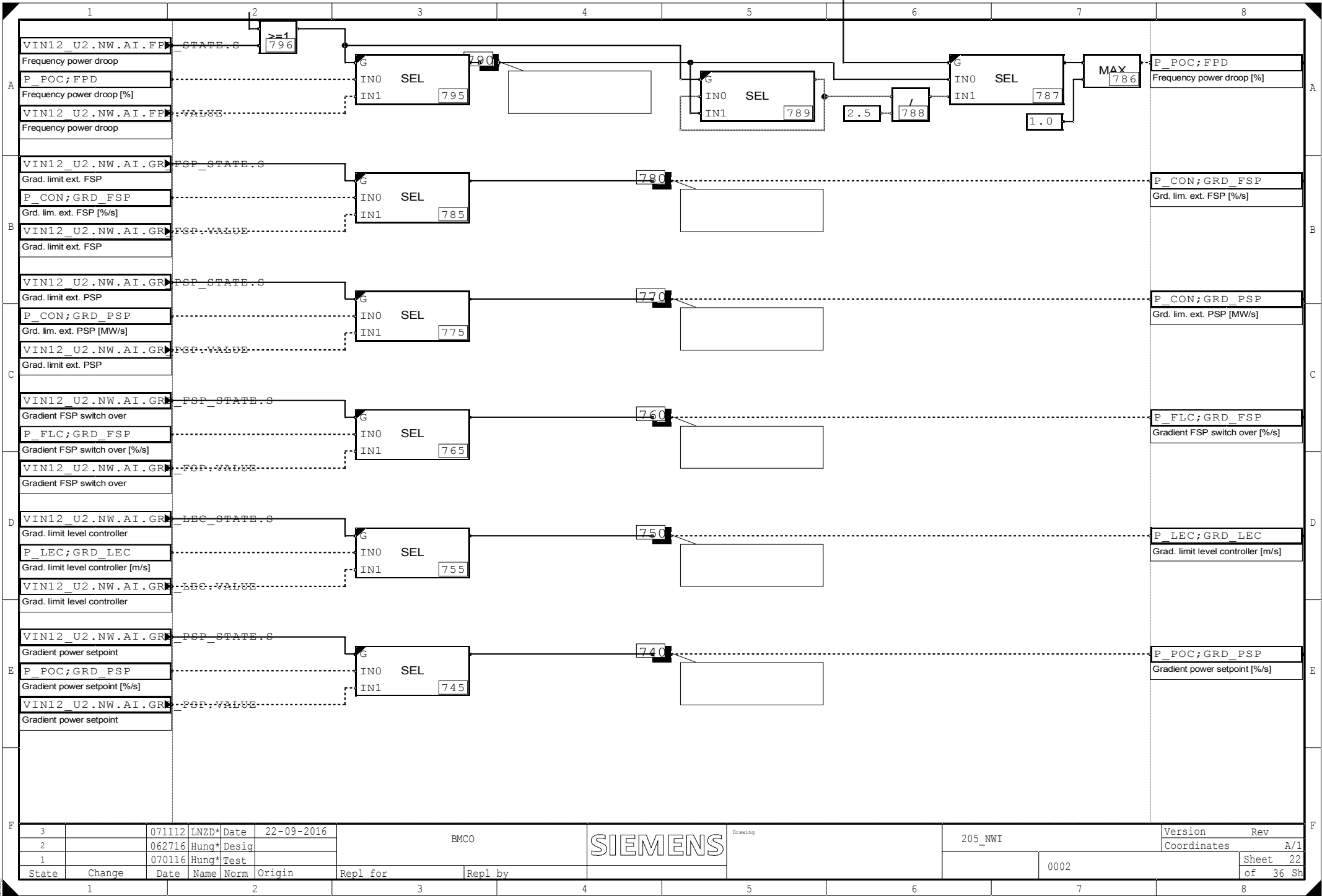
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C								C
D								D
E								E
F								F

DO10

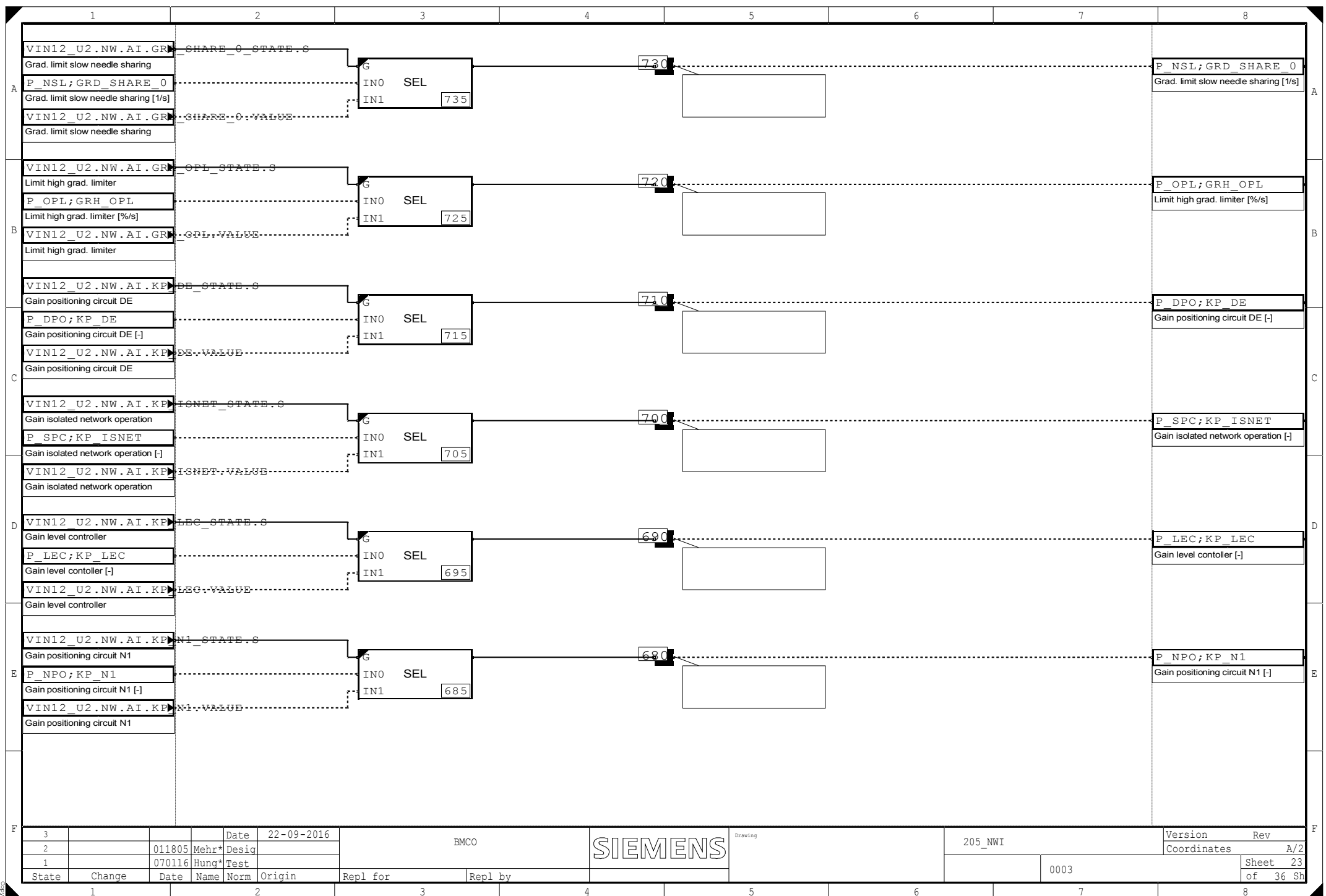
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State	Change	Date	Name	Norm	Origin	Repl for	Repl by				0007		Sheet 18 of 19 Sh

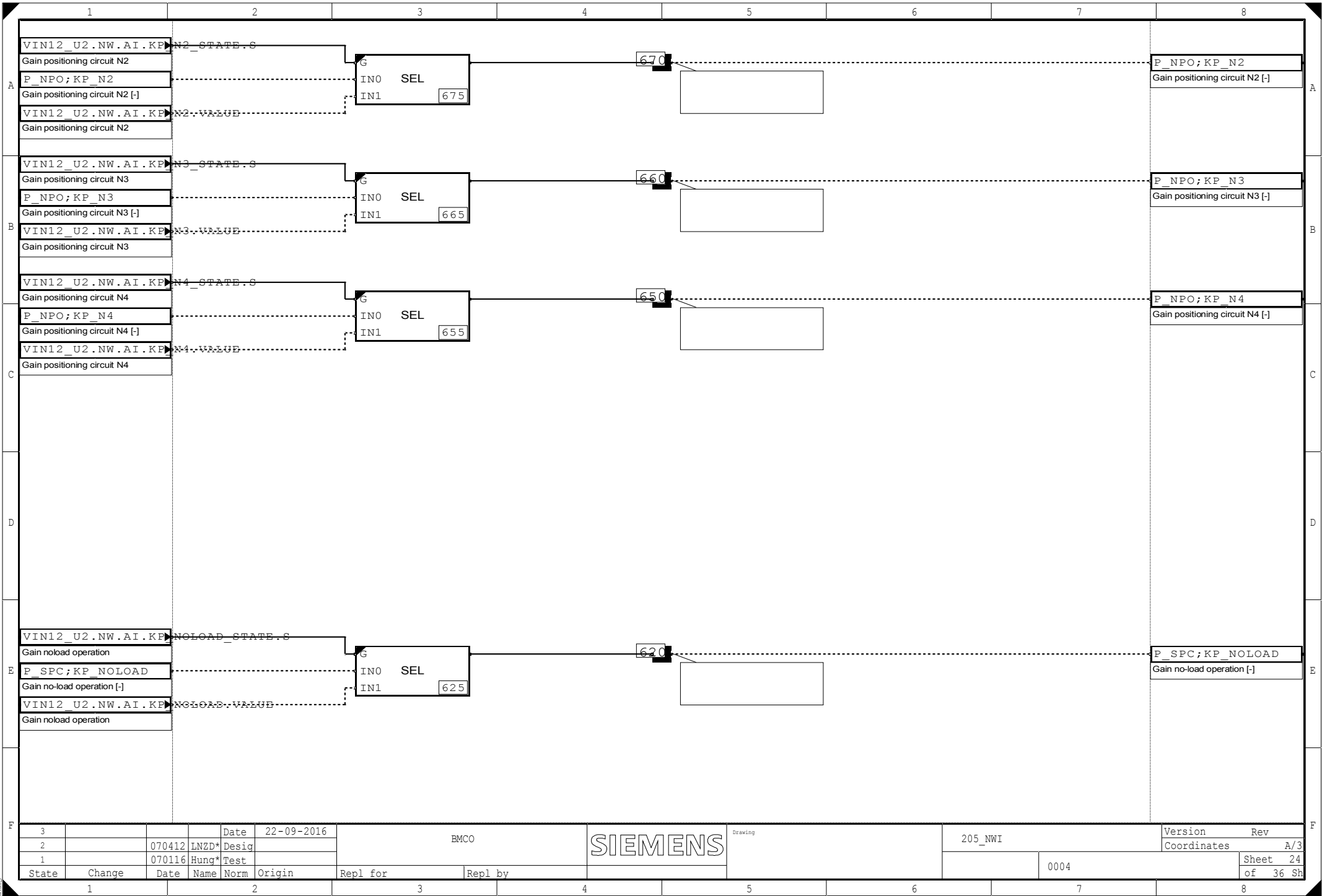
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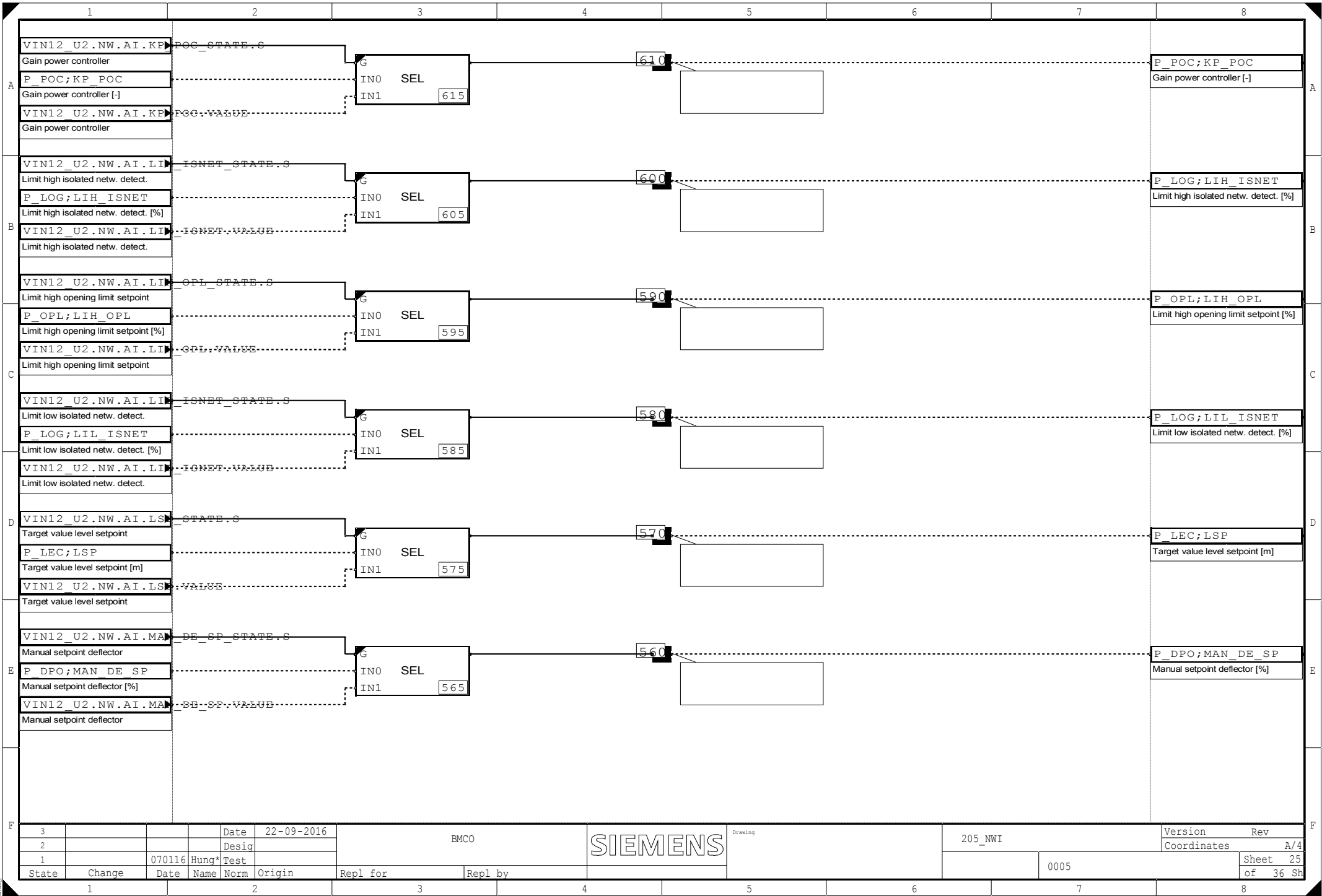


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State	Change	Date	Name	Norm	Origin	Repl for	Repl by			0002	Sheet	22	of 36 Sh

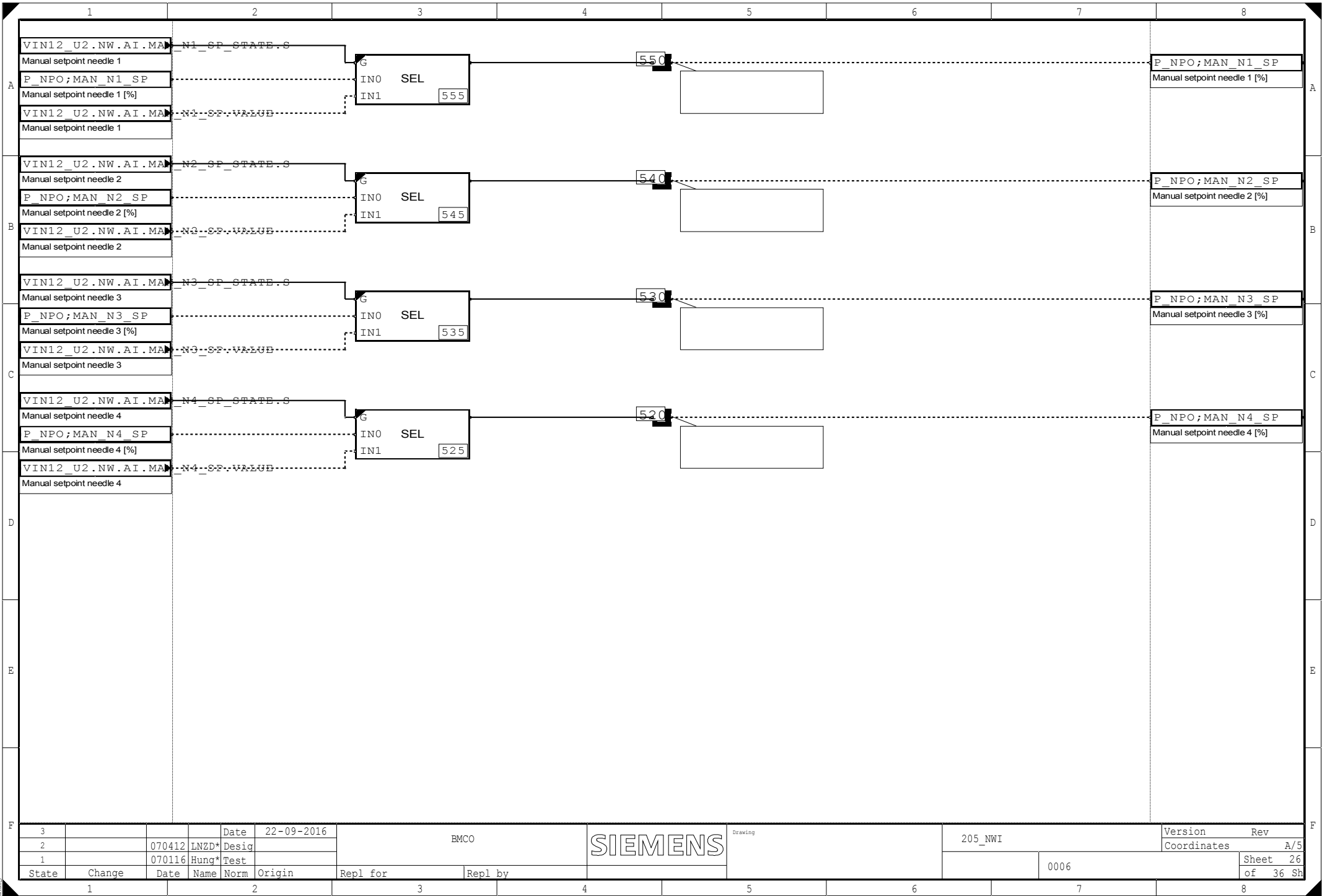




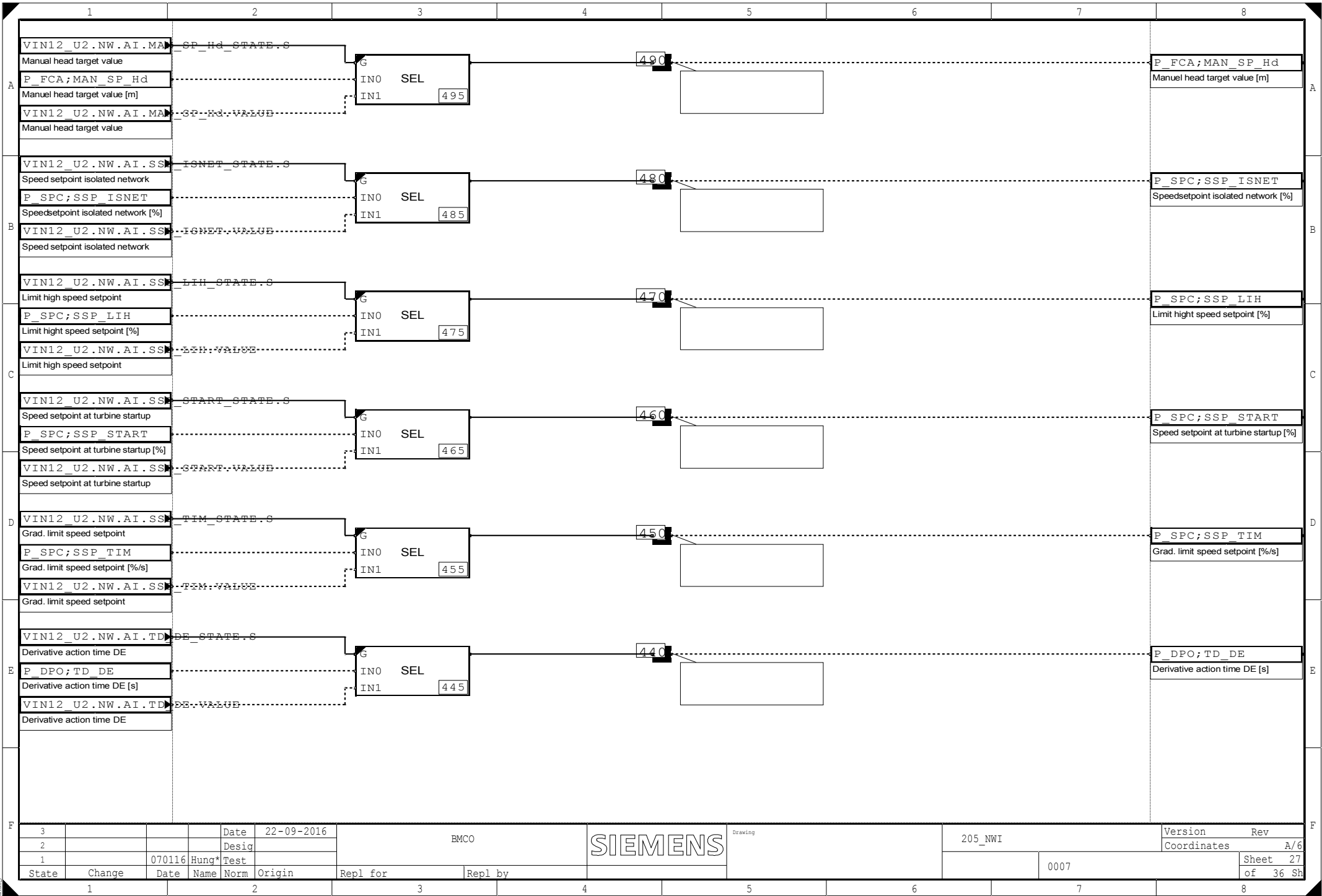
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1		070116	Hung*	Test	Repl for		Repl by	0004		Sheet	24
State	Change	Date	Name	Norm						of	36 Sh

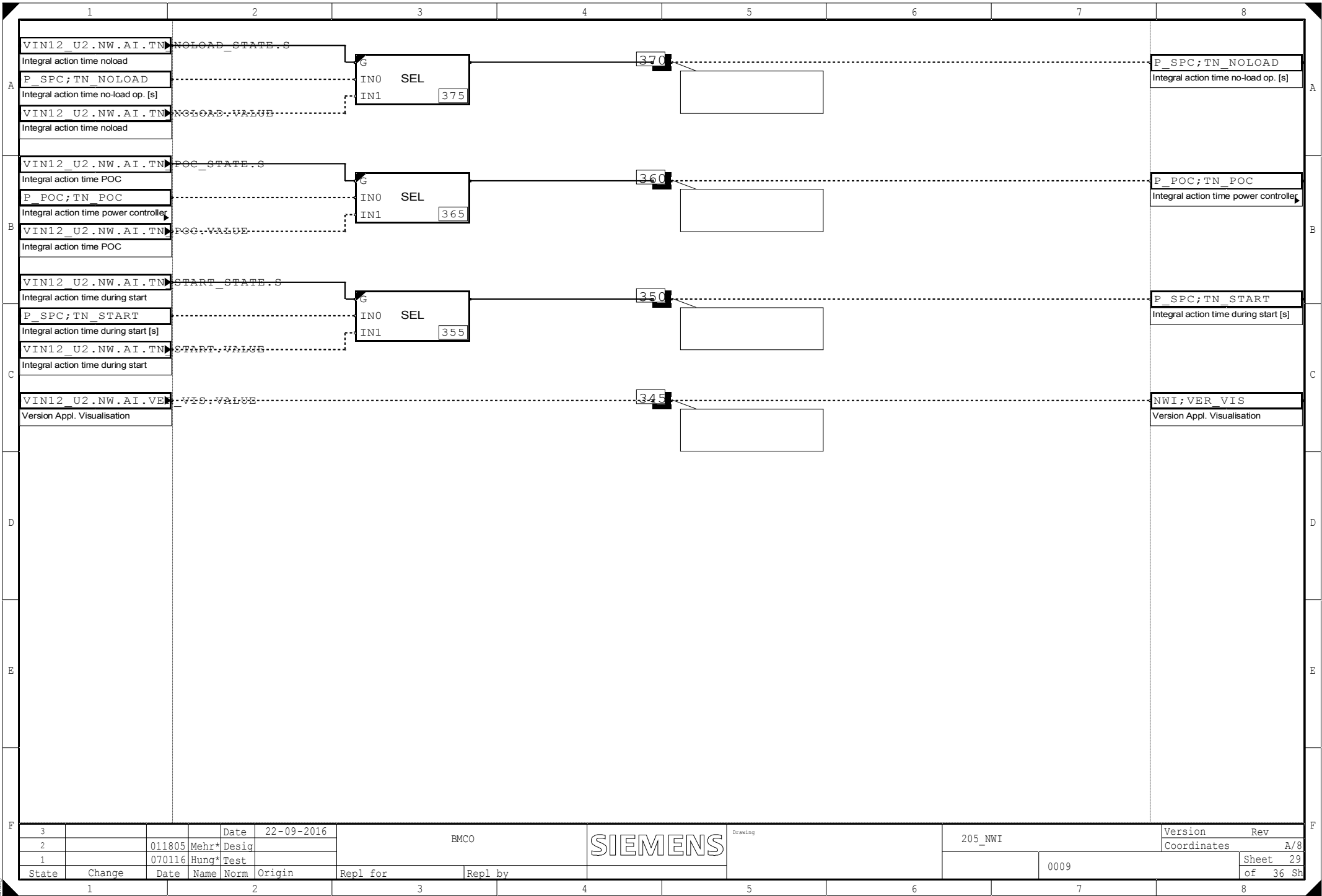


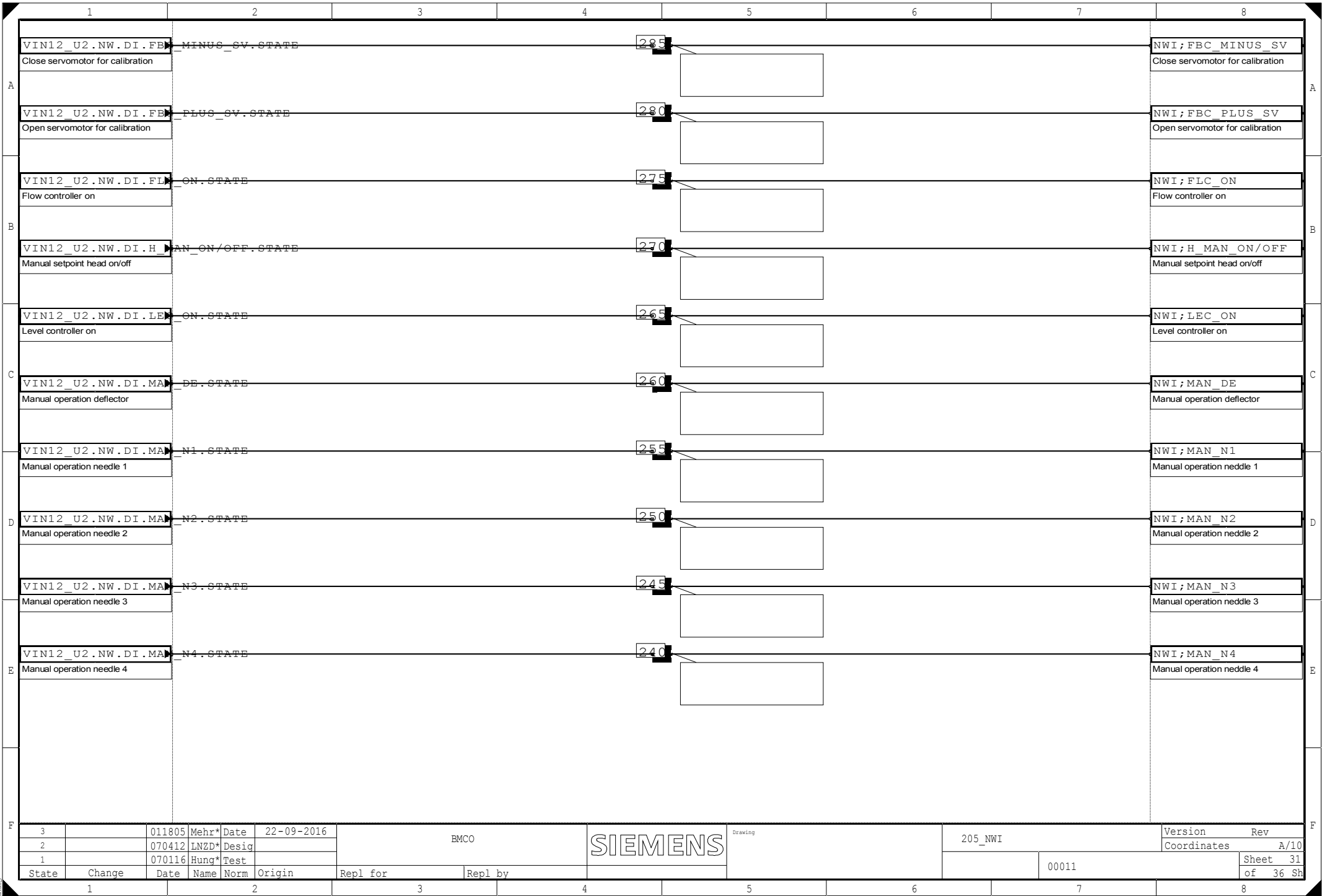
3				Date	22-09-2016	EMCO		SIEMENS	Drawing	205_NWI		Version	Rev	
2				Desig								Coordinates	A/4	
1		070116	Hung*	Test								Sheet	25	
State	Change	Date	Name	Norm	Origin	Repl for	Repl by			0005		of 36 Sh		



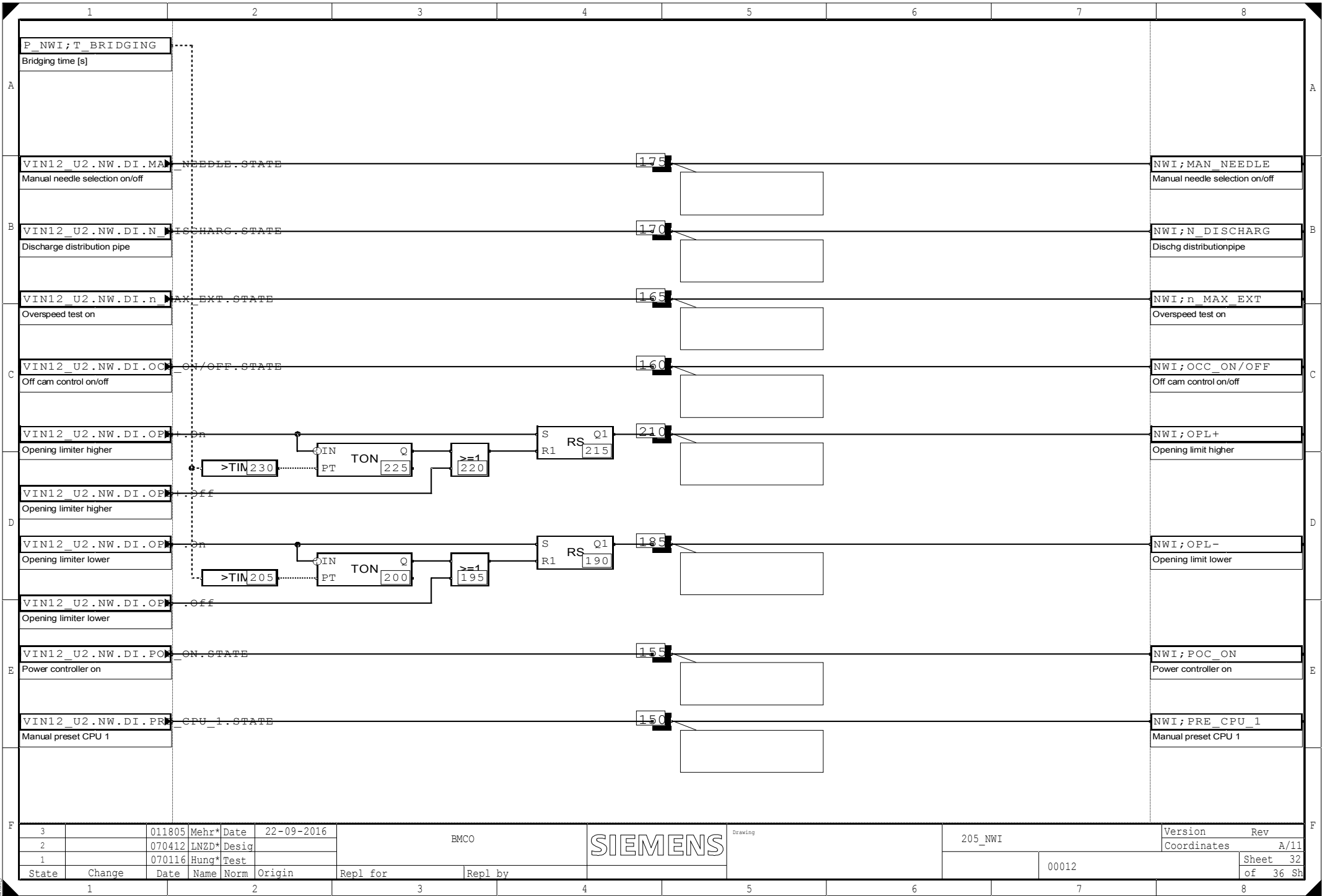
3			Date	22-09-2016	BMC0		SIEMENS		Drawing		205_NWI		Version	Rev
2		070412	LNZD*	Desig									Coordinates	A/5
1		070116	Hung*	Test										
State	Change	Date	Name	Norm	Origin	Repl for	Repl by					0006	Sheet	26
1													of	36 Sh



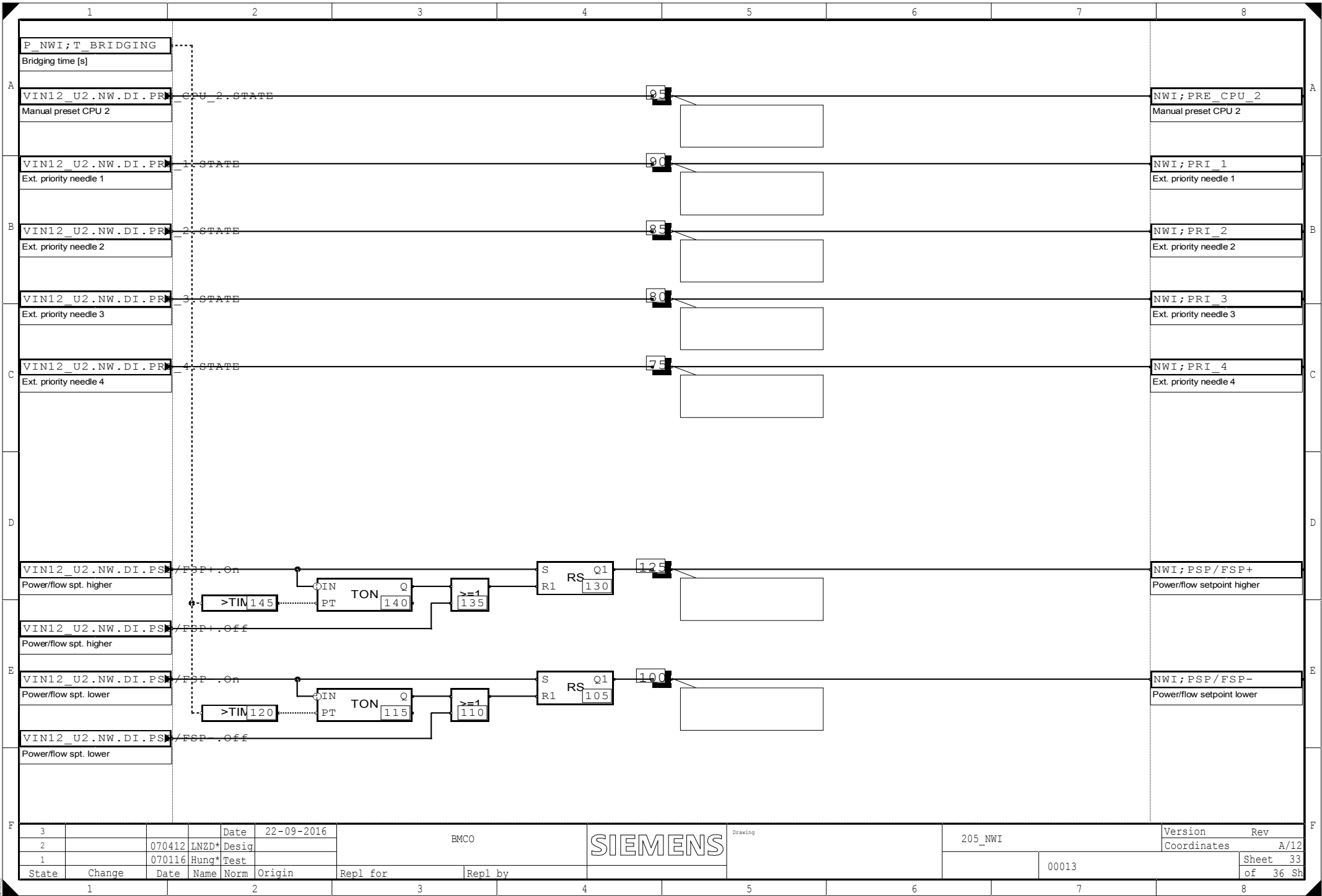


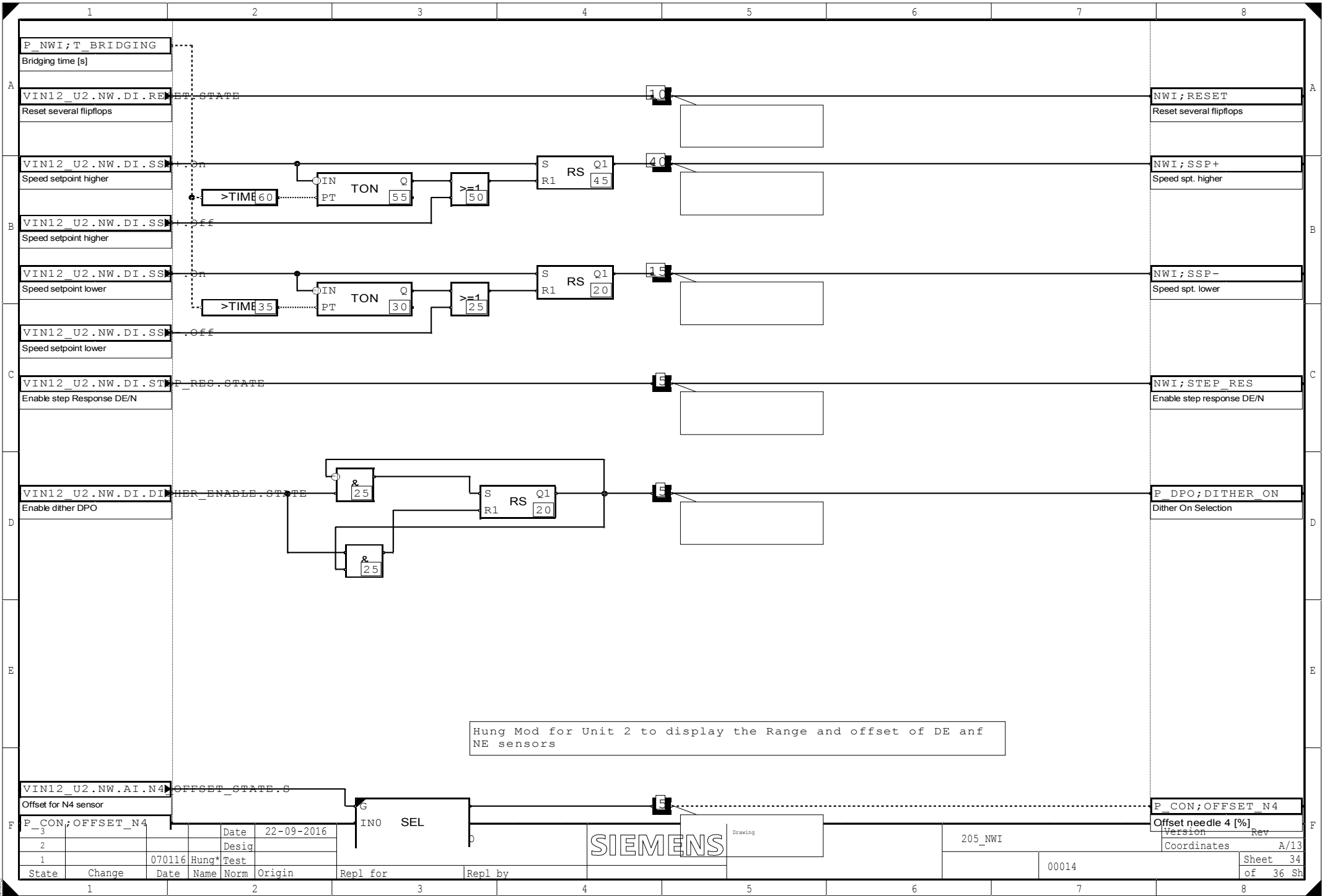


3		011805	Mehr*	Date	22-09-2016	BMC0		Drawing	205_NWI		Version		Rev
2		070412	LNZD*	Desig							Coordinates		A/10
1		070116	Hung*	Test					00011		Sheet 31		of 36 Sh
State	Change	Date	Name	Norm	Origin								

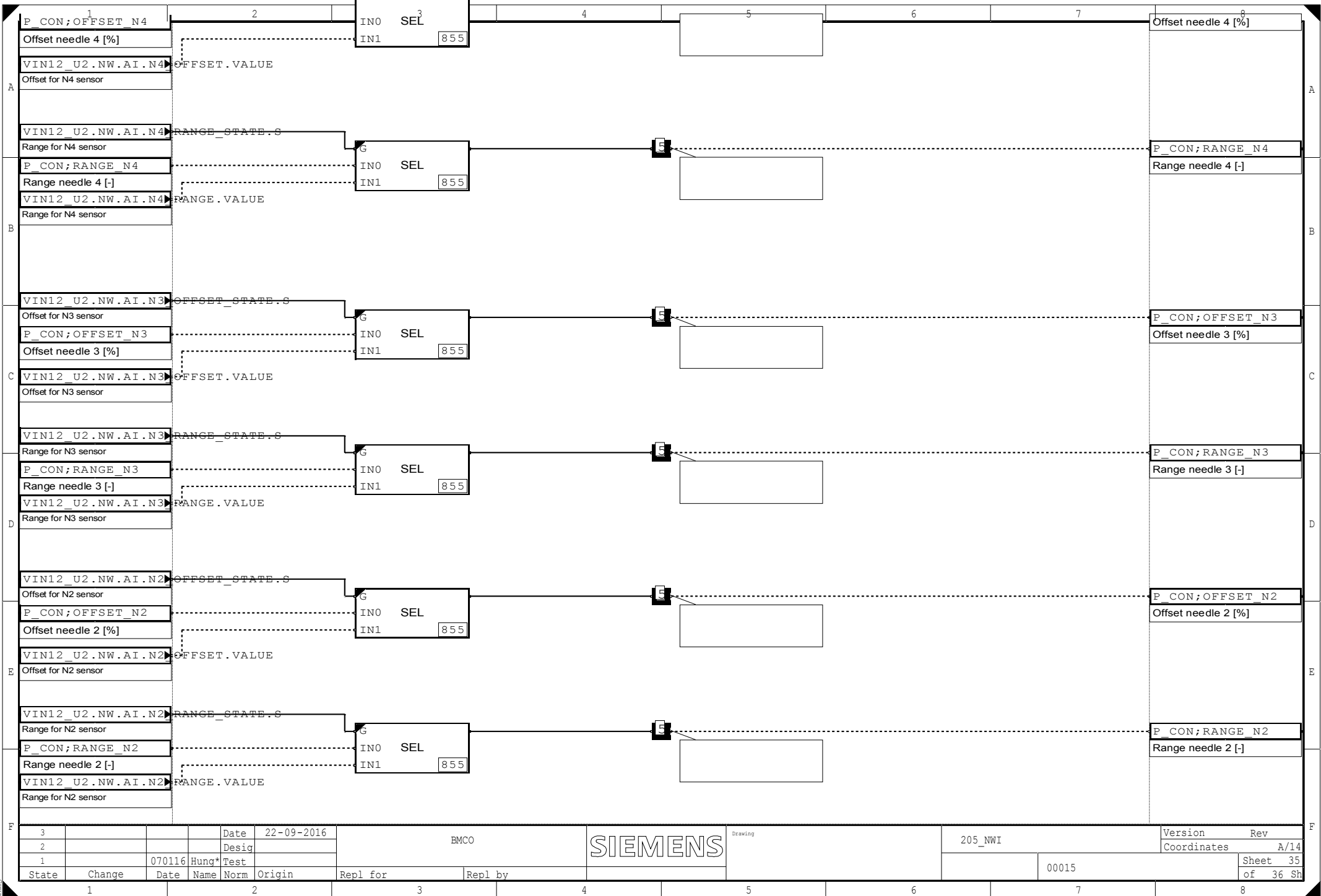


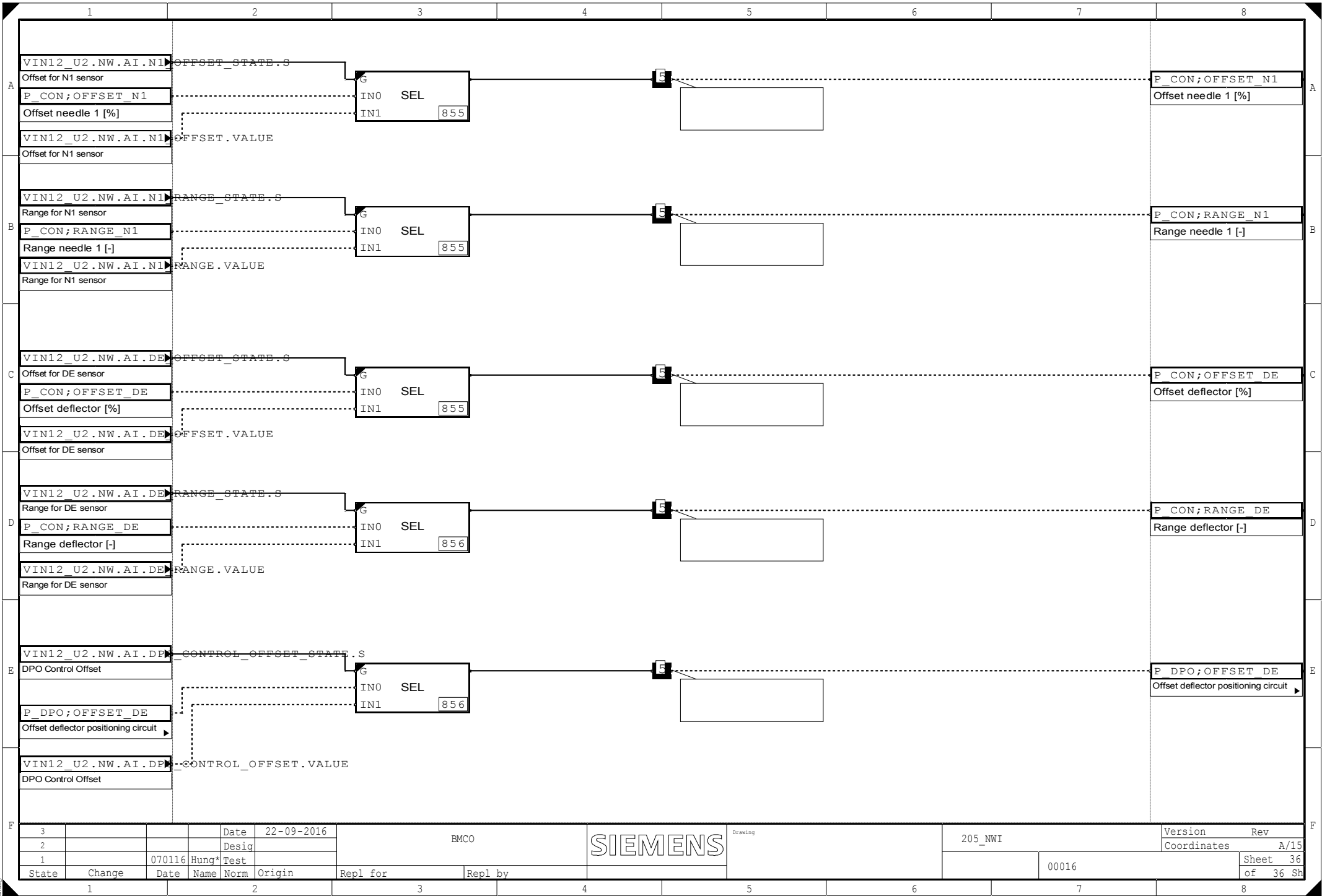
3		011805	Mehr*	Date	22-09-2016	BMC0		SIEMENS	Drawing	205_NWI		Version	Rev
2		070412	LNZD*	Desig								Coordinates	A/11
1		070116	Hung*	Test									
State	Change	Date	Name	Norm	Origin	Repl for	Repl by				00012	Sheet	32
1												of	36 Sh

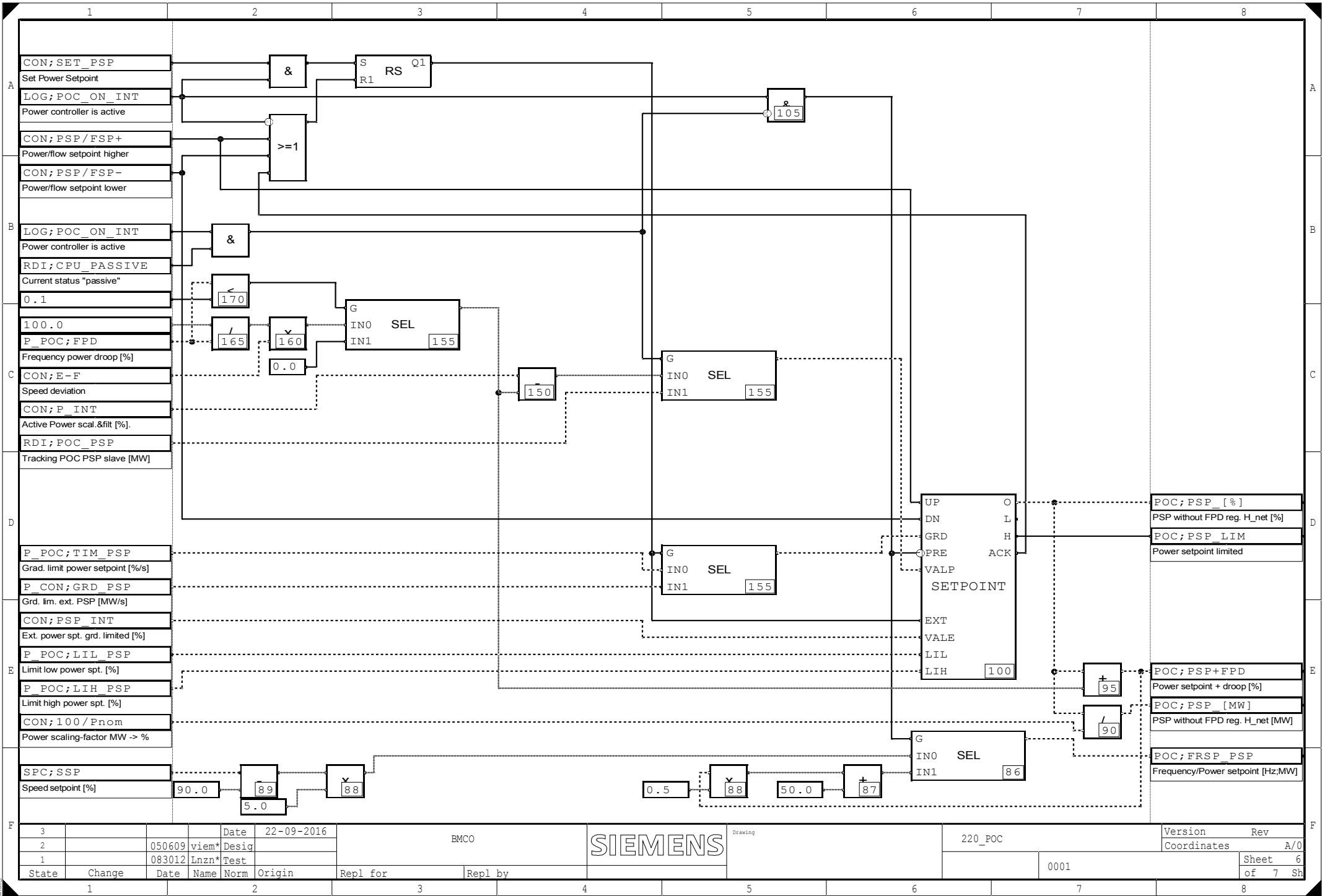




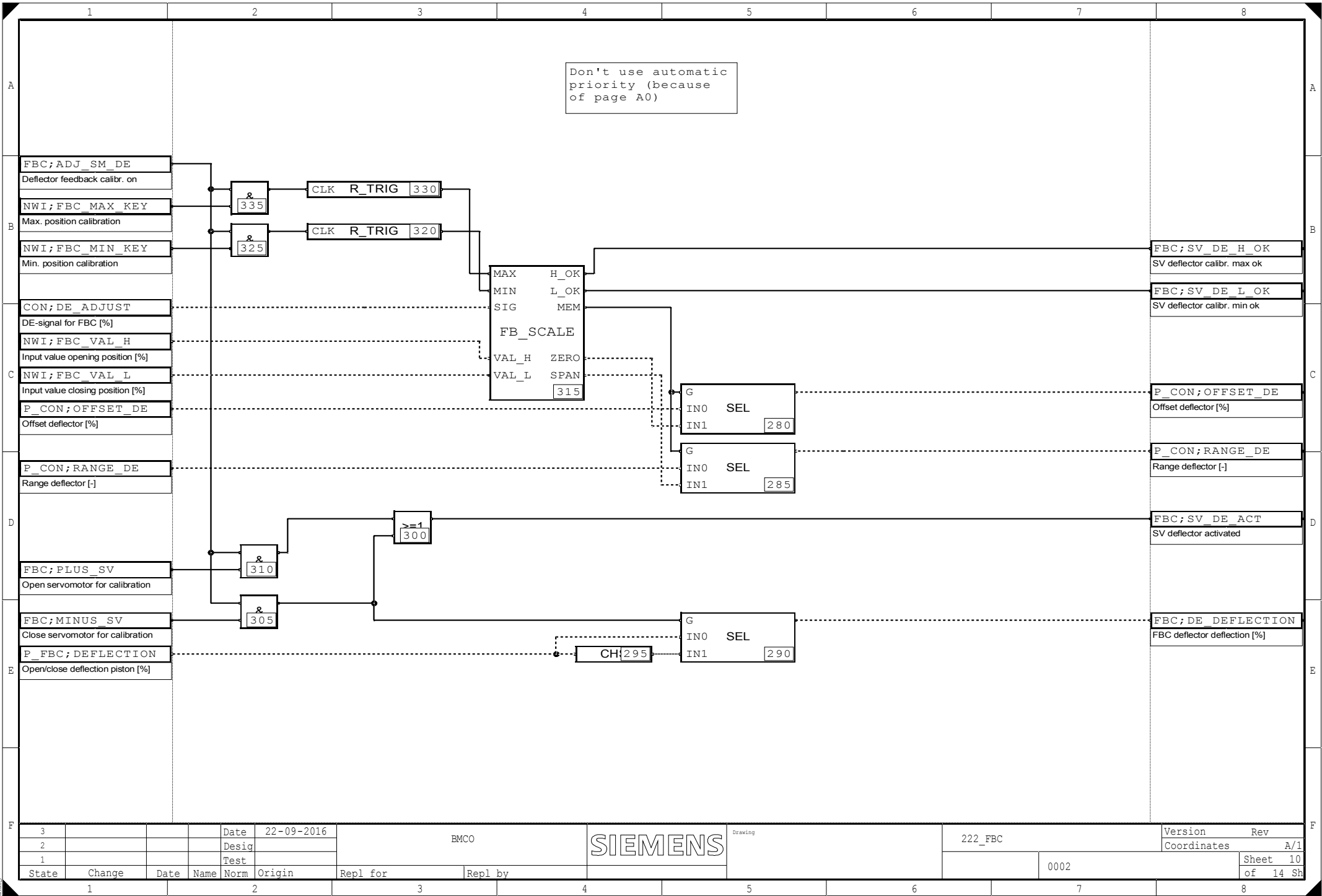
P_CON; OFFSET_N4		Date		22-09-2016		IN0		SEL		Drawing		205_NWI		Offset needle 4 [%]		Version		Rev		Coordinates		A/13	
2				Desig																		Sheet 34	
1		070116		Hung* Test																		of 36 Sh	
State		Change		Date		Name		Norm		Origin		Repl for		Repl by				00014					



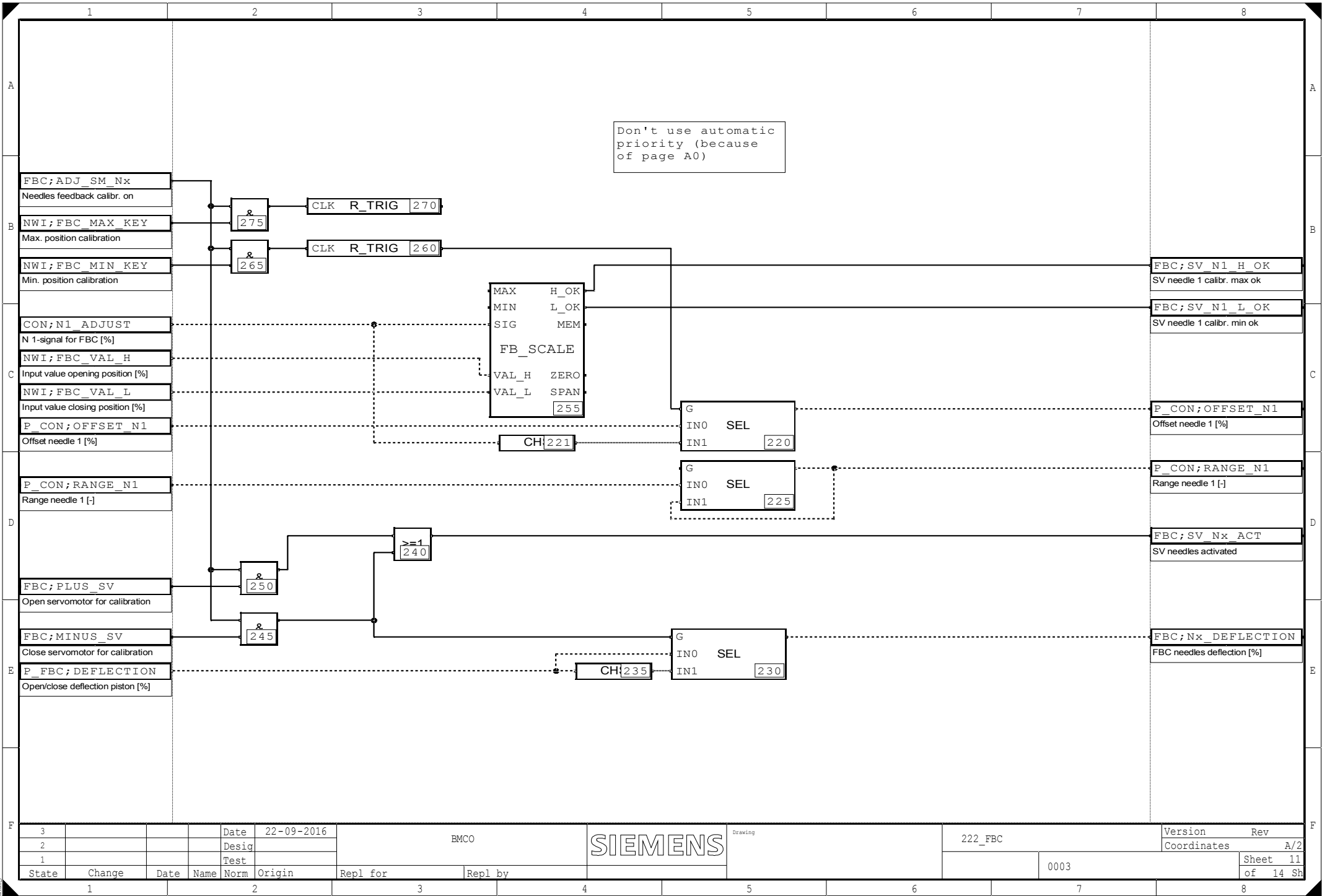


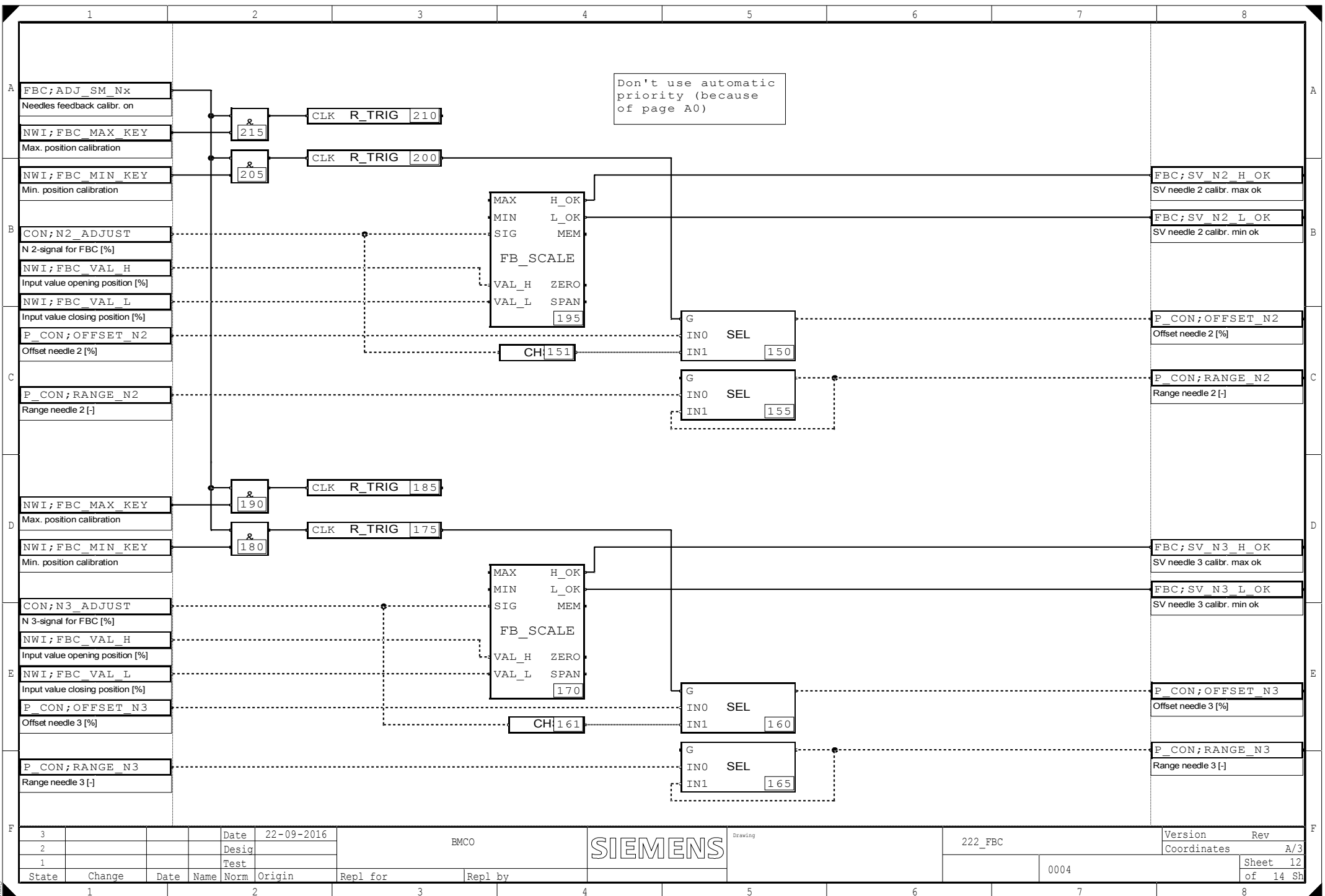


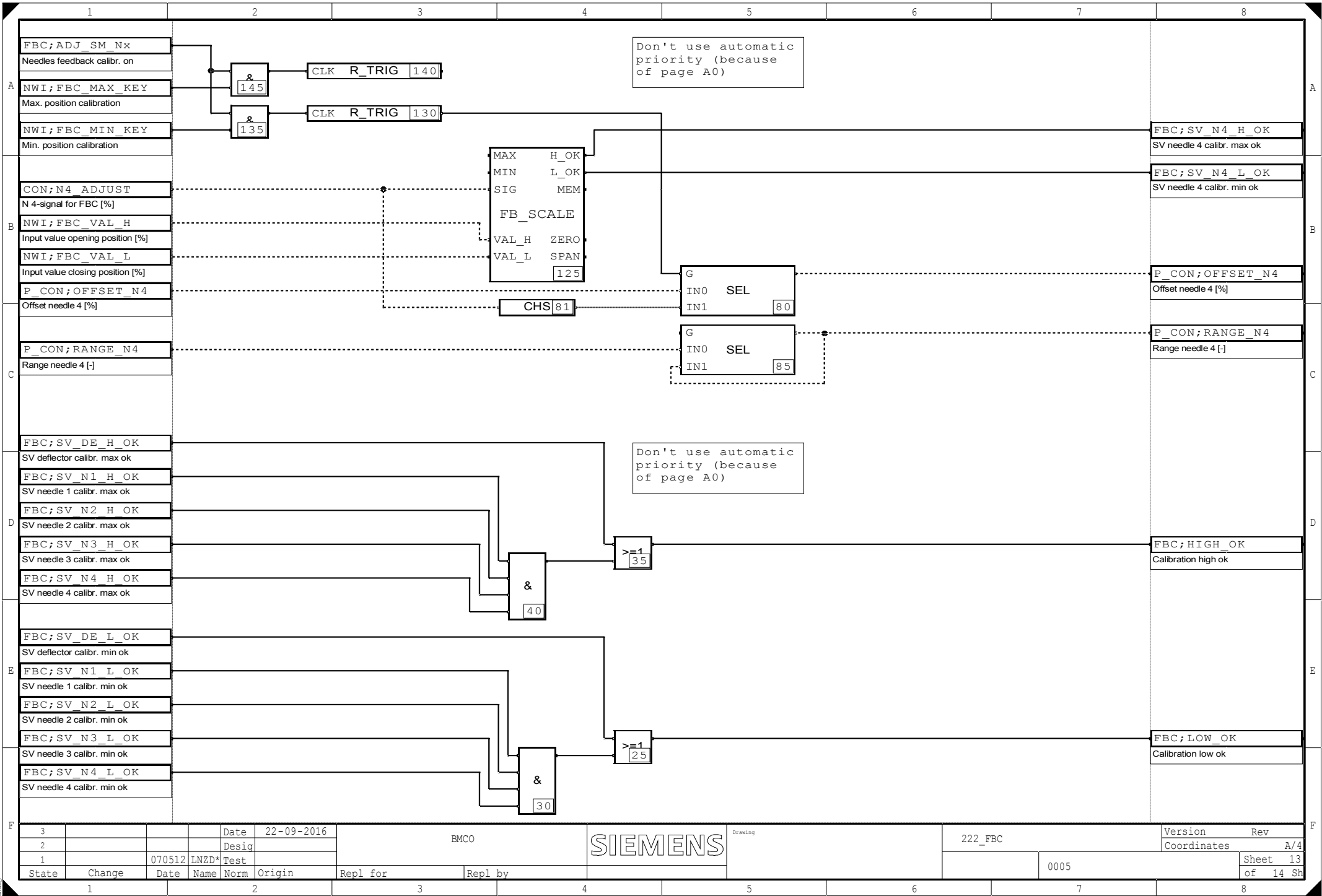
3			Date	22-09-2016	BMC0		Drawing	220_POC		Version	Rev
2		050609	viem*	Desig						Coordinates	A/0
1		083012	Lnzn*	Test						Sheet	6
State	Change	Date	Name	Norm	Origin	Repl for		0001		of	7 Sh
1											
2											
3											
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6											
7											
8											



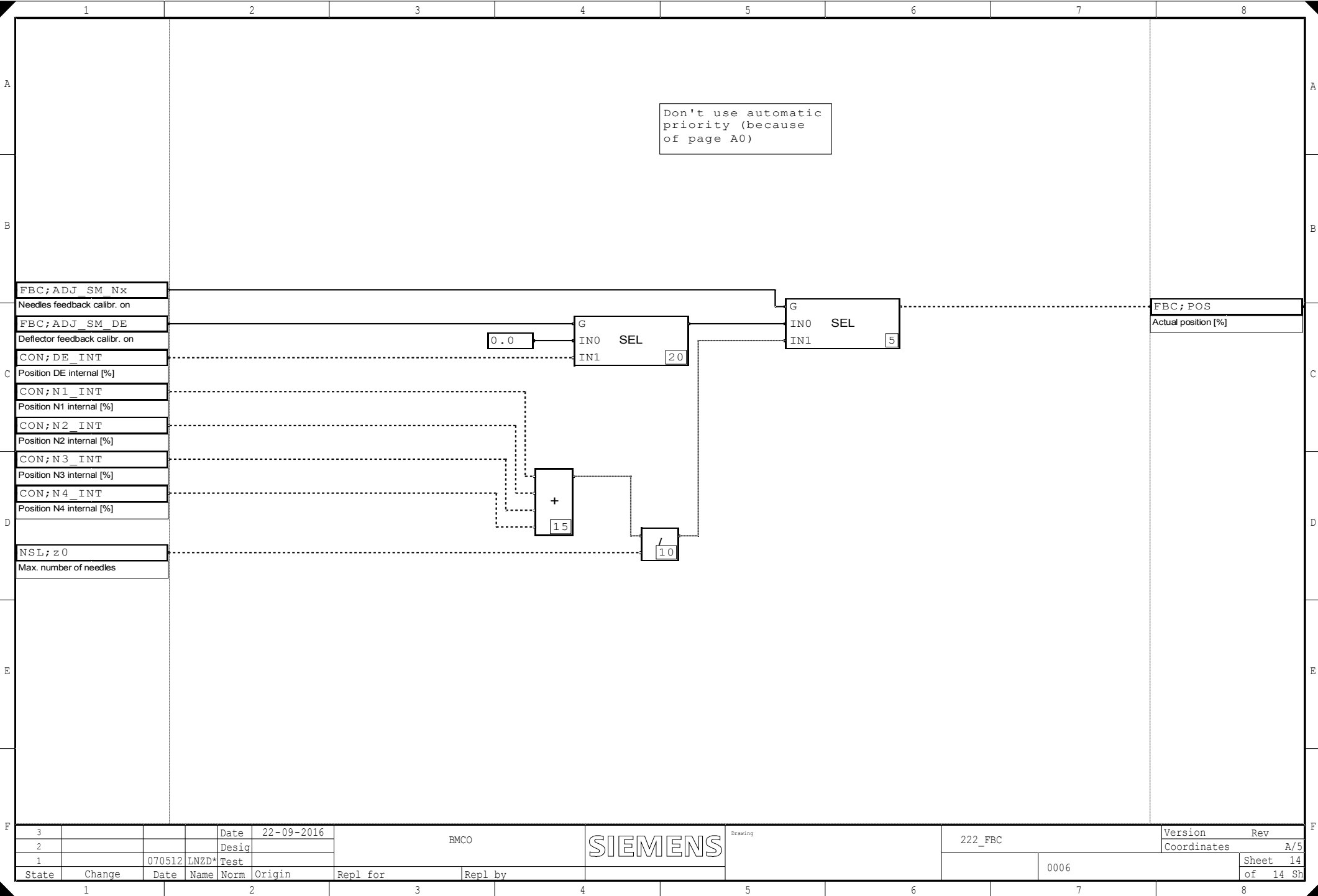
3			Date	22-09-2016	BMCO		SIEMENS		Drawing		222_FBC		Version	Rev
2			Desig										Coordinates	A/1
1			Test											
State	Change	Date	Name	Norm	Origin	Repl for	Repl by					0002	Sheet	10
1													of	14 Sh

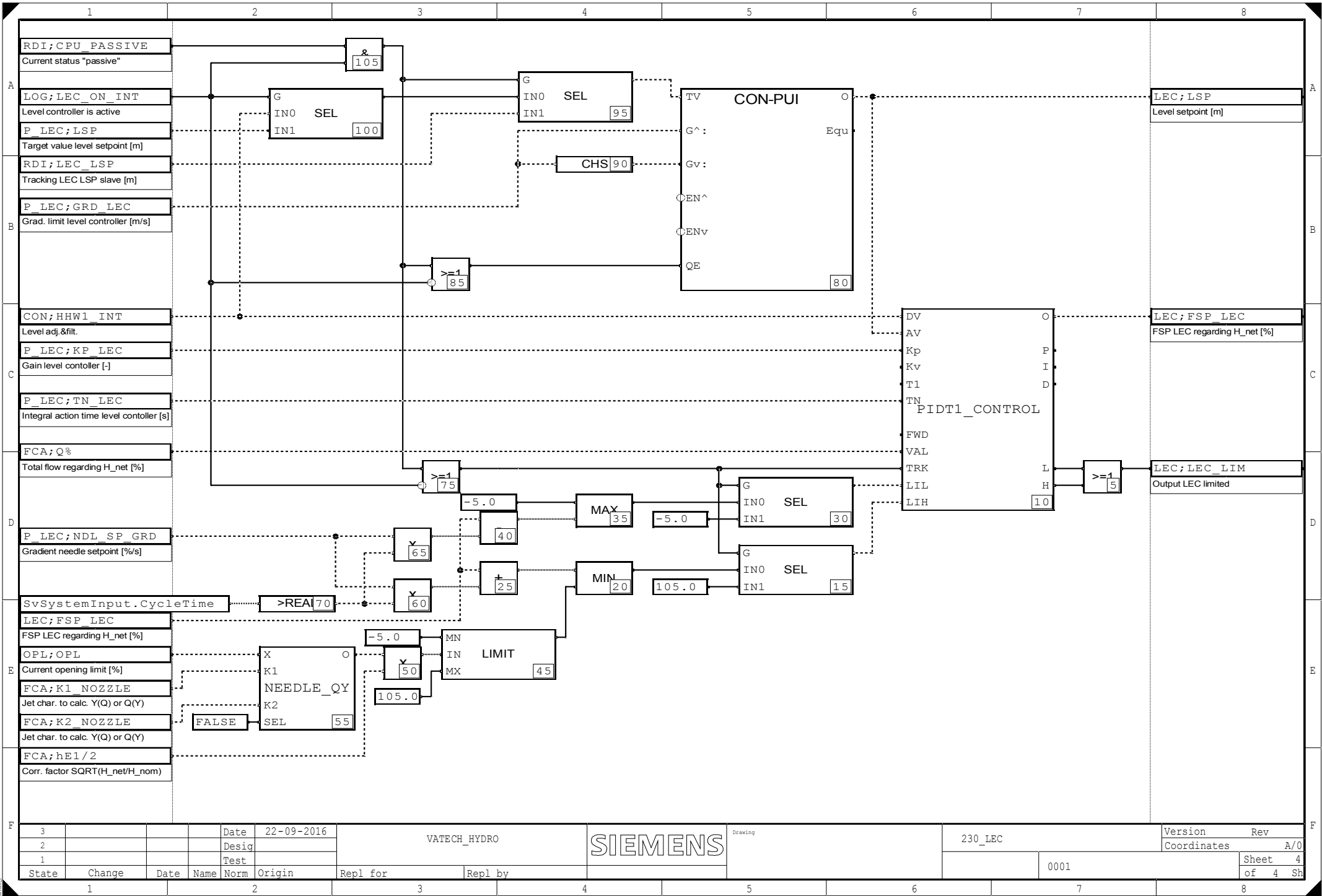




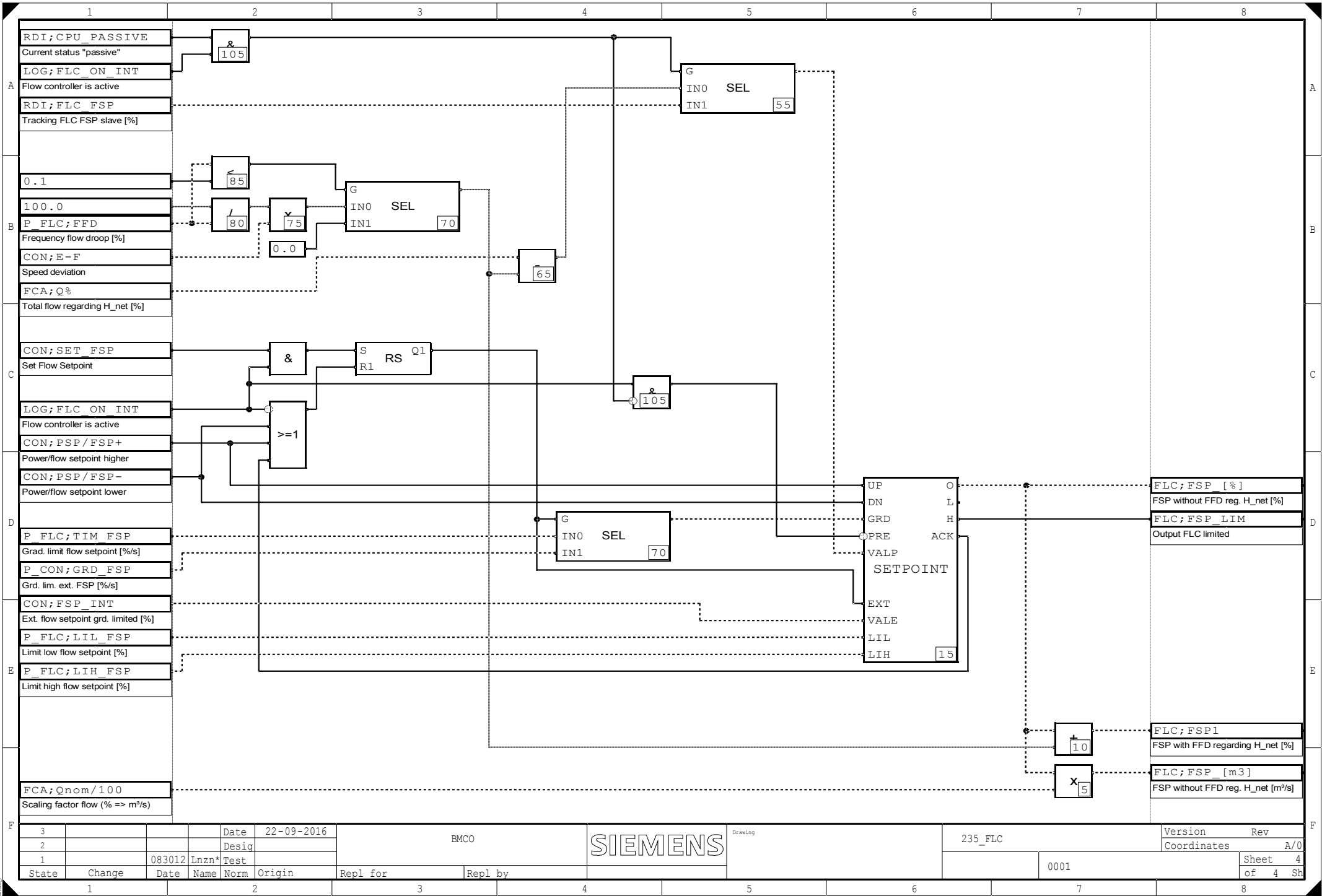


3			Date	22-09-2016	BMC0		SIEMENS		Drawing		222_FBC		Version	Rev
2			Desig										Coordinates	A/4
1			070512	LNZD*	Test									Sheet 13
State	Change	Date	Name	Norm	Origin	Repl for	Repl by					0005	of 14 Sh	

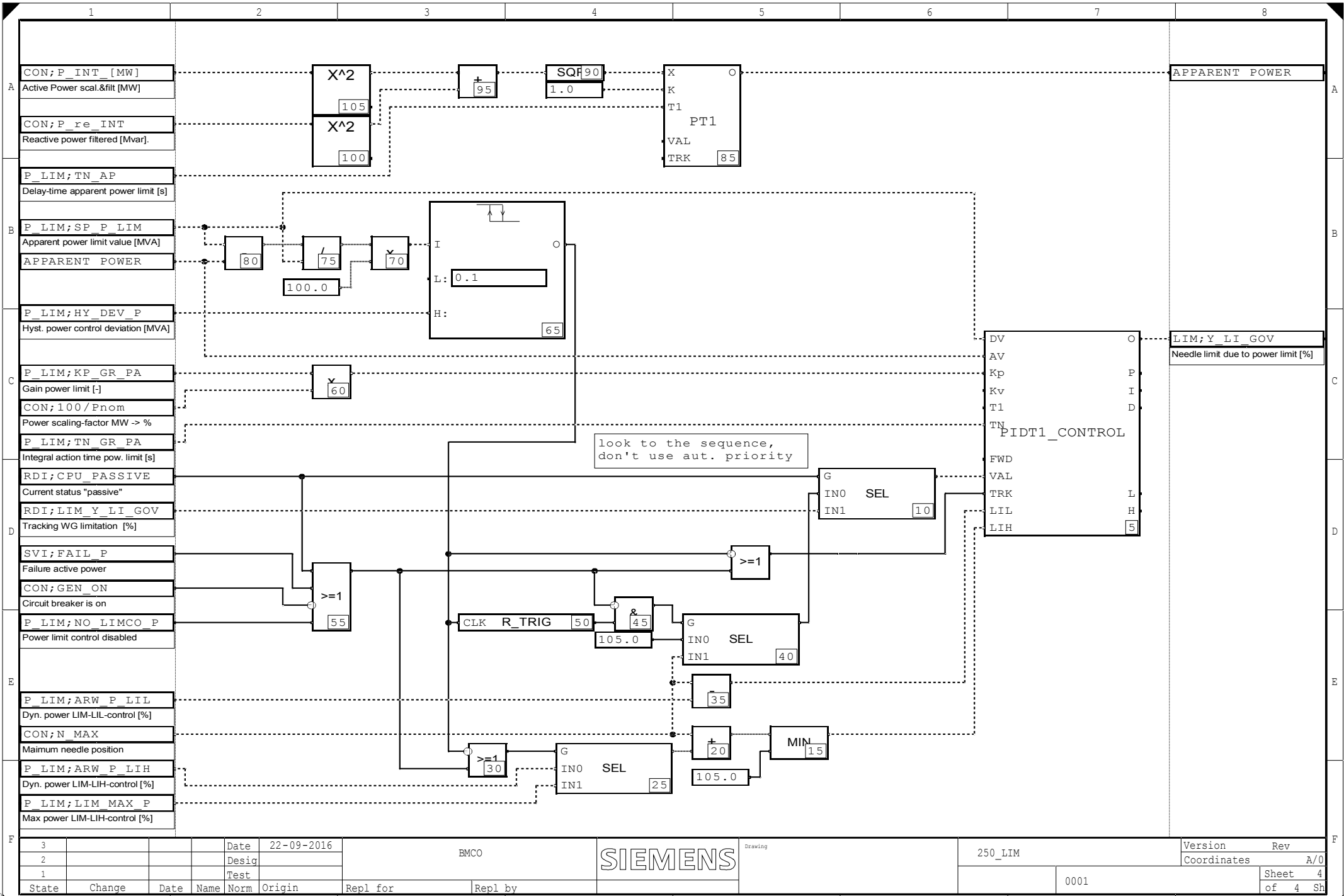




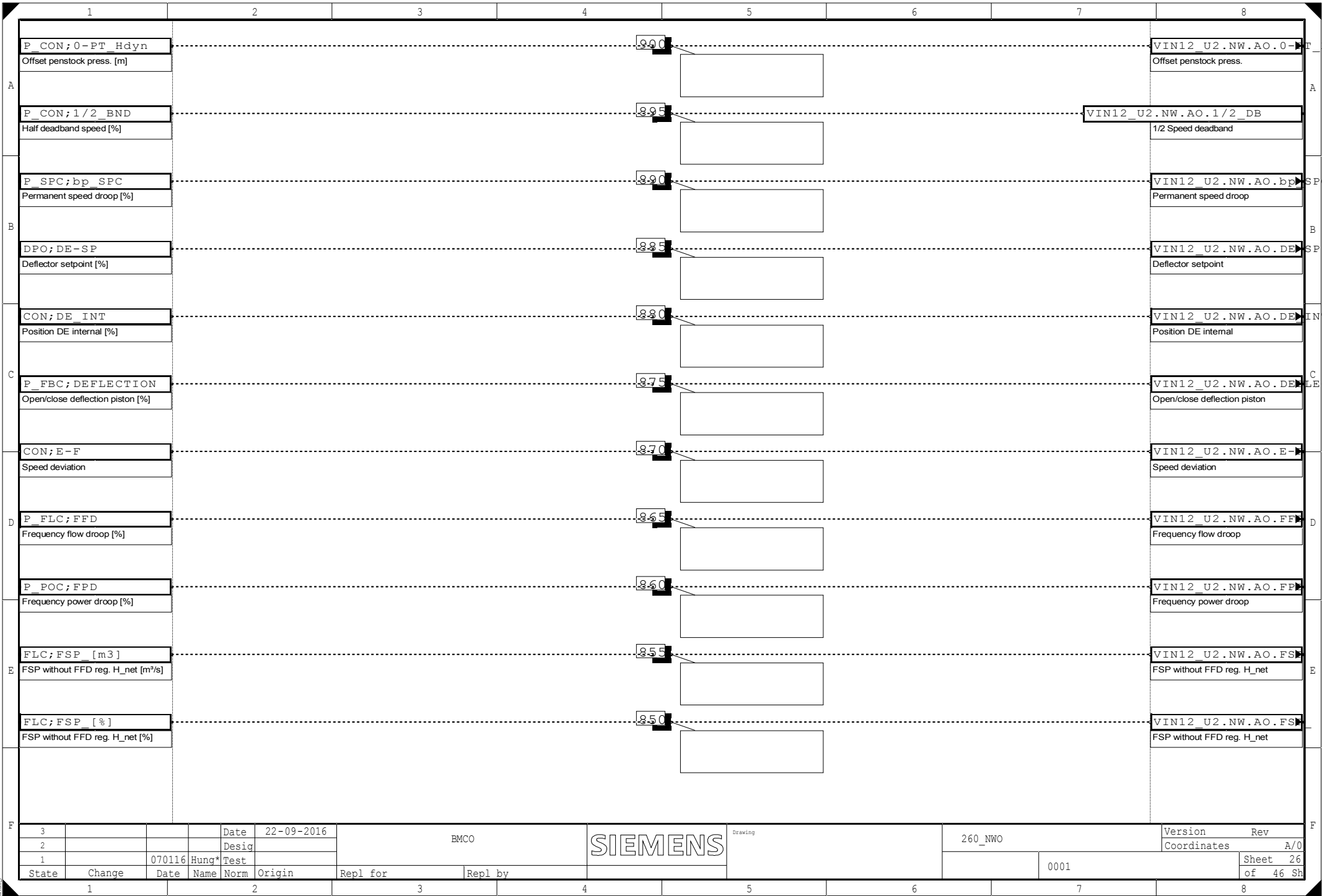
3				Date	22-09-2016		VATECH_HYDRO		SIEMENS	Drawing	230_LEC		Version		Rev			
2				Desig									Coordinates		A/0			
1				Test									Sheet		4			
State	Change	Date	Name	Norm	Origin	Repl for	Repl by				0001		of		4		Sh	



3			Date		22-09-2016		EMCO		SIEMENS		Drawing		235_FLC		Version		Rev		
2			Desig												Coordinates		A/0		
1			083012		Lnzn* Test												Sheet		4
State		Change		Date		Name		Norm		Origin		Repl for		Repl by				of 4 Sh	

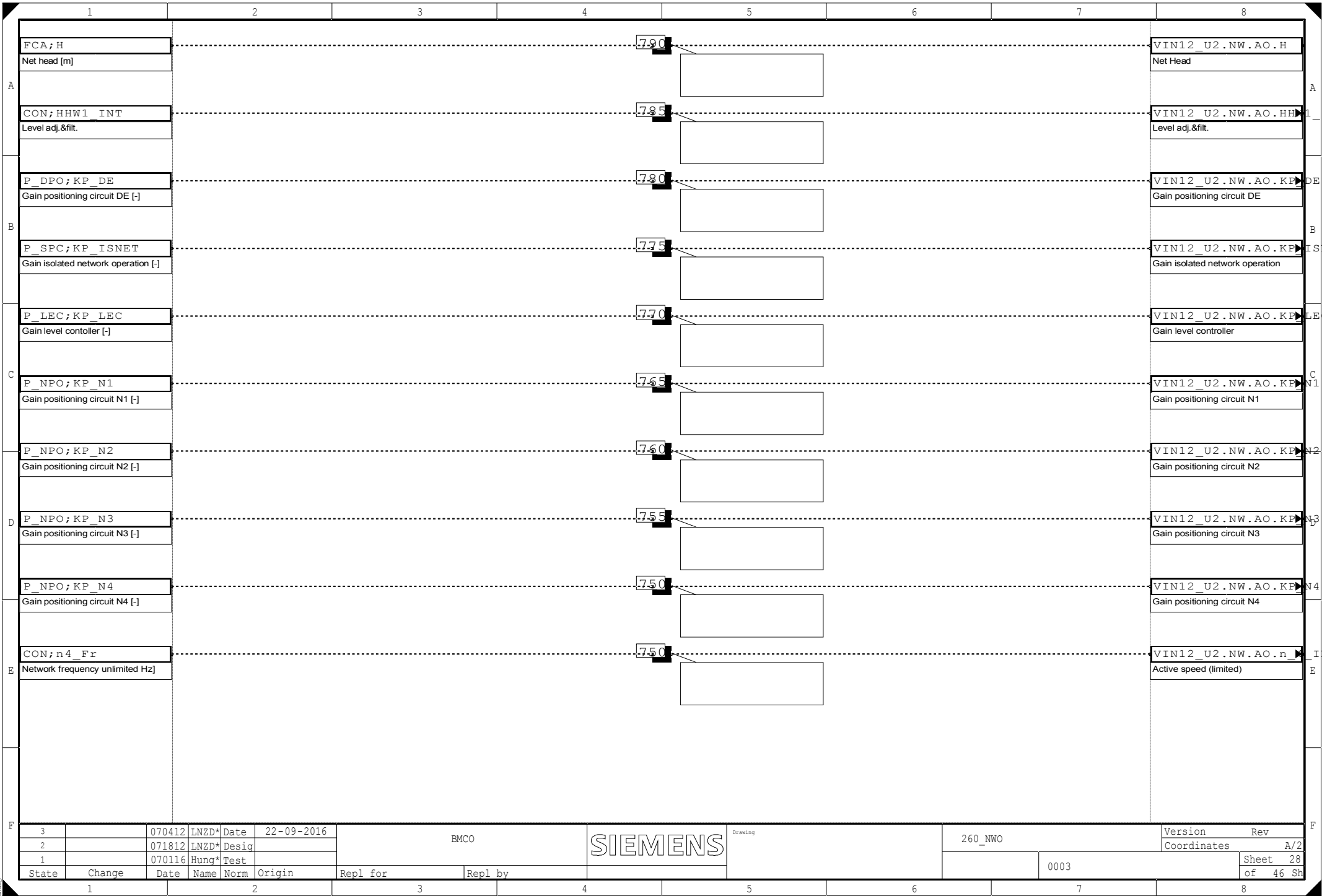


3			Date	22-09-2016	EMCO		SIEMENS	Drawing	250_LIM		Version	Rev
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1			Test								Sheet	4
State	Change	Date	Name	Norm	Origin	Repl for	Repl by		0001	of 4 Sh		



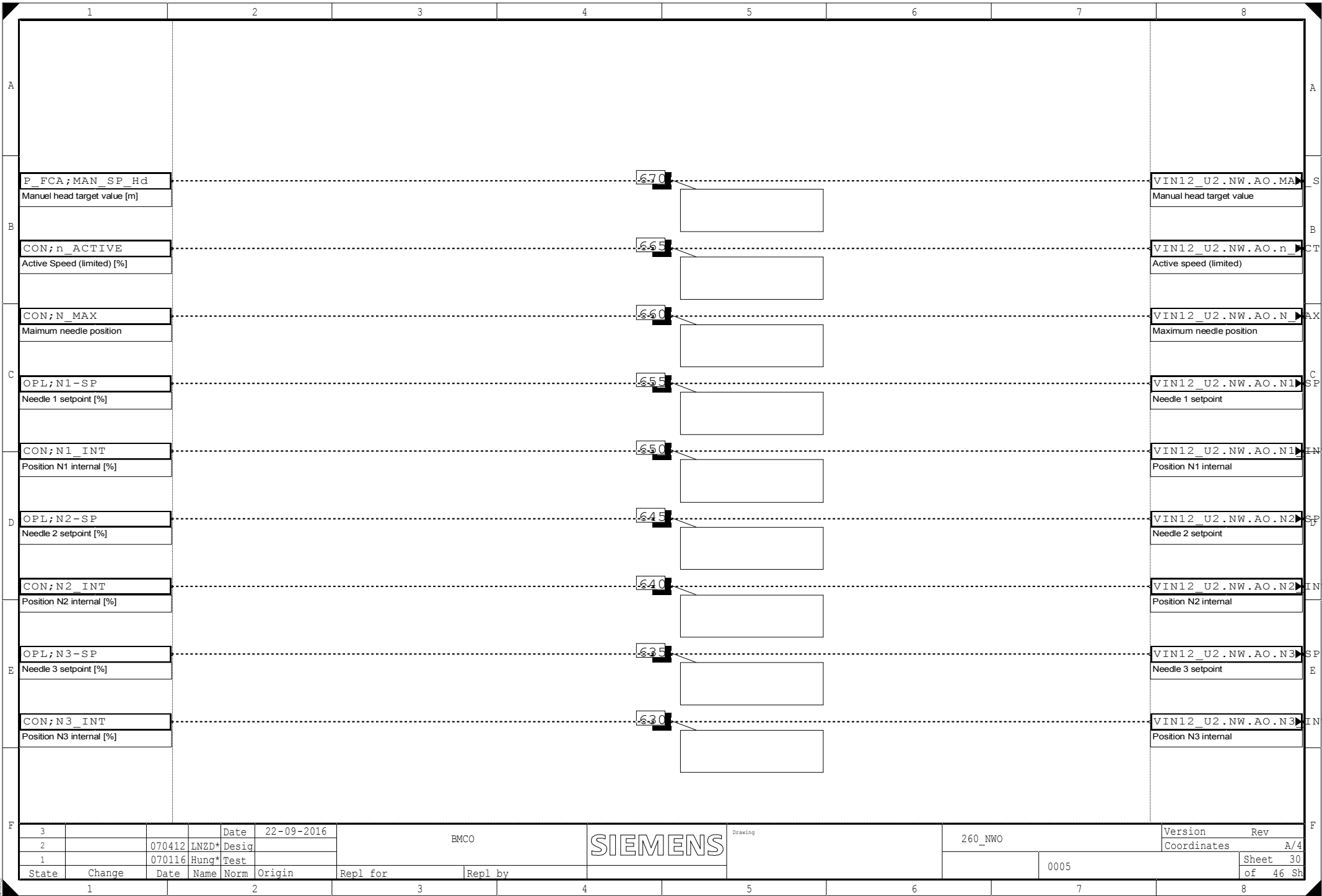
3			Date	22-09-2016	EMCO		SIEMENS	Drawing	260_NWO		Version	Rev
2			Desig								Coordinates	A/0
1		070116	Hung*	Test							Sheet	26
State	Change	Date	Name	Norm	Origin	Repl for	Repl by			0001	of 46 Sh	

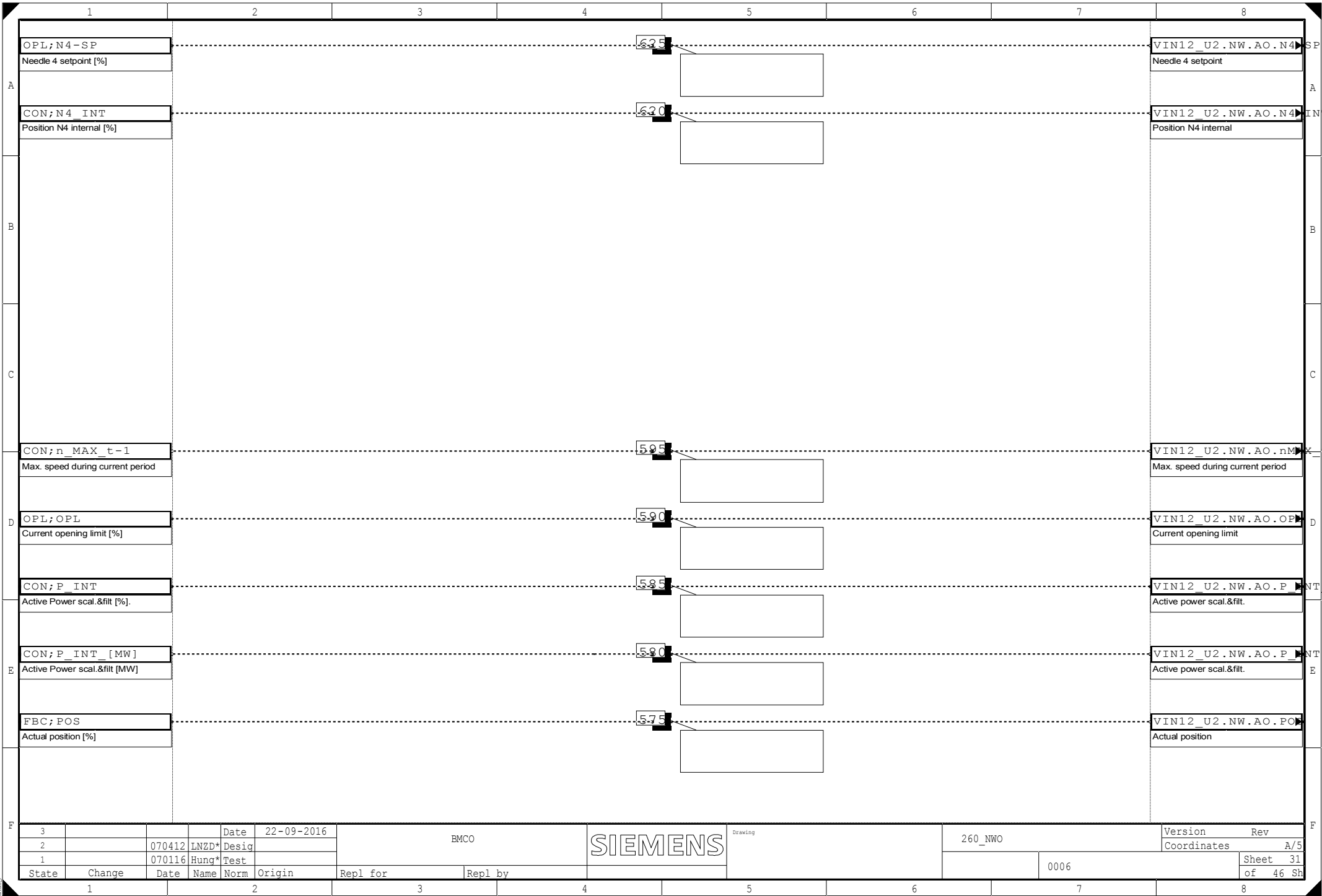




3		070412	LNZD*	Date	22-09-2016	BMCO		SIEMENS		Drawing		260_NWO		Version		Rev
2		071812	LNZD*	Desig										Coordinates		A/2
1		070116	Hung*	Test												
State	Change	Date	Name	Norm	Origin	Repl for	Repl by						0003		Sheet	28
1		2		3		4		5		6		7		8		of 46 Sh

	1	2	3	4	5	6	7	8																																																																												
A	P_SPC;KP_NOLOAD				735				VIN12_U2.NW.AO.KP	NOLOAD																																																																										
	Gain no-load operation [-]								Gain noload operation																																																																											
	P_POC;KP_POC				730				VIN12_U2.NW.AO.KP	POC																																																																										
	Gain power controller [-]								Gain power controller																																																																											
B	P_LOG;LIH_ISNET				735				VIN12_U2.NW.AO.LI	ISNET																																																																										
	Limit high isolated netw. detect. [%]								Limit high isolated netw. detect.																																																																											
	P_OPL;LIH_OPL				730				VIN12_U2.NW.AO.LI	OPL																																																																										
	Limit high opening limit setpoint [%]								Limit high opening limit setpoint																																																																											
C	P_LOG;LIL_ISNET				715				VIN12_U2.NW.AO.LI	ISNET																																																																										
	Limit low isolated netw. detect. [%]								Limit low isolated netw. detect.																																																																											
	P_LEC;LSP				710				VIN12_U2.NW.AO.LS	LSP																																																																										
	Target value level setpoint [m]								Target value level setpoint																																																																											
D	P_DPO;MAN_DE_SP				705				VIN12_U2.NW.AO.MA	DE_SP																																																																										
	Manual setpoint deflector [%]								Manual setpoint deflector																																																																											
	P_NPO;MAN_N1_SP				700				VIN12_U2.NW.AO.MA	N1_SP																																																																										
	Manual setpoint needle 1 [%]								Manual setpoint needle 1																																																																											
E	P_NPO;MAN_N2_SP				695				VIN12_U2.NW.AO.MA	N2_SP																																																																										
	Manual setpoint needle 2 [%]								Manual setpoint needle 2																																																																											
	P_NPO;MAN_N3_SP				690				VIN12_U2.NW.AO.MA	N3_SP																																																																										
	Manual setpoint needle 3 [%]								Manual setpoint needle 3																																																																											
F	P_NPO;MAN_N4_SP				685				VIN12_U2.NW.AO.MA	N4_SP																																																																										
	Manual setpoint needle 4 [%]								Manual setpoint needle 4																																																																											
<table><tr><td>3</td><td></td><td></td><td></td><td>Date</td><td>22-09-2016</td><td colspan="3" rowspan="3">BMCO</td><td rowspan="3">SIEMENS</td><td rowspan="3">Drawing</td><td colspan="2">260_NWO</td><td>Version</td><td>Rev</td></tr><tr><td>2</td><td></td><td></td><td></td><td>Desig</td><td></td><td colspan="2">Coordinates</td><td>A/3</td></tr><tr><td>1</td><td></td><td></td><td>070116</td><td>Hung*</td><td>Test</td><td colspan="2"></td><td></td></tr><tr><td>State</td><td>Change</td><td>Date</td><td>Name</td><td>Norm</td><td>Origin</td><td>Repl for</td><td>Repl by</td><td colspan="2"></td><td></td><td></td><td>0004</td><td>Sheet</td><td>29</td></tr><tr><td colspan="2"></td><td>1</td><td colspan="2"></td><td>2</td><td colspan="2"></td><td>3</td><td colspan="2"></td><td>4</td><td colspan="2"></td><td>5</td><td colspan="2"></td><td>6</td><td colspan="2"></td><td>7</td><td colspan="2"></td><td>8</td><td colspan="2"></td></tr></table>											3				Date	22-09-2016	BMCO			SIEMENS	Drawing	260_NWO		Version	Rev	2				Desig		Coordinates		A/3	1			070116	Hung*	Test				State	Change	Date	Name	Norm	Origin	Repl for	Repl by					0004	Sheet	29			1			2			3			4			5			6			7			8		
3				Date	22-09-2016	BMCO			SIEMENS	Drawing	260_NWO		Version	Rev																																																																						
2				Desig							Coordinates		A/3																																																																							
1			070116	Hung*	Test																																																																															
State	Change	Date	Name	Norm	Origin	Repl for	Repl by					0004	Sheet	29																																																																						
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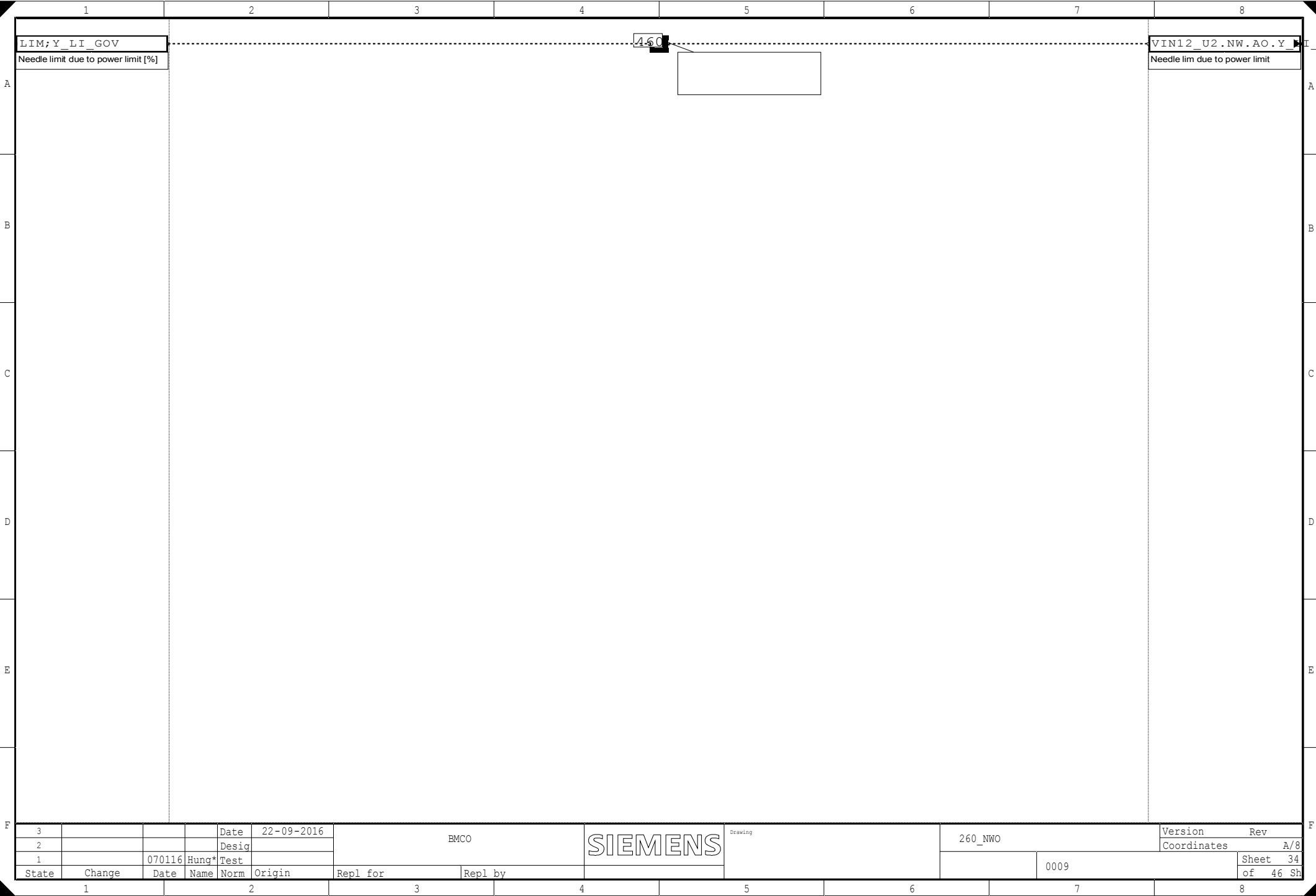


3			Date	22-09-2016	BMC0			Drawing		260_NWO		Version	Rev
2		070412	LNZD*	Desig								Coordinates	A/5
1		070116	Hung*	Test						0006		Sheet	31
State	Change	Date	Name	Norm	Origin	Repl for	Repl by					of	46 Sh

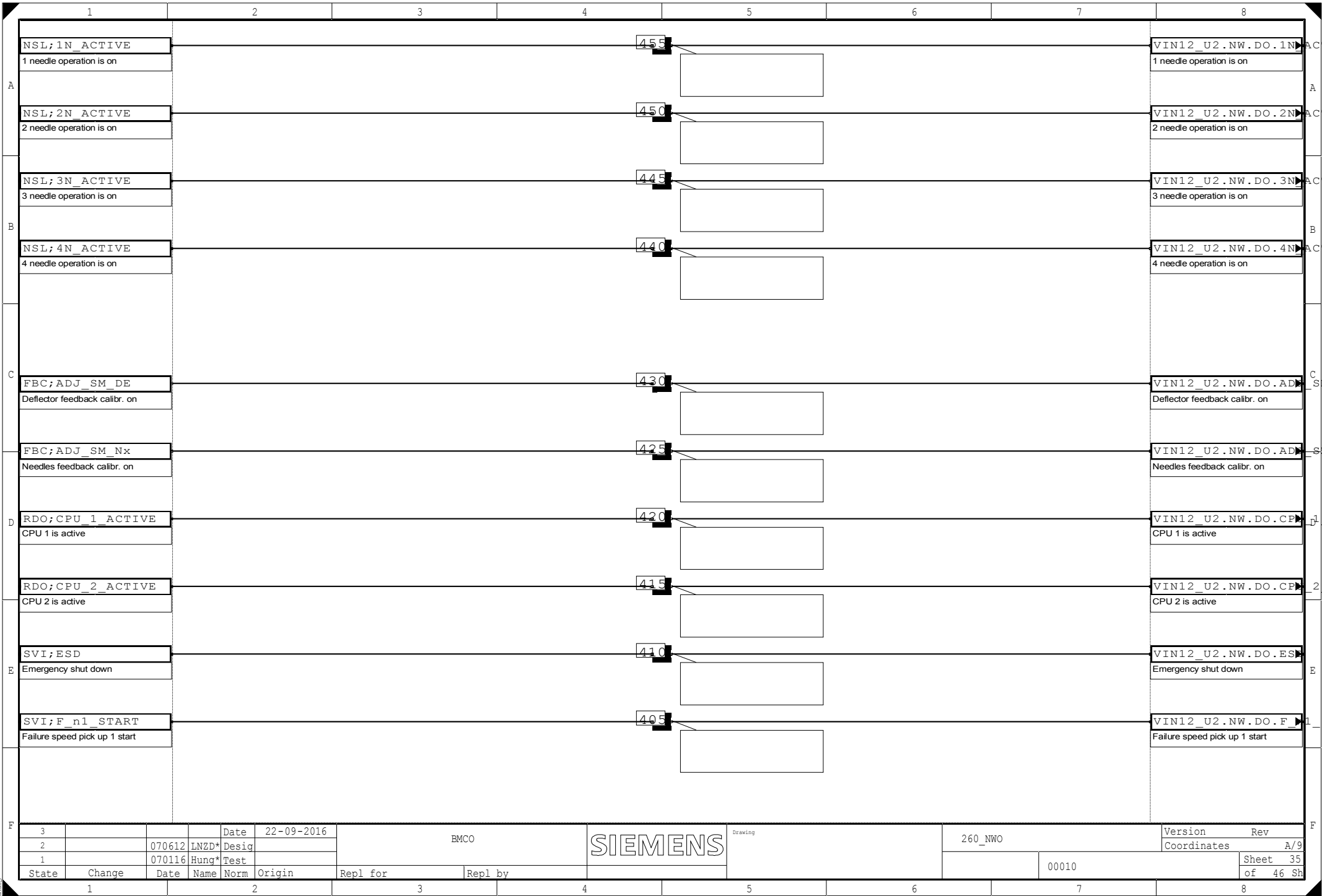


3				Date	22-09-2016	EMCO		SIEMENS	Drawing	260_NWO		Version	Rev	
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1		070116	Hung*	Test								Sheet	32	
State	Change	Date	Name	Norm	Origin	Repl for	Repl by			0007		of 46 Sh		

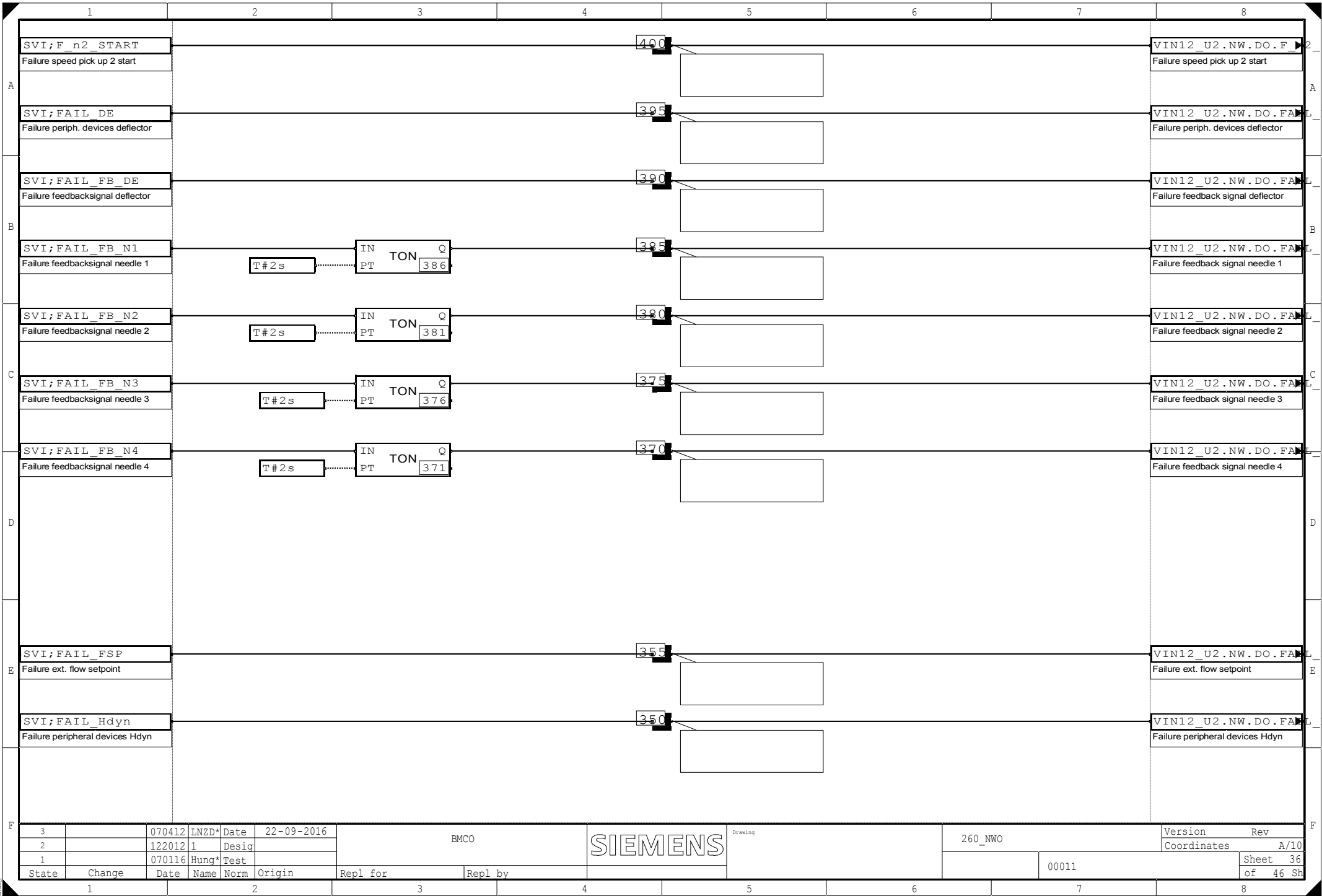
	1	2	3	4	5	6	7	8																																																								
A	P_DPO;TD_DE			515				VIN12_U2.NW.AO.TD	DE																																																							
	Derivative action time DE [s]							Derivative action time DE																																																								
B	P_SPC;TD_SPC			510				VIN12_U2.NW.AO.TD	SPC																																																							
	Derivative action time Td [s]							Derivative action time Td																																																								
B	P_FLC;TIM_FSP			505				VIN12_U2.NW.AO.TI	FSP																																																							
	Grad. limit flow setpoint [%/s]							Gradient limit flow setpoint																																																								
B	P_OPL;TIM_OPL			500				VIN12_U2.NW.AO.TI	OPL																																																							
	Opening limiter ramp time [%/s]							Opening limiter ramp time																																																								
C	P_POC;TIM_PSP			495				VIN12_U2.NW.AO.TI	PSP																																																							
	Grad. limit power setpoint [%/s]							Grad. limit power setpoint																																																								
C	P_SPC;TN_ISNET			490				VIN12_U2.NW.AO.TN	ISNET																																																							
	Integral action time isol. netw. op. [s]							Integral action time isol. netw.																																																								
D	P_LEC;TN_LEC			485				VIN12_U2.NW.AO.TN	LEC																																																							
	Integral action time level controller [s]							Integral action time LEC																																																								
D	P_SPC;TN_NOLOAD			480				VIN12_U2.NW.AO.TN	NOLOAD																																																							
	Integral action time no-load op. [s]							Integral action time noload																																																								
E	P_POC;TN_POC			475				VIN12_U2.NW.AO.TN	POC																																																							
	Integral action time power controller [s]							Integral action time POC																																																								
E	P_SPC;TN_START			470				VIN12_U2.NW.AO.TN	START																																																							
	Integral action time during start [s]							Integral action time during start																																																								
F	SVI;VER_CAEx			465				VIN12_U2.NW.AO.VE	CAEx																																																							
	Version Appl. CAEx plus							Version Appl. CAEx plus																																																								
<table><tr><td>3</td><td></td><td></td><td>Date</td><td>22-09-2016</td><td colspan="2" rowspan="2">BMC0</td><td colspan="2" rowspan="2">Drawing</td><td rowspan="4">260_NWO</td><td>Version</td><td>Rev</td></tr><tr><td>2</td><td></td><td></td><td>Desig</td><td></td><td>Coordinates</td><td>A/7</td></tr><tr><td>1</td><td></td><td>070116</td><td>Hung*</td><td>Test</td><td colspan="2"></td><td colspan="3"></td><td rowspan="2">0008</td><td>Sheet 33</td></tr><tr><td>State</td><td>Change</td><td>Date</td><td>Name</td><td>Norm</td><td>Origin</td><td>Repl for</td><td>Repl by</td><td colspan="3"></td><td>of 46 Sh</td></tr><tr><td>1</td><td></td><td></td><td></td><td>2</td><td></td><td>3</td><td></td><td>4</td><td></td><td>7</td><td>8</td></tr></table>										3			Date	22-09-2016	BMC0		Drawing		260_NWO	Version	Rev	2			Desig		Coordinates	A/7	1		070116	Hung*	Test						0008	Sheet 33	State	Change	Date	Name	Norm	Origin	Repl for	Repl by				of 46 Sh	1				2		3		4		7	8
3			Date	22-09-2016	BMC0		Drawing		260_NWO	Version	Rev																																																					
2			Desig							Coordinates	A/7																																																					
1		070116	Hung*	Test						0008	Sheet 33																																																					
State	Change	Date	Name	Norm	Origin	Repl for	Repl by				of 46 Sh																																																					
1				2		3		4		7	8																																																					



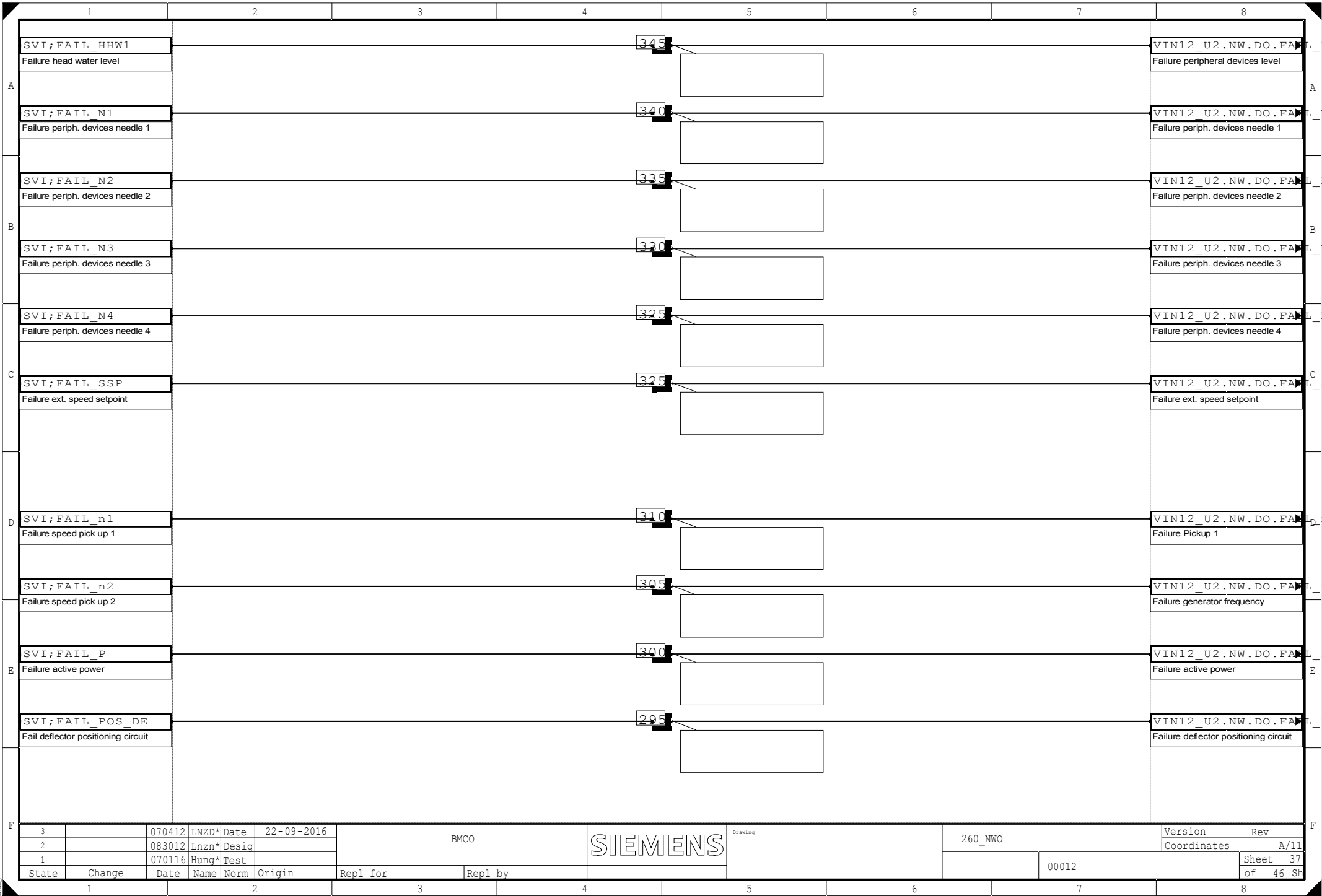
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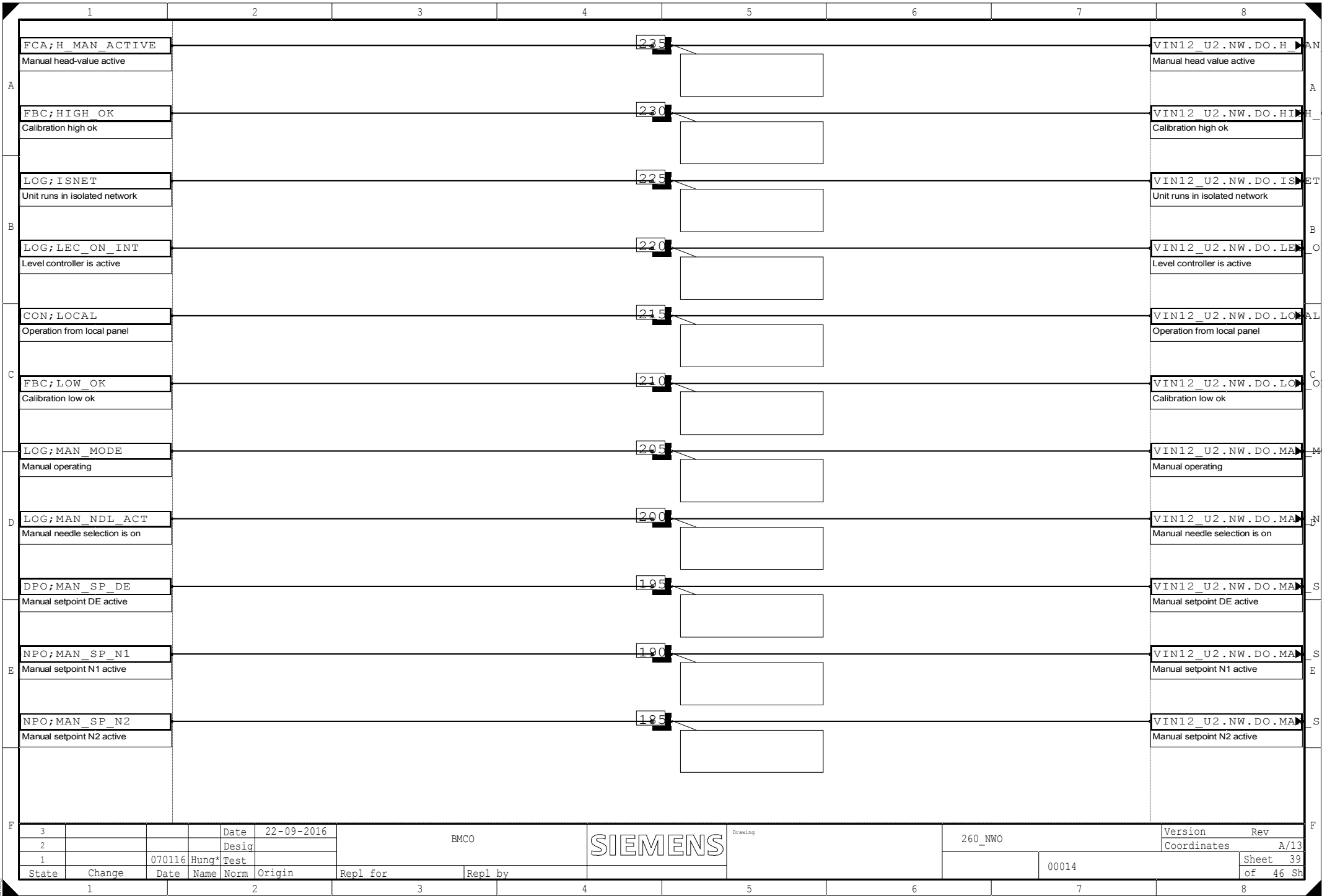
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1		070116	Hung*	Test				00010		Sheet	35
State	Change	Date	Name	Norm	Repl for	Repl by				of	46 Sh

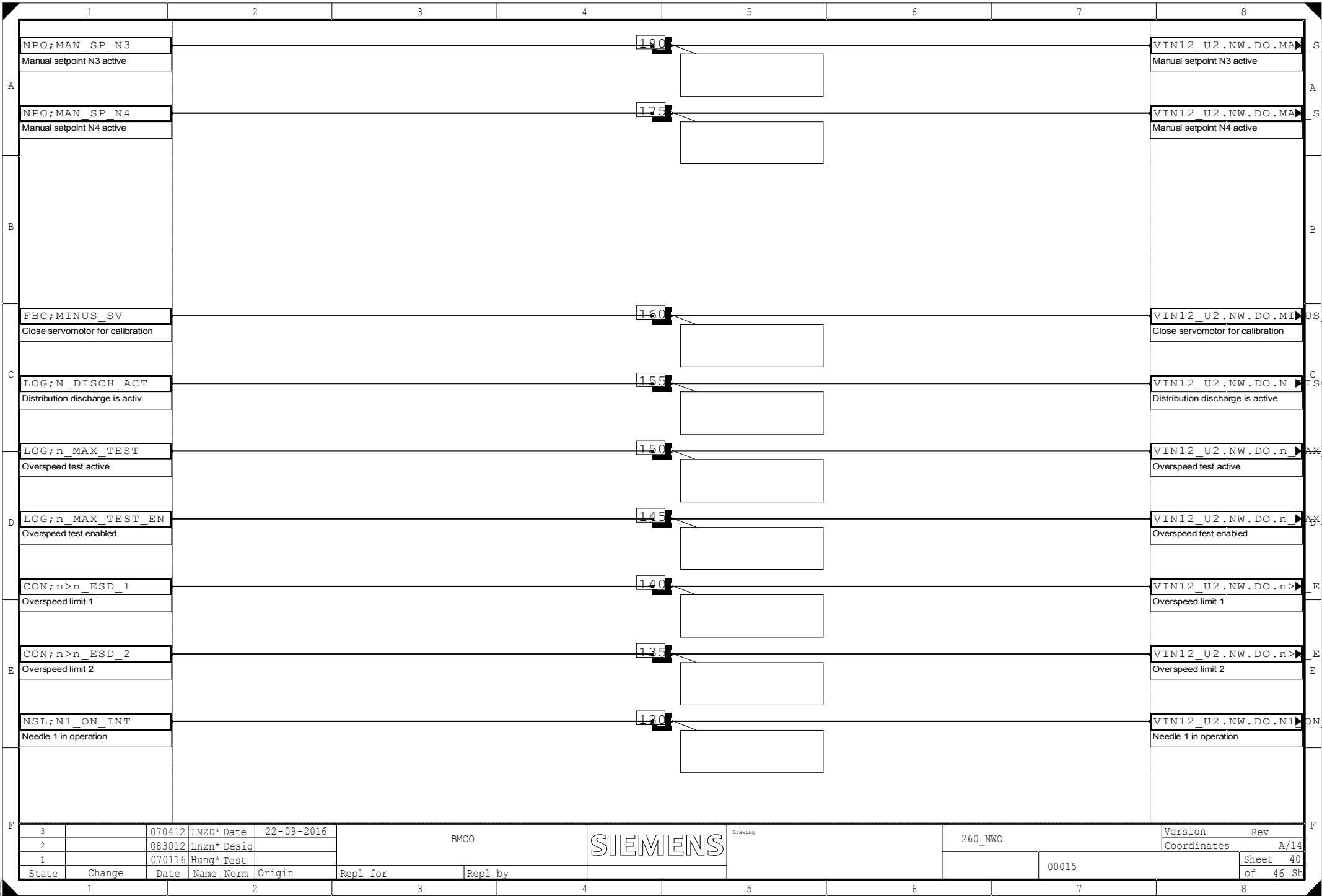


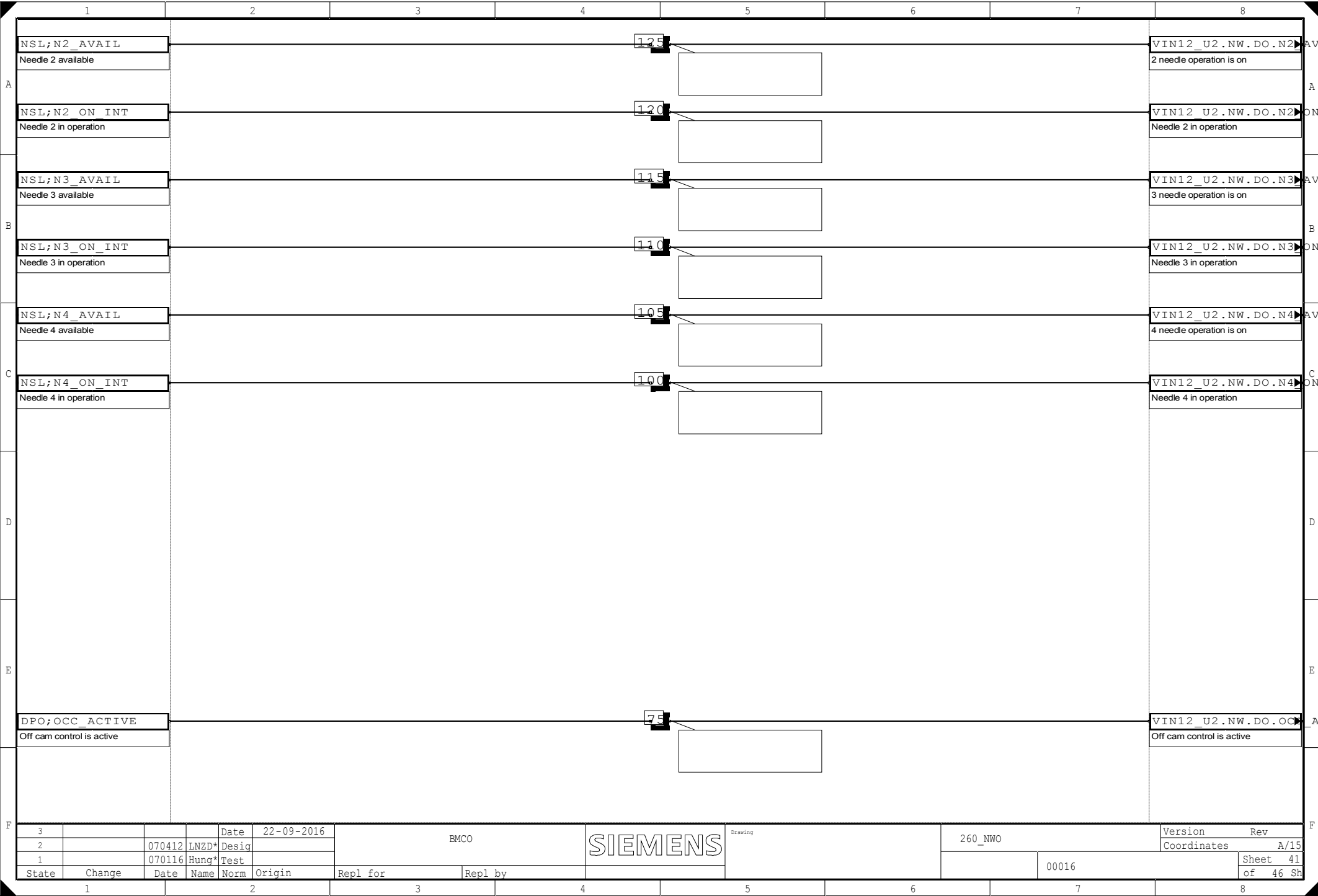
3		070412	LNZD*	Date	22-09-2016	EMCO		SIEMENS	Drawing	260_NWO		Version	Rev		
2		122012	1	Desig								Coordinates	A/10		
1		070116	Hung*	Test							00011	Sheet 36 of 46 Sh			
State	Change	Date	Name	Norm	Origin									Repl for	Repl by

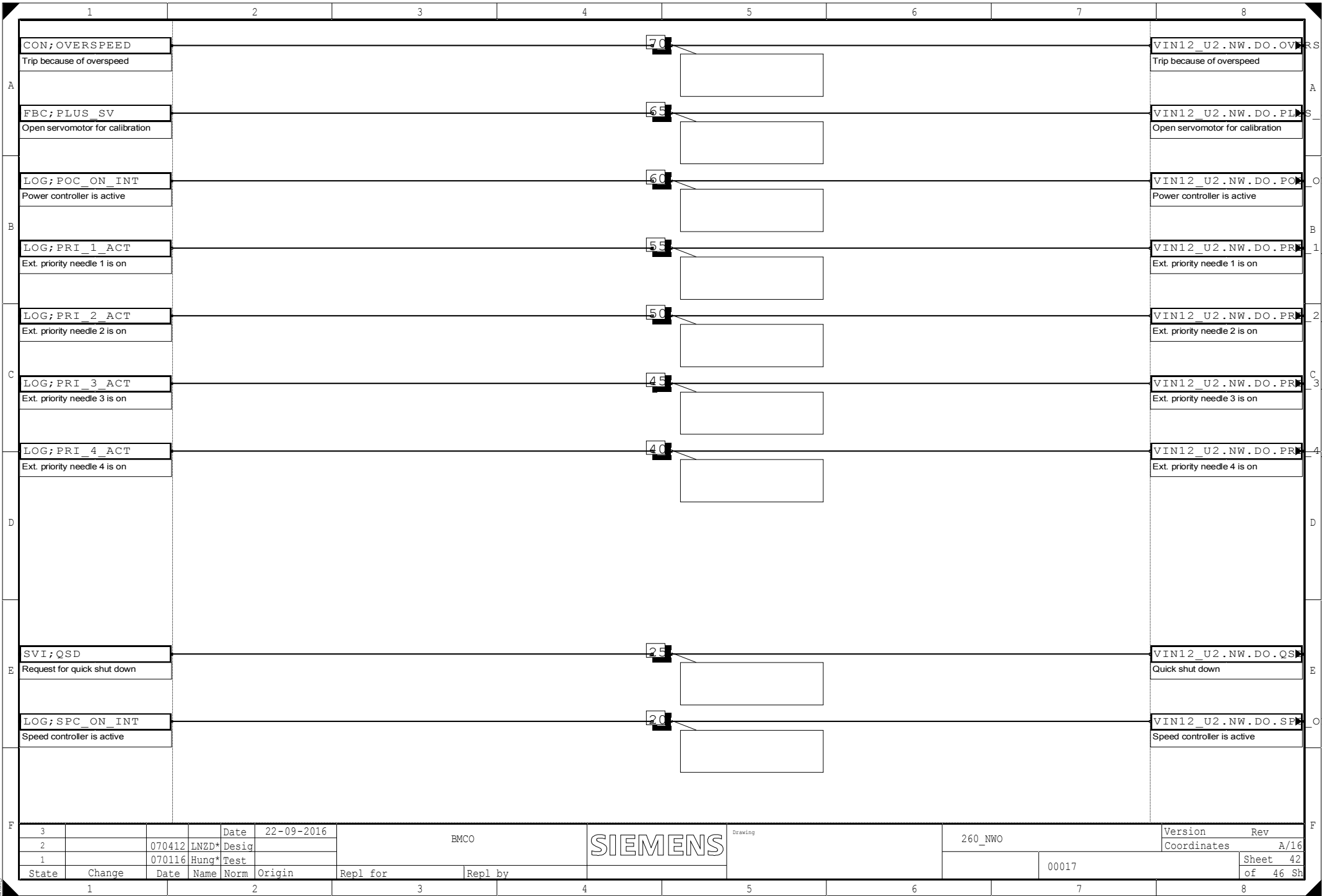


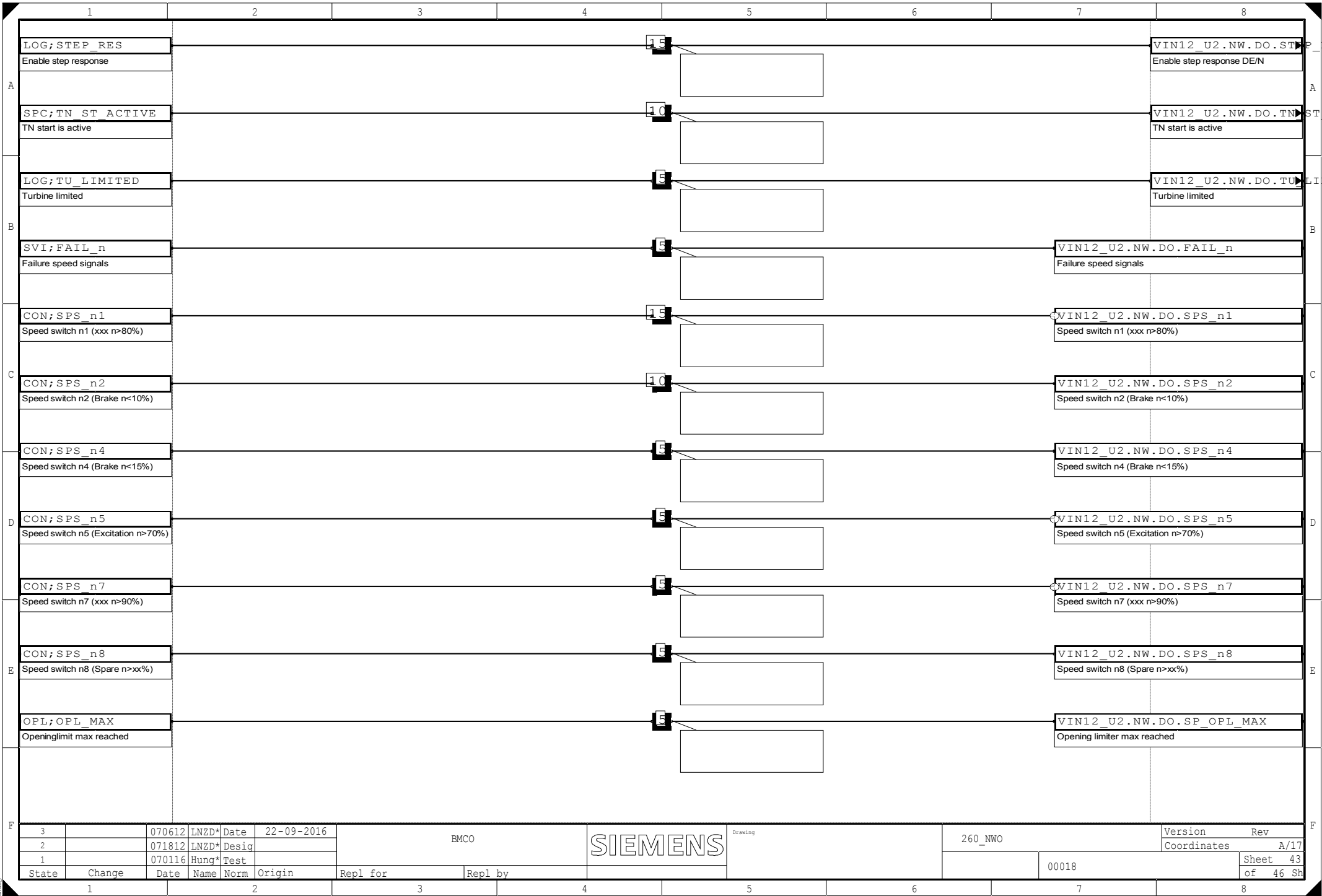




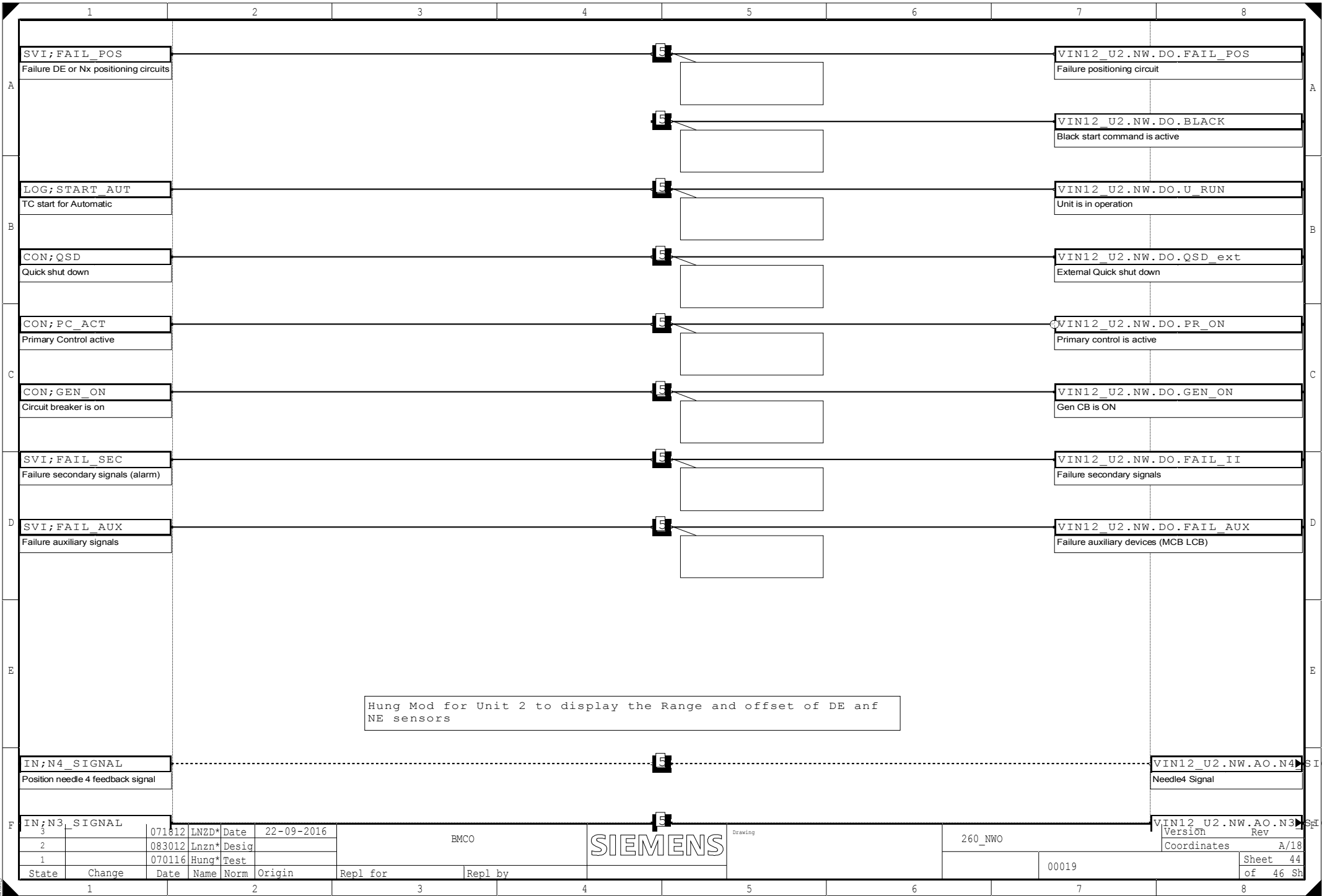








3		070612	LNZD*	Date	22-09-2016	EMCO		SIEMENS	Drawing	260_NWO		Version		Rev
2		071812	LNZD*	Desig								Coordinates		A/17
1		070116	Hung*	Test										Sheet 43
State	Change	Date	Name	Norm	Origin	Repl for	Repl by			00018	of 46 Sh			



Hung Mod for Unit 2 to display the Range and offset of DE and NE sensors

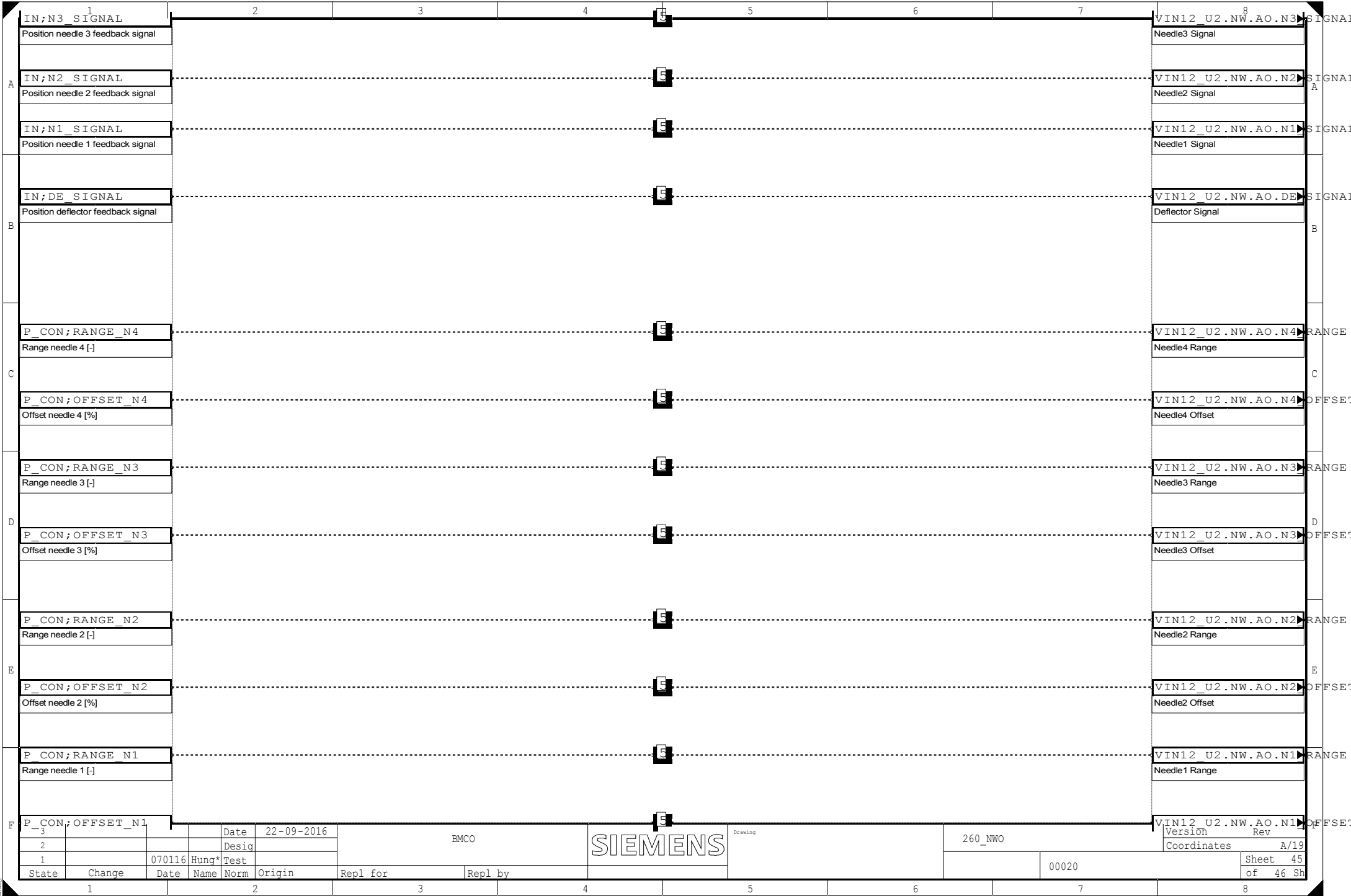
IN; N4_SIGNAL
Position needle 4 feedback signal

VIN12_U2.NW.AO.N4_SIGNAL
Needle4 Signal

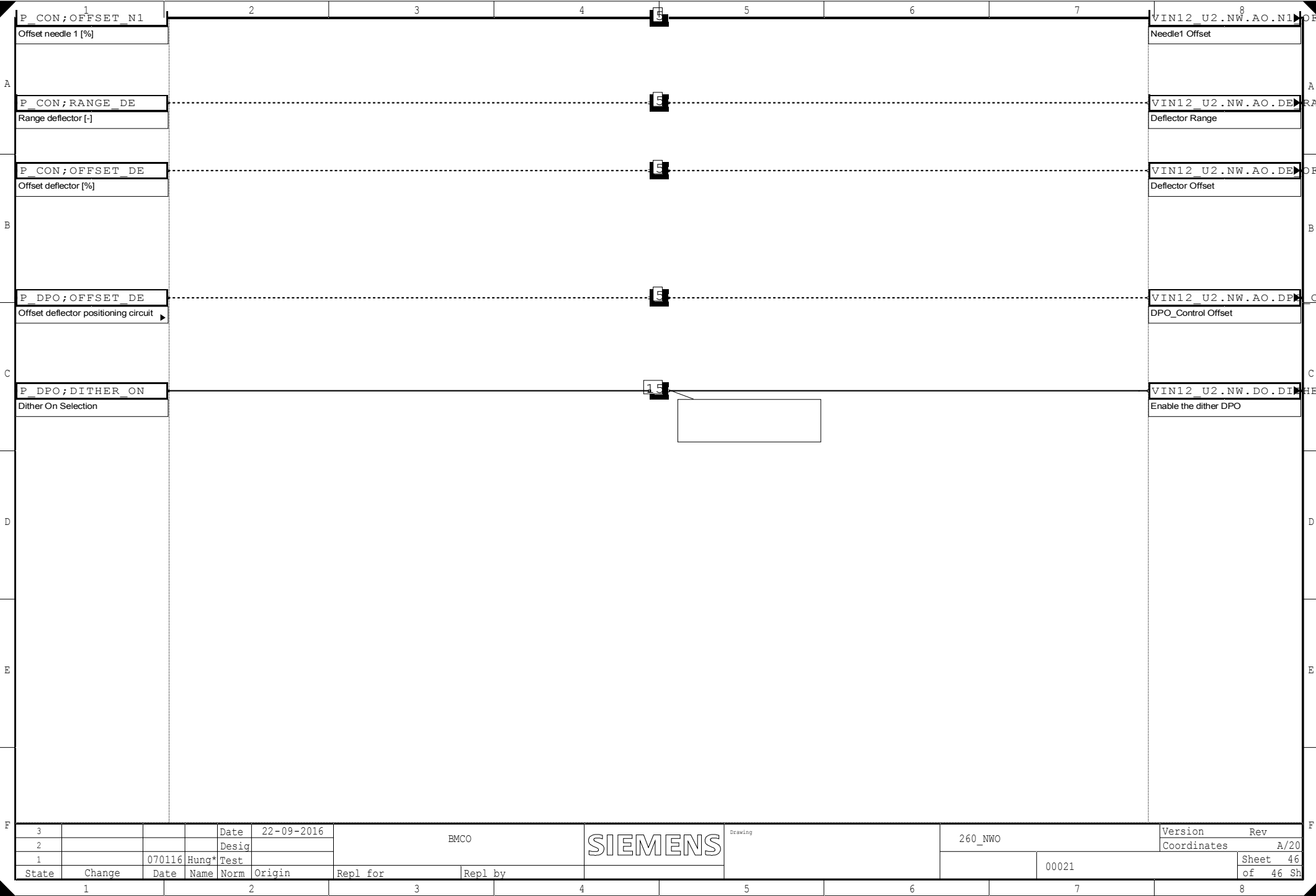
IN; N3_SIGNAL

VIN12_U2.NW.AO.N3_SIGNAL

3		071812	LN2D*	Date	22-09-2016	BMC0		SIEMENS	Drawing	260_NWO		version		Rev	
2		083012	Lnzn*	Desig								Coordinates		A/18	
1		070116	Hung*	Test											
State	Change	Date	Name	Norm	Origin	Repl for	Repl by				00019			Sheet 44	
														of 46 Sh	



P_CON, OFFSET_N1			Date	22-09-2016	BMC0			SIEMENS	Drawing	260_NWO		VIN12 U2.NW.AO.N1		OFFSET		
2			Desig							Version	Rev	Coordinates		A/19		
1		070116	Hung*	Test										Sheet	45	
State	Change	Date	Name	Norm	Origin	Repl for	Repl by					00020			of	46 Sh



	1	2	3	4	5	6	7	8																																																									
A	VARIABLE-DECLARATION Global-variable list "GLOBAL_PAR"								A																																																								
	Name	Var-Type	Attrib	Init. Value	Long Name	Phys. Adr.	Usage																																																										
	P_CON;0-PT_HHW1	VAR_GLOBAL	RETAIN	0.0	Offset levelmeasurement [m]																																																												
	P_CON;0-PT_Hdyn	VAR_GLOBAL	RETAIN	0.8673	Offset penstock press. [m]																																																												
	P_CON;1/2_BND	VAR_GLOBAL	RETAIN	0.4	Half deadband speed [%]																																																												
	P_CON;1/2_BND_PR	VAR_GLOBAL	RETAIN	0.1	Half deadband speed PC [%]																																																												
	P_CON;DELAY_n0	VAR_GLOBAL	RETAIN	30.0	On-delay speed switch 0 [s]																																																												
B	P_CON;ENG_DE	VAR_GLOBAL	RETAIN	1.0	Deflector positioing SSI ->% [-]																																																												
	P_CON;ENG_N	VAR_GLOBAL	RETAIN	1.0	Needle positioing SSI ->% [-]																																																												
	P_CON;FSP_min	VAR_GLOBAL	RETAIN	2.0	Minimum Flow Setpoint																																																												
	P_CON;GRD_FSP	VAR_GLOBAL	RETAIN	2.0	Grd. lim. ext. FSP [%/s]																																																												
	P_CON;GRD_PSP	VAR_GLOBAL	RETAIN	2.0	Grd. lim. ext. PSP [MW/s]																																																												
	P_CON;HY_NEEDLE	VAR_GLOBAL	RETAIN	0.5	Hysteresis needle closed [%]																																																												
	P_CON;HY_n	VAR_GLOBAL	RETAIN	0.5	Hysteresis speed switch [%]																																																												
	P_CON;HY_n2_ACT	VAR_GLOBAL	RETAIN	1.0	Hysteresis n2 -> n_ACTIVE [%]																																																												
	P_CON;LIM_n2_ACT	VAR_GLOBAL	RETAIN	2.0	Limit n2 -> n_ACTIVE [%]																																																												
C	P_CON;LI_N	VAR_GLOBAL	RETAIN	20.0	Limit needle closed [%]																																																												
	P_CON;MODEL_ON	VAR_GLOBAL	RETAIN	0	Turbine model on																																																												
	P_CON;OFFSET_DE	VAR_GLOBAL	RETAIN	-29.91	Offset deflector [%]																																																												
	P_CON;OFFSET_N1	VAR_GLOBAL	RETAIN	-83.1	Offset needle 1 [%]																																																												
	P_CON;OFFSET_N2	VAR_GLOBAL	RETAIN	-84.1	Offset needle 2 [%]																																																												
	P_CON;OFFSET_N3	VAR_GLOBAL	RETAIN	-86.1	Offset needle 3 [%]																																																												
	P_CON;OFFSET_N4	VAR_GLOBAL	RETAIN	-85.3	Offset needle 4 [%]																																																												
	P_CON;PSP_min	VAR_GLOBAL	RETAIN	2.0	Minimum Power Setpoint																																																												
	P_CON;PULSE_n3	VAR_GLOBAL	RETAIN	1.0	Pulses to detect																																																												
D	P_CON;Pnom	VAR_GLOBAL	RETAIN	34.0	Active power nominal [MW, KW]																																																												
	P_CON;RANGE_DE	VAR_GLOBAL	RETAIN	2.62	Range deflector [-]																																																												
	P_CON;RANGE_N1	VAR_GLOBAL	RETAIN	-1.589	Range needle 1 [-]																																																												
	P_CON;RANGE_N2	VAR_GLOBAL	RETAIN	-1.577	Range needle 2 [-]																																																												
	P_CON;RANGE_N3	VAR_GLOBAL	RETAIN	-1.5444	Range needle 3 [-]																																																												
	P_CON;RANGE_N4	VAR_GLOBAL	RETAIN	-1.611	Range needle 4 [-]																																																												
	P_CON;SPS_n1	VAR_GLOBAL	RETAIN	80.0	Speed switch 1-limit [%]																																																												
	P_CON;SPS_n2	VAR_GLOBAL	RETAIN	10.0	Speed switch 2-limit [%]																																																												
	P_CON;SPS_n4	VAR_GLOBAL	RETAIN	12.0	Speed switch 4-limit [%]																																																												
E	P_CON;SPS_n5	VAR_GLOBAL	RETAIN	60.0	Speed switch 5-limit [%]																																																												
	P_CON;SPS_n7	VAR_GLOBAL	RETAIN	90.0	Speed switch 7-limit [%]																																																												
	P_CON;SPS_n8	VAR_GLOBAL	RETAIN	200.0	Speed switch 8-limit [%] SPARE																																																												
	P_CON;T+_n2_ACT	VAR_GLOBAL	RETAIN	10.0	On-delay n2 -> n_ACTIVE [s]																																																												
	P_CON;TA	VAR_GLOBAL	RETAIN	6.04	Unit acceleration constant [s]																																																												
	P_CON;TN_HHW1	VAR_GLOBAL	RETAIN	10.0	Delay-time filter level intake gate [s]																																																												
	P_CON;TN_Hdyn	VAR_GLOBAL	RETAIN	1.2	Delay-time filter net head [s]																																																												
	P_CON;TN_Pa	VAR_GLOBAL	RETAIN	0.5	Delay-time filter active power [s]																																																												
	P_CON;TN_Pre	VAR_GLOBAL	RETAIN	1.0	Delay-time filter reactive power [s]																																																												
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	P_CON;T_n_ESD_1	VAR_GLOBAL	RETAIN	20.0	Delay time ESD overspeed 1																																																						
	P_CON;T_n_ESD_2	VAR_GLOBAL	RETAIN	0.0	Delay time ESD overspeed 2																																																						
	P_CON;WIDTH_SYNC	VAR_GLOBAL	RETAIN	1.0	Half width speed switch synch. [%]																																																						
	P_CON;WIDTH_n0	VAR_GLOBAL	RETAIN	2.5	Half width speed switch n0 [%]																																																						
	P_CON;f_NOM_n1	VAR_GLOBAL	RETAIN	40.0	Nominal frequence pick up 1 [Hz]																																																						
B	P_CON;f_NOM_n2	VAR_GLOBAL	RETAIN	50.0	Nominal frequence generator frequency *																																																						
	P_CON;f_NOM_n4	VAR_GLOBAL	RETAIN	50.0	Nominal frequence network frequency [Hz]																																																						
	P_CON;n0	VAR_GLOBAL	RETAIN	2.0	Standstill limit [%]																																																						
	P_CON;n1	VAR_GLOBAL	RETAIN	25.0	Speed n1-limit [%]																																																						
	P_CON;n_ESD_1	VAR_GLOBAL	RETAIN	120.0	Speed switch ESD 1: overspeed [%]																																																						
	P_CON;n_ESD_2	VAR_GLOBAL	RETAIN	120.0	Speed switch ESD2: overspeed [%]																																																						
	P_CON;n_SYNC	VAR_GLOBAL	RETAIN	100.0	Speed switch noload [%]																																																						
	P_DPO;A_DE	VAR_GLOBAL	RETAIN	1.0	Factor positioning circuit DE [-]																																																						
	P_DPO;DITHER_ON	VAR_GLOBAL	RETAIN	0	Dither ON selection																																																						
C	P_DPO;DIT_AMP	VAR_GLOBAL	RETAIN	5.0	Dither Amplitude																																																						
	P_DPO;DIT_FREQ	VAR_GLOBAL	RETAIN	5.0	Dither Frequency																																																						
	P_DPO;GRD_MSP_DE	VAR_GLOBAL	RETAIN	25.0	Gradient for manual SP DE [%/s]																																																						
	P_DPO;KP_DE	VAR_GLOBAL	RETAIN	1.5	Gain positioning circuit DE [-]																																																						
	P_DPO;MAN_DE_SP	VAR_GLOBAL	RETAIN	0.0	Manual setpoint deflector [%]																																																						
	P_DPO;OFFSET_CURVE	VAR_GLOBAL	RETAIN	0.0	Offset deflector positioning circuit [%]																																																						
	P_DPO;OFFSET_DE	VAR_GLOBAL	RETAIN	67.5	Offset deflector positioning circuit [%]																																																						
	P_DPO;TD_DE	VAR_GLOBAL	RETAIN	0.02	Derivative action time DE [s]																																																						
	P_DPO;TN_DE	VAR_GLOBAL	RETAIN	50.0	Intergrator action time																																																						
D	P_DPO;TN_DE_ON	VAR_GLOBAL	RETAIN	1	DPO Intergator ON selection																																																						
	P_DPO;T_GRD	VAR_GLOBAL	RETAIN	5.0	Delay time GRD-low to GRD-high [s]																																																						
	P_FBC;DEFLECTION	VAR_GLOBAL	RETAIN	5.0	Open/close deflection piston [%]																																																						
	P_FCA;H_NOM	VAR_GLOBAL	RETAIN	588.0	Nominal head [m]																																																						
	P_FCA;K1_NOZZLE	VAR_GLOBAL	RETAIN	1.597	Jet char. to calc. Y(Q) or Q(Y)																																																						
	P_FCA;K2_NOZZLE	VAR_GLOBAL	RETAIN	0.00597	Jet char. to calc. Y(Q) or Q(Y)																																																						
	P_FCA;MAN_SP_Hd	VAR_GLOBAL	RETAIN	606.0	Manuel head target value [m]																																																						
	P_FCA;Q_NOM	VAR_GLOBAL	RETAIN	6.48	Nominal flow [m³/s]																																																						
	P_FCA;c5	VAR_GLOBAL	RETAIN	1.0	For head correction																																																						
E	P_FCA;h_dyn	VAR_GLOBAL	RETAIN	0.0	Factor for head correction																																																						
	P_FLC;FFD	VAR_GLOBAL	RETAIN	5.0	Frequency flow droop [%]																																																						
	P_FLC;GRD_FSP	VAR_GLOBAL	RETAIN	4.0	Gradient FSP switch over [%/s]																																																						
	P_FLC;LIH_FSP	VAR_GLOBAL	RETAIN	105.0	Limit high flow setpoint [%]																																																						
	P_FLC;LIL_FSP	VAR_GLOBAL	RETAIN	-5.0	Limit low flow setpoint [%]																																																						
	P_FLC;TIM_FSP	VAR_GLOBAL	RETAIN	1.0	Grad. limit flow setpoint [%/s]																																																						
	P_FLC;T_MINLOAD	VAR_GLOBAL	RETAIN	2.0	Pulse width after GEN_ON [s]																																																						
	P_LEC;GRD_LEC	VAR_GLOBAL	RETAIN	0.05	Grad. limit level controller [m/s]																																																						
	P_LEC;KP_LEC	VAR_GLOBAL	RETAIN	1.0	Gain level contoller [-]																																																						
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B	P_LEC;LSP	VAR_GLOBAL	RETAIN	655.0	Target value level setpoint [m]																																																										
	P_LEC;NDL_SP_GRD	VAR_GLOBAL	RETAIN	1.0	Gradient needle setpoint [%/s]																																																										
	P_LEC;TN_LEC	VAR_GLOBAL	RETAIN	100.0	Integral action time level contoller [s]																																																										
	P_LIM;ARW_P_LIH	VAR_GLOBAL	RETAIN	10.0	Dyn. power LIM-LIH-control [%]																																																										
	P_LIM;ARW_P_LIL	VAR_GLOBAL	RETAIN	10.0	Dyn. power LIM-LIL-control [%]																																																										
	P_LIM;HY_DEV_P	VAR_GLOBAL	RETAIN	0.4	Hyst. power control deviation [MVA]																																																										
	P_LIM;KP_GR_PA	VAR_GLOBAL	RETAIN	2.0	Gain power limit [-]																																																										
	P_LIM;LIM_MAX_P	VAR_GLOBAL	RETAIN	105.0	Max power LIM-LIH-control [%]																																																										
	P_LIM;NO_LIMCO_P	VAR_GLOBAL	RETAIN	1	Power limit control disabled																																																										
	P_LIM;SP_P_LIM	VAR_GLOBAL	RETAIN	40.0	Apparent power limit value [MVA]																																																										
C	P_LIM;TN_AP	VAR_GLOBAL	RETAIN	1.0	Delay-time apparent power limit [s]																																																										
	P_LIM;TN_GR_PA	VAR_GLOBAL	RETAIN	3.0	Integral action time pow. limit [s]																																																										
	P_LOG;BIG_ISNET	VAR_GLOBAL	RETAIN	0	Big isolated network remain																																																										
	P_LOG;HYH_ISNET	VAR_GLOBAL	RETAIN	0.3	Hysteresis Isol.netw.-limit high [%]																																																										
	P_LOG;HYL_ISNET	VAR_GLOBAL	RETAIN	0.3	Hysteresis isol.netw.-limit low [%]																																																										
	P_LOG;ISNET_MOD	VAR_GLOBAL	RETAIN	0	Isolated network model is used																																																										
	P_LOG;LIH_ISNET	VAR_GLOBAL	RETAIN	104.8	Limit high isolated netw. detect. [%]																																																										
	P_LOG;LIL_ISNET	VAR_GLOBAL	RETAIN	92.0	Limit low isolated netw. detect. [%]																																																										
	P_LOG;N_DISCHAR	VAR_GLOBAL	RETAIN	3.0	Needle spt. distrib. pipe discharge [%]																																																										
	P_LOG;T_DISCHAR	VAR_GLOBAL	RETAIN	10.0	Off-delay distributionpipe discharge [s]																																																										
D	P_LOG;T_ISOL	VAR_GLOBAL	RETAIN	1.0	Timepulse tracking SPC [s]																																																										
	P_LOG;T_ON_ISNET	VAR_GLOBAL	RETAIN	0.2	On delay detection [s]																																																										
	P_MOD;A_TW	VAR_GLOBAL	RETAIN	1.0	Factor PTL-element																																																										
	P_MOD;C0_eg	VAR_GLOBAL	RETAIN	1.0	Self-regulation-factor																																																										
	P_MOD;C1_eg	VAR_GLOBAL	RETAIN	0.0	Self-regulation-factor																																																										
	P_MOD;C2_eg	VAR_GLOBAL	RETAIN	0.0	Self-regulation-factor																																																										
	P_MOD;DE_SYM	VAR_GLOBAL	RETAIN	0.0	Offset deflector [%]																																																										
	P_MOD;GRL_Q	VAR_GLOBAL	RETAIN	1.0	Rate of change head-water-basin inflow *																																																										
	P_MOD;HTW_OFFS	VAR_GLOBAL	RETAIN	0.0	Offset tailwater [m]																																																										
	P_MOD;H_MAN	VAR_GLOBAL	RETAIN	0.0	Manually operated head-value [m]																																																										
E	P_MOD;H_MAN_SWITCH	VAR_GLOBAL	RETAIN	0	Manually operated head																																																										
	P_MOD;LIH_HHW1	VAR_GLOBAL	RETAIN	700.0	Limit high level calculation [m]																																																										
	P_MOD;LIL_HHW1	VAR_GLOBAL	RETAIN	600.0	Limit low level calculation [m]																																																										
	P_MOD;N1_SYM	VAR_GLOBAL	RETAIN	0.0	Offset needle 1 [p.u.]																																																										
	P_MOD;N2_SYM	VAR_GLOBAL	RETAIN	0.0	Offset needle 2 [p.u.]																																																										
	P_MOD;N3_SYM	VAR_GLOBAL	RETAIN	0.0	Offset needle 3 [p.u.]																																																										
	P_MOD;N4_SYM	VAR_GLOBAL	RETAIN	0.0	Offset needle 4 [p.u.]																																																										
	P_MOD;P_ISNET	VAR_GLOBAL	RETAIN	50.0	Isolated network power [MW, KW]																																																										
	P_MOD;Q_RUN_IN	VAR_GLOBAL	RETAIN	10.0	Head-water-basin inflow [m³/s]																																																										
	P_MOD;TI_HHW1	VAR_GLOBAL	RETAIN	10.0	Time const. equal the basin surface [m²]																																																										
P_MOD;TI_N/DE	VAR_GLOBAL	RETAIN	1.0	Integral action time N/DE servom. [s]																																																											
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VARIABLE-DECLARATION
Global-variable list "GLOBAL_PAR"

Name	Var-Type	Attrib	Init. Value	Long Name	Phys. Adr.	Usage
P_OPL;OPL_START	VAR_GLOBAL	RETAIN	8.0	Opening limit during start [%]		
P_OPL;TIM_OPL	VAR_GLOBAL	RETAIN	2.0	Opening limiter ramp time [%/s]		
P_POC;A_PPENS	VAR_GLOBAL	RETAIN	1.0	Factor of a PT1-element [-]		
P_POC;DFSP_MAX	VAR_GLOBAL	RETAIN	1.2	Max. limit of the flow deviation [-]		
P_POC;DFSP_MIN	VAR_GLOBAL	RETAIN	0.8	Min. limit of the flow deviation [-]		
P_POC;FPD	VAR_GLOBAL	RETAIN	5.0	Frequency power droop [%]		
P_POC;GRD_PSP	VAR_GLOBAL	RETAIN	4.0	Gradient power setpoint [%/s]		
P_POC;KP_POC	VAR_GLOBAL	RETAIN	0.4	Gain power controller [-]		
P_POC;LIH_PSP	VAR_GLOBAL	RETAIN	104.944	Limit high power spt. [%]		
P_POC;LIL_PSP	VAR_GLOBAL	RETAIN	-2.3	Limit low power spt. [%]		
P_POC;P_MIN	VAR_GLOBAL	RETAIN	-0.3	Minimum power value [%]		
P_POC;TD_DFSP	VAR_GLOBAL	RETAIN	0.5	Deviate action time [s]		
P_POC;TD_PPENS	VAR_GLOBAL	RETAIN	0.5	Derivate action time (penstock) [s]		
P_POC;TIM_PSP	VAR_GLOBAL	RETAIN	0.8	Grad. limit power setpoint [%/s]		
P_POC;TN_DFSP	VAR_GLOBAL	RETAIN	5.0	Delay time PT1filter [s]		
P_POC;TN_POC	VAR_GLOBAL	RETAIN	9.0	Integral action time power controller [*		
P_POC;T_PT1_PSP	VAR_GLOBAL	RETAIN	2.0	Filter constant power setpoint [s]		
P_RDI;HY_DE_INT	VAR_GLOBAL	RETAIN	0.5	Hysteresis deflector comp. [%]		
P_RDI;HY_HHW_INT	VAR_GLOBAL	RETAIN	0.03	Hysteresis level comp. [m]		
P_RDI;HY_H_INT	VAR_GLOBAL	RETAIN	0.05	Hysteresis net head comparison [%]		
P_RDI;HY_P_INT	VAR_GLOBAL	RETAIN	0.5	Hysteresis active pow. comp. [MW]		
P_RDI;HY_n_Active	VAR_GLOBAL	RETAIN	0.5	Hysteresis speed comparison [%]		
P_SEL;A2_mPL	VAR_GLOBAL	RETAIN	3.0	Noload flow [%]		
P_SEL;N_=-5%	VAR_GLOBAL	RETAIN	-5.0	Needle setpoint = -5% (closed)		
P_SPC;+xW_START	VAR_GLOBAL	RETAIN	5.0	Limit deviation start-detection [%]		
P_SPC;A_dn/dt	VAR_GLOBAL	RETAIN	0.2	Factor speed derivation [-]		
P_SPC;Dmax=+20%	VAR_GLOBAL	RETAIN	20.0	D-element limit positive [%]		
P_SPC;Dmin=-20%	VAR_GLOBAL	RETAIN	-20.0	D-element limit negative [%]		
P_SPC;HY_+xW	VAR_GLOBAL	RETAIN	0.5	Hyst. deviation start-detection [%]		
P_SPC;KP_ISNET	VAR_GLOBAL	RETAIN	4.0	Gain isolated network operation [-]		
P_SPC;KP_NOLOAD	VAR_GLOBAL	RETAIN	1.3	Gain no-load operation [-]		
P_SPC;K_dn/dt	VAR_GLOBAL	RETAIN	6.0	Gain speed-derivation [-]		
P_SPC;SPC_LIL	VAR_GLOBAL	RETAIN	0.0	Speed-controller limit low [%]		
P_SPC;SSP_ISNET	VAR_GLOBAL	RETAIN	101.5	Speedsetpoint isolated network [%]		
P_SPC;SSP_LIH	VAR_GLOBAL	RETAIN	110.0	Limit hight speed setpoint [%]		
P_SPC;SSP_LIL	VAR_GLOBAL	RETAIN	90.0	Limit low speed setpoint [%]		
P_SPC;SSP_START	VAR_GLOBAL	RETAIN	100.0	Speed setpoint at turbine startup [%]		
P_SPC;SSP_TIM	VAR_GLOBAL	RETAIN	0.1	Grad. limit speed setpoint [%/s]		
P_SPC;TD_SPC	VAR_GLOBAL	RETAIN	1.3	Derivative action time Td [s]		
P_SPC;TD_dn/dt	VAR_GLOBAL	RETAIN	1.0	Derivate act. time speed-deriva. [s]		
P_SPC;TN_ISNET	VAR_GLOBAL	RETAIN	10.0	Integral action time isol. netw. op. [s]		

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A	VARIABLE-DECLARATION Global-variable list "GLOBAL_PAR"								A				
	Name	Var-Type	Attrib	Init. Value	Long Name	Phys. Adr.	Usage						
B	P_SPC;TN_NOLOAD	VAR_GLOBAL	RETAIN	12.0	Integral action time no-load op. [s]								
	P_SPC;TN_START	VAR_GLOBAL	RETAIN	1.0	Integral action time during start [s]								
	P_SPC;bp_SPC	VAR_GLOBAL	RETAIN	5.0	Permanent speed droop [%]								
	P_SVI;DB_POS	VAR_GLOBAL	RETAIN	5.0	Deadband for pos. monit [%]								
	P_SVI;HY_DE	VAR_GLOBAL	RETAIN	5.0	Hysteresis deflector moni. [%]								
	P_SVI;HY_N1-6	VAR_GLOBAL	RETAIN	5.0	Hysteresis needle mon. needle 1-6 [%]								
	P_SVI;LIM_DE	VAR_GLOBAL	RETAIN	20.0	Limit deflector monitoring [%]								
	P_SVI;LIM_N1-6	VAR_GLOBAL	RETAIN	20.0	Limit needle monitoring needle 1-6 [%]								
	P_SVI;TIM_DE	VAR_GLOBAL	RETAIN	20.0	Delay-time deflector moni. [s]								
	P_SVI;TIM_N1-6	VAR_GLOBAL	RETAIN	40.0	Delay-time needle mon. needle 1-6 [s]								
C	P_SVI;T_F_n1_OP	VAR_GLOBAL	RETAIN	0.05	On-delay fail SMS pickup 1 [s]								
	P_SVI;T_F_n1_ST	VAR_GLOBAL	RETAIN	15.0	On-delay fail SMS pickup 1 start [s]								
	P_SVI;T_F_n2_OP	VAR_GLOBAL	RETAIN	0.05	On-delay fail SMS pickup 2 start [s]								
	P_SVI;T_F_n2_ST	VAR_GLOBAL	RETAIN	25.0	On-delay fail SMS pickup 2 start [s]								
	P_SVI;T_OFF	VAR_GLOBAL	RETAIN	0.0	Deadtime t_off feedback signals [s]								
	P_SVI;tc_DE	VAR_GLOBAL	RETAIN	1.6	Closing time deflector [s]								
	P_SVI;tc_Nx	VAR_GLOBAL	RETAIN	55.0	Closing time needles [s]								
	P_SVI;to_DE	VAR_GLOBAL	RETAIN	4.5	Opening time deflector [s]								
	P_SVI;to_Nx	VAR_GLOBAL	RETAIN	30.0	Opening time needles [s]								
D									D				
E									E				
F									F				
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A	VARIABLE-DECLARATION Global-variable list "GLOBAL_VAR"							
	Name	Var-Type	Attrib	Init. Value	Long Name	Phys. Adr.	Usage	
	CON;P_INT_[MW]	VAR_GLOBAL			Active Power scal.&filt [MW]			
	CON;P_re_INT	VAR_GLOBAL			Reactive power filtered [Mvar].			
	CON;Pnom	VAR_GLOBAL			Active power nominal [MW, KW]			
	CON;QSD	VAR_GLOBAL			Quick shut down			
	CON;QSD_PAN	VAR_GLOBAL			Quick shut down from panel			
B	CON;RDI_ISN_INT	VAR_GLOBAL			Status Active CPU "ISNET_INT"			
	CON;RESET	VAR_GLOBAL			Reset several FLIP FLOP			
	CON;RES_ISO	VAR_GLOBAL			Reset Isolated Operation			
	CON;SET_FSP	VAR_GLOBAL			Set Flow Setpoint			
	CON;SET_PSP	VAR_GLOBAL			Set Power Setpoint			
	CON;SNL_POS	VAR_GLOBAL			Speed NO Load Position			
	CON;SPS_n1	VAR_GLOBAL			Speed switch n1 (xxx n>80%)			
	CON;SPS_n2	VAR_GLOBAL			Speed switch n2 (Brake n<10%)			
	CON;SPS_n4	VAR_GLOBAL			Speed switch n4 (Brake n<15%)			
C	CON;SPS_n5	VAR_GLOBAL			Speed switch n5 (Excitation n>70%)			
	CON;SPS_n7	VAR_GLOBAL			Speed switch n7 (xxx n>90%)			
	CON;SPS_n8	VAR_GLOBAL			Speed switch n8 (Spare n>xx%)			
	CON;SP_SET	VAR_GLOBAL			Set Setpoint (enable Setpoint)			
	CON;SSP+	VAR_GLOBAL			Speed setpoint higher			
	CON;SSP-	VAR_GLOBAL			Speed setpoint lower			
	CON;STOP_n1	VAR_GLOBAL			Standstill pickup 1 detected			
	CON;STOP_n2	VAR_GLOBAL			Standstill pickup 2 detected			
	CON;STOP_n4	VAR_GLOBAL			Standstill pickup 4 detected			
D	CON;TA	VAR_GLOBAL			Unit acceleration constant [s]			
	CON;TC_START	VAR_GLOBAL			Turbinecontroller start			
	CON;TC_STOP	VAR_GLOBAL			Turbinecontroller stop			
	CON;TC_STOP_EX	VAR_GLOBAL			Stop from external			
	CON;TRK_FSP	VAR_GLOBAL			Tracking flow setpoint			
	CON;TRK_PSP	VAR_GLOBAL			Tracking power setpoint			
	CON;n1	VAR_GLOBAL			Speed pick up 1 unlimited [%]			
	CON;n2	VAR_GLOBAL			Generator frequency unlimited [%]			
	CON;n4	VAR_GLOBAL			Network frequency unlimited [%]			
E	CON;n4_Fr	VAR_GLOBAL			Network frequency unlimited Hz]			
	CON;n<n0	VAR_GLOBAL			Speed-switch standstill			
	CON;n=n_SYNCHRON	VAR_GLOBAL			Synchron speed			
	CON;n>n1	VAR_GLOBAL			Speed-switch start			
	CON;n>n_ESD_1	VAR_GLOBAL			Overspeed limit 1			
	CON;n>n_ESD_2	VAR_GLOBAL			Overspeed limit 2			
	CON;n_1_INT	VAR_GLOBAL			Speedmes. 1 grd. limited			
	CON;n_2_INT	VAR_GLOBAL			Speedmes. 2 grd. limited			
	CON;n_4_INT	VAR_GLOBAL			Line Frequency grd. limited			
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A	VARIABLE-DECLARATION Global-variable list "GLOBAL_VAR"							
	Name	Var-Type	Attrib	Init. Value	Long Name	Phys. Adr.	Usage	
	FCA;H_int_H_man	VAR_GLOBAL			Head or manual Head [m]			
	FCA;K1_NOZZLE	VAR_GLOBAL			Jet char. to calc. Y(Q) or Q(Y)			
	FCA;K2_NOZZLE	VAR_GLOBAL			Jet char. to calc. Y(Q) or Q(Y)			
	FCA;Q%	VAR_GLOBAL			Total flow regarding H_net [%]			
	FCA;Q_N1-6:z0	VAR_GLOBAL			Total avg. flow regarding Hnom [%]			
B	FCA;Q_NOM	VAR_GLOBAL			Nominal flow [m³/s]			
	FCA;Q_m3	VAR_GLOBAL			Total flow regarding H_net [m³/s]			
	FCA;Qnom/100	VAR_GLOBAL			Scaling factor flow (% => m³/s)			
	FCA;he	VAR_GLOBAL			Correction factor (H_net/H_nom)			
	FCA;he1/2	VAR_GLOBAL			Corr. factor SQRT(H_net/H_nom)			
	FLC;FSP1	VAR_GLOBAL			FSP with FFD regarding H_net [%]			
	FLC;FSP_LIM	VAR_GLOBAL			Output FLC limited			
	FLC;FSP_[%]	VAR_GLOBAL			FSP without FFD reg. H_net [%]			
	FLC;FSP_[m3]	VAR_GLOBAL			FSP without FFD reg. H_net [m³/s]			
C	FLC;MINLOAD	VAR_GLOBAL			Pulse after GEN_ON			
	FLC;T_MINLOAD	VAR_GLOBAL			Time pulse after GEN_ON [s]			
	IN;BLACK	VAR_GLOBAL			Black Start			
	IN;BP00_01	VAR_GLOBAL			Droop change			
	IN;DE_SIGNAL	VAR_GLOBAL			Position deflector feedback signal			
	IN;DIGPOS_DE_ON	VAR_GLOBAL			Digital Positioner of Deflector			
	IN;DIGPOS_N1_ON	VAR_GLOBAL			Digital Positioner of Needle 1			
	IN;DIGPOS_N2_ON	VAR_GLOBAL			Digital Positioner of Needle 2			
	IN;DIGPOS_N3_ON	VAR_GLOBAL			Digital Positioner of Needle 3			
D	IN;DIGPOS_N4_ON	VAR_GLOBAL			Digital Positioner of Needle 4			
	IN;FLC_ON	VAR_GLOBAL			Flowcontroller on			
	IN;FREEZE	VAR_GLOBAL			Needle Freezing			
	IN;FSP	VAR_GLOBAL			Ext. Flow setpoint [%]			
	IN;F_FUSE	VAR_GLOBAL			MCB Tripped			
	IN;F_LCC	VAR_GLOBAL			Load Circuit Controls Fault			
	IN;F_PS1	VAR_GLOBAL			Power Supply 1 Fault			
	IN;F_PS2	VAR_GLOBAL			Power Supply 2 Fault			
	IN;GEN_ON	VAR_GLOBAL			Circuit breaker on			
E	IN;HHW1	VAR_GLOBAL			Level intake gate [m]			
	IN;Hdyn	VAR_GLOBAL			Penstock pressure [m]			
	IN;ISNET_EXT	VAR_GLOBAL			Isolated network ext.			
	IN;LEC_ON	VAR_GLOBAL			Levelcontroller on			
	IN;LINE_CH	VAR_GLOBAL			Line charging			
	IN;LOCAL	VAR_GLOBAL			Operation from local panel			
	IN;N1_SIGNAL	VAR_GLOBAL			Position needle 1 feedback signal			
	IN;N2_SIGNAL	VAR_GLOBAL			Position needle 2 feedback signal			
	IN;N3_SIGNAL	VAR_GLOBAL			Position needle 3 feedback signal			
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A	VARIABLE-DECLARATION Global-variable list "GLOBAL_VAR"							
	Name	Var-Type	Attrib	Init. Value	Long Name	Phys. Adr.	Usage	
	LOG;4N_ON	VAR_GLOBAL			4 needle operation on			
	LOG;FLC_ON_INT	VAR_GLOBAL			Flow controller is active			
	LOG;GRID	VAR_GLOBAL			interconnected operation			
	LOG;ISNET	VAR_GLOBAL			Unit runs in isolated network			
	LOG;ISNET_INT	VAR_GLOBAL			Isolated network internal detection			
B	LOG;ISNET_MOD	VAR_GLOBAL			Variable isolated network			
	LOG;ISNET_MOD_SEL	VAR_GLOBAL			Variable isolatet network selected			
	LOG;LEC_ON_INT	VAR_GLOBAL			Level controller is active			
	LOG;MAN_DE	VAR_GLOBAL			Manual setpoint deflector on			
	LOG;MAN_MODE	VAR_GLOBAL			Manual operating			
	LOG;MAN_N1	VAR_GLOBAL			Manual setpoint needle 1 on			
	LOG;MAN_N2	VAR_GLOBAL			Manual setpoint needle 2 on			
	LOG;MAN_N3	VAR_GLOBAL			Manual setpoint needle 3 on			
	LOG;MAN_N4	VAR_GLOBAL			Manual setpoint needle 4 on			
C	LOG;MAN_NDL_ACT	VAR_GLOBAL			Manual needle selection is on			
	LOG;NO_GRID	VAR_GLOBAL			Non interconnected operation			
	LOG;N_DISCH_ACT	VAR_GLOBAL			Distribution discharge is activ			
	LOG;N_DISCH_SP	VAR_GLOBAL			Distribution discharge spt. [%]			
	LOG;POC_ON_INT	VAR_GLOBAL			Power controller is active			
	LOG;PRI_1_ACT	VAR_GLOBAL			Ext. priority needle 1 is on			
	LOG;PRI_2_ACT	VAR_GLOBAL			Ext. priority needle 2 is on			
	LOG;PRI_3_ACT	VAR_GLOBAL			Ext. priority needle 3 is on			
	LOG;PRI_4_ACT	VAR_GLOBAL			Ext. priority needle 4 is on			
D	LOG;ROT_BACKW	VAR_GLOBAL			Rotation backward			
	LOG;ROT_FORW	VAR_GLOBAL			Rotation forward			
	LOG;SPC_ON_INT	VAR_GLOBAL			Speed controller is active			
	LOG;START_AUT	VAR_GLOBAL			TC start for Automatic			
	LOG;START_INT	VAR_GLOBAL			Internal start-order			
	LOG;STEP_RES	VAR_GLOBAL			Enable step response			
	LOG;STOP_INT	VAR_GLOBAL			Internal stop-order			
	LOG;TC_ACT	VAR_GLOBAL			Turbine Controller active			
	LOG;TRK_ISNET	VAR_GLOBAL			Islolated network model on			
E	LOG;TU_LIMITED	VAR_GLOBAL			Turbine limited			
	LOG;n_MAX_TEST	VAR_GLOBAL			Overspeed test active			
	LOG;n_MAX_TEST_EN	VAR_GLOBAL			Overspeed test enabled			
	MOD;DE	VAR_GLOBAL			Model deflector pos. [%]			
	MOD;HHW1	VAR_GLOBAL			Model water level intake gate [m]			
	MOD;HTW	VAR_GLOBAL			Tailwater calculated [m]			
	MOD;Hdyn	VAR_GLOBAL			Head calculated			
	MOD;N1	VAR_GLOBAL			Model needle 1 pos. [%]			
	MOD;N2	VAR_GLOBAL			Model needle 2 pos. [%]			
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A	VARIABLE-DECLARATION Global-variable list "GLOBAL_VAR"								A																																																															
	Name	Var-Type	Attrib	Init. Value	Long Name	Phys. Adr.	Usage																																																																	
	MOD;N3	VAR_GLOBAL			Model needle 3 pos. [%]																																																																			
	MOD;N4	VAR_GLOBAL			Model needle 4 pos. [%]																																																																			
	MOD;Pa	VAR_GLOBAL			Active power model [MW]																																																																			
	MOD;Q	VAR_GLOBAL			Flow model [m³/s]																																																																			
	MOD;h_stat	VAR_GLOBAL			Head statically calculated [m]																																																																			
	MOD;n	VAR_GLOBAL			Model speed [%]																																																																			
B	NPO;MAN_SP_N1	VAR_GLOBAL			Manual setpoint N1 active			B																																																																
	NPO;MAN_SP_N2	VAR_GLOBAL			Manual setpoint N2 active																																																																			
	NPO;MAN_SP_N3	VAR_GLOBAL			Manual setpoint N3 active																																																																			
	NPO;MAN_SP_N4	VAR_GLOBAL			Manual setpoint N4 active																																																																			
	NPO;N1_ON_RDY	VAR_GLOBAL			N1 Ready for Operation																																																																			
	NPO;N1_SP	VAR_GLOBAL			Needle 1 setpoint [%]																																																																			
	NPO;N2_ON_RDY	VAR_GLOBAL			N2 Ready for Operation																																																																			
	NPO;N2_SP	VAR_GLOBAL			Needle 2 setpoint [%]																																																																			
C	NPO;N3_ON_RDY	VAR_GLOBAL			N3 Ready for Operation			C																																																																
	NPO;N3_SP	VAR_GLOBAL			Needle 3 setpoint [%]																																																																			
	NPO;N4_ON_RDY	VAR_GLOBAL			N4 Ready for Operation																																																																			
	NPO;N4_SP	VAR_GLOBAL			Needle 4 setpoint [%]																																																																			
	NPO;SIG_N1	VAR_GLOBAL			Output needle 1 pos. circuit																																																																			
	NPO;SIG_N1_MOD	VAR_GLOBAL			Output needle 1 pos. circuit model																																																																			
	NPO;SIG_N2	VAR_GLOBAL			Output needle 2 pos. circuit																																																																			
	NPO;SIG_N2_MOD	VAR_GLOBAL			Output needle 2 pos. circuit model																																																																			
	NPO;SIG_N3	VAR_GLOBAL			Output needle 3 pos. circuit																																																																			
D	NPO;SIG_N3_MOD	VAR_GLOBAL			Output needle 3 pos. circuit model			D																																																																
	NPO;SIG_N4	VAR_GLOBAL			Output needle 4 pos. circuit																																																																			
	NPO;SIG_N4_MOD	VAR_GLOBAL			Output needle 4 pos. circuit model																																																																			
	NSL;1N_ACTIVE	VAR_GLOBAL			1 needle operation is on																																																																			
	NSL;2N_ACTIVE	VAR_GLOBAL			2 needle operation is on																																																																			
	NSL;3N_ACTIVE	VAR_GLOBAL			3 needle operation is on																																																																			
	NSL;4N_ACTIVE	VAR_GLOBAL			4 needle operation is on																																																																			
	NSL;N1_ON_INT	VAR_GLOBAL			Needle 1 in operation																																																																			
	NSL;N2_AVAIL	VAR_GLOBAL			Needle 2 available																																																																			
E	NSL;N2_ON_INT	VAR_GLOBAL			Needle 2 in operation			E																																																																
	NSL;N3_AVAIL	VAR_GLOBAL			Needle 3 available																																																																			
	NSL;N3_ON_INT	VAR_GLOBAL			Needle 3 in operation																																																																			
	NSL;N4_AVAIL	VAR_GLOBAL			Needle 4 available																																																																			
	NSL;N4_ON_INT	VAR_GLOBAL			Needle 4 in operation																																																																			
	NSL;SHARE_N1	VAR_GLOBAL			Switching-factor needle 1 [p.u.]																																																																			
	NSL;SHARE_N2	VAR_GLOBAL			Switching-factor needle 2 [p.u.]																																																																			
	NSL;SHARE_N3	VAR_GLOBAL			Switching-factor needle 3 [p.u.]																																																																			
	NSL;SHARE_N4	VAR_GLOBAL			Switching-factor needle 4 [p.u.]																																																																			
F	<table><tr><td>3</td><td></td><td>072716</td><td>Hung*</td><td>Date</td><td>22-09-2016</td><td colspan="2">BMC0</td><td rowspan="4">SIEMENS</td><td rowspan="4">Variable-declaration</td><td colspan="2">GLOBAL_VAR</td><td>Version</td><td>Rev</td></tr><tr><td>2</td><td></td><td>081516</td><td>1</td><td>Desig</td><td></td><td colspan="2"></td><td colspan="2">Coordinates</td><td colspan="2"></td></tr><tr><td>1</td><td></td><td>081516</td><td>1</td><td>Test</td><td></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td></tr><tr><td>State</td><td>Change</td><td>Date</td><td>Name</td><td>Norm</td><td>Origin</td><td>Repl for</td><td>Repl by</td><td colspan="2"></td><td colspan="2"></td></tr><tr><td>1</td><td></td><td></td><td></td><td></td><td>2</td><td></td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr></table>								3		072716	Hung*	Date	22-09-2016	BMC0		SIEMENS	Variable-declaration	GLOBAL_VAR		Version	Rev	2		081516	1	Desig				Coordinates				1		081516	1	Test								State	Change	Date	Name	Norm	Origin	Repl for	Repl by					1					2		3	4	5	6	7	8	F
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A	VARIABLE-DECLARATION Global-variable list "GLOBAL_VAR"								A					
	Name	Var-Type	Attrib	Init. Value	Long Name	Phys. Adr.	Usage							
	NSL;z0	VAR_GLOBAL			Max. number of needles									
	NSL;z0/SHARE	VAR_GLOBAL			Ratio num. needl./num. needl. in op.									
	NWI;1N_ON/OFF	VAR_GLOBAL			1 needle operation on/off									
	NWI;2N_ON/OFF	VAR_GLOBAL			2 needle operation on/off									
	NWI;3N_ON/OFF	VAR_GLOBAL			3 needle operation on/off									
B	NWI;4N_ON/OFF	VAR_GLOBAL			4 needle operation on/off									
	NWI;ADJ_SM_DE	VAR_GLOBAL			Deflector feedback-calibr. on									
	NWI;ADJ_SM_Nx	VAR_GLOBAL			Needles feedback-calibr. on									
	NWI;CLOSE_FBC	VAR_GLOBAL			Closing page FBC									
	NWI;CLOSE_SPF	VAR_GLOBAL			Closing page SPF									
	NWI;FBC_MAX_KEY	VAR_GLOBAL			Max. position calibration									
	NWI;FBC_MINUS_SV	VAR_GLOBAL			Close servomotor for calibration									
	NWI;FBC_MIN_KEY	VAR_GLOBAL			Min. position calibration									
	NWI;FBC_PLUS_SV	VAR_GLOBAL			Open servomotor for calibration									
C	NWI;FBC_VAL_H	VAR_GLOBAL		100.0	Input value opening position [%]									
	NWI;FBC_VAL_L	VAR_GLOBAL		0.0	Input value closing position [%]									
	NWI;FLC_ON	VAR_GLOBAL			Flow controller on									
	NWI;H_MAN_ON/OFF	VAR_GLOBAL			Manual setpoint head on/off									
	NWI;LEC_ON	VAR_GLOBAL			Level controller on									
	NWI;MAN_DE	VAR_GLOBAL			Manual operation deflector									
	NWI;MAN_N1	VAR_GLOBAL			Manual operation neddle 1									
	NWI;MAN_N2	VAR_GLOBAL			Manual operation neddle 2									
	NWI;MAN_N3	VAR_GLOBAL			Manual operation neddle 3									
D	NWI;MAN_N4	VAR_GLOBAL			Manual operation neddle 4									
	NWI;MAN_NEEDLE	VAR_GLOBAL			Manual needle selection on/off									
	NWI;N_DISCHARG	VAR_GLOBAL			Dischg distributionpipe									
	NWI;OCC_ON/OFF	VAR_GLOBAL			Off cam control on/off									
	NWI;OPL+	VAR_GLOBAL			Opening limit higher									
	NWI;OPL-	VAR_GLOBAL			Opening limit lower									
	NWI;POC_ON	VAR_GLOBAL			Power controller on									
	NWI;PRE_CPU_1	VAR_GLOBAL			Manual preset CPU 1									
	NWI;PRE_CPU_2	VAR_GLOBAL			Manual preset CPU 2									
E	NWI;PRI_1	VAR_GLOBAL			Ext. priority needle 1									
	NWI;PRI_2	VAR_GLOBAL			Ext. priority needle 2									
	NWI;PRI_3	VAR_GLOBAL			Ext. priority needle 3									
	NWI;PRI_4	VAR_GLOBAL			Ext. priority needle 4									
	NWI;PSP/FSP+	VAR_GLOBAL			Power/flow setpoint higher									
	NWI;PSP/FSP-	VAR_GLOBAL			Power/flow setpoint lower									
	NWI;RESET	VAR_GLOBAL			Reset several flipflops									
	NWI;RES_ISO	VAR_GLOBAL			Reset Isolated Operation									
	NWI;SSP+	VAR_GLOBAL			Speed spt. higher									
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A	VARIABLE-DECLARATION Global-variable list "GLOBAL_VAR"							
	Name	Var-Type	Attrib	Init. Value	Long Name	Phys. Adr.	Usage	
	RDI;DELTA_H_OK	VAR_GLOBAL			Comparison net head ok			
	RDI;DELTA_P_OK	VAR_GLOBAL			Comparison active power ok			
	RDI;DELTA_n_OK	VAR_GLOBAL			Comparison active speed ok			
	RDI;DE_INT	VAR_GLOBAL			Internal DE pos. for comparison [%]			
	RDI;DPO_CTRL_DEV	VAR_GLOBAL			Tracking control deviation DE [%]			
B	RDI;FLC_FSP	VAR_GLOBAL			Tracking FLC FSP slave [%]			
	RDI;HHW_INT	VAR_GLOBAL			Water level intake gate comp. [m]			
	RDI;Hdyn_INT	VAR_GLOBAL			Head of water for comparison [m]			
	RDI;LEC_LSP	VAR_GLOBAL			Tracking LEC LSP slave [m]			
	RDI;LIM_Y_LI_GOV	VAR_GLOBAL			Tracking WG limitation [%]			
	RDI;LOG_ISN_INT	VAR_GLOBAL			Status Active CPU "ISNET_INT"			
	RDI;NSL_N1_IS_ON	VAR_GLOBAL			Active CPU needle 1 in operation			
	RDI;NSL_N2_IS_ON	VAR_GLOBAL			Active CPU needle 2 in operation			
	RDI;NSL_N3_IS_ON	VAR_GLOBAL			Active CPU needle 3 in operation			
C	RDI;NSL_N4_IS_ON	VAR_GLOBAL			Active CPU needle 4 in operation			
	RDI;OPL_SPT	VAR_GLOBAL			Tracking opening limiter setpoint [%]			
	RDI;POC_PSP	VAR_GLOBAL			Tracking POC PSP slave [MW]			
	RDI;P_INT	VAR_GLOBAL			Active Power for comparison [MW].			
	RDI;SPC_N-SP	VAR_GLOBAL			Tracking SPC setpoint slave [%]			
	RDI;SPC_SSP	VAR_GLOBAL			Tracking SPC SSP slave [%]			
	RDI;n_ACTIVE	VAR_GLOBAL			Active speed for comparison [%]			
	RDO;CPU_1_ACTIVE	VAR_GLOBAL			CPU 1 is active			
	RDO;CPU_2_ACTIVE	VAR_GLOBAL			CPU 2 is active			
D	SEL;FSP_SELm3	VAR_GLOBAL			FSP regarding H_net [m³/s]			
	SEL;N1POS_FSP	VAR_GLOBAL			Pos. N 1 after POC/LEC/FLC sel. [%]			
	SEL;N1POS_SEL	VAR_GLOBAL			Pos. N 1 after SPC/FSP sel. [%]			
	SEL;N1POS_SPC	VAR_GLOBAL			Needle 1 SPC setpoint [%]			
	SEL;N2POS_FSP	VAR_GLOBAL			Pos. N 2 after POC/LEC/FLC sel. [%]			
	SEL;N2POS_SEL	VAR_GLOBAL			Pos. N 2 after SPC/FSP sel. [%]			
	SEL;N2POS_SPC	VAR_GLOBAL			Needle 2 SPC setpoint [%]			
	SEL;N3POS_FSP	VAR_GLOBAL			Pos. N 3 after POC/LEC/FLC sel. [%]			
	SEL;N3POS_SEL	VAR_GLOBAL			Pos. N 3 after SPC/FSP sel. [%]			
	SEL;N3POS_SPC	VAR_GLOBAL			Needle 3 SPC setpoint [%]			
E	SEL;N4POS_FSP	VAR_GLOBAL			Pos. N 4 after POC/LEC/FLC sel. [%]			
	SEL;N4POS_SEL	VAR_GLOBAL			Pos. N 4 after SPC/FSP sel. [%]			
	SEL;N4POS_SPC	VAR_GLOBAL			Needle 4 SPC setpoint [%]			
	SEL;N_SP_MAX	VAR_GLOBAL			Maximum needle setpoint [%]			
	SEL;PSP	VAR_GLOBAL			Power setpoint reg. H_net [%]			
	SEL;SEL_FSP	VAR_GLOBAL			FSP regarding Hnom [%]			
	SPC;K*dn/dt	VAR_GLOBAL			Derivated active speed amplified			
	SPC;KP_SPC	VAR_GLOBAL			Gain speed controller [-]			
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A	VARIABLE-DECLARATION Global-variable list "GLOBAL_VAR"							
	Name	Var-Type	Attrib	Init. Value	Long Name	Phys. Adr.	Usage	
	SPC;SPC_LIM	VAR_GLOBAL			Output SPC limited			
	SPC;SPC_N-SP	VAR_GLOBAL			Output PID speed controller [%]			
	SPC;SSP	VAR_GLOBAL			Speed setpoint [%]			
	SPC;SSP_PRE	VAR_GLOBAL			Speed setpoint preset			
	SPC;SSP_PRE_VAL	VAR_GLOBAL			Speed setpoint preset [%]			
B	SPC;SSP_isol	VAR_GLOBAL			SSP Variable Isolated Network [%]			
	SPC;TN_SPC	VAR_GLOBAL			Integral action time SPC [s]			
	SPC;TN_ST_ACTIVE	VAR_GLOBAL			TN start is active			
	SPC;Y_ISNET	VAR_GLOBAL			Isol. network preset value [%]			
	SPC;Y_isol	VAR_GLOBAL			Y variable isolated network [%]			
	SPC;bp*dWG	VAR_GLOBAL			Speed droop noload-correction			
	SVI;ESD	VAR_GLOBAL			Emergency shut down			
	SVI;ESD_RED	VAR_GLOBAL			ESD supervision for redundancy			
	SVI;FAIL_AUX	VAR_GLOBAL			Failure auxiliary signals			
C	SVI;FAIL_DE	VAR_GLOBAL			Failure periph. devices deflector			
	SVI;FAIL_FB_DE	VAR_GLOBAL			Failure feedbacksignal deflector			
	SVI;FAIL_FB_N1	VAR_GLOBAL			Failure feedbacksignal needle 1			
	SVI;FAIL_FB_N2	VAR_GLOBAL			Failure feedbacksignal needle 2			
	SVI;FAIL_FB_N3	VAR_GLOBAL			Failure feedbacksignal needle 3			
	SVI;FAIL_FB_N4	VAR_GLOBAL			Failure feedbacksignal needle 4			
	SVI;FAIL_FSP	VAR_GLOBAL			Failure ext. flow setpoint			
	SVI;FAIL_HHW1	VAR_GLOBAL			Failure head water level			
	SVI;FAIL_Hdyn	VAR_GLOBAL			Failure peripheral devices Hdyn			
D	SVI;FAIL_N1	VAR_GLOBAL			Failure periph. devices needle 1			
	SVI;FAIL_N1-N4	VAR_GLOBAL			Failure of needle 1 to 4			
	SVI;FAIL_N1-N6	VAR_GLOBAL			Failure of needle 1 to 6			
	SVI;FAIL_N2	VAR_GLOBAL			Failure periph. devices needle 2			
	SVI;FAIL_N3	VAR_GLOBAL			Failure periph. devices needle 3			
	SVI;FAIL_N4	VAR_GLOBAL			Failure periph. devices needle 4			
	SVI;FAIL_P	VAR_GLOBAL			Failure active power			
	SVI;FAIL_PE00_TM5	VAR_GLOBAL			Dropout module PE00 TM5			
	SVI;FAIL_POS	VAR_GLOBAL			Failure DE or Nx positioning circuits			
E	SVI;FAIL_POS_DE	VAR_GLOBAL			Fail deflector positioning circuit			
	SVI;FAIL_POS_N1	VAR_GLOBAL			Failure needle 1 positioning circuit			
	SVI;FAIL_POS_N2	VAR_GLOBAL			Failure needle 2 positioning circuit			
	SVI;FAIL_POS_N3	VAR_GLOBAL			Failure needle 3 positioning circuit			
	SVI;FAIL_POS_N4	VAR_GLOBAL			Failure needle 4 positioning circuit			
	SVI;FAIL_PR	VAR_GLOBAL			Fail reactive power			
	SVI;FAIL_PSP	VAR_GLOBAL			Failure ext. power setpoint			
	SVI;FAIL_SEC	VAR_GLOBAL			Failure secondary signals (alarm)			
	SVI;FAIL_SSP	VAR_GLOBAL			Failure ext. speed setpoint			
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	State	Change	Date	Name	Norm	Origin	Repl for	Repl by
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A	VARIABLE-DECLARATION Global-variable list "GLOBAL_VAR"								A		
	Name	Var-Type	Attrib	Init. Value	Long Name	Phys. Adr.	Usage				
B	SVI;FAIL_VER	VAR_GLOBAL			Version check failed						
	SVI;FAIL_n	VAR_GLOBAL			Failure speed signals						
	SVI;FAIL_n1	VAR_GLOBAL			Failure speed pick up 1						
	SVI;FAIL_n2	VAR_GLOBAL			Failure speed pick up 2						
	SVI;FAIL_n3	VAR_GLOBAL			Failure speed pick up 2 (Creep)						
	SVI;FAIL_n4	VAR_GLOBAL			Failure network frequency						
	SVI;F_ROT	VAR_GLOBAL			Failure rotation monitoring						
	SVI;F_n1_START	VAR_GLOBAL			Failure speed pick up 1 start						
	SVI;F_n2_START	VAR_GLOBAL			Failure speed pick up 2 start						
	SVI;NOT_ESD	VAR_GLOBAL			Not emergency shut down						
C	SVI;NOT_QSD	VAR_GLOBAL		1	No request for quick shut down						
	SVI;QSD	VAR_GLOBAL			Request for quick shut down						
	SVI;QSD_RED	VAR_GLOBAL			QSD supervision for redundancy						
	SVI;START_OK	VAR_GLOBAL			TC ready for start						
	SVI;SYS_START	VAR_GLOBAL			Reboot CP6003 is concluded						
	SVI;VER_CAEx	VAR_GLOBAL			Version Appl. CEAx plus						
	SVI;n1_BIT_0	VAR_GLOBAL			Speed pickup1 Bit0						
	SVI;n1_BIT_1	VAR_GLOBAL			Speed pickup1 Bit1						
	SVI;tc_DE	VAR_GLOBAL			Closing time deflector [s]						
	SVI;tc_Nx	VAR_GLOBAL			Closing time needles [s]						
D	SVI;to_DE	VAR_GLOBAL			Opening time deflector [s]						
	SVI;to_Nx	VAR_GLOBAL			Opening time needles [s]						
E									E		
F									F		
	3	072716	Hung*	Date	22-09-2016	Variable-declaration		GLOBAL_VAR	Version	Rev	
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